



► FOR SOCIETY

Teaching and Examination Regulations 2025-2026 of the full-time and part-time programme B HBO-ICT (ISAT 30020) of Fontys ICT.

Adopted by the director of Fontys ICT on 26-05-2025 after receiving the Degree Programme Committee's consent on 22-05-2025 and the Institute Participation Council's consent on 22-05-2025.

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General information

These Teaching and Examination Regulations (TER) apply to the following bachelor study programme:

name: B HBO-ICT

enrolment variant: full-time and part-time study programme

isat: 30020

institute: Fontys ICT

degree: Bachelor of Science

and applies to the 2025-2026 academic year. These TER were adopted by the director of Fontys ICT on 26-05-2025 after receiving the Degree Programme Committee's consent on 22-05-2025 and the Institute Participation Council's consent on 22-05-2025.

Changes and additions

The TER may be amended during the academic year, for example where this is required due to a change in the law or a change in the study programme. You will be explicitly informed of any change to the TER.

There have been no changes or additions.

What are Teaching and Examination Regulations?

If you are studying or wish to study at a Dutch university of applied sciences, you will have both rights and obligations. These rights and obligations apply for you and for Fontys. These TER describe all that which is associated with the study programme in which you have enrolled or wish to enrol. What does the study programme offer you, what are you being trained for, what requirements do you have to meet and what does the study programme ask of you? You can find the answer to many questions here. The [Students' Charter](#) contains rights and obligations as well – e.g. with regard to the house rules, the participation councils, tuition fees or complaints procedures.

These TER describe the nature of the study programme and the assessment process and contain information on where to turn to if you have questions or complaints. We have included a separate chapter to explain the most important terms, as well as a chapter on everything you will have to deal with when you apply.

Who is involved in drawing up the TER?

Part of the TER consists of arrangements that apply to the whole of Fontys, in which the Central Participation Council has a say.

In addition, the TER contain arrangements that are specific to a certain study programme, in which the Degree Programme Committee and the Institute Participation Council have a say. The TER are established for each academic year.

We cannot provide for every eventuality in advance

Sometimes, rules can lead to unintended situations. In these cases, the TER allow for an exception. More information about that can be found in [article 40](#).

Questions?

If you have any questions about the TER, please contact your student coach. The student coach can also refer you if

necessary.

1. What are the admission requirements for the 2026-2027 academic year?

We can admit you to the study programme if you have the right diploma, which may be a Dutch diploma or a foreign diploma. Some study programmes set requirements for your subject cluster in secondary school. We call these subject cluster requirements. Additional admission requirements may apply as well. Part-time study programmes may set requirements for your working environment.

This chapter outlines the admission requirements.

Article 1 If you have a Dutch diploma

MBO, HAVO or VWO-level diploma

For this study programme, you need a secondary vocational education (MBO) level 4, senior general secondary education (HAVO) or pre-university education (VWO) diploma. This study programme sets no requirements for your subject cluster in secondary school.

We can also admit you if you already completed an Associate degree- or Bachelor study programme, or if you can provide evidence of having completed the propaedeutic phase of a study programme at a university of applied sciences or a university.

The institute director has declared a HAVO or VWO diploma with an old subject cluster or no subject cluster to be at least equivalent to a diploma that complies with the new subject cluster requirements. You can be admitted with this diploma; no requirements apply with regard to your subject cluster or subject combination in secondary school.

Other diplomas

If you have a different diploma, the director of the institute will decide whether your diploma is equivalent.

Article 2 If you have a foreign diploma

If you have a foreign diploma that allows you to enrol in higher education in the country in which you obtained that diploma, we can admit you to the study programme.

Command of the English language

For international students, the following language requirement applies for admission to an English-language programme.

Language tests taken in a test centre	Overall Score	Testpart	One lower testpart score allowed
IELTS academic	≥ 6,0		
• IELTS reading		≥ 6,0	≥ 5,5
• IELTS listening		≥ 6,0	≥ 5,5
• IELTS speaking		≥ 6,0	≥ 5,5
• IELTS writing		≥ 6,0	≥ 5,5
Cambridge English	≥ 169		
• Cambridge reading		≥ 169	≥ 160
• Cambridge reading Use of English		≥ 169	≥ 160
• Cambridge listening		≥ 169	≥ 160
• Cambridge speaking		≥ 169	≥ 160
• Cambridge writing		≥ 169	≥ 160
TOEFL IBT (Traditional & Paper Edition)	≥ 80		
• TOEFL reading		≥ 18	-
• TOEFL listening		≥ 17	-
• TOEFL speaking		≥ 20	-
• TOEFL writing		≥ 17	-
TOEIC listening and reading	≥ 865		
• TOEIC listening		≥ 400	-
• TOEIC reading		≥ 385	-
TOEIC speaking and writing	≥ 335		
• TOEIC speaking		≥ 160	-
• TOEIC writing		≥ 150	-
LanguageCert Academic	≥ 65		
Pearson PTE Academic	≥ 61		

A previously completed language test, with the exception of Cambridge, may be no more than two years old at the time the study commences, which can be either on 1 September or on 1 February.

Exemption

Exemption may be granted for this requirement if the international student can submit a diploma from prior education that was obtained in a country in which English is the official language of instruction. If the diploma was granted more than two years ago, the student will be asked to demonstrate their language skills again, unless English is the only language of instruction in the country where the student obtained the diploma.

Furthermore, an international student may be exempted from having to take a mandatory language test in case the student obtained a diploma of secondary education in a country listed in the [diploma list](#) drawn up on behalf of the associations of institutions of higher education, as published on the [website](#) of the Code of Conduct.

Article 3 If you do not have the right diploma

If you do not have the right diploma and are 21 years old or over when you enrol for the study programme, you can participate in a Fontys admission assessment (the '21+ admission assessment'). This enables us to determine whether you have the right level to enrol in the study programme and achieve a diploma and this also involves a check of whether you have a sufficient command of the English language. If you pass the admission assessment, the result will remain valid for five years. If you do not pass the admission assessment, you can try again a year later. Click [here](#) for more practical information on the admission assessment (Section 7.29 of the WHW).

The admission assessment consists of two components:

1. an investigation into your professional and intellectual ability. We use this component to determine whether this matches the level of higher professional education;
2. a language test to assess your command of the English language. You must at least achieve level B2 (level 3F). If you have at least one of the certificates in the table in [article 2](#), you can apply for an exemption from the language test.

You will receive the results within two weeks after the assessment.

Article 4 Additional requirements for enrolment

There are no additional requirements for this study programme.

Article 5 Requirements concerning your work for part-time study programmes

This article does not apply to your study programme.

Article 6 Requirements concerning your work for dual study programmes

This article does not apply to your study programme.

Article 7 Maximum number of students

There is no maximum number of students that can be admitted.

Article 8 Students from outside the European Union

If you are from a country outside the European Union (EU) and are 18 years or older on the date you first enrol in the study programme, you will be required to have a valid residence permit (Section 7.32 of the WHW) and will need to obtain 50% of the credits to be obtained each year. If you fail to obtain enough credits, Fontys will have to report this to the Immigration and Naturalisation Service (IND), except when special circumstances prevented you from obtaining enough credits. Fontys may decide not to report an insufficient number of credits obtained to the IND once per special circumstance.

2. What happens after you apply?

After applying you can take part in a study choice check to find out whether the study programme suits you. In addition, you can request an interview to see if you can make use of previous degrees or other learning experiences.

Article 9 Study choice advice

To obtain a study choice advice, you can take part in a study choice check at Fontys. The study choice check has been detailed in the [Study Choice Check Rules](#). The study choice check consists of a digital questionnaire and a contact moment with the study programme. After submitting your application, you will receive a link to the digital questionnaire. Once you have completed it, you will receive an invitation for the contact moment.

The contact moment consists of:

If you already live in the Netherlands, you will receive an invitation for an afternoon of activities consisting of information and trial lessons after your registration. If you still feel the need, you can have an intake interview on request. If you do not take part in the activity afternoon, you will always get an individual intake interview. If you did not register until after 1 May, you will receive binding study advice based on a questionnaire and an interview.

For full-time:

- there are various information activities;
- you can attend workshops and taster classes at the teaching location;
- and you will receive a personal interview afterwards upon request.

For part-time, no additional activities are offered.

The contact moment and any additional activities take place from 1 March to 31 August for students starting in September and from 1 December to 31 January for students starting halfway through the academic year..

For international students as referred to in the Study Choice Check Rules, the study choice check is different.

International students will receive information about the study choice check within four weeks after their application.

If you are from abroad, you are invited to a Study Choice Check Webinar. This is broadcast live monthly and is also available via the website. The Webinar provides information about the programme and about studying and facilities in Eindhoven. You can also attend a counselling session online.

For international students, the contact moment takes place from 1 January to 31 July for students starting in September from 1 November to 31 December for students starting halfway through the academic year.

If you start your study programme in September, you can complete the digital questionnaire at least until 1 September. If you start your study programme in February, you can complete the digital questionnaire at least until 1 February, and the contact moment with the study programme and study choice activities for international students will be planned in this period as well.

After the contact moment with the study programme, you will be notified of the study choice advice by email within ten working days.

Article 10 Previous learning experiences and exemptions

If you have previous learning experiences or previously passed interim examinations and obtained diplomas, your application for a study programme may involve an interview with a curriculum owner or other subject matter expert. During this interview, you will gain insight into your future learning pathway.

You can use evidence of previous learning experiences or interim examinations you have passed and diplomas you have obtained in the past to demonstrate learning outcomes. Such evidence of interim examinations you have passed and diplomas you obtained in the past can also be used to substantiate a request for exemption for the [Examination Board](#). This is also possible during your studies.

In order to qualify for an exemption, you must demonstrate that, in terms of their content, scope (credits) and level, the interim examinations you have passed are at least equivalent to the interim examinations from which you wish to be exempted. You may also show that, in terms of their content and level, the knowledge and skills you acquired outside higher education are at least equal to the interim examinations from which you wish to be exempted. If the exemption is granted, you will receive a decision from the Examination Board stating the term of validity of the exemption.

In some cases, you can receive an exemption for the entire propaedeutic phase (Section 7.30 of the WHW). This is referred to as an exemption from the propaedeutic examination. An exemption is possible if you have an equivalent Dutch or foreign diploma or have – for example – successfully completed the propaedeutic phase of the same study programme. You can submit your application for an exemption from the propaedeutic examination to the examination board.

3. What are you being trained for?

Article 11 What does this programme train you for?

The study programme trains you for the professions listed in the (national) programme profile of this study programme. The programme profile describes what you should know and be able to do at the end of the study programme.

You can find the study programme profile in the appendix Exit qualifications.

Once you have completed the study programme, you will receive the degree associated with your study programme. For your study programme, that is the degree Bachelor of Science.

Article 12 Specialisations

The study programme does not offer specialisations.

Article 13 Advancement after your studies

Once you complete your study programme, you can choose to continue your studies. For some study programmes, separate arrangements apply to this, to ensure proper or accelerated advancement.

For your study programme, we have made the following arrangements with the following universities of applied sciences or universities:

1. To advance to a master's program at Eindhoven University of Technology (TU/e) or the Jheronimus Academy of Data Science (JADS), you can choose Academic Preparation in your first free space (the 4th semester). Passing that semester gives access to a minor of courses from the TU/e pre-master's program or a minor consisting of the JADS pre-master's program. Completion of that program and obtaining the Bachelor's degree gives you direct access to a number of TU/e or JADS master's programs.
2. To advance to a Master's programme at Tilburg University, you can make use of a free minor consisting of subjects from Tilburg University's pre-master programme.
3. To transfer to the Master of Applied IT at Fontys ICT, you must meet the admission requirements and selection criteria of that master's programme. For these, see Section 2 of the Fontys ICT OER 2025-2026 Master of Applied IT.

4. What does your study programme look like?

Your study programme has a proposed learning pathway for you to follow. You can also make your own choices, for example on how to use the free choice space that is offered as part of your study programme. Over the course of your study programme, you will be guided by - among others - a student coach. In addition, there are also various facilities you can use if you require extra support.

Article 14 How your study programme is set up?

Your study programme consists of units of study with credits. A unit of study may be a unit of learning outcomes with a facilitated learning pathway. The Interim Examination and Test Schedule annex provides an overview of how the learning pathway is structured and whether it uses units of learning outcomes.

The study load of your study programme is 240 credits, and it consists of:

- a propaedeutic phase – the first year of study with 60 credits;
- a post-propaedeutic phase; the final three years with 180 credits.

The study load of the full-time and part-time study programme is 60 credits per year.

Article 15 The courses and educational activities you can participate in

Fontys offers various courses and educational activities in a proposed learning pathway, which will help you learn and demonstrate learning outcomes or learning objectives. Examples include: workshops, master classes, working groups, practicals and internships, challenges, coaching and tests.

The courses and educational activities offered can be found [in the programme guide](#). The overview indicates which courses and educational activities facilitate the unit of study and the language in which the education is provided, and whether you need to register for the relevant course, educational activity or test.

The timetable of courses and educational activities is announced [Fontys ICT portal](#), at least three calendar weeks before the start of the teaching period. If participation in courses and educational activities is compulsory, this will be indicated in the digital learning environment.

Article 16 Free choice space

In the post-propaedeutic phase, your study programme offers a Free choice space with a study load of 30 credits. You are free how to fill this part of your study programme. You can also choose from minors and other courses within or outside Fontys. What you learn in your Free choice space may not overlap with what you learn in the rest of your study programme. You may use this part of your study programme to deepen or broaden your studies, or do something completely different at – at least – a higher professional education level.

The full-time study programme at Fontys ICT allows you to do a minor, provided there is not too much overlap with your study programme. Unfortunately, there is no free choice in the part time study programme.

In your study programme, you may not take the below Fontys minor(s) due to an overlap with the rest of your study programme:

You cannot choose both Digital Experience Design and Immersive Experiences in one study programme.

The study programme also has minors that specifically match your programme. The minor regulations of these study programme minors can be found in the annexes. It concerns the following minor(s):

- Global Acting in IT (minor GA)
- Signals & Embedded Systems (minor SES)
- Smart Industry Tehnologies (minor SIT)

If you take an extra minor on top of your normal study load, this is only allowed if you take an entire minor. It is not allowed for separate parts of minors.

If you wish to decide on how to use the Free choice space yourself or wish to use it for a foreign minor or an external minor, you have to ask the Examination Board's permission in advance. If you want to take an additional minor, this will be on top of your normal study load. This is only allowed if you have not completed the study programme yet. Minors can be found via [Kies op Maat](#), Fontys' [minors page](#) and [the Study Guide Fontys ICT](#). For some minors, there are requirements you need to meet in order to take them.

If you have participated in and successfully completed the Fontys Empower programme, you can list this as the Free choice space.

Article 17 Entry requirements

Before being allowed to start on courses and educational activities of the post-propaedeutic phase, you must meet the following standard:

You must have successfully completed the propaedeutic phase of your study programme.

The Examination Board may make an exception. It may decide that you may start the post-propaedeutic phase despite not meeting the standard yet. This is only possible in special circumstances. You can submit your application for the exception to the [Examination Board](#) (Section 7.30(3) of the WHW).

The entry requirements are set out in an annex to [article 14](#).

Article 18 The guidance you will receive

As you learn, you will receive feedback from various teachers, experts and fellow students on everything you have worked on professional products you have created and skills you have demonstrated. You record feedback in a portfolio and you use the feedback along with other evidence to demonstrate learning outcomes or learning goals.

Your student coach

You will have a student coach who will guide you in making choices and shaping your personal learning pathway and desired study pace. At the start of the study programme, you will explore your opportunities and ambitions with your student coach. If you have done something outside of Fontys from which you learned, or can provide evidence of interim examinations you have passed and diplomas you have obtained in the past, you can discuss with your student coach whether you can use that as evidence to demonstrate learning outcomes. Evidence of previously passed interim examinations or obtained diplomas that you include in a portfolio must always meet the requirements for evidence, for example in terms of how current these are.

In addition, the student coach will support you in your personal and professional development. If you are not

comfortable with your student coach, you can submit a request to the education manager to be assigned another coach.

Article 19 Study advice

In the first year of your study programme, you can familiarise yourself with the study programme and the profession you are being trained for. Halfway through your first year of study, you will receive a message about your study progress and discuss your study progress with your student coach. At the end of your first year of study, you will receive a non-binding study advice. If this study advice is positive, you can continue your studies. If you do not receive a positive advice, you will consult with your student coach about what to do next. If the study programme does not suit you, Fontys will help you make another choice.

Your study advice will be positive in the following circumstances:

- you passed the start semester or
- you got an exemption for the start semester and you passed semester 2.

The start semester of the full-time AD-ICT is equal to that of the full-time Bachelor ICT.

The start semester of the part-time AD-ICT is equal to the that of the part-time Bachelor ICT.

The study advice you receive is in all cases valid for both the AD-ICT and the Bachelor ICT programmes and for both full-time and part-time programmes.

The study advice also applies to these study programmes.

Your study advice will be negative if you do not meet the conditions for a positive study advice.

Article 20 The facilities that are available to you

If you have a request for support

If you require support, for example due to neurodiversity, a temporary or permanent physical or mental disability or a chronic illness, you can apply for additional facilities, such as adjustments to the curriculum, timetables, teaching methods, testing or learning resources. To do so, contact your student coach or a [student counsellor](#) as soon as possible for assistance. They can help you request the necessary facilities from the [Examination Board](#).

Whether the study programme can provide these facilities, and whether you can use these for the duration of your study programme, is at the discretion of the Examination Board. You will review whether the facilities are still appropriate on an annual basis, in consultation with your student coach.

According to the law, you are entitled to these kinds of facilities, except if they impose too excessive a burden on the study programme (Section 7.13 of the WHW and Section 2a of the Equal Treatment of Disabled and Chronically Ill People Act (Wet gelijke behandeling op grond van handicap of chronische ziekte)).

More information can be found at [Fontys Helps](#).

If you want to improve your command of the Dutch language

If you want to improve your Dutch-language reading and writing skills, you can get help with that. You can find more information at [Fontys Taalportaal](#).

Your study programme also offers additional support. You can find more information at [in this Canvas course](#).

If you are an elite athlete

If you have Elite Athlete or Talent status, you can make use of the facilities provided under the [Athlete sport policy and Elite Athlete Scheme](#). These include – for example – adjustment of tests or test schedules, adapted rules for the attendance requirement or a modified internship. You can submit the relevant application to the Elite sport contact person associated with your study programme. More information can be found [here](#).

If you are an entrepreneur

If you have entrepreneur status, you can make use of the facilities provided under the [Student Entrepreneur Scheme](#). These include – for example – adjustment of tests or test schedules, adapted rules for the attendance requirement or a modified internship. You make arrangements about this with your student coach, which you can subsequently get approved by the [Examination Board](#). More information can be found [here](#).

If you hold an administrative position within Fontys or in a student association

You can use administrative activities in a portfolio. You can coordinate with your student coach on how these administrative activities can contribute to the acquisition of learning outcomes of your program.

Financial support in connection with administrative activities is provided for in the 'Regulations on the Participation Councils and Degree Programme Committees', the 'Regulations on Board Membership Grants' and the 'Remuneration Scheme for committees and steering groups'. All these sets of regulations can be found [here](#). You can have your administrative activities listed in the [diplomasupplement](#) as well.

Additional assistance in special circumstances

If you need extra support due to special circumstances that hinder your studies you can contact [Student+](#)

Article 21 Evaluation of the teaching

Halfway through the semester, you will receive an e-mail invitation to complete a survey about your semester and your teachers. A time will also be scheduled with your class to discuss the results of the survey with. If necessary, an additional evaluation can be requested with reasons given. For internships and graduation, the evaluation is done at the end of the semester. Results of all evaluations are published on [the student square](#). Improvements as a result of the results are described on Canvas for the respective unit of study. Teacher evaluations are included in discussions between teachers and their supervisors. For questions or complaints, please contact: fontysict-kwaliteitszorg@fontys.nl

5. How will you be tested?

The following describes how your study programme has set up its assessment and what rules and guidelines apply.

Article 22 How to complete a unit of study

Each unit of study is concluded with an examination. In the Appendix Interim examination and test schedule you will find an overview of all examinations for all units of study in your program. Here you will also find what the exam looks like. For the exam you always submit a portfolio. Sometimes the exam also includes an interview, presentation, performance, and so on. The contents of the exams and assessment criteria can be found in the digital learning environment of your course. There you will also find the criteria that the evidence must meet. Examiners decide and substantiate whether you have passed or failed the exam. The decision is always made by at least two examiners appointed by the examination board. You will also receive feedforward, on how to continue your learning. You will receive the credits for the entire unit of study all at once.

Article 23 How to participate in an interim examination or test?

You will have at least two opportunities per academic year to participate in an interim examination or test or to submit your portfolio for the interim examination. Results of an interim examination or test will be received within 15 working days, via the programme administration of your study programme. For some interim examinations and tests, such as an internship, the resit or adjustment may be scheduled for the following academic year. You may resit an interim examination or test for which you obtained a satisfactory result once. In such case, the highest result will apply.

The annex 'Interim examination and test schedule' indicates whether you have to register for interim examinations and tests and, if so, which interim examinations and tests this applies to. How and when you can register for an interim examination or test will be announced at the beginning of the academic year. At least three calendar weeks before the interim examination or test, the digital learning environment will contain information on when you can sit the interim examination or test or hand in your portfolio.

Oral interim examinations are held in public. Those interested in attending an oral interim examination must submit a request to that end to the examiner(s) at least two weeks in advance, and the examiner will inform you of these requests. If you object to it, an interested party's request will be rejected.

Article 24 How to provide proof of your identity when sitting an interim examination or test?

If you have to be present for an interim examination or test, you must always be able to provide proof of your identity. You can use a valid legal identification document: a passport, driving licence, Dutch identity card, ID card or passport from an EEA country or Dutch foreign national's document.

If you have no identification document due to loss or theft, you may take part in an interim examination or test if you can show official proof of application for a new proof of identity and your Fontys student card.

Article 25 What to do if you cannot participate in an interim examination or test?

If you cannot participate in an interim examination or test, you can participate in it at the next test opportunity. If the timing of the next test opportunity causes an unreasonable delay in your studies, you can submit a substantiated request to have the test opportunity brought forward to the [Examination Board](#).

Article 26 What to do if you fail an interim examination or test?

If you fail an examination or test, you can participate in the next test opportunity. In the annex 'Interim examination and test schedule' you can see if you have registered for this (see also article 23). If the timing of the next test opportunity causes an unreasonable delay, you can submit a substantiated request to have the test opportunity brought forward to the [Examination Board](#).

Inspection

Your study programme will also allow you to inspect your completed interim examinations and tests. Additional information on inspection can be found [on our policy wiki](#).

Article 27 What to do if you disagree with your assessment?

If you disagree with an assessment, you can discuss it with the examiner, the assessor or your student coach.

If you do not agree with the assessment of an interim examination or test, you can submit a request for review of your assessment to the [Examination Board](#). You must do so within 20 working days of receiving the assessment. The Examination Board will respond within 20 working days of submission of your request.

If you disagree with the way an interim examination or test has been assessed, you can also lodge an appeal with the Examination Appeals Board. You must do so within six calendar weeks of receiving the assessment.

More information can be found in the [Students' Charter](#) and via the [Student Desk](#).

Article 28 How Fontys deals with irregularities and fraud

If the Examination Board cannot guarantee the quality or result of a test, it may decide that the test will not be assessed or that the test result will be declared invalid.

If the irregularity falls under the responsibility of the study programme, the Examination Board will ensure that the study programme offers a new test opportunity in the short term.

If an examiner doubts the authenticity of work you submit for assessment, the examiner may conduct an additional interview with you to check and establish that authenticity.

Not following the rules that apply for an interim examination or test may be considered fraud. If you have committed fraud, the Examination Board can deny you the right to participate in one or more interim examinations and tests for a maximum of one year. If the interim examination or test has already been assessed, the result will be declared invalid.

In case of serious fraud, the Examination Board may propose that the Executive Board permanently terminates your enrolment (Section 7.12b of the WHW).

If the irregularity or fraud is only discovered after the completion of your study programme, the Examination Board may determine that you will not be awarded a certificate. If you have already received the certificate, it may be declared invalid and you may be asked to hand it in. In such case, the Examination Board may determine that you must pass new interim examinations or new tests before receiving your certificate after all.

Before making a decision, the Examination Board will hear you and any interested parties. A report will be drawn up, of which you will receive a copy. You are not obligated to answer questions in a meeting with the Examination Board. However, this may impact their decision. The Examination Board will communicate its decision to you, in writing, as soon as possible. The options for appeal will be indicated in the relevant communication.

More information can be found in the [Fraud policy](#) of your study programme's Examination Board.

Article 29 Changes in learning outcomes or learning objectives during your programme of study

The content of interim examinations or tests may change or interim examinations or tests may no longer be offered. In such case, a transitional scheme will apply. This means that, in addition to the normal offer, the old form of any interim examination or test that has changed will be offered at least two more times. The annex 'Transitional Scheme for Article 29' indicates which interim examination or test you can take to replace the 'old' component.

If you are working on a unit of learning outcomes, you will always have at least 12 months to pass the exam. If the learning outcomes change after those 12 months, the new learning outcomes will apply to you.

1. It is recommended that when restarting a Main semester you switch to a different Main semester, as this will allow you to fully participate in class. Consult with your Semester Coach to determine which Main semester is going suit you best;
2. If you want to retake Media 3 with the old set of learning outcomes, you can do so. You will then be assigned to the Main semester "Media Creation" but will be assessed and supervised by the teachers on the set of learning outcomes as defined in Media 3;
3. If you have passed Media 3, you need to get permission from the Examination Board to choose the Main semester "Front End Development" or "Media Creation".

Article 30 Term of validity of passed interim examinations and tests

Passed interim examinations can expire if the knowledge and skills to which these interim examinations relate are demonstrably outdated. Fontys considers knowledge and skills assessed more than ten years ago to be demonstrably outdated. Therefore, the term of validity of successfully completed interim examinations is ten years. The Examination Board may extend this term at your request.

If you have received assistance based on the [Student Support fund scheme](#), the term of validity of interim examinations will be extended by at least the duration of the assistance provided.

6. How do you complete your studies?

Below is a description of the rules that apply when you complete your study programme or if you want to quit early.

Article 31 When will you have completed the course?

You will have passed the examination of the propaedeutic phase or the study programme if you have passed all the units of study (Section 7.10 of the WHW). The Executive Board will notify the Education Executive Agency (DUO) when you have passed the final examination of your study programme.

Article 32 When have you graduated cum laude?

Cum laude is an additional annotation that will be added to your certificate if you complete your degree with an exceptionally good result. This annotation is subject to the following conditions:

Based on the study progress information in Progress, the Examination Board determines if and which distinction will be mentioned on the certificate. The Examination Board is authorized in individual cases to withhold a distinction when the Board has established a form of fraud by the student.

The distinction 'cum laude' is the highest degree possible. The requirements for the distinction 'cum laude' are:

- the student has passed all components of semesters 6, 7 and 8 uninterrupted (without restart) **AND**
- one of your advanced topics in semester 6 or 7 has been marked Outstanding **AND**
- the graduation project has been marked Outstanding.

Article 33 When will you receive your certificate?

You will receive your certificate after the Examination Board has determined that you have passed your propaedeutic examination or final examination. Fontys will organise a graduation ceremony for the handing out of the certificates. Alternatively, you may collect your certificate from the programme administration of your study programme at an earlier or later date. You can do so until a maximum of one year after completion of your studies. You will receive your certificate after it has been established that you have been enrolled and have paid the tuition fees for all the years of your enrolment (Section 7.11 of the WHW).

The certificate states the date on which you completed the study programme. This is the date on which you passed your last interim examination. If you want the date on your certificate to be a later date, you will have to postpone completing your last test activity.

Article 34 Contents of your diploma supplement?

Along with your certificate, you will receive a diploma supplement. This contains important information on your study programme, such as the degree you have obtained, the number of credits and the results you have achieved

The diploma supplement may also indicate whether you have performed administrative activities or been a student member of the Examination Appeals Board. You can apply for such an annotation yourself. The application form and

protocol can be found under [Useful documents & formats \(fontys.nl\)](#).

Article 35 What happens if you drop out?

If you drop out of your study programme, you can ask the Examination Board of your study programme to provide a statement of the interim examinations you have passed (Section 7.11 of the WHW). The statement may state how long the interim examination results are valid, subject to significant changes in the study programme.

7. Who to turn to with questions or complaints?

Article 36 Where to ask questions?

The first point of contact for questions about your study programme is your student coach. Your student coach can answer many questions and – where necessary – can refer you to the student counsellor, the education manager or the [Examination Board](#). For questions about the content of programme components, please contact the relevant lecturer. For information on practical matters such as how to register for tests, timetables and tuition fees, please contact the programme administration of your study programme.

Any further information, for example on facilities, switching study programmes and personal circumstances, can be found at [Fontys Helps](#).

Article 37 Where to submit complaints?

If you feel that a Fontys staff member or a student has done something that harms you, or you disagree with organisational matters within your study programme, the first point of contact is your student coach, the education manager or a [confidential advisor](#). If you believe certain aspects of your study programme are not properly organised, you can report this to the Degree Programme Committee. You also have the right to lodge a complaint about this with the Executive Board. You can submit complaints via the [Student Desk](#). If the complaint has not yet been investigated by your study programme or the institute at which you study, the Student Desk may first forward the complaint to the director of your institute.

The institute your study programme falls under has a complaints procedure in place as well. This can be found [here](#).

Article 38 Duties of the Examination Board

The Examination Board is independent and monitors the quality of tests and interim examinations. It also determines whether all units of study have been passed at the end of the study programme and awards the certificate.

In addition, the Examination Board decides on individual requests from students concerning – for example – exemptions and facilities, such as extra time for tests. Furthermore, the Examination Board looks into possible [irregularities and fraud](#). The annexes include a list of who is on the Examination Board.

If you want to submit a request to the Examination Board, you can do so in the following way:
via the [Examboard portal](#)

Article 39 How to get into contact?

You can reach the study programme's administration as follows:

- Programme administration of your study programme: Up to and including internship: fontysict-r10@fontys.nl.
After internship: fontysict-tq@fontys.nl

- Examination Board: <https://examboard.fhict.nl/>
- Intake coordinator: fontysict-r10@fontys.nl
- Elite sport contact person: d.deman@fontys.nl
- Centre for Entrepreneurship: [Centrum voor Ondernemerschap | Fontys](#)
- Student Desk: <https://www.fontys.nl/en/Appeals-objections-and-complaints.htm>
- Student counsellor and student psychologists: <https://www.fontys.nl/en/Fontys-Helps/Contact.htm>
- Central Participation Council: [website](#) / [email](#)
- Institute Participation Council: fontysict-ipc@fontys.nl
- Degree Programme Committee: fontysict-jdpc@fontys.nl

Article 40 Hardship clause and unforeseen circumstances

If you feel that you are unreasonably disadvantaged in the application of these TER, the Examination Board may take a decision on this at your request. When it concerns admission to the study programme, the institute director makes the decision. To invoke this hardship clause, you will need to send a written request to the Examination Board, which must include substantiation of why you feel you are unreasonably disadvantaged.

In deciding whether this hardship clause must be applied, the Examination Board or the institute director must weigh your interests and those of the study programme. The Examination Board or the institute director will decide on your request and communicate the decision to you in writing, with substantiation, and will include your options for appeal in the communication. See also Article 44 of the [Students' Charter](#).

Where the TER do not provide for a specific situation, the Examination Board will decide, unless it concerns subjects that fall under the authority of the institute director.

If an immediate decision is required, the chair or deputy chair of the Examination Board or may decide on behalf of the Examination Board.

8. What do we mean by...?

Article 41 Terminology

Fontys terminology

Academic year	The period that begins on 1 September and ends on 31 August the following year.
Assessment	The process of making your learning visible in relation to learning outcomes.
Assessment portfolio	The evidence you selected that you believe demonstrates the learning outcomes of a unit of study.
Assessor	Examiner who decides and substantiates whether you have demonstrated the learning outcomes of a unit of study.
Associate Degree study programme	An Associate Degree study programme is a study programme in higher professional education, with a study load of 120 credits. According to the European Qualifications Framework (EQF), it is a level 5 study programme.
Bachelor's programme	A Bachelor's study programme is a study programme in higher education, with a study load of 240 credits. According to the European Qualifications Framework (EQF), it is a level 6 study programme.
Centre for Participation	The team within Fontys that helps participation councils, departmental committees and their discussion partners (Executive Board or directors) to do their work well.
Certificate	Written statement showing that you have certain knowledge and skills.
Competences	Competences are a combination of knowledge, skills, behaviour and attitude. These are required to function in the profession you are being trained for.
Courses and educational activities	Activities organised as part of the study programme that can help you learn.
CPC	Central Participation Council
Credit	A credit is the unit used to express the study load of a unit of study. Each credit is equal to 28 standardised study load hours.
Criterion-based interview	An interview between you and two examiners with the aim of assessing whether you have demonstrated the learning outcomes or learning objectives.
Curriculum	A coherent set of courses and educational activities in which students can participate in the context of the study programme.
Deficiency	The knowledge that is missing if you did not take the right subject cluster.
Degree Programme Committee	An elected committee of teachers and students that monitors the implementation of the TER and the quality of the study programme.
Diploma supplement	Official annex to your diploma, which is designed according to a European format. This lists the level, form, content and status of your study programme.
Diploma with subject combination	Old HAVO or VWO diploma (from before 1998), with a subject combination rather than a subject cluster.
Dual study programme	A dual study programme is one in which you combine studying and professional practice. Such professional practice is part of the curriculum of the study programme.
DUO	Education Executive Agency (DUO)
Elite Athletes Scheme	Scheme describing how Fontys students with elite athlete status are supported in combining their sport and their studies.
Evidence	The material, appropriate to the learning outcomes, through which you demonstrate your own development and your level in a portfolio.
Examination	Investigation by the Examination Board to determine whether you have passed all the units of study of the study programme (or the propaedeutic phase thereof).

Examination Appeals Board	The Board that will hear your appeal. The Board consists of a chair (legal expert), a lecturer member and a student member.
Examination Board	The statutory independent board that monitors the quality of study programme assessment.
Examiner	A lecturer or an expert from outside Fontys appointed by the Examination Board to assess your knowledge and skills.
Executive Board	The central governing body of Fontys University of Applied Sciences
Exemption	There are two forms of exemption: 1) for all or some of the admission requirements or enrolment requirements for the study programme; 2) for sitting interim examinations or subtests.
Feedback	Feedback to improve your learning. This feedback covers the quality of your work, your understanding of the subject, your commitment and study skills, and other aspects of your development.
Feedback for learning	Information about process and products during a student's learning journey. Development is requested by students to gain insight into where they are at regarding the demonstration of the learning outcomes, and what next steps they can undertake to achieve mastery.
Fraud	Everything you do, or in some cases fail to do, that prevents the examiner from forming a reliable opinion of your knowledge and ability.
Full-time study programme	A full-time study programme is a study programme whose structure is such that students are assumed not to participate in any activities other than courses and educational activities.
Hardship clause	Provision in a law or regulation that allows you to derogate from what is in that law or regulation if you are unreasonably disadvantaged.
IELTS	The International English Language Testing System (IELTS) is a tool to test your proficiency in the English language.
Institute	An institute offers study programmes. Each institute is headed by a director.
Institute director	A director in charge of a Fontys institute.
Institution	Fontys University of Applied Sciences
Interim examination	An examination to establish your knowledge, ability and professional attitude as the conclusion of a unit of study. An interim examination may consist of a portfolio assessment with an oral component if necessary.
	Mandatory evidence may be part of your assessment portfolio. If you pass your interim examination, credits will be awarded.
Interim examination and test schedule	Annex to the TER showing which interim examinations and tests are offered by the study programme.
IPC	Institute Participation Council
Isat code	Each study programme is listed with a unique code in the RIO register. This code is used for each study programme by each university recognised by the Ministry of Education, Culture and Science.
Learning objective	A learning objective clearly and concretely describes what you will learn in terms of knowledge, skills and professional attitude.
Learning outcome	A learning outcome describes the knowledge, skill and professional attitude you demonstrate at the end of a unit of study.
Mandatory evidence	Evidence that you are required to include in the portfolio.
Master's study programme	A Master's study programme is a study programme in higher education, with a study load of 60–240 credits. According to the European Qualifications Framework (EQF), it is a level 7 study programme.
Minor	Elective curriculum of 15 or 30 credits that allows you to deepen or broaden your knowledge.
Minor regulations	Regulations that describe the content, courses, educational activities, assessment and completion of a minor.
National programme profile	The profile that describes what the graduate should know and be able to do in broad terms.
Nt2 diploma	Diploma of the official Nt2 State Examination in Dutch as a second language, of which programme II is considered to be the guideline for admittance to higher education.

Occupational requirements	If you are being trained for a profession that is subject to statutory requirements (occupational requirements), the training will enable you to meet these.
Part-time study programme	A study programme that takes into account the possibility of you having other commitments in addition to your studies.
Portfolio	A portfolio is a collection of evidence/material created by you that reflects your progress and development.
Portfolio assessment	An integrated assessment moment that determines whether you have demonstrated the learning outcomes based on your assessment portfolio and possibly an oral component.
Post-propaedeutic phase	Second phase of the Bachelor's study programme.
Professional product	Products you will soon need to be able to deliver as a professional in the exercise of your profession.
Profiling Fund Scheme	Scheme that provides student support in the form of granting graduation support from the Profiling Fund Committee.
Propaedeutic year	First phase of a Bachelor's study programme.
Proposed learning pathway	A pathway proposed by the study programme with courses and educational activities to demonstrate the learning outcomes.
Registration	Registering for courses, educational activities and tests.
RIO	The RIO national register for institutes and programmes [Registratie Instellingen en Opleidingen – RIO] is a national register listing all study programmes leading to an official higher education certificate.
Specialisation	A specific specialisation in the curriculum of a study programme.
Student Charter	The regulations stating students' rights and obligations.
Student coach	Coach to guide you in making choices and shaping your personal learning pathway. In addition, the student coach will support you in your personal and professional development.
Student counsellor	A student counsellor is there to help students apply for facilities (aids you need to be able to study, such as extra time on a test) and help you find solutions to obstacles in your studies.
Student Entrepreneur Scheme	Scheme describing how Fontys students who have been recognised as student entrepreneurs are supported in combining entrepreneurship and their studies.
Student support fund Committee	Committee implementing the Student support fund Scheme. The Student support fund provides financial support for students who are faced with special circumstances.
Study advice	A recommendation given by your study programme at the end of your first year. The advice is about your chances and opportunities to successfully complete this study programme.
Study choice advice	Advice before you start the study programme, on whether it suits you.
Study choice check	The activity offered by Fontys before the start of a study programme to find out whether the study programme suits you.
Study load	The number of hours required by the average student to pass the study programme or an interim examination. The study load is expressed in credits, and 1 credit represents 28 study load hours.
Study plan	Plan in which you set out the arrangements for your personal learning pathway.
Study programme	A coherent set of units of study that allows you to meet the study programme profile. Each study programme is registered in the RIO and leads to a certificate.
Study programme form	Indicates whether the study programme is offered on a full-time, part-time or dual basis.
Study programme minor	A minor that can only be taken by students from a specific domain or study programme.
Subject cluster requirements	These are the requirements some study programmes set for the subject cluster you chose for your HAVO or VWO diploma.
Subtest	If an interim examination consists of several tests, we refer to these as subtests.
Teaching period	Fontys has divided the academic year into four 10-week teaching periods. You can find the period schedule in the Fontys annual agenda.
TER	Teaching and Examination Regulations

Test	A form of evidence, showing what you know or are able to do.
unit of learning outcomes	Educational unit with a coherent set of knowledge and skills, in which you have the freedom to choose how you work on it and to provide your own evidence to demonstrate this knowledge and skills.
Unit of study	Part of a study programme that concludes with an interim examination. Students are awarded the relevant credits upon passing the interim examination for the unit of study.
WEB	Dutch Adult and Vocational Education Act [Wet Educatie en Beroepsonderwijs].
WHW	Dutch Higher Education and Research Act [Wet op het hoger onderwijs en wetenschappelijk onderzoek].
Working day	Monday to Friday, except holidays and holiday days, which are included in the Fontys annual calendar.

Canvas	Electronic learning environment where among other things you will find learning materials, submit products, and communicate with teachers. Also, your formative feedback is saved there.
Formative indication	Development-oriented feedback and feedforward moment.
Open Learning	An open learning pathway, in which you determine the learning outcomes and the road to those outcomes by yourself. Students choose their own learning outcomes, study materials (Open Educational Resources) and educational activities in dialogue with a coach. The criteria within which the learning outcomes can be determined are specified in the Programme guide Fontys ICT. From semester 2 onwards, students can opt for the "Personalised Programme" unit of study, within which the teaching format is open learning. The Personalised Programme can be taken for one or more semesters.
Programma Guide Fontys ICT	The programme guide contains all the information of all educational activities. The programme guide shall be digitally published 1 week before the start of the semester through the curriculumdatabase .
Progress Dashboard	Progress is the Fontys-wide study progress programme that records the results achieved that ultimately lead to the award of credits.
Semester coach	A teacher in the role of student coach for a semester.
Teaching method	An implementation form of education. Education uses different forms of learning to meet the different needs of students and teachers (especially in the degree of predictability and flexibility).

Appendix 1 Exit qualifications

Exit qualifications

The exit qualifications of the master are based on the HBO-i competence framework (see www.hbo-i.nl). This model splits the field of ICT in 5 architectural layers: user interaction, organisational processes, infrastructure, software and hardware interfacing. Also, it defines 5 engineering activities: analysis, advise, design, realisation and manage & control. Since every activity can be done in any architectural layer, the model defines 25 cells. For each cell there are 4 levels of proficiency. Level 1 is an introductory level. Level 2 corresponds to Associate Degree level, level 3 corresponds to Bachelor level and level 4 defines the Master level. For each cell at each level a number of example performance indicators are defined, all of which need to be shown by students in order to prove their proficiency at that level.

The HBO-i model covers all Dublin descriptors. The model is also discussed with and endorsed by the work field.

In addition, the student works on professional development. The student demonstrates professional development within the educational units according to the criteria described in the HBO-I framework. This model is split in different levels as described on [the Fontys ICT policy wiki](#).

The second level is the desired final Associate degree level. An Associate degree graduate at Fontys ICT has a T-shaped profile. This means that for all competences (Advise - Analyse - Design - Realise and Manage & Control) level 2 has been demonstrated in at least one architectural layer and at least level 1 has been demonstrated in 4 other cells. Also, all professional skills have been demonstrated at level 2.

The third level is the desired final Bachelor level. A Bachelor graduate at Fontys ICT has a T-shaped profile. This means that for all competences (Advise - Analyse - Design - Realise and Manage & Control) level 3 has been demonstrated in at least one architectural layer and at least level 2 has been demonstrated in 4 other cells. Also, all professional skills have been demonstrated at level 3.

The fourth level is the desired final master level. A master graduate at Fontys ICT has a T-shaped profile. This means that for all competences (Advise - Analyse - Design - Realise and Manage & Control) level 4 has been demonstrated in at least one architectural layer and at least level 3 has been demonstrated in 4 other cells. Also, all professional skills have been demonstrated at level 4.

Interim examination and test schedule

1. In Eindhoven, the full-time programme is offered in English. An exception to this are the first-year study units. These are offered in both English and Dutch. In Tilburg, the programme is offered in Dutch. The teaching material is largely in English.

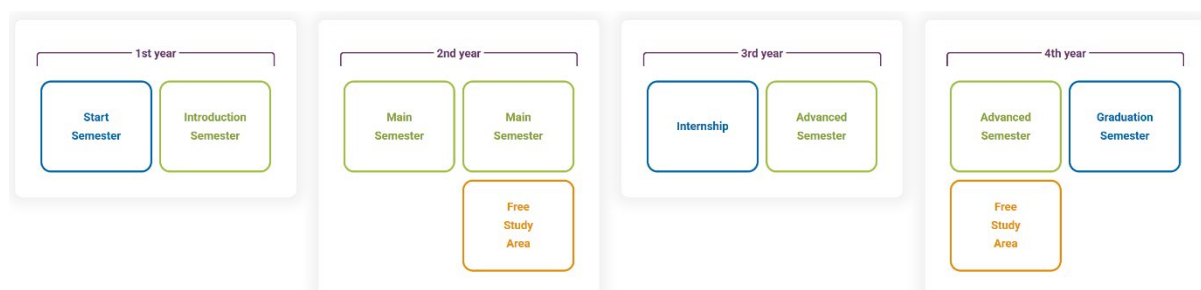
In part-time, the programme is offered in Dutch. The teaching material is largely in English.

2. A shortened programme is offered to students who enter the full-time bachelor's programme with a completed ICT associate degree programme. This two-year study programme consists of semesters 4, 6, 7 and 8 of the regular bachelor's programme. The student is exempted from semesters 1, 2, 3 and 5.

A shortened programme is offered to students who enter the part-time programme with a completed ICT associate degree programme. They start in semester 5 of the part-time programme. The student is exempted from semesters 1 to 4.

3. A teaching unit is assessed by examiners at the end of the semester based on a portfolio assessment. The portfolio can consist of submitted professional products, written tests on knowledge, given and received feedback and the like. During the semester, the lecturers provide feedback on this. The knowledge, insight and skills relating to the learning outcomes of the relevant teaching unit must be demonstrated by completing the portfolio. The content and requirements of a portfolio can be found in the Fontys ICT training guide and on Canvas.
4. The progression is determined by whether or not the teaching unit has been passed. If the teaching unit has not been passed, the student must retake the entire teaching unit.
5. The overview below shows the distribution of credits.
Where the assessment scale is stated U/ S/ G/ O, this always means Unsatisfactory/ Satisfactory/ Good/ Outstanding.
6. The entry requirements are stated in a separate appendix.

The bachelor's program as a whole consists of eight semesters that can be completed according to the schedule below (see also [studyguide.fontysict.nl](https://www.fontys.nl/studyguide/fontysict/)). In one of the Free Study Areas, you may choose a minor:



The APA teaching units in the table below are taken alongside regular semesters for the Accelerated ICT & Software Engineering Program. They can also all be used together to fulfill the Free Study Area in the second year.

Tentamen- en toetsprogramma 2025-2026

Opleiding HBO-ICT
Afstudeerrichting n.v.t.
Cohort n.v.t.
Inschrijfvariant vt

Opleidings- fase	Level	Code	Naam onderwijseenheid	Studiepunten	Naam tentamen of naam deelttoets	-tentamen -deelttoets	Vorm	Weging (alleen bij deelttoetsen)	Beoordelings- schaal	Minimaal te ingangsensen Ja / Nee	Taal Ja / Nee	Intekenen Ja / Nee	
Propedeuse	Startsemester	STA	Introduction to IT	30	Introduction to the field of IT	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Propedeuse	Introduction	IN-AIB	AI Basics	30	Data Preparation & Analysis, Machine Learning, Large Language Models & AI Ethics	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Propedeuse	Introduction	IN-BDA	Business IT & Data Analysis - Introduction	30	Business Analysis, Process Modelling & Low Code Development, Data Handling & Statistics	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Propedeuse	Introduction	IN-GV	Game Development	30	Game Engine Exploration, Playful Interaction & Scripting	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Propedeuse	Introduction	IN-IM	Interactive Media	30	Experience Design, Creative Thinking, Basics of Media Creation & Front End Development	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Propedeuse	Introduction	IN-IT	Intelligent Technologies	30	Embedded software, Microcontrollers and Hardware interfacing	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Propedeuse	Introduction	IN-NCB	Network & Cloud Basics	30	Connecting, Provisioning & Managing IT Services, DevOps	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Propedeuse	Introduction	IN-OLB	Open Learning	30	Personalised Learning Plan Semester 2	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Propedeuse	Introduction	IN-SDE	Software Design & Engineering	30	Software Life Cycle, Software Development, Design & Testing	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Main	MA-AAI	Applied Generative AI	30	Generative AI Solution Design, Responsible Implementation, Quality & Validation	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Main	MA-AMD	AI, Machine Learning & Data	30	Domain Analysis, Exploratory Data Analysis, Data Understanding, Prediction Modelling	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Main	MA-BDA	Business IT & Data Analysis - Main	30	Business Process Optimization, Dashboarding, Low Code Systems Development & Implementation	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Main	MA-CSE	Cyber Security Essentials	30	Hacking & Defending Against Cyber Security Threats	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Main	MA-FED	Front End Development	30	Ensuring Quality Experiences, Web Frameworks, Coding Together with AI	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Main	MA-FSD	Full Stack Development	30	Agile Development, Quality Assurance & Software Process Automation	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Main	MA-GD	Game Design	30	Player Experience, Game Programming & Prototype Development	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Main	MA-ID	Intelligent Devices	30	Distributed Embedded Software, Object Oriented Design, Edge AI, Internet of Things	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Main	MA-II	Industry 4.0 & Internet of Things	30	Machine to Machine Interaction, Data Analytics & Machine Learning, Business Data Processing	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Main	MA-MAD	Mobile Apps Development	30	User Interaction Design, iOS & Android Programming of Mobile Apps	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Main	MA-MC	Media Creation	30	Design Mastery, Media Production & Digital Art Direction, AI Assisted Realization	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Main	MA-NCA	Network & Cloud Automation	30	Deploying & Automating Intelligent IT Services, DevSecOps	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Main	MA-OLC	Open Learning	30	Personalised Learning Plan Semester 3	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Main	MA-OLD	Open Learning	30	Personalised Learning Plan Semester 4	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Internship	ICTS	Internship ICT	30	Bachelor Internship	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Free study area	APA1a	Logic & Set Theory	4	Logic & Set Theory	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Free study area	APA1b	Data Science Mathematics	4	Data Science Mathematics	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Free study area	APA2a	Data Structures & Algorithms 1	4	Data Structures & Algorithms 1	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Free study area	APA2b	Automata	3	Automata	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Free study area	APA3a	Statistics	4	Statistics	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Free study area	APA3b	Decision Theory	4	Decision Theory	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Free study area	APA4a	Functional Programming	2	Functional Programming	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Free study area	APA4b	Synchronization	2	Synchronization	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Free study area	APA4c	Data Structures & Algorithms 2	3	Data Structures & Algorithms 2	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Advanced	AV-BDA	Business IT & Data Analysis - Advanced	30	IT Governance, IT Architecture, Predictive Modelling	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Advanced	AV-CSP	Cyber Security Professional	30	Security Innovation & Red, Blue or Yellow Specialisation	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Advanced	AV-CSS	Complex Software Systems	30	Development of Complex Software Environments, Non-Functional & Data Requirements	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Advanced	AV-EAI	Enhanced AI Techniques	30	Artificial & Convolutional Neural Networks, Deep Learning, Reinforcement Learning & NLP	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Advanced	AV-GP	Game Production	30	Advanced Prototyping, Studio Collaboration & Game Development Pipelines	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Advanced	AV-HC	Human Centered Design	30	Progressive Web Development, User Experience Research & Design	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Advanced	AV-HIT	Hi-Tech Embedded Software Engineering	30	Hi-Tech Embedded Software Design & Solutions	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Advanced	AV-IE	Immersive Experiences	30	Showcase Installations, Experimental Digital & Analog Crafts, Interactive Art & HCI	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Advanced	AV-IV	Industry 5.0	30	AI & Robotics, Digital Twins, Swarm Computing, Smart Cities, Supply Chain Ecosystems	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Advanced	AV-NCI	Network & Cloud Architecture Innovation	30	Interoperability, Agility & Innovation of Enterprise Architecture, DevSecOps	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Advanced	AV-OLF	Open Learning	30	Personalised Learning Plan Semester 6	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Advanced	AV-OLG	Open Learning	30	Personalised Learning Plan Semester 7	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Minor	GA	Global Acting in IT	30	HBO-ICT Minor	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Minor	SES	Signals & Embedded Systems	30	HBO-ICT Minor	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Minor	SIT	Smart Industry Technologies	30	HBO-ICT Minor	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Graduation	GRAa	Graduation	30	Bachelor Graduation	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee

Opleiding HBO-ICT
Afstudeerrichting n.v.t.
Cohort n.v.t.

Inschrijfvariant dt

Opleidings- fase	Level	Code	Naam onderwijsseenheid	Studiepunten	Naam tentamen of naam deelttoets	-tentamen -deelttoets	Vorm	Weging (alleen bij deelttoetsen)	Beoordelings- schaal	Minimaal t/ ingangseisen Ja / Nee	Taal	Intekenen Ja / Nee	
Propedeuse	Startsemester	O-STA	Introduction to IT	30	Introduction to the field of IT	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	nl	Nee
Propedeuse	Introduction	O-IN-BDA	Business IT & Data Analysis - introduction	30	Business Analysis, Process Modelling & Low Code Development, Data Handling & Statistics	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	nl	Nee
Propedeuse	Introduction	O-IN-IM	Interactive Media	30	Experience Design, Creative Thinking, Basics of Media Creation & Front End Development	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Propedeuse	Introduction	O-IN-IT	Intelligent Technologies	30	Embedded software, Microcontrollers and Hardware interfacing	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Propedeuse	Introduction	O-IN-NCB	Network & Cloud Basics	30	Connecting, Provisioning & Managing IT Services, DevOps	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Propedeuse	Introduction	O-IN-SDE	Software Design & Engineering	30	Software Life Cycle, Software Development, Design & Testing	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	nl	Nee
Postpropedeuse	Main	O-MA-OLC	Open Learning	30	Personalised Learning Plan Semester 3	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Main	O-MA-OLD	Open Learning	30	Personalised Learning Plan Semester 4	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Main	O-MA-OLE	Open Learning	30	Personalised Learning Plan Semester 5	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Advanced	O-AV-OLF	Open Learning	30	Personalised Learning Plan Semester 6	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Advanced	O-AV-OLG	Open Learning	30	Personalised Learning Plan Semester 7	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee
Postpropedeuse	Graduation	GRAa	Graduation	30	Bachelor Graduation	Tentamen	Portfoliobeoordeling	nvt	U/S/G/O	S	Nee	en	Nee

Appendix 3 Entry Requirements

Entry requirements

The entry requirements are formulated per teaching unit. In general, the requirement is that a teaching unit has been successfully completed if the summative final assessment is graded with S(atisfactory), G(ood) or O(utstanding). In the event that the assessment is U(nsatisfactory), the teaching unit must be completely retaken (restarted). This leads to a delay of 1 semester. The reasons for this are:

- Each teaching unit contains one or more larger projects that are different for each execution. These projects consist of a number of required phases. You cannot skip phases in a project and, for example, demonstrate realization without first making an analysis and design.
- One of the requirements for an ICT engineer is that he is able to complete a specified workload within a certain time. If repairs were allowed, you would be given more time than your fellow students and would therefore be graded differently. If you do not complete your product(s) on time due to personal circumstances, you can go to a student counselor and request an arrangement with the examination board.

The entry requirements for the semesters are described in a table below.

For the full-time programme:

Educational unit	Entry requirement	Explanation
Start semester	None.	This is the first semester of your study programme
Introduction semesters	The Start semester is completed.	
Main semesters	At least one introduction semester is completed.	
Free study area	At least one introduction semester is completed and you participated in at least one main semester.	
Internship	A half year ago, at least one main semester was completed.	To get permission for internship (halfway through a semester), at least one main semester must be completed.
Advanced semester	The internship is completed and you have at least 150 study points.	
Graduation	All previous semesters have been completed.	

Appendix 4 Transitional Scheme for Article 29

From September, the educational units of Fontys ICT will have new names and codes. A number of new educational units will also be added. Students who previously failed an educational unit can restart it under the new name. There are no drastic changes in content other than those stated in article 29. Students of each cohort can freely choose from all educational units as described in the appendix entry requirements.

Topic Semesters Level Introduction, Main & Advanced Fontys ICT September 2025

Level	Former semester	New semester (topic)	Remark
Start	Start Semester	Start Semester	
Introduction	Business S2	Business IT & Data Analysis	
Introduction	Infrastructure S2	Network & Cloud Basics	
Introduction	Media Design S2	Interactive Media	
Introduction	Software Engineering S2	Software Design & Engineering	
Introduction	Technology S2	Intelligent Technologies	
Introduction	Open Learning S2	Open Learning S2	
Introduction	<i>none</i>	AI Basics	
Introduction	<i>none</i>	Game Development	
Main	Business S3	Business IT & Data Analysis	
Main	Infrastructure S3	Network & Cloud Automation	
Main	Media Design S3	<i>canceled</i>	
Main	<i>none</i>	Front End Development	
Main	<i>none</i>	Media Creation	
Main	Software Engineering S3	Full Stack Development	
Main	Technology S3	Intelligent Devices	
Main	Open Learning S3	Open Learning S3	
Main	Artificial Intelligence S4	AI, Machine Learning & Data	
Main	Artificial Intelligence S4	Applied Generative AI	
Main	Cyber Security S4	Cyber Security Essentials	
Main	Game Design S4	Game Design	
Main	Smart Industry S4	Industry 4.0 & Internet of Things	
Main	Smart Mobile S4	Mobile Apps Development	
Main	Open Learning S4	Open Learning S4	
Free Study Area	Educatie S4	Education	
Free Study Area	Academic Preparation S4	Academic Preparation	
Advanced	Business S6/S7	Business IT & Data Analysis	
Advanced	Infrastructure S6/S7	Network & Cloud Architecture Innovation	
Advanced	Cyber Security S6/S7	Cyber Security Professional	
Advanced	Media Design S6/S7	Human Centered Design	
Advanced	Game Design S6/S7	Game Production	
Advanced	Smart Mobile S6/S7	<i>canceled</i>	
Advanced	<i>none</i>	Immersive Experiences	
Advanced	Software Engineering S6/S7	Complex Software Systems	
Advanced	Artificial Intelligence S6/S7	Enhanced AI Techniques	
Advanced	Technology S6/S7	Hi-Tech Embedded Software Engineering	
Advanced	Smart Industry S6/S7	Industry 5.0	
Advanced	Open Learning S6	Open Learning S6	
Advanced	Open Learning S7	Open Learning S7	

Free Study Area	Educatie S6/S7	Education	
Free Study Area	Pre-Masters	Pre-Masters	
Minor	Global Acting in IT	Global Acting in IT	
Minor	Signals & Embedded Systems	Signals & Embedded Systems	
Minor	AI for Society	AI for Society	
Minor	Cyberstars	Cyberstars	<i>In Tilburg</i>
Minor	Data Driven Business Lab	Data Driven Business Lab	
Minor	Digital Experience Design DXD	Immersive Experience Design IxD	
Minor	Digital Marketing	Digital Marketing	<i>In Tilburg</i>
Minor	EdTech	Smart Education with AI	
Minor	XR Design	<i>merges with Immersive Experience Design IxD</i>	<i>In Tilburg</i>
Minor	<i>none</i>	Smart Industry Technologies	
Minor	<i>none</i>	Virtual Production (starts February 2026)	<i>In Tilburg</i>

For the new name of the semester that you want to do, check the new semester on same line. E.g. Smart Industry S6/S7 is now advanced Industry 5.0.

Once you have completed a former semester, you cannot do the new semester on the same line, as it is similar in both content and learning outcomes.

If you have completed former semester Media Design S3 you can do Front End Development or Media Creation only with prior approval of the exam board.

Examination Board composition of Fontys ICT

Bachelor, Master and Associate Degree

Version 01-03-2026

Policy chamber

Mr. drs. N.P.M. Welman	representative chamber A
Mr. ing. X.B.C. Hover	representative chamber B, deputy chair
Mr. V.Y. Kabzar, MSc	representative chamber C
Ms. F. Saçan-Ilhan, LL.M.	representative chamber D, chair
Mr. K. Suilen, MSc	representative chamber F
Mr. ir. R.J. Steins	representative chamber G
Mr. T.A. Gerbrands MSc	external member
Ms. P.C.G. Huijden	official secretary, deputy secretary
Mr. ir. J.P.A.M.T. van den Heuvel	secretary

Executive chambers

Examination chamber A

Mr. drs. N.P.M. Welman	Chair
Mr. ing. M.W. Ederveen	Secretary

Examination chamber B

Mr. ing. X.B.C. Hover	Chair
Mr. E.P.H.H. Bell, BAT	Secretary

Examination chamber C

Mr. V.Y. Kabzar, MSc	Chair
Ms. C.M. van Vucht, BSc Hons	Secretary

Examination chamber D

Ms. F. Saçan-Ilhan, LL.M.	Chair
Mr. ing. B.J.C.A.H.M. Hamels	Secretary

Examination chamber E (only administrative)

Ms. P.C.G. Huijden	Official secretary
Ms. Y.D.M. Derksen, MSc BEd	Secretary

Examination chamber F

Mr. K. Suilen, MSc	Chair
Mr. ing. P.I. Willems	Secretary

Examination chamber G

Mr. ir. R.J. Steins	Chair
Mr. drs. S.R.P. Vrijssen	Secretary

In each case of illness or absence of a member of an executive chamber, the Chair or Secretary shall act as an alternate for that role. Examination chamber A is alternate for examination chamber B, etc. Examination Chamber G is alternate for A.

Appendix 6 Minor regulation Global Acting

1. Name of minor: Global Acting in ICT

2. English name: Global Acting in ICT

3. Content of minor

The objective of the minor 'Global Acting in ICT' is to combine international IT aspects and trends with intercultural aspects. To that end, students visit multiple universities abroad. The focus of the minor is on intercultural and international competencies and global citizenship. This is applied and experienced in an IT-setting with an innovation project in which current trends are studied and applied.

Summary

In the minor 'Global Acting in ICT', international IT aspects and intercultural aspects in general are studied and experimentally experienced in an international setting. Collaboration with and visits to various international universities is part of the programme.

The name of the minor and the corresponding number of credits shall be included on the diploma supplement. If a minor is completed with a positive assessment, the student will receive a certificate for it.

4. (Admission) restrictions of the minor

Only 12 Fontys-students can participate in this minor. The deadline for application for the autumn is 15 May 2025. Selection will take place in order of application. Students who apply will be told whether they have been placed by 1 June 2025. The minor - subject to change - will also be conducted in the spring of 2026. The deadline for application for spring is 15 October 2025. The student applying will be told whether the student is placed by 1 November 2025.

5. Overview of educational activities of the minor

The minor consists of 6 modules: a kickoff week at the beginning of the minor and 5 modules provided by the 5 participating universities and colleges. The kickoff includes intercultural and internationalisation aspects as well as planning an international project. The modules focus on a specific area in IT such as: User Experience Design, Data Science, AI, Immersive Technology and IoT (autumn 2025 - other topics may be offered for spring 2026). These specific subject areas are studied and applied in an innovative project related to the UN Sustainable Development Goals. Students choose to go to all 5 locations of the participating universities and colleges to follow the modules and thus experience international and intercultural aspects. The 6th module is spread over the beginning for start-up and at the end for completion of project and minor. Students may have prior knowledge on one or more/some of the IT themes in the modules. The international setting in which the themes are covered will then provide a broader knowledge and experience and of the relevant theme. See also the programme guide included as an appendix to the OER.

6. Enrolment for educational activities of the minor

Not applicable

7. Minor testing and enrolment for testing

Assessment is described in the programme guide included as an appendix to the OER.

8. Completion of the minor

Students will be creating professional products, which will be assessed. The programme guide provides an outline of the completion criteria of the professional products and of the overall assessment. The assessment scale is U/ S/ G/ O.

9. Examination Board

The Fontys ICT Examination Board is authorised to adjudicate on requests from students regarding the testing in the minor. The names of the members of the Examination Board can be found in the appendix of the OER Composition Examination Board.

10. Validity

This information shall be valid for the 2025-2026 academic year.

11. Admission requirements minor

The propaedeutic phase has been completed and the student attended at least one main semester.

12. Not accessible to:

Students who do not meet the admission requirements.

No further requirements shall apply to students in relation to participation and completion of the minor other than those stated in these minor regulations.

Appendix 7 Minor regulation Signals & Embedded Systems

1. Name of minor: Signals & Embedded Systems (minor SES)

2. English name: Signals & Embedded Systems

3. Content of minor

The minor in Signals & Embedded Systems is aimed at Fontys ICT students interested in how embedded systems work; how signals from the world around us can be measured and used in applications, and how in the real world signals can be generated by these applications.

The most obvious participants for this minor are students following more software-related and/or embedded system-related pathways within Fontys ICT. But in addition, this minor explicitly invites students from other profiles to follow this minor.

This semester focuses on research and design. This means that the semester focuses on performing authentic, methodically structured, professional tasks, which contribute to new insights for the professional field. The focus is on learning, by carrying out research and development activities, within practice-based research. Students are (mainly) responsible for setting up and monitoring their own learning path.

The research projects of this minor are based on real problems from our Partners in Innovation and from the professorships. You will work on one of these research projects, and on a project of your own

Summary

In this minor, students gained knowledge and skills in signals, sensors and actuators, embedded systems and filtering techniques.

The diploma supplement will include the name and the number of credits of the minor. If a minor is completed with a positive grade, the student will receive a certificate for it.

4. Restrictions on (admission to) the minor

Not applicable.

5. Overview of educational activities of the minor

Educational activities are listed in the programme guide included as an appendix to the OER.

6. Enrolment for educational activities of the minor

Not applicable.

7. Minor testing and enrolment for testing

Assessment is described in the programme guide included as an appendix to the OER.

8. Completion of the minor

Students will be creating professional products, which will be assessed. The programme guide provides an outline of the completion criteria of the professional products and of the overall assessment. The assessment scale is U/ S/ G/ O.

9. Examination Board

The Fontys ICT Examination Board is authorised to adjudicate on requests from students regarding the testing in the minor. The names of the members of the Examination Board can be found in the appendix of the OER Composition Examination Board.

10. Validity

This information shall be valid for the 2025-2026 academic year.

11. Admission requirements minor

The propaedeutic phase has been completed and the student attended at least one main semester.

12. Not accessible to:

Students who do not meet the admission requirements.

No further requirements shall apply to students in relation to participation and completion of the minor other than those stated in these minor regulations.

Appendix 8 Minor regulation Smart Industry Technologies

1. Name of minor: Smart Industry Technologies

2. English name: Smart Industry Technologies

3. Content of minor

Smart Industry is a revolutionary approach that integrates Information Technologies (IT) with Operational Technologies (OT) to create adaptive, efficient, and digitally connected industrial systems. It ensures that production processes, products, and services are flexible and adjustable in real time by utilizing data. These data are collected, communicated, and applied across all components of the production network, enabling automation and optimization at unprecedented levels.

The field of Smart Industry spans over multiple enabling technologies and requires professionals who can operate at the intersection of ICT and Engineering. The demand for such experts is growing rapidly, especially in the High-Tech Brainport region of Eindhoven and Noord-Brabant. This minor at Fontys is a unique collaboration between the ICT and Engineering departments at Fontys, offering a cross-disciplinary, research-based learning environment.

This minor is perfect for students who want to enhance existing skills or explore new technologies in Smart Industry. The program offers a self-regulated learning path, allowing students to take charge of their own development with a high level of self-management.

By the end of the minor, students will obtain expertise in the following areas:

- A solid understanding of Smart Industry enabling technologies.
- Practical experience in applications of relevant technologies, as for instance industrial automation, IoT, vision, industry data management and AI, and their integration in innovative solutions.
- A deep awareness of the importance of data in Smart Industry and the way it is used for optimisation.
- Hands-on project experience in solving real industry challenges.

Summary

This interdisciplinary minor integrates ICT and Engineering, equipping students with the skills needed to meet the growing demand for Smart Industry professionals. Through real industry cases, students gain practical experience in IT/OT technologies, their application and integration in novel solutions. The self-regulated learning approach fosters technical and professional skills in a multidisciplinary environment, emphasizing real-world problem-solving.

4. Restrictions on (admission to) the minor

Not applicable

5. Overview of educational activities of the minor

The minor follows the research-based learning methodology with research projects being the core of all educational activities. The research projects focus on various use case where different technologies need to be combined. The minor offers flexibility so that students can choose own learning pathway and follow it with a high degree of self-management.

The minor activities are roughly divided in two phases: *Exploration* and *Application*. In the *Exploration* phase students focus on learning the basics of the technologies on their own choice. They learn by working on smaller-scope steered research projects supported by workshops, instruction sessions, experts' sessions and feedback sessions. There are various predefined projects for every technology offered in the minor, but students have freedom to propose own project. Exploration spans over the first 6 weeks of the semester and, if needed, can extend over several more weeks but with lower intensity.

In the *Application* phase students work on a real-world smart industry related problem, together with a group of fellow students, experts from the client company, Fontys researchers, experts and coaches.

These projects have broader scopes and higher complexity and require integration of multiple technologies. In the multidisciplinary group every student takes the role of an expert in the technology the student wants to develop further.

The minor focuses on the following topics:

- **Machine vision**
Image processing with OpenCV, Vision AI with e.g. YOLO.
- **Industrial automation & remote commissioning**
Programming and use of PLCs and industrial control systems e.g. Siemens TIA Portal, CodeSys, EtherCAT for operating production machinery; Integration of production machines into a smart factory with OPC-UA.
- **Internet of Things & Connected Systems & Infrastructure**
Use of smart sensors, actuators, and network technologies e.g. BLE, LoRa, MESH, MQTT for data collection, and for monitoring and control of industrial systems; Cloud integration and cloud services e.g. AWS, Azure; Data and network security.
- **Data Analysis & Data Visualisation & Machine Learning for Process Optimisation and Predictive maintenance**
Application of Python and Pandas for data analysis and machine learning; Visualisation and simulation using interactive dashboards with e.g. Node-Red and Digital Twins.

Other topics and tools can also be explored depending on the research that needs to be conducted for a certain project.

6. Enrolment for educational activities of the minor

n.v.t.

7. Minor testing and enrolment for testing

Toets	Toetsvorm	Individueel/groep	Beoordelingsschaal
Formative assessments	Portfolio	Individual	USGO

Students are assessed by appointed assessors on the intended growth in the learning outcomes. Assessment is longitudinally based on the portfolio development and performance of the individual student. The student determines the components that will be included in the portfolio that demonstrate the learning progress with respect to the learning outcomes. More information about the assessment can be found in the programme guide included as an appendix to the TER.

8. Completion of the minor

Students are assessed based on the evidences provided in the portfolio. In week 20, the integral summative assessment of the semester is carried out, in the assessors meeting, for the 30 ECTS of the study unit. The assessment scale is Unsatisfactory, Satisfactory, Good, or Outstanding. The programme guide describes the norms for completing the minor in more detail.

9. Examination Board

The Fontys ICT Examination Board is authorized to adjudicate on requests from students regarding the assessment in the minor. The names of the members of the Examination Board can be found in the appendix of the ICT TER Composition Examination Board.

10. Validity

This information shall be valid for the 2025-2026 academic year.

11. Admission requirements minor

Students from the ICT and the Engineering study programs at Fontys can directly participate in the minor under the condition that they have completed 1st year of their education. To be eligible to participate in the minor, the students must fulfill the requirements imposed by their program to be eligible to follow a minor or otherwise have permission from the examination board of their program to enroll in the minor.

12. Not accessible to:

Students who do not meet the admission requirements.

No further requirements shall apply to students in relation to participation and completion of the minor other than those stated in these minor regulations.

FONTYS ICT PROGRAMME GUIDE

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47. [AV-CSP]	Cyber Security Professional
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50. [GRAa]	Graduation Fall
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53. [AV-IE]	Immersive Experiences
54. [AV-IV]	Industry 5.0
55. [AV-NCI]	Network & Cloud Architecture Innovation
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- 59. [O-AV-OLG] Open Learning
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1. INFORMATION ABOUT [IN-AIB] AI BASICS (30EC)

Data Preparation & Analysis, Machine Learning, Large Language Models & AI Ethics

1.1. Content

In this semester you'll dive headfirst into AI's exciting world. You'll master fundamental techniques like classification algorithms, bias detection, and model evaluation. You explore concepts including training data balance, decision boundaries, and prediction accuracy—the building blocks of effective AI systems. You'll implement improvements through hyperparameter tuning.

The course covers practical applications: from clustering customer data and visualizing complex relationships to using sentiment analysis for content recommendation systems work and explore the creative side of AI through generative models.

Through hands-on projects like analyzing satellite imagery, optimizing delivery routes, and predicting customer behavior, you'll gain experience with real-world AI and sustainability in urban environments.

By semester's end, you'll be equipped to translate client requests into AI solutions and make smart, responsible decisions in your future career. Let the journey begin!

1.2. Learning outcomes

1. **Learning outcome:** Professional standard
You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory research, and data analysis.
2. **Learning outcome:** Personal leadership
You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional development.
3. **Learning outcome:** Analysing
You can analyze data, prepare and use it for AI applications within a specific domain, where you assess the data quality, use visualizations to present findings, and communicate results.
- Explanation:** In real-world AI, raw data is rarely ready to use. You'll act as both detective—questioning biases, origins, and flaws—and storyteller. The outcome stresses that all data decisions have consequences, making impact evaluation critical to building reliable AI foundations.
4. **Learning outcome:** Advising
You can make recommendations based on your analysis of AI applications, analyze their societal and ethical impact, and provide advice on this to stakeholders.
- Explanation:** This outcome highlights your role as both technical interpreter and ethical guide. Beyond what AI can do, you'll explore what it should do for diverse audiences, maintaining accuracy and ethical rigour throughout.
5. **Learning outcome:** Designing
You can design AI applications that meet the needs of both stakeholders and users, where you take into consideration model choice, feasibility, quality, and integration.
- Explanation:** This is about crafting AI solutions that are not just clever but useful. You'll balance stakeholder goals with user needs, choosing more appropriate algorithms to structuring system architecture—that align with resource limits, long-term performance, and integration needs. Designing for user experience is key.
6. **Learning outcome:** Realising
You can implement existing AI models and services in practical applications, where you can recognize and justify their possibilities and limitations.
- Explanation:** Here, theory becomes action. You'll bring AI systems to life, making informed decisions about which models to use and why. It's about practical implementation and evaluation.
7. **Learning outcome:** Managing & Controlling
You can monitor and manage the performance and quality of AI systems, define appropriate indicators, manage model versions, and take into account feedback loops.
- Explanation:** Managing AI means continuous attention to its reliability and relevance. You'll monitor key metrics, define what success looks like for different stakeholders, and ensure transparency ensures that AI systems deliver lasting value.

1.3. Expected entry profile

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1		

1.4. Outflow profile

	Analysis	Advise	Design	Realise	Manage & control
User interaction	1	1	1	1	1
Organisational processes	1	1	1	1	1
Infrastructure					
Software	1	1	1	1	1
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

1.5. Examination and final grading

1.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work you do throughout the semester.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning path. Throughout the entire semester, the assessors monitor your progress.

1.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record these conversations in your portfolio.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue to work on it.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. This gives you the opportunity to address and process the feedback received.

1.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory' or better.

1.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

1.6. Learning activities

Three-Track Learning Journey

Three tracks work together will provide you with a balanced AI education; your integrated learning experience. This carefully designed structure ensures you become a well-rounded AI practitioner ready to contribute meaningfully in this rapidly evolving field.

Industry Project

Grounds your learning in practical application with real-world industry partners.

Throughout the semester, you'll work with a team on a real AI challenge provided by one of our industry partners. This forms the backbone of your learning. You'll collect and analyze data, implement AI models, address ethical considerations, and conduct practical experiments. The iterative nature of this challenge mirrors real-world semester experience.

AI Explorations

Help you master specific technical and ethical aspects of AI through focused challenges.
During the semester, you'll engage in several short, hands-on experiments designed by your teachers. These explorations allow you to investigate spe

Personal Challenge

Encourages autonomy and deeper engagement with topics that resonate with you personally.
The Personal Challenge marks a crucial transition in your professional development. This is where you move from **consuming research** to **producing**
develop the core competencies that will serve you throughout your career.

1.7. Learning materials

A Canvas course featuring the necessary learning resources

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2. INFORMATION ABOUT [APA2B] AUTOMATA (3EC)

2.1. Content

Algorithms are at the heart of any computer application solving non-trivial problems in transportation, healthcare, education, social media etc. Writing e
In this course you learn the basics of formal languages and automata: regular languages, regular expressions, deterministic finite automata, non-deter
Compilers systematically translate high-level languages designed for humans into a program written in low-level assembly more suited to machines or
language (programming language or a domain specific language) by grammars and how to use them in combination with a parser generator, to parse c

2.2. Learning outcomes

2.3. Examination and final grading

2.3.1. Examination

The teachers will provide the students with formative feedback on deliverables and demonstrations. It is expected that this feedback will be processed
The student shows the work, alone or in a group context, but will be given feedback on an individual base.
The student work will be assessed on analysis, design and technical aspects, where the student will be informed whether the learning goals are met co
At these assessments the learning outcomes are determined based on state of the deliverables at that point. Ideally the learning outcomes are proficie
assessments. It is expected that the student keeps on progressing on the learning outcomes, not doing so might lead to a lower grade of beginning or c

2.3.2. Repairs and resits

During the course, students are facilitated to process formative feedback on their learning goals in their deliverables and activities that are to be record

2.3.3. Tools allowed

Except for paper & pen, and a laptop, no additional equipment is needed. The regular rules about plagiarism apply.

2.3.4. Final grading

A student's growth is maintained in his portfolio. The student's growth on each learning outcome is registered according to one of the five levels of mas

1. undefined,
2. orienting,
3. beginning,
4. proficient,
5. advanced.

After each formative assessment, the level of mastery for all learning outcomes is assessed for the student's deliveries.

At the end of the semester the student will be assessed on his performance of the semester. During the assessment meeting the teacher will use the st
advanced, the grade will be at least Satisfactory. This does not mean that obtaining proficient is the minimum requirement to achieve a satisfactory gra

2.4. Learning activities

Based on the characteristics of an academic premaster and based on a highly knowledge-based scientific world, the course is structured according the
the study program.

The education is designed according to the principles of the 4C/ID model (four-component instructional design model). Education is pre-structured in te
Knowledge, competences and professional skills are further gained and applied in practical assignments, which increase in size, abstraction and comp
Depending on the subjects, a variety of side-roads will be offered for in-depth knowledge and broader research.

2.5. Learning materials

All course content is in Canvas and Sharepoint. All software (compilers, IDEs, libraries, tools) is available for free.

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3. INFORMATION ABOUT [IN-BDA] BUSINESS IT & DATA ANALYSIS - INTRODUCTION

Business Analysis, Process Modelling & Low Code Development, Data Handling & Statistics

3.1. Content

In the introduction semester Business IT and Data Analysis, you will work on a variety of topics. You will become proficient in business themes, data and bring these themes together. You will work on these in the context of a professional task which will be performed for an external stakeholder.

3.2. Learning outcomes

- Learning outcome:** Professional standard
You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory
- Learning outcome:** Personal leadership
You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional
- Learning outcome:** Business analysis
You show that you can systematically analyze organizational processes and advise how these processes can be optimized. You include relevant
- Learning outcome:** Data modelling and design
You model, design, prepare, store and question a dataset using proper and meaningful designing techniques.
- Learning outcome:** Dataset analysis
You prepare data and perform statistical analysis on data to create meaningful information using data analysis and reporting tools.
- Learning outcome:** Advising the customer
You provide an advice on implementation and management of an IT solution with explicit attention to the KPIs and the business objectives to be achieved

3.3. Expected entry profile

There are no prerequisites for this semester so all are welcome. Students with natural sense curiosity as well as a willingness to interact with and help

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1		

3.4. Outflow profile

Domain description HBO-i 2023

	Manage & control	Realise	Design	Advise	Analysis
User interaction	1	1	1	1	1
Organisational processes	1	1	1	1	1
Infrastructure					
Software	1	1	1	1	1
Hardware-interfacing					

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

3.5. Examination and final grading

3.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work you do, the quality of your work, and the progress you make.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning path. Throughout the entire semester, the assessors monitor your progress.

3.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record these conversations in your portfolio.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome with good results.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue to work on it.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale, giving you the opportunity to address and process the feedback received.

3.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory' or better.

3.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

3.6. Learning activities

Your daily learning activities will consist of:

Learning:

You will learn new skills through a combination of attending workshops, self-study of the provided course materials, online trainings, and researching the literature.

Practicing:

You will practice and hone your newly acquired skills by applying them to various assignments.

Implementing:

Once proficient, you will individually apply your skills to your assigned professional task.

Combining:

You will each introduce your individual work to your professional task team. This is then combined to a single deliverable which you all will present to your client.

Next to these you will receive a series of **Challenges** which you must complete.

3.7. Learning materials

All necessary materials will be provided. No extra materials are needed for this semester.

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4. INFORMATION ABOUT [O-IN-BDA] BUSINESS IT & DATA ANALYSIS - INTRODUCTORY

Business Analysis, Process Modelling & Low Code Development, Data Handling & Statistics

4.1. Inhoud

In semester 2 Business ga je werken aan veel verschillende thema's. Je bekwaamt je in bedrijfskundige thema's, statistische basisvaardigheden, data-analyse, processen, en low code development.

In semester 1 heb je al kennis gemaakt met de verschillende thema's bij Business. In semester twee bouwen we hiërarchisch voort.

Leerinhoud:

OE2 PRO Business		
Organisatie	ICT	PO
<i>Organisatie en informatie</i> Strategische analyse: - Balanced scorekaart - BMC, Porter Informatiebeheer: - Requirements - MoSCoW-methode - KPI's - Dataset genereren Organisaties begrijpen: - Processen beschrijven, ontwerpen, verbeteren, ist / soll / knelpuntanalyse, BPMN 2.0 - Interne controle en beheersingsmaatregelen met functiescheiding Doel = voorwaarden opgesteld om IT-organisaties te ondersteunen.	<i>Programmeren R/SQL of Python/Power-BI</i> Datamodelleren en SQL - ERD-modellering - Query op basis van datamodel: - Volledige SQL selecteren - Subquery's voldoen aan Common Table Expressions R/tidyverse en SQL - Query's op basis van het omgekeerde omgekeerde - Valideren van opgeloste query's met SQL en vervalsing - Visualiseer van queryresultaten met ggplot Statistiek - Variabelen en meetniveaus - Gegevens presenteren - Centrummaten - Spreidingsmaten - Normaalverdeling Doel = relatie zien tussen data en informatie (basis). Focus op visualiseren.	Professionele standaard: Je past individueel en in teamverband een professionele werkwijze toe op het gebied van projectorganisatie, communicatie met belanghebbenden, verkennend onderzoek en rapportage. Persoonlijk Leiderschap: Je neemt initiatief om feedback te vragen en te reflecteren en benoemt jouw kernwaarden als vertrekpunt voor je studieloopbaan en je professionele ontwikkeling.

Professionele vaardigheden

In semester 2 komen veel verschillende onderwerpen aan bod die allemaal essentieel zijn bij het begrijpen, verbeteren en ontwikkelen van bedrijfsproces. Kracht is van een goede ICT & Business professional, staat toepassing van op toegevoegde kennis en kunde centraal in dit semester.

In het startsemester heb je je kunnen oriënteren op ICT én jezelf. Ook heb je ervaren en geleerd wat er allemaal komt kijken bij het samenwerken in een semester 2 deur. Je toont niet alleen wat je kunt, maar ook hoe je dat bereikt hebt. Je werkt op een prettige manier samen, je neemt niet alles klakkeloos aan.

4.2. Leernutkomsten

1. Learning outcome: Professionele standaard

Je past individueel en in teamverband een professionele werkwijze toe op het gebied van projectorganisatie, communicatie met belanghebbende

Explanation: Je herkent ethische en maatschappelijk kwesties. Je herkent eigen en andermans grenzen en handelt daarnaar. Je faseert -onderzoek in de uitwerking na over het gevolg van de oplossing voor betrokken mensen en organisaties. Je bent je bewust van de impact die culturele verschillen hebben op jou en op andere mensen met wie je omgaat.

Je houdt rekening met directe belanghebbenden bij de opdracht. Je hebt aandacht voor wat je wil communiceren en in welke vorm. Je neemt je verantwoordelijkheid op jou en op andere mensen met wie je omgaat.

Je blijft gedurende het hele oplosproces nieuwsgierig en vragen stellen. Je beantwoordt vragen met een passende aanpak: pragmatisch, kritisch

2. Learning outcome: Persoonlijk leiderschap

Je neemt initiatief om feedback te vragen en te reflecteren en benoemt jouw kernwaarden als vertrekpunt voor je studieloopbaan en je professionele ontwikkeling.

Explanation: Je ziet en grijpt kansen. Je motiveert jezelf. Je neemt verantwoordelijkheid voor jouw handelen. Je werkt resultaatgericht aan je ontwikkeling. Je begrijpt wat cultuur is. Je herkent je eigen cultuur en een aantal dimensies van cultuurverschillen.

3. Learning outcome: (3) Bedrijfsanalyse; Je laat zien dat je een organisatie in kaart kan brengen. Je stelt een advies op voor de optimalisatie daarvan.

Explanation: De organisatie is een kleine of middelgrote organisatie. Organisaties bestaan met een reden. Dit bepaalt uiteindelijk inrichting van de organisatie.

4. Learning outcome: (4) Data prepareren; Je modelleert, ontwerpt, bewaart en bevraagd een dataset met behulp van correcte en zinvolle ontwerptechnieken.

Explanation: Modelleren en ontwerpen van een dataset bestaat uit het creëren van een datamodel binnen de context. Gebaseerd op dit datamodel implementeer je management rapportages in de database.

5. Learning outcome: (5) Dataset analyseren; Je bereidt gegevens voor en analyseert deze data statistisch zodat je zinvolle informatie creëert op basis van de data.

Explanation: Prepareren van data het gebruiken van opschoon- en transformatie technieken om de data om te zetten naar een gestructureerde, resulteren in een rapport dat past bij de opdrachtgever.

4.3. Verwacht ingangprofiel

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes	1	1	1		
Infrastructure					
Software					
Hardware-interfacing					

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1		

4.4. Uitstroomprofiel

	Analysis	Advise	Design	Realise	Manage & control
User interaction	1	1			
Organisational processes	1	1	1	1	1
Infrastructure					
Software			1	1	
Hardware-interfacing					

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

4.5. Toetsing en beoordeling

4.5.1. Toetsing: Holistisch en Longitudinaal

De beoordeling in dit semester is holistisch en longitudinaal. Holistisch betekent dat alles wat je als student doet meetelt voor de eindbeoordeling: de h

De leeruitkomsten beschrijven het gewenste niveau, maar omdat deze onafhankelijk van het leertraject kunnen worden aangetoond, is er ruimte om ee dingen doet. Dit is het longitudinale aspect: gedurende het hele semester volgen de beoordelaars je vooruitgang.

4.5.2. Feedback: Bevestiging en begeleiding

Docenten begeleiden en beoordelen je vooral door middel van feedback. Je hebt wekelijks gesprekken met een of meer docenten over je werk. Deze f

Daarnaast vinden op vaste momenten in het semester portfolio opleveringen plaats. Hierop krijg je feedback van je docenten op basis van de developr

Ontwikkelingsgerichte feedbackschaal

- *Onbepaald*: Je hebt nog geen activiteiten ondernomen voor het aantonen van de leeruitkomst.
- *Oriënterend*: Je hebt een begin gemaakt en de mogelijkheden verkend om de leeruitkomst aan te tonen.
- *Beginnend*: Je hebt eerste stappen gezet en deze uitgevoerd welke bijdragen aan het aantonen van de leeruitkomst.
- *Geoefend*: Je hebt meerdere keren laten zien dat je een basis gecreëerd hebt om de leeruitkomst aan te tonen. Je zal de leeruitkomst op een vol
- *Gevorderd*: Je hebt meerdere keren laten zien dat je met deze leeruitkomst bezig bent geweest met goed resultaat. Je hebt boven verwachting g blijft voortzetten.

Deze feedback is een momentopname van je huidige groei in de richting van het leerresultaat. Het is mogelijk dat je achteruit gaat in je ontwikkeling als

Binnen het semester zijn er minstens drie van deze formatieve evaluatiemomenten waarbij de beoordelaars feedback geven per leerresultaat aan de h
beoordelaarsvergadering. Dit geeft je de gelegenheid om de ontvangen feedback te behandelen en te verwerken.

4.5.3. Assessment Richtlijnen

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discr

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achiev

4.5.4. Herkansingen en reparaties

Herkansingen of reparaties zijn niet van toepassing op dit semester. Er wordt van je verwacht dat je je werk een onbeperkt aantal keren opnieuw indier

4.6. Leeractiviteiten

Inspiratiecolleges, werkcolleges, gastcolleges, workshops, groepsgewijs werken aan een casusopdracht, onderzoeksopdracht bij (of in samenwerking

4.7. Leermaterialen

(Status: X = verplicht, O = aanschaffen na overleg)

ISBN	Titel	Druk	Auteur	Uitgever	Prijs	Status
9789001831592	Basisvaardigheden toegepaste statistiek HO	2e	Buuren, H. en Reus, G.J.	Noordhoff Uitgevers	€24,95	X
9781615473021	Slaying Excel Dragons (e-book)		Mike Girvin	Holy Macro Books	€18,00	O
9781491910399	R for Data Science	2e	Hadley Wickham & Garrett Golemund	O'Reilly	€34,99	O

Diverse bronnen worden beschikbaar gesteld via de Canvas course.

Professional Skills (gedurende de hele opleiding):

- Rapportagetechniek; Bas Andeweg, Sjaak Baars, Jaap de Jong, Rien Elling, ISBN 9789001841744 (gedurende de hele opleiding)
- Onderzoeksvaardigheden, aan de hand van triangulatieframework, zie site: ictresearchmethods.nl
- Licentie "Hogeschooltaal", met Fontys account inloggen, kosten €70,= voor 5 studiejaar.

Software (alle zijn freeware):

- Voor Excel: MS professional
- Voor R: R Studio, gratis
- Datacamp course (inlogaccount startsemester)

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5. INFORMATION ABOUT [APA1B] DATA SCIENCE MATHEMATICS (4EC)

5.1. Content

Data Science, responsible for transforming (big) data into useful information, has become an unmissable link in almost every industry, from entertainme
examples of data science applications.

The subjects of the course are:

- vector and matrix operations
- linear regression and gradient descent
- bias & variance
- classification
- K-nearest neighbour
- perceptrons
- forward & backward propagation
- (deep) neural networks

5.2. Learning outcomes

5.3. Examination and final grading

5.3.1. Examination

The teachers will provide the students with formative feedback on deliverables and demonstrations. It is expected that this feedback will be processed

The student shows the work, alone or in a group context, but will be given feedback on an individual base.

The student work will be assessed on analysis, design and technical aspects, where the student will be informed whether the learning goals are met co

At these assessments the learning outcomes are determined based on state of the deliverables at that point. Ideally the learning outcomes are proficient assessments. It is expected that the student keeps on progressing on the learning outcomes, not doing so might lead to a lower grade of beginning or c

5.3.2. Repairs and resits

During the course, students are facilitated to process formative feedback on their learning goals in their deliverables and activities that are to be record

5.3.3. Tools allowed

Except for paper & pen, and a laptop, no additional equipment is needed. The regular rules about plagiarism apply.

5.3.4. Final grading

A student's growth is maintained in his portfolio. The student's growth on each learning outcome is registered according to one of the five levels of mas

1. undefined,
2. orienting,
3. beginning,
4. proficient,
5. advanced.

After each formative assessment, the level of mastery for all learning outcomes is assessed for the student's deliveries.

At the end of the semester the student will be assessed on his performance of the semester. During the assessment meeting the teacher will use the st advanced, the grade will be at least Satisfactory. This does not mean that obtaining proficient is the minimum requirement to achieve a satisfactory gra

5.4. Learning activities

Based on the characteristics of an academic premaster and based on a highly knowledge-based scientific world, the course is structured according the the study program.

The education is designed according to the principles of the 4C/ID model (four-component instructional design model). Education is pre-structured in te

Knowledge, competences and professional skills are further gained and applied in practical assignments, which increase in size, abstraction and comp

Depending on the subjects, a variety of side-roads will be offered for in-depth knowledge and broader research.

5.5. Learning materials

All course content is in Canvas and Sharepoint. All software (compilers, IDEs, libraries, tools) is available for free.

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6. INFORMATION ABOUT [APA2A] DATA STRUCTURES & ALGORITHMS 1 (4EC)

6.1. Content

Algorithms are at the heart of any computer application solving non-trivial problems in transportation, healthcare, education, social media etc. Writing e

This course provides a comprehensive introduction to fundamental data structures and algorithms used in computer science. You'll learn how to design

This course will help you train these skills by examining several classic algorithms for sorting, searching and optimization problems. You will learn to re optimization strategies discussed are greedy, backtracking, dynamic programming.

It explores algorithmic techniques including searching, sorting, graph traversal, dynamic programming, and divide-and-conquer. Topics also include alg programming assignments, you'll develop a solid foundation in data structures and algorithms that are essential for solving real-world computational ch

6.2. Learning outcomes

1. Learning outcome: Analysis

Explanation: You learn to estimate algorithm complexity, to recognize appropriate algorithmic strategies for solving real problems etc.

2. Learning outcome: Design

Explanation: You design efficient algorithms and reason about their correctness and complexity.

3. Learning outcome: Realization

Explanation: You implement your algorithmic designs into actual working solutions.

4. Learning outcome: Testing

Explanation: You design and apply unit tests to guarantee correctness of the solutions, and apply performance tests as well.

5. Learning outcome: Research

Explanation: Throughout the course, you adopt a curious attitude and take charge of your learning process. You are invited to investigate in more

6. Learning outcome: Interaction

Explanation: There will be enough opportunity for interaction with your teachers and colleagues.

7. Learning outcome: Communication & leadership

Explanation: You describe and justify, mostly in writing, the correctness and complexity of your algorithms.

6.3. Examination and final grading

6.3.1. Examination

The teachers will provide the students with formative feedback on deliverables and demonstrations. It is expected that this feedback will be processed

The student shows the work, alone or in a group context, but will be given feedback on an individual base.

The student work will be assessed on analysis, design and technical aspects, where the student will be informed whether the learning goals are met or

At these assessments the learning outcomes are determined based on state of the deliverables at that point. Ideally the learning outcomes are proficient assessments. It is expected that the student keeps on progressing on the learning outcomes, not doing so might lead to a lower grade of beginning or

6.3.2. Repairs and resits

During the course, students are facilitated to process formative feedback on their learning goals in their deliverables and activities that are to be recorded

6.3.3. Tools allowed

Except for paper & pen, and a laptop, no additional equipment is needed. The regular rules about plagiarism apply.

6.3.4. Final grading

A student's growth is maintained in his portfolio. The student's growth on each learning outcome is registered according to one of the five levels of mastery

1. undefined,
2. orienting,
3. beginning,
4. proficient,
5. advanced.

After each formative assessment, the level of mastery for all learning outcomes is assessed for the student's deliverables.

At the end of the semester the student will be assessed on his performance of the semester. During the assessment meeting the teacher will use the student's growth. If the student is advanced, the grade will be at least Satisfactory. This does not mean that obtaining proficient is the minimum requirement to achieve a satisfactory grade

6.4. Learning activities

Based on the characteristics of an academic pre-master and based on a highly knowledge-based scientific world, the course is structured according to the study program.

The education is designed according to the principles of the 4C/ID model (four-component instructional design model). Education is pre-structured in terms of

Knowledge, competences and professional skills are further gained and applied in practical assignments, which increase in size, abstraction and complexity

Depending on the subjects, a variety of side-roads will be offered for in-depth knowledge and broader research.

6.5. Learning materials

All course content is in Canvas and Sharepoint. All software (compilers, IDEs, libraries, tools) is available for free.

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7. INFORMATION ABOUT [APA4C] DATA STRUCTURES & ALGORITHMS 2 (3EC)

7.1. Content

Explore the fundamental question of P versus NP in this course on computational complexity. Delve into the distinction between the complexity classes P and NP, their impact on cryptography, optimization, and practical computing. Gain insights into current developments and research directions in the field.

Through lectures, discussions, and problem-solving exercises, students will develop a solid foundation in computational complexity theory and gain the ability to

7.2. Learning outcomes

1. **Learning outcome:** In addition to the general Academic Preparations learning outcomes, the student is able to: • don't focus only on your current

7.3. Examination and final grading

7.3.1. Examination

The teachers will provide the students with formative feedback on deliverables and demonstrations. It is expected that this feedback will be processed

The student shows the work, alone or in a group context, but will be given feedback on an individual base.

The student work will be assessed on analysis, design and technical aspects, where the student will be informed whether the learning goals are met co

At these assessments the learning outcomes are determined based on state of the deliverables at that point. Ideally the learning outcomes are proficie assessments. It is expected that the student keeps on progressing on the learning outcomes, not doing so might lead to a lower grade of beginning or c

7.3.2. Repairs and resits

During the course, students are facilitated to process formative feedback on their learning goals in their deliverables and activities that are to be record

7.3.3. Tools allowed

Except for paper & pen, and a laptop, no additional equipment is needed. The regular rules about plagiarism apply.

7.3.4. Final grading

A student's growth is maintained in his portfolio. The student's growth on each learning outcome is registered according to one of the five levels of mas

1. undefined,
2. orienting,
3. beginning,
4. proficient,
5. advanced.

After each formative assessment, the level of mastery for all learning outcomes is assessed for the student's deliveries.

At the end of the semester the student will be assessed on his performance of the semester. During the assessment meeting the teacher will use the st advanced, the grade will be at least Satisfactory. This does not mean that obtaining proficient is the minimum requirement to achieve a satisfactory gra

7.4. Learning activities

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The education is designed according to the principles of the 4C/ID model (four-component instructional design model). Education is pre-structured in te

Knowledge, competences and professional skills are further gained and applied in practical assignments, which increase in size, abstraction and comp

Depending on the subjects, a variety of side-roads will be offered for in-depth knowledge and broader research.

7.5. Learning materials

All course content is in Canvas and Sharepoint. All software (compilers, IDEs, libraries, tools) is available for free.

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8. INFORMATION ABOUT [APA3B] DECISION THEORY (4EC)

8.1. Content

This course focuses on studying the theoretical foundations and practical applications of decision-making under uncertainty.

It covers Reinforcement Learning (RL) to solve MDPs where an agent learns through interactions with its environment. You'll explore various RL algorit

Throughout the course, you can expect a mix of theoretical concepts, algorithmic techniques, and practical implementations. Assignments and projects

8.2. Learning outcomes

1. **Learning outcome:** Students will be able to identify and formulate basic decision problems as Markov decision processes (mini project).
2. **Learning outcome:** Students will be able to implement basic automated decision-making algorithms to solve decision problems (mini project).

8.3. Examination and final grading

8.3.1. Examination

The teachers will provide the students with formative feedback on deliverables and demonstrations. It is expected that this feedback will be processed

The student shows the work, alone or in a group context, but will be given feedback on an individual base.

The student work will be assessed on analysis, design and technical aspects, where the student will be informed whether the learning goals are met or not.

At these assessments the learning outcomes are determined based on state of the deliverables at that point. Ideally the learning outcomes are proficient assessments. It is expected that the student keeps on progressing on the learning outcomes, not doing so might lead to a lower grade of beginning or not.

8.3.2. Repairs and resits

During the course, students are facilitated to process formative feedback on their learning goals in their deliverables and activities that are to be recorded.

8.3.3. Tools allowed

Except for paper & pen, and a laptop, no additional equipment is needed. The regular rules about plagiarism apply.

8.3.4. Final grading

A student's growth is maintained in his portfolio. The student's growth on each learning outcome is registered according to one of the five levels of mastery.

1. undefined,
2. orienting,
3. beginning,
4. proficient,
5. advanced.

After each formative assessment, the level of mastery for all learning outcomes is assessed for the student's deliverables.

At the end of the semester the student will be assessed on his performance of the semester. During the assessment meeting the teacher will use the student's growth. If the student is advanced, the grade will be at least Satisfactory. This does not mean that obtaining proficient is the minimum requirement to achieve a satisfactory grade.

8.4. Learning activities

Based on the characteristics of an academic pre-master and based on a highly knowledge-based scientific world, the course is structured according to the study program.

The education is designed according to the principles of the 4C/ID model (four-component instructional design model). Education is pre-structured in terms of knowledge, competences and professional skills.

Knowledge, competences and professional skills are further gained and applied in practical assignments, which increase in size, abstraction and complexity.

Depending on the subjects, a variety of side-roads will be offered for in-depth knowledge and broader research.

8.5. Learning materials

All course content is in Canvas and Sharepoint. All software (compilers, IDEs, libraries, tools) is available for free.

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9. INFORMATION ABOUT [APA4A] FUNCTIONAL PROGRAMMING (2EC)

9.1. Content

Functional Programming is a programming style where mathematical functions (that is, expressions with input parameters and an output result) are fun much faster to develop and easier to test/debug, and is also very suitable for parallel applications. Lisp, Haskell, Scala, Elm are examples of purely functional programming languages.

This course aims at exposing you to the functional way of thinking and introducing FP concepts and techniques that have been proven useful in practice. The FP style in your programming practice, "mixed" languages like F# and Python can be investigated.

9.2. Learning outcomes

1. Learning outcome: Analysis

Explanation: You learn to use the functional programming style confidently, understand the pros and cons and recognize, in a programming situation, when to use it.

2. Learning outcome: Design

Explanation: You design programs using functional programming paradigms (modular, reusable, without side-effects).

3. Learning outcome: Realization

Explanation: You program with basic functional concepts like filter, map, reduce, recursion.

4. Learning outcome: Testing

Explanation: You reason about the correctness and time/space complexity of your programs, and measure their practical performance.

5. Learning outcome: Research

Explanation: Throughout the course, you adopt a curious attitude and take charge of your learning process. In the FP module, you will also work on research projects.

6. Learning outcome: Interaction

Explanation: There will be enough opportunity for interaction with your teachers and colleagues.

7. **Learning outcome:** Communication and leadership

Explanation: There will be enough opportunity to describe and present your work, for instance in the research assignment.

9.3. Examination and final grading

9.3.1. Examination

The teachers will provide the students with formative feedback on deliverables and demonstrations. It is expected that this feedback will be processed

The student shows the work, alone or in a group context, but will be given feedback on an individual base.

The student work will be assessed on analysis, design and technical aspects, where the student will be informed whether the learning goals are met or

At these assessments the learning outcomes are determined based on state of the deliverables at that point. Ideally the learning outcomes are proficient assessments. It is expected that the student keeps on progressing on the learning outcomes, not doing so might lead to a lower grade of beginning or

9.3.2. Repairs and resits

During the course, students are facilitated to process formative feedback on their learning goals in their deliverables and activities that are to be recorded

9.3.3. Tools allowed

Except for paper & pen, and a laptop, no additional equipment is needed. The regular rules about plagiarism apply.

9.3.4. Final grading

A student's growth is maintained in his portfolio. The student's growth on each learning outcome is registered according to one of the five levels of mastery

1. undefined,
2. orienting,
3. beginning,
4. proficient,
5. advanced.

After each formative assessment, the level of mastery for all learning outcomes is assessed for the student's deliveries.

At the end of the semester the student will be assessed on his performance of the semester. During the assessment meeting the teacher will use the student's growth. If the student is advanced, the grade will be at least Satisfactory. This does not mean that obtaining proficient is the minimum requirement to achieve a satisfactory grade

9.4. Learning activities

Based on the characteristics of an academic pre-master and based on a highly knowledge-based scientific world, the course is structured according to the study program.

The education is designed according to the principles of the 4C/ID model (four-component instructional design model). Education is pre-structured in terms of

Knowledge, competences and professional skills are further gained and applied in practical assignments, which increase in size, abstraction and complexity

Depending on the subjects, a variety of side-roads will be offered for in-depth knowledge and broader research.

9.5. Learning materials

All course content is in Canvas and Sharepoint. All software (compilers, IDEs, libraries, tools) is available for free.

[\[Return to table of content\]](#)

10. INFORMATION ABOUT [IN-GV] GAME DEVELOPMENT (30EC)

Game Engine Exploration, Playful Interaction & Scripting

10.1. Content

Game Development lets you apply your ICT skills in the most fun and interactive field of all: games!

This semester introduces you to software development using industry-standard game engines. You build confidence in core programming principles with

After starting with the best game development practices and patterns, you'll then get to choose between many of the facets of game development, such as

10.2. Learning outcomes

1. **Learning outcome:** Analysis

You can research basic principles of game development in a methodical fashion and analyze technical and design-related issues to make well-founded

Explanation: Basic principles: game engines and techniques that fit your assignment.

Methodical: use appropriate research methods such as the DOT framework.

Well-founded choices: substantiate your choices with context (DOT framework), examples and link your analysis to your design.

2. Learning outcome: Advice

You can make substantiated recommendations about game development choices, based on research and/or analysis.

Explanation: Substantiated: make sure to base your advice on your knowledge and analyses.

3. Learning outcome: Design

You can design and structure the applied techniques using existing game development frameworks, theories, and models, and substantiate your

Explanation: Techniques: this includes techniques that you use for game mechanisms, but also, for example, audiovisual or aesthetic componen

Frameworks, theories and models: demonstrate that you know how to use for example, UML, ERD, and Design Patterns.

Substantiated: justify your design choices based on your knowledge and analyses.

4. Learning outcome: Realization

You can develop a functional game prototype with the use of a game engine by applying your technical skills based on, and linked to, your design

Explanation: Functional game prototype: a working version of your game that is ready to be tested or played.

Individually and in groups: during the semester you are expected to realize both individual products and to make a substantial contribution to the

5. Learning outcome: Management & Control

You can collaborate effectively on a game development project by using, for example, an iterative process, version control, and basic project man

Explanation: Collaborate effectively: you make a meaningful contribution to the group process and actively participate in group meetings.

Iterative process: work systematically using a cycle of design, development, testing and collecting and analyzing feedback, so that the design car

Version control and project management methods: make visible use of tools such as Git, and methods such as Agile/Scrum.

6. Learning outcome: Professional standard

You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory

Explanation: You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholde

7. Learning outcome: Personal leadership

You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professi

Explanation: You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study caree

10.3. Expected entry profile

No previous knowledge or skills levels are required for entry into this semester. Therefore, all semester 1 profiles are welcome, although a keen interes

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1		

10.4. Outflow profile

This semester is part of a three-semester track on Game-related topics. We suggest directly following up with the main-level semester on Game Design end of your studies.

Game Development is both technically challenging and reliant on good insights into the user experience. Therefore, any user experience of software-or

Below is an explanation of the the expected outflow levels in relation to the HBO-i framework.

	Analysis	Advise	Design	Realise	Manage & control
User interaction	Level: 1 Explanation: students analyse the group assignments to identify the core elements by analysing the wishes of the external client and translating them into a fitting, interactive game design.	Level: 1 Explanation: students advise their project groups on their area of expertise and the client about the best fitting game engine and other technical options.	Level: 1 Explanation: students use an iterative design process, including prototyping, to design the necessary technical building blocks to realize the desired interaction in the game.	Level: 1 Explanation: students realise and user-test at least two game prototypes on their effectiveness in terms of user interaction (i.e., motivation)	Level: 1 Explanation: students demonstrate their project management skills by displaying their products and results, as well as the process leading to them through devlogs, reflections, and presentations in an online portfolio.
Software	Level: 1 Explanation: students link the 'Pizza' assignment to the appropriate game	Level: 1 Explanation: students formulate and discuss specific requirements for the desired game,	Level: 1 Explanation: students design the appropriate game mechanics to go with the overall game,	Level: 1 Explanation: students combine the game mechanics they realise into a full game	Level: 1 Explanation: students manage their projects by making use of industry standard

	engine and analyse the best techniques to realize the wishes of the external client.	based on the available game engines, which they use to formulate technical their advise to the group and their stakeholders.	including the technology on how to make and test them.	prototype, applying multiple advanced software development techniques, which they test with potential end users.	version control systems appropriate to the assignment.
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Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	1	1	1	1	1
Organisational processes					
Infrastructure					
Software	1	1	1	1	1
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

10.5. Examination and final grading

10.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount c

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape p aspect: throughout the entire semester, the assessors monitor your progress.

10.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You recor

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the dev

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning o
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectat development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't c

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using th gives you the opportunity to address and process the feedback received.

10.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discre

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achiev

10.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

10.6. Learning activities

During the semester, lectures and practicals will be offered in the form of workshops. These workshops will deal with subjects related to various game t

Students will report to a dedicated coach/teacher to present and discuss progress on a regular basis, and teachers will provide students with regular fe

The first phase of this semester will consist of 4 weeks of getting to know each other, the semester, as well as the game engines and other commonly u

The second phase consists of a 7-week group project, where you will create your first game, based on an industry case and feedbacked on by a number of peers.

The final phase consists of another 7-week project where you will tackle a new, but similar group challenge.

Each phase ends with a showcase (or playtest) event, and throughout the semester, multiple portfolio reviews will be conducted by the tutors.

10.7. Learning materials

Canvas course including all linked information.

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11. INFORMATION ABOUT [O-IN-IT] INTELLIGENT TECHNOLOGIES (30EC)

Embedded software, Microcontrollers and Hardware interfacing

11.1. Content

This semester, you'll learn to design, analyze, build, and test intelligent solutions that seamlessly integrate hardware and software. You'll develop problem detectors and environmental sensors and various types of motors. You'll also explore the possibilities and limitations of embedded systems. Beyond that, you'll collaborate to achieve a common goal.

Equally important, you'll refine your interpersonal skills in Agile teams, working on a solution for a partner company. You'll apply structured development processes to manage your work.

Semester Coach

The semester coach provides individual guidance and group process guidance. The semester coach is partly responsible for the guidance and assessment of your work, acting situationally by choosing the right role and approach at the appropriate moment.

Points of Contact

The lecturers of this semester are your first point of contact for the topics they are involved with. The semester project leader is available for matters with the semester project.

The semester coach is often a point of contact for personal conversations, such as private situations that impact your studies.

11.2. Learning outcomes

1. Learning outcome: Professional standard

You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory work, and problem solving.

2. Learning outcome: Personal leadership

You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional development.

3. Learning outcome: Applied Procedural Programming for Embedded Systems:

The student applies procedural programming techniques by developing software applications across embedded systems that are fully tested, functional, and documented.

4. Learning outcome: Structured Software Design and Documentation:

The student demonstrates the ability to describe a system and develop a software design using a specified development methodology. This includes a system architecture document and a test plan that adhere to applicable quality standards.

5. Learning outcome: Embedded Systems with Actuators and Sensors:

The student demonstrates the ability to build their own embedded system based on given requirements. The system has multiple actuators and sensors, measurement tools, report on them, and deliver a functioning and tested system. The student is also able to describe the architecture and operation of the system.

6. Learning outcome: System Communication using Transport Protocols:

The student demonstrates that different systems can communicate with each other based on existing transport layers such as TCP/IP and a custom protocol. The student delivers a functioning and tested system along with the corresponding documentation.

11.3. Expected entry profile

Basic programming skills at semester 1 level are advised. Experience with the Arduino platform is an advantage. Example projects include small embedded systems, a pool access based on a card reader, a simple thermostat for controlling an electric heater, etc.

Professional Development BSc 2020

Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
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Professional development	1	1		
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11.4. Outflow profile

After completing this semester students are encouraged to further develop their expertise in:

- Intelligent devices
- Industry 4.0 & the Internet of Things

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes					
Infrastructure			1	1	
Software			1	1	1
Hardware-interfacing	1	1	1	1	1

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

11.5. Examination and final grading

11.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning path. Throughout the entire semester, the assessors monitor your progress.

11.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental

Developmental Feedback Scale

- *Undefined:* You have not yet undertaken any activities to demonstrate the learning outcome.
- *Orienting:* You have made a start and explored possibilities to demonstrate the learning outcome.
- *Beginning:* You have taken initial steps that contribute to demonstrating the learning outcome.
- *Proficient:* You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome.
- *Advanced:* You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using this scale, giving you the opportunity to address and process the feedback received.

11.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a

11.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

11.6. Learning activities

We use a Blended Learning approach, with two contact sessions each week. Students must prepare every week before attending these sessions. The material, identify problems or difficulties, and turn these into clear, focused questions.

One evening per week, there is an in-person session of 4 lesson hours at a physical location. On another evening, there is an online session of 2 lessons preparing on their own. The in-person session is mainly for working on assignments. During this session, the teacher gives feedback (written and/or spoken).

11.7. Learning materials

Required:

1. Canvas course. The course contains various learning materials such as task descriptions, workshops and other resources.
2. Laptop. You need a laptop on which to do your work. Linux, Mac or Windows.
3. C Compiler, Git, Code editor, Documentation authoring tools. The required tools are either free/open source or supplied by Fontys

Recommend:

1. An Arduino R3 or R4, or another microcontroller
2. A multimeter (e.g. UNI-T UT132D).
3. Logic Analyzer (e.g. IKALogic SQ series (SQ25,SQ50) or Analog Devices ADALM2000 or Analog Discovery version 2 or 3).
4. Portable power supply or +5v wall plug

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12. INFORMATION ABOUT [IN-IT] INTELLIGENT TECHNOLOGIES (30EC)

Embedded software, Microcontrollers and Hardware interfacing

12.1. Content

This semester, you'll learn to design, analyze, build, and test intelligent solutions that seamlessly integrate hardware and software. You'll develop problem detectors and environmental sensors and various types of motors. You'll also explore the possibilities and limitations of embedded systems. Beyond that, you'll collaborate to achieve a common goal.

Equally important, you'll refine your interpersonal skills in Agile teams, working on a solution for a partner company. You'll apply structured development

Semester Coach

The semester coach provides individual guidance and group process guidance. The semester coach is partly responsible for the guidance and assessment discussions, acting situationally by choosing the right role and approach at the appropriate moment.

Points of Contact

The lecturers of this semester are your first point of contact for the topics they are involved with. The semester project leader is available for matters with

The semester coach is often a point of contact for personal conversations, such as private situations that impact your studies.

12.2. Learning outcomes

1. **Learning outcome:** Applied Procedural Programming for Embedded Systems:

The student applies procedural programming techniques by developing software applications across embedded systems that are fully tested, fun

2. **Learning outcome:** Structured Software Design and Documentation:

The student demonstrates the ability to describe a system and develop a software design using a specified development methodology. This includes a document and a test plan that adhere to applicable quality standards.

3. **Learning outcome:** Embedded Systems with Actuators and Sensors:

The student demonstrates the ability to build their own embedded system based on given requirements. The system has multiple actuators and sensors, measurement tools, report on them, and deliver a functioning and tested system. The student is also able to describe the architecture and operation

4. **Learning outcome:** System Communication using Transport Protocols:

The student demonstrates that different systems can communicate with each other based on existing transport layers such as TCP/IP and a custom layers. The student delivers a functioning and tested system along with the corresponding documentation.

5. **Learning outcome:** Personal leadership

You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional

6. **Learning outcome:** Professional standard

You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory

12.3. Expected entry profile

Basic programming skills at semester 1 level are advised. Experience with the Arduino platform is an advantage. Example projects include small embedded pool access based on a card reader, a simple thermostat for controlling an electric heater, etc.

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1		

12.4. Outflow profile

After completing this semester students are encouraged to further develop their expertise in:

- Intelligent devices
- Industry 4.0 & the Internet of Things

Domain description HBO-i 2023

	Manage & control	Realise	Design	Advise	Analysis
User interaction					
Organisational processes					
Infrastructure		1	1		
Software		1	1	1	1
Hardware-interfacing	1	1	1	1	1

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

12.5. Examination and final grading

12.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape the learning path. Throughout the entire semester, the assessors monitor your progress.

12.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the development

Developmental Feedback Scale

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- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectation

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the feedback scale. This gives you the opportunity to address and process the feedback received.

12.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a

12.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times. Page: 67/396

12.6. Learning activities

There are four main lines.

1. Individual project: You work on a project individually. You can choose from a predefined project or define your own. Each sprint you'll add function
2. Industry project: You work in a group on a project in cooperation with a company partner. The partner company sets the domain context of the project
3. Personal Leadership: You work on personal leadership individually and put it in practice during the semester. You do this in repeating cycles by self-reflection
4. Workshops: Teachers will give various kinds of workshops such as inspirational, explanations, demo's or practice.

12.7. Learning materials

Required:

1. Canvas course. The course contains various learning materials such as task descriptions, workshops and other resources.
2. Laptop. You need a laptop on which to do your work. Linux, Mac or Windows.
3. C Compiler, Git, Code editor, Documentation authoring tools. The required tools are either free/open source or supplied by Fontys

Recommend:

1. An Arduino R3 or R4, or another microcontroller
2. A multimeter (e.g. UNI-T UT132D).
3. Logic Analyzer (e.g. IKALogic SQ series (SQ25,SQ50) or Analog Devices ADALM2000 or Analog Discovery version 2 or 3).
4. Portable power supply or +5v wall plug

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13. INFORMATION ABOUT [O-IN-IM] INTERACTIVE MEDIA (30EC)

13.1. Content

The theme of the semester is Interactive Media. The semester requires students to explore the future field of Media Design in its full scope. Several projects documenting all the material from the project in portfolio forms the basis of the assessment.

The semester offers students the freedom to make their own choices and therefore requires the student to act as a professional-to-be. This means fulfilling feedback, and complying with agreements in the student group.

13.2. Learning outcomes

1. **Learning outcome:** Professional standard
You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory
2. **Learning outcome:** Personal leadership
You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional
3. **Learning outcome:** Learning outcome: Interactive Media: You iteratively create interactive media products that are based on exploratory research

Explanation: Explanation: Iteratively

This means you don't make your product all at once and hope it works. Instead, you:

- Get to know the context of the project
- Create a first version (a prototype).
- Show it to clients, teachers and end users.
- Get feedback from them.
- Use that feedback to improve your product in the next version.
- Repeat this process several times—each version is called an iteration.

The goal is to improve your product step-by-step, based on what you learn along the way. Show these steps in your portfolio.

Create

In this context create not just making things but also applying the theoretical knowledge you will acquire during the semester and using the insights

Interactive media products

These are not final, fully working products but interactive prototypes. They can vary in how detailed they are:

- Low-fidelity prototypes: simple wireframes or sketches.
- Mid-fidelity prototypes: clickable digital mockups.
- High-fidelity prototypes: nearly complete versions that look and feel close to the final product.

Each version helps you and others understand what the product might look like and how it will work.

Exploratory user research

This means your design choices are based on research, not just on guesses or personal opinions. That includes:

- Starting with a clear research question.
- Choosing the right research methods (like interviews, surveys, usability tests, etc.).
- Collecting real insights from users and/or stakeholders.
- Applying those insights in your project to make better decisions in your iterations.

By doing this, you make sure the product you're creating actually fits the needs of real users.

4. **Learning outcome:** Learning outcome: Front-end development: You develop several interactive media products using front-end development languages

Explanation: Explanation: Develop

You write code that results in products that can be used by others. The code itself is well structured.

Front-end development languages

This means you're learning how to build the visual and interactive part of a website — the part users actually see and use. You'll be using:

- HTML – to structure the content (like headings, paragraphs, images)
- CSS – to style your website (like colors, fonts, layout)
- JavaScript – to make your site interactive (like buttons that do things, forms that respond)

Commit code in a version control environment

This is about keeping your code organized, understandable, and trackable using Git. Instead of just throwing all your code online at once, you:

- Create a README file to explain your project and how to use it
- Write comments in your code to explain what it does
- Use commits to save your progress step-by-step, with clear messages (like "added homepage", "fixed form bug")
- You start using version control in group projects

5. Learning outcome: Learning outcome: Visual Design:

You design products that are substantiated and validated by making use of the visual design process and through design research.

Explanation: Explanation: Visual design process

The visual design process consists of the following steps

- Learn from work that others made by looking at it and formulate what makes the work interesting/good (inspiration).
- Start from uncertainty then go from quantity to quality, by making use of design research. Which will lead to the best visual, dynamic and interac
- From analog sketches on paper, to applications created with software like adobe creative suite and Figma.

Design research

- Many variations and iterations based on design elements are needed in all phases of the project
- Get feedback from clients, teachers and end-users
- Select your work
- Present your work to teachers and clients
- Reflect on your work and the process.

Components of visual design

The following design elements must be considered in the design process

- Composition
- Layout
- Typography
- Animation
- Form
- Color

13.3. Expected entry profile

Students who have finished their first semester successfully. An interest in several or at least one of the following topics is of course helpful namely, fro

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	

13.4. Outflow profile

By the end of the semester, you worked on many different topics and products. So you have a background in the topics of visual design, user experien mobile, game design and open learning.

Domain description HBO-i 2023

	Manage & control	Realise	Design	Advise	Analysis
User interaction	1	1	1	1	1
Organisational processes					
Infrastructure					
Software	1	1		1	
Hardware-interfacing					

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	

13.5. Examination and final grading

13.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning path. Throughout the entire semester, the assessors monitor your progress.

13.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record these conversations.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue to grow.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. This gives you the opportunity to address and process the feedback received.

13.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory' grade.

13.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

13.6. Learning activities

1. The semester includes several group projects with external clients. The clients can be a Partner in Education or a collaborating Fontys Institute. The projects are designed to be challenging and to provide you with practical experience.
2. The semester includes an individual project. The student chooses his or her own subject and actively approaches one or more lecturers for individual supervision.
3. Throughout the semester, students work on their portfolio in Portflow. The material for this comes from the group project, the individual project and the personal leadership activities.

The implementation of educational activities may differ in terms of planning from the implementations in Tilburg and Eindhoven. The information in Canvas provides an overview of the activities.

13.7. Learning materials

Canvas course including all linked information.

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14. INFORMATION ABOUT [IN-IM] INTERACTIVE MEDIA (30EC)

14.1. Content

The theme of the semester is Interactive Media. The semester requires students to explore the future field of Media Design in its full scope. Several projects will be undertaken, documenting all the material from the project in portflow forms the basis of the assessment.

The semester offers students the freedom to make their own choices and therefore requires the student to act as a professional-to-be. This means fulfilling responsibilities, giving and receiving feedback, and complying with agreements in the student group.

14.2. Learning outcomes

1. **Learning outcome:** Interactive Media: You iteratively create interactive media products that are based on exploratory research that leads to a goal.

Explanation: Iteratively

This means you don't make your product all at once and hope it works. Instead, you:

- Get to know the context of the project
- Create a first version (a prototype).
- Show it to clients, teachers and end users.
- Get feedback from them.
- Use that feedback to improve your product in the next version.
- Repeat this process several times—each version is called an iteration.

The goal is to improve your product step-by-step, based on what you learn along the way. Show these steps in your portfolio.

Create

In this context create not just making things but also applying the theoretical knowledge you will acquire during the semester and using the insight

Interactive media products

These are not final, fully working products but interactive prototypes. They can vary in how detailed they are:

- Low-fidelity prototypes: simple wireframes or sketches.
- Mid-fidelity prototypes: clickable digital mockups.
- High-fidelity prototypes: nearly complete versions that look and feel close to the final product.

Each version helps you and others understand what the product might look like and how it will work.

Exploratory user research

This means your design choices are based on research, not just on guesses or personal opinions. That includes:

- Starting with a clear research question.
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- Collecting real insights from users and/or stakeholders.
- Applying those insights in your project to make better decisions in your iterations.

By doing this, you make sure the product you're creating actually fits the needs of real users.

2. Learning outcome: Front-end development: You develop several interactive media products using front-end development languages and commit

Explanation: Develop

You write code that results in products that can be used by others. The code itself is well structured.

Front-end development languages

This means you're learning how to build the visual and interactive part of a website — the part users actually see and use. You'll be using:

- HTML – to structure the content (like headings, paragraphs, images)
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- JavaScript – to make your site interactive (like buttons that do things, forms that respond)

Commit code in a version control environment

This is about keeping your code organized, understandable, and trackable using Git. Instead of just throwing all your code online at once, you:

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3. Learning outcome: Visual Design:

You design products that are substantiated and validated by making use of the visual design process and through design research.

Explanation: Visual design process

The visual design process consists of the following steps

- Learn from work that others made by looking at it and formulate what makes the work interesting/good (inspiration).
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- Many variations and iterations based on design elements are needed in all phases of the project
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- Present your work to teachers and clients
- Reflect on your work and the process.

Components of visual design

The following design elements must be considered in the design process

- Composition
- Layout
- Typography
- Animation
- Form
- Color

4. Learning outcome: Professional standard

You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory

5. Learning outcome: Personal leadership

You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional

14.3. Expected entry profile

Students who have finished their first semester successfully. An interest in several or at least one of the following topics is of course helpful namely, from

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1		

14.4. Outflow profile

By the end of the semester, you worked on many different topics and products. So you have a background in the topics of visual design, user experience, mobile, game design and open learning.

Domain description HBO-i 2023

	Manage & control	Realise Design	Advise Analysis		
User interaction	1	1	1	1	1
Organisational processes					
Infrastructure					
Software	1	1		1	
Hardware-interfacing					

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

14.5. Examination and final grading

14.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work you do and the quality of your work.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning path. Throughout the entire semester, the assessors monitor your progress.

14.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record these conversations in your portfolio.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
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- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue to work on it.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. This gives you the opportunity to address and process the feedback received.

14.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory' grade.

14.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times. Page: 72/396

14.6. Learning activities

1. The semester includes several group projects with external clients. The clients can be a Partner in Education or a collaborating Fontys Institute. T
2. The semester includes an individual project. The student chooses his or her own subject and actively approaches one or more lecturers for individ
3. Throughout the semester, students work on their portfolio in Portflow. The material for this comes from the group project, the individual project and

The implementation of educational activities may differ in terms of planning from the implementations in Tilburg and Eindhoven. The information in Can

14.7. Learning materials

Canvas course including all linked information.

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15. INFORMATION ABOUT [ICT5A] INTERNSHIP ICT FALL (30EC)

15.1. Content

NOTE: This description is only for student that leave on internship in the FALL semester. There is a separate description for the spring seme

You will do an internship in the 5th semester of your studies. During that period, you demonstrate in (professional) practice that you have integrated the and thus get a better idea of what it has to offer.

Internships are about working and learning in professional practice. It is an assignment/work that:

- you perform at an institution or company;
- matches the choice of study and what you can do with it later;
- asks you to make the action very clear, so that it can be understood by others, why and for what purpose you did or did not do something;
- results in a portfolio with professional products.

Duration internship

The internship lasts a whole semester: from Fontys week 1 to Fontys week 18. The assessment takes place at the end of the semester (Fontys week 1 accumulate (see internship agreement) can be taken in consultation with the company. Days off and/or company closures, which are mandatory within

An internship must have a minimum of 90 days. You perform these 90 days between Monday Fontys week 1 and Friday Fontys week 18.

Sick leave

Let's hope not, but it is of course possible that you become ill during a (prolonged) period of your internship or miss days on your internship for some of problem to achieve the minimum number of days.

What are internship days?

Internship days are all days on which you work on your internship and/or on your internship assignment. You can include visits that you make in the cor

Internship permission

In order to start arranging an internship, you must have internship permission from the examination board. Please note: before you can actually start th completed and handed in yourself.

Permission to go on an internship is given when you have completed one of the semesters of the second year of study. You will then first follow the oth semester). Hence, you will always have one semester to find a suitable internship. It is recommended to start searching in time. After your internship, y

15.2. Learning outcomes

Learning outcome 1:

[Professional duties] You carry out the professional duties on a junior bachelor level resulting in professional products in line with the IT-area you are w

Learning outcome 2:

[Situation-orientation] You apply your previously acquired knowledge and skills in an authentic context to deliver relevant results for the project and con

Learning outcome 3:

[Professional Standard] You take responsibility for solving an IT issue. You conduct research using a relevant method you select and advise your stake

Learning outcome 4:

[Personal Leadership] You are entrepreneurial around your projects and personal development, you pay attention to your own learning ability and keep

15.3. Examination and final grading

15.3.1. Examination

Key moments of examination

Week 3-5	Projectplan/projectdocument. Formative assessment; the first assessor provides feedback during the first company
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	visit in week 3-5. The second assessor provides online feedback on assignment only.
Week 9/10	Midterm review; formative assessment.
Week 10-17	Hand in concept portfolio with a request for feedback; formative assessment.
Week 17	Portfolio including proces description and achievements. Hand in at the beginning of week 17.
Week 18	Formative assessment during the second company visit in week 18. Final presentation and/or demonstration of the final product, combined with accountability about the execution and evaluation of the process. Feedback and advised grading from internship coach and company coach.
Week 19/20	Summative assessment bij first and second assessor. Criterion-based Interview on learning outcomes taking into account all aspects of your internship.

15.3.2. Repairs and resits

According to the TER this is a practice based semester. Hence there are no options for repair or retake. If you fail your internship, you should either rec

15.3.3. Tools allowed

Not applicable. The regular rules about plagiarism apply.

15.3.4. Final grading

The assessment will take place in week 19. The assessment is taken by two assessors: the first assessor, which is the teacher that guided you during ; based interview (CGI) to determine if all learning outcomes have been sufficiently achieved.

If a student has at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the asse

15.4. Learning activities

The internship is largely independently acquired by you as a student and carried out at an external client. This may be both at home and abroad. This r suitable assignment and internship position. Failing to do so may force you to put you studies on hold until you found a suitable position, since you will

During your internship you will be supervised by someone from the company; your company supervisor. In addition, you will receive guidance from sch This is a co-assessor at the final assessment. During your internship, your internship teacher is your first point of contact from school and also acts as :

15.5. Learning materials

There is no standard teaching material available for the internship. You can, however, use the documents that you can find on Canvas. Here you will fir

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16. INFORMATION ABOUT [ICT5B] INTERNSHIP ICT SPRING (30EC)

16.1. Content

NOTE: This description is only for student that leave on internship in the SPRING semester. There is a separate description for the fall seme

You will do an internship in the 5th semester of your studies. During that period, you demonstrate in (professional) practice that you have integrated the and thus get a better idea of what it has to offer.

Internships are about working and learning in professional practice. It is an assignment/work that:

- you perform at an institution or company;
- matches the choice of study and what you can do with it later;
- asks you to make the action very clear, so that it can be understood by others, why and for what purpose you did or did not do something;
- results in a portfolio with professional products.

Duration internship

The internship lasts a whole semester: from Fontys week 1 to Fontys week 18. The assessment takes place at the end of the semester (Fontys week 1 accumulate (see internship agreement) can be taken in consultation with the company. Days off and/or company closures, which are mandatory within

An internship must have a minimum of 90 days. You perform these 90 days between Monday Fontys week 1 and Friday Fontys week 18.

Sick leave

Let's hope not, but it is of course possible that you become ill during a (prolonged) period of your internship or miss days on your internship for some other problem to achieve the minimum number of days.

What are internship days?

Internship days are all days on which you work on your internship and/or on your internship assignment. You can include visits that you make in the company.

Internship permission

In order to start arranging an internship, you must have internship permission from the examination board. Please note: before you can actually start the internship, the permission must be completed and handed in yourself.

Permission to go on an internship is given when you have completed one of the semesters of the second year of study. You will then first follow the other semester. Hence, you will always have one semester to find a suitable internship. It is recommended to start searching in time (i.e. as soon as you know you want to do an internship).

16.2. Learning outcomes

Learning outcome 1:

[Professional duties] You carry out professional duties at a junior bachelor level, producing professional-grade work aligned with the IT subdomain (IT-a).

Learning outcome 2:

[Situation-orientation] You apply your previously acquired knowledge and skills in a new, authentic context to deliver valuable and relevant results for the company.

Learning outcome 3:

[Professional Standard] Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, plan, execute, and evaluate a project, taking into account ethical, intercultural, and sustainable aspects.

Learning outcome 4:

[Personal Leadership] You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose to act accordingly.

16.3. Examination and final grading

16.3.1. Examination

Key moments of examination

Week 3-5	Project plan/project document. Formative assessment; the first assessor provides feedback during the first company visit in week 3-5.
Week 9/10	Additional feedback moment at school; formative assessment.
Week 10-17	Hand in concept portfolio with a request for feedback; formative assessment.
Week 17	Portfolio including process description and achievements. Hand in at the beginning of week 17.
Week 18	Formative assessment during the second company visit in week 18. Final presentation and/or demonstration of the final product, combined with accountability about the execution and evaluation of the process. Feedback and advised grading from internship coach and company coach.
Week 19/20	Summative assessment by first and second assessor. Criterion-based Interview on learning outcomes taking into account all aspects of your internship.

16.3.2. Repairs and resits

According to the TER this is a practice based semester. Hence there are no options for repair or retake. If you fail your internship, you should either re-enroll in the next year or choose another degree program.

16.3.3. Tools allowed

Not applicable. The regular rules about plagiarism apply.

16.3.4. Final grading

The assessment will take place in week 19. The assessment is taken by two assessors: the first assessor, which is the teacher that guided you during the internship, and the second assessor, which is the teacher that guides you during the criterion-based interview (CGI) to determine if all learning outcomes have been sufficiently achieved.

If a student has at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

16.4. Learning activities

The internship is largely independently acquired by you as a student and carried out at an external client. This may be both at home and abroad. This is a suitable assignment and internship position. Failing to do so may force you to put your studies on hold until you found a suitable position, since you will

During your internship you will be supervised by someone from the company; your company supervisor. In addition, you will receive guidance from school. This is a co-assessor at the final assessment. During your internship, your internship teacher is your first point of contact from school and also acts as a

16.5. Learning materials

There is no standard teaching material available for the internship. You can, however, use the documents that you can find on Canvas. Here you will find

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17. INFORMATION ABOUT [O-STA] INTRODUCTION TO IT (30EC)

17.1. Inhoud

Introductie

Welkom bij het Startsemester van Fontys ICT PRO.

Je staat aan het begin van een nieuwe stap in je carrière. Dit is het begin van een innoverende studie aan Fontys ICT. Een hogeschool waarbij je vana

Voor de één is het wellicht al even geleden dat je met studeren bezig bent geweest, voor de ander is er wellicht een directe overgang van MBO naar HOGESCHOOL. Je combineert je studie met je werk- en privéleven. Die privé-omgeving kan er voor iedereen anders uitzien. Thuiswonend, samenwonend, samenwonen, rekening moeten houden met het feit dat je hier aan begonnen bent.

Dit betekent dat je in het begin op zoek moet gaan naar een goed studieritme. Wat zijn geschikte momenten om te studeren? Hoe zorg je ervoor dat je kunt studeren met vragen of verhalen.

Ons onderwijs is gebaseerd op de visie "student nabij onderwijs en toetsen". Hierbij staat ontwikkelingsgerichte feedback voorop, en die feedback dient de goede kant op te gaan. Wanneer je moet samenwerken in groepen zal iedereen rekening moeten houden met elkaar. Mocht het niet soepel lopen, ga dan

Je leert veelal in de context van levende projecten (challenges). Tijdens het semester moet je eigenaarschap nemen van je eigen leerproces. Je wordt gestimuleerd om gefundeerde keuzes te maken voor je eigen leerpad.

De focus van dit semester ligt op oriëntatie op het gebied van ICT en wat ons onderwijs te bieden heeft, en het maken van een keuze voor je tweede semester. Je wilt inschrijven en bereid je je daarop ook voor.

Daarnaast oriënteer je je op de juiste opleiding: Associate degree of Bachelor. Aan het einde van dit semester kan hier ook nog tussen overgestapt worden

De leeromgeving is zo georganiseerd dat je nauw contact hebt met je docenten en medestudenten. De fysieke leeromgeving, ook wel Fontys Innovative Challenges, leren en ontmoeten van medestudenten en docenten.

We wensen je een succesvolle opleiding toe in een inspirerende omgeving!

17.2. Leeruitkomsten

1. Learning outcome: Professionele standaard

Je past individueel en in teamverband een professionele werkwijze toe op het gebied van projectorganisatie, communicatie met belanghebbende

Explanation: Professionele standaard

Je past de beroepspraktijk toe, zowel individueel als in teamverband, op het gebied van projectorganisatie, communicatie met belanghebbenden,

Projectorganisatie:

Voor een bepaalde opdracht maak je -met begeleiding- een planning (timing & deeltaken) met je teamleden. Je maakt afspraken over de verwachte

Communicatie en rapportage:

Je houdt rekening met de directe belanghebbende wanneer je aan een opdracht werkt. Je let op wat je wilt communiceren en in welke vorm je dit kunt hebben.

Verkenkend onderzoek:

Je onderzoekt en stelt vragen tijdens het oplossingsproces. Je beantwoordt vragen door een passende aanpak te gebruiken: pragmatisch, kritisch

2. Learning outcome: Persoonlijk leiderschap

Je neemt initiatief om feedback te vragen en te reflecteren en benoemt jouw kernwaarden als vertrekpunt voor je studieloopbaan en je professioneel

Explanation: Persoonlijk leiderschap

Je neemt initiatief in het vragen om en reflecteren op feedback. Je identificeert je eigen kernwaarden als basis voor je studieloopbaan en professioneel

Feedback en reflectie:

Je zoekt actief feedback (van collega's en coaches) op je leeractiviteiten die gericht zijn op zowel technische als professionele leeruitkomsten.

Je denkt na over de ontvangen feedback om leermogelijkheden te identificeren en je inspanningen te richten op professionele, persoonlijke en academische. Je herkent je eigen leerbehoeften en staat open voor feedback. Je gebruikt feedback als een hulpmiddel om je vaardigheden en kennis te verbeteren

Kernwaarden:

Door zelfevaluatie en reflectie identificeer je wat je drijft en motiveert en herken je je persoonlijke kwaliteiten en talenten in de context van je persoonlijke

Je maakt weloverwogen keuzes voor je ontwikkeling en studieprogramma, waarbij je ervoor zorgt dat deze aansluiten bij de uitkomst van je reflectie

Je beschrijft je motieven en kwaliteiten in de context van je toekomstige rol als ICT-professional, waarbij je begrijpt hoe ze je carrièrepad en professioneel

3. Learning outcome: Oriëntatie

Je oriënteert je op het brede IT werkveld door je te verdiepen in verschillende onderdelen van het leerplan. Je kiest een vervolgstap die bij je pas

Explanation: Verdiepen:

Door leeractiviteiten in verschillende contexten uit te voeren, maak je kennis met verschillende onderdelen van ons curriculum.

Onderbouwen:

Op basis van deze activiteiten leg je uit waarom je voor een bepaalde volgende stap kiest.

4. Learning outcome: Maken van professionele IT-producten

Je creëert één of meer IT beroepsproducten. Hiervoor analyseer je de probleemcontext, adviseer je over te maken keuzes, ontwerp je een passe

Explanation: Analyseren:

Je onderzoekt de opdracht en begrijpt de randvoorwaarden van je oplossing. Je legt de verzamelde requirements en kwaliteitscriteria vast op een
Voorbeelden van professionele producten: analysedocument, testplan, use cases, user stories, product backlog, procesanalyse.

Advies:

Je legt uit waarom de professionele producten en de keuzes die je hebt gemaakt bijdragen aan het oplossen van het probleem.

Voorbeelden van professionele producten: technische documentatie, productvergelijkingen, adviesrapportage, verbetervoorstellen.

Ontwerp:

Je bedenkt een oplossing op basis van de eisen en kwaliteitscriteria en legt je keuzes vast. Je gebruikt hiervoor standaardmethoden.

Voorbeelden van beroepsproducten: klassendiagram, stroomdiagram, databaseschema, architectuurontwerp, systeemontwerp, hardwareontwerp, gegevensstromen, communicatieprotocol.

Realiseren:

Je maakt je oplossing op basis van je ontwerp, met aandacht voor kwaliteitscriteria.

Voorbeelden van professionele producten: broncode, geïnstalleerde software, screenshots of opnames van werkende software, high fidelity inter

Valideren:

Je test je producten met standaardtechnieken en volgens een standaardwerkwijze en bepaalt of de oplossing voldoet aan de vereisten en kwalite

Voorbeelden van professionele producten: feedback van belanghebbenden, testrapportage, opname van demosessie, gebruikerstests, gebruikst

17.3. Verwacht ingangspatroon

17.4. Uitstroomprofiel

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1		

17.5. Toetsing en beoordeling

17.5.1. Toetsing: Holistisch en Longitudinaal

De beoordeling in dit semester is holistisch en longitudinaal. Holistisch betekent dat alles wat je als student doet meetelt voor de eindbeoordeling: de h

De leeruitkomsten beschrijven het gewenste niveau, maar omdat deze onafhankelijk van het leertraject kunnen worden aangetoond, is er ruimte om ee dingen doet. Dit is het longitudinale aspect: gedurende het hele semester volgen de beoordelaars je vooruitgang.

17.5.2. Feedback: Bevestiging en begeleiding

Docenten begeleiden en beoordelen je vooral door middel van feedback. Je hebt wekelijks gesprekken met een of meer docenten over je werk. Deze f

Daarnaast vinden op vaste momenten in het semester portfolio opleveringen plaats. Hierop krijg je feedback van je docenten op basis van de developr

Ontwikkelingsgerichte feedbackschaal

- *Onbepaald:* Je hebt nog geen activiteiten ondernomen voor het aantonen van de leeruitkomst.
- *Oriënterend:* Je hebt een begin gemaakt en de mogelijkheden verkend om de leeruitkomst aan te tonen.
- *Beginnend:* Je hebt eerste stappen gezet en deze uitgevoerd welke bijdragen aan het aantonen van de leeruitkomst.
- *Geoefend:* Je hebt meerdere keren laten zien dat je een basis gecreëerd hebt om de leeruitkomst aan te tonen. Je zal de leeruitkomst op een vol
- *Gevorderd:* Je hebt meerdere keren laten zien dat je met deze leeruitkomst bezig bent geweest met goed resultaat. Je hebt boven verwachting g blijft voortzetten.

Deze feedback is een momentopname van je huidige groei in de richting van het leerresultaat. Het is mogelijk dat je achteruit gaat in je ontwikkeling als

Binnen het semester zijn er minstens drie van deze formatieve evaluatiemomenten waarbij de beoordelaars feedback geven per leerresultaat aan de h
beoordelaarsvergadering. Dit geeft je de gelegenheid om de ontvangen feedback te behandelen en te verwerken.

17.5.3. Assessment Richtlijnen

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discre

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achiev

17.5.4. Herkansingen en reparaties

Herkansingen of reparaties zijn niet van toepassing op dit semester. Er wordt van je verwacht dat je je werk een onbeperkt aantal keren opnieuw indier

17.6. Leeractiviteiten

Je leert eerst de basis van ICT aan de hand van voorbeelden en opdrachten die je later als ICT-professional tegenkomt. Van daaruit word je gestimuleer
bieden heeft, en het maken van een keuze voor je tweede semester. Daarbij ontwikkel je ook professionele vaardigheden die relevant zijn als ICT-profi

Je kunt ook nog kiezen om te switchen tussen Associate degree of Bachelor.

De leeromgeving is zo ingericht dat je nauw samenwerkt met je docenten en medestudenten.

Gedurende het semester wordt aan drie challenges gewerkt:

- Eerste challenge:
Gedurende de eerste vier weken ga je aan de slag met de eerste challenge van dit semester. Dit voer je individueel uit en heeft een vaste contex
- Tweede challenge:
Gedurende vijf weken ga je individueel aan de slag met een challenge waarbij je enkele keuzemogelijkheden hebt.
- Derde challenge:
In deze laatste challenge van negen weken ga je aan de slag met een grotere challenge waarbij je met een groep van vier tot zes studenten aan
zullen bijdragen. Ook hierbij kan gekozen uit een lijst met challenges.

17.7. Leermaterialen

Canvas bevat al het lesmateriaal.

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18. INFORMATION ABOUT [STA] INTRODUCTION TO IT (30EC)

18.1. Content

Welcome to Startsemester at Fontys ICT.

Our education is based on the vision "student centered education". You will learn in the context of real-life projects. During the semester you will need t
later work as an ICT professional. From this base, you will be stimulated to make well-founded choices for your own learning path.

The focus of this semester is on orientation on the field of ICT and what our education has to offer and making a choice for your second semester. In d
would like to enroll and prepare for it as well:

- Semester 2 topics to choose from
 - AI Basics
 - Business IT & Data Analysis
 - Game Development
 - Intelligent Technologies
 - Interactive Media
 - Network & Cloud Basics
 - Software Design & Engineering
 - Open Learning

You can also choose to switch to

- Associate Degree
- Accelerated program

The learning environment is organised in such a way that you interact closely with your teachers and fellow students. The physical learning environmer
on exercises & projects, learning and meeting up with fellow-students and teachers.

18.2. Learning outcomes

1. Learning outcome: Professional standard

You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory

Explanation: Project organisation

For a given assignment you - with guidance - make a planning (timing & part-tasks) with your team members. You make agreements about the
expected quality.

Communication and reporting

You take the direct stakeholder into account when working on an assignment. You pay attention to what you want to communicate and in which shape you do this. You recognize tasks within the group work, and take on your role within the group. You are aware of bias and the consequences bias could have.

Exploratory research

You explore and ask questions throughout the solution process. You answer questions by using an appropriate approach: pragmatic, critical and based on multiple sources.

2. Learning outcome: Personal leadership

You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional development.

Explanation: Feedback and reflection

You plan your assignments efficiently and realistically. You work result-oriented on your assignments. You recognize learning needs in yourself, are open to feedback, and reflect on this.

Core values

You identify your own core values. You make considered choices for your development and your study program.

3. Learning outcome: Orientation

You orient yourself to the broad IT work field by immersing yourself in various parts of the curriculum. You choose a next step that suits you and your study program.

Explanation: Immersing

By carrying out learning activities in different contexts, you get acquainted with various parts of our curriculum.

Substantiate

You explain, on the basis of these activities, why you choose a particular next step.

4. Learning outcome: Creating professional IT products

You create one or more IT professional products. To do so, you analyse the problem context, advise on choices to be made, design a suitable solution, and realise and validate the solution.

Explanation: Analyse

You research the assignment and understand the preconditions of your solution. You record the collected requirements and quality criteria in a structured way. Examples of professional products: analysis document, test plan, use cases, user stories, product backlog, process analysis.

Advise

You explain why the professional products and the choices you made contribute to solving the problem.

Examples of professional products: technical documentation, product comparisons, advisory reporting, improvement proposals.

Design

You devise a solution based on the requirements and quality criteria and record your choices. You use standard ways to do so.

Examples of professional products: class diagram, flow chart, database diagram, architecture design, system design, hardware design, pseudo-code, communication protocol.

Realise

You make your solution based on your design, paying attention to quality criteria.

Examples of professional products: source code, installed software, screenshots or recording of working software, high fidelity interactive prototype.

Validate

You test your products using standard techniques and according to a standard operating procedure and determine whether the solution meets the requirements.

Examples of professional products: stakeholder feedback, test reporting, recording of demo session, user tests, usability tests, validated professional products.

18.3. Expected entry profile

18.4. Outflow profile

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1		

18.5. Examination and final grading

18.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work you do, the quality of your work, and the progress you make.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning path. Throughout the entire semester, the assessors monitor your progress.

18.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record your progress in a portfolio. In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale.

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome with good results.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue to work on it.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale, giving you the opportunity to address and process the feedback received.

18.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory' or better.

18.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

18.6. Learning activities

Project one:	During the first four weeks you will start working on the first project of this semester. The first project is an individual project and has a fixed context.
Project two:	The next four weeks you will be executing a second individual project. Within this Canvas module three possible projects are being displayed. You are free to choose one of these projects.
Project three and four	For the next five weeks you will work on project three and four simultaneously. The third project is an individual one. The fourth and fifth projects will be a group project. A group exists out of 6 students. You are free to choose any project out of the project list, as long as you take a different one for each project phase.
Project five	During the last five weeks of this semester you will execute the fifth project. Project five is a group project and will be the last project of this semester.

18.7. Learning materials

Canvas contains all the learning materials.

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19. INFORMATION ABOUT [APA1A] LOGIC & SET THEORY (4EC)

19.1. Content

Explore the fundamental principles of logic and set theory in this course. Learn propositional and predicate logic, proof techniques. Study set theory, inductive and mathematical foundations.

Furthermore, it introduces constraint programming and Satisfiability Modulo Theories (SMT) solvers. You'll learn how to model real-world problems using constraints and program verification, and practical applications in areas such as scheduling and resource allocation. The course combines lectures, practicals, and assignments.

19.2. Learning outcomes

19.3. Examination and final grading

19.3.1. Examination

The teachers will provide the students with formative feedback on deliverables and demonstrations. It is expected that this feedback will be processed and used for improvement.

The student shows the work, alone or in a group context, but will be given feedback on an individual base.

The student work will be assessed on analysis, design and technical aspects, where the student will be informed whether the learning goals are met or not.

At these assessments the learning outcomes are determined based on state of the deliverables at that point. Ideally the learning outcomes are proficient assessments. It is expected that the student keeps on progressing on the learning outcomes, not doing so might lead to a lower grade of beginning or not yet proficient.

19.3.2. Repairs and resits

During the course, students are facilitated to process formative feedback on their learning goals in their deliverables and activities that are to be recorded.

19.3.3. Tools allowed

Except for paper & pen, and a laptop, no additional equipment is needed. The regular rules about plagiarism apply.

19.3.4. Final grading

A student's growth is maintained in his portfolio. The student's growth on each learning outcome is registered according to one of the five levels of mastery.

1. undefined,
2. orienting,
3. beginning,
4. proficient,
5. advanced.

After each formative assessment, the level of mastery for all learning outcomes is assessed for the student's deliveries.

At the end of the semester the student will be assessed on his performance of the semester. During the assessment meeting the teacher will use the student's growth. If the student is advanced, the grade will be at least Satisfactory. This does not mean that obtaining proficient is the minimum requirement to achieve a satisfactory grade.

19.4. Learning activities

Based on the characteristics of an academic pre-master and based on a highly knowledge-based scientific world, the course is structured according to the study program.

The education is designed according to the principles of the 4C/ID model (four-component instructional design model). Education is pre-structured in terms of knowledge, competences and professional skills are further gained and applied in practical assignments, which increase in size, abstraction and complexity. Depending on the subjects, a variety of side-roads will be offered for in-depth knowledge and broader research.

19.5. Learning materials

All course content is in Canvas and Sharepoint. All software (compilers, IDEs, libraries, tools) is available for free.

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20. INFORMATION ABOUT [O-IN-NCB] NETWORK & CLOUD BASICS (30EC)

Connecting, Provisioning & Managing IT Services, DevOps

20.1. Content

In today's digital world, IT infrastructure plays a crucial role in business operations. In this semester, you will explore the fundamentals of networking, cloud computing, and DevOps. You will learn how networks are designed, deployed, and secured, covering key concepts such as perimeter defense security and device management. DevOps practices will be introduced to help you automate and streamline IT processes, ensuring efficient collaboration between development and operations. Throughout the semester, you will not only develop technical expertise but also gain insights into real-world IT challenges, preparing you for a career in IT.

20.2. Learning outcomes

- 1. Learning outcome:** Professional standard
You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory research, and documentation.
- 2. Learning outcome:** Personal leadership
You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional development.
- 3. Learning outcome:** Basic Interoperability
After conducting research, you are able to design and implement a sustainable infrastructure based on the specified requirements of a small SME and implement a sustainable infrastructure based on the specified requirements of a small SME, taking into account the different types of data, cloud services, and security measures.
- 4. Learning outcome:** Basic Securing
After conducting research, you are able to design and implement modern and traditional basic security measures across all infrastructure domain.
- 5. Learning outcome:** Basic Automation
After conducting research, you are able to automate the deployment of an infrastructure, application, and business process by programming scripts.

6. Learning outcome: Manage & Control basics

After conducting research, you are able to design an infrastructure based on the specified requirements of a small SME that considers certain ma

20.3. Expected entry profile

A basic understanding of networking and some hands-on experience with an operating system will help you get started in this semester. However, curic knowledge and develop practical expertise.

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1		

20.4. Outflow profile

Domain description HBO-i 2023

	Manage & control	Realise	Design	Advise	Analysis
User interaction					
Organisational processes					
Infrastructure	1	1	1	1	1
Software		1	1	1	
Hardware-interfacing					

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

20.5. Examination and final grading

20.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount c

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape p aspect: throughout the entire semester, the assessors monitor your progress.

20.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You recor

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the deve

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning o
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectat development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't c

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using th gives you the opportunity to address and process the feedback received.

20.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessor.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory' or better.

20.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

20.6. Learning activities

During this semester, you and your team will analyze, design, and realize an IT solution for a real-world company. You will apply your knowledge of networking and cloud technologies to design, implement, and manage the solution.

Throughout the project, you will manage and control your solution, ensuring it meets business requirements while maintaining security and performance.

Your learning journey will be supported by a mix of practical exercises, workshops, and challenges, helping you develop hands-on skills in perimeter defense security and device management. Tutors and coaches will guide you in refining your technical and professional skills.

By the end of the semester, you will have built a solid foundation in networking and cloud technologies, enabling you to design, implement, and manage the solution.

20.7. Learning materials

A Canvas course featuring the necessary learning resources

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21. INFORMATION ABOUT [IN-NCB] NETWORK & CLOUD BASICS (30EC)

Connecting, Provisioning & Managing IT Services, DevOps

21.1. Content

In today's digital world, IT infrastructure plays a crucial role in business operations. In this semester, you will explore the fundamentals of networking, cloud computing, and DevOps practices.

You will learn how networks are designed, deployed, and secured, covering key concepts such as perimeter defense security and device management.

DevOps practices will be introduced to help you automate and streamline IT processes, ensuring efficient collaboration between development and operations.

Throughout the semester, you will not only develop technical expertise but also gain insights into real-world IT challenges, preparing you for a career in IT.

21.2. Learning outcomes

- Learning outcome:** Professional standard
You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory research, and problem-solving.
- Learning outcome:** Personal leadership
You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional development.
- Learning outcome:** Basic Interoperability
After conducting research, you are able to design and implement a sustainable infrastructure based on the specified requirements of a small SME and implement a sustainable infrastructure based on the specified requirements of a small SME, taking into account the different types of data, cloud services, and network topologies.
- Learning outcome:** Basic Securing
After conducting research, you are able to design and implement modern and traditional basic security measures across all infrastructure domain.
- Learning outcome:** Basic Automation
After conducting research, you are able to automate the deployment of an infrastructure, application, and business process by programming scripts.
- Learning outcome:** Manage & Control basics
After conducting research, you are able to design an infrastructure based on the specified requirements of a small SME that considers certain management and control aspects.

21.3. Expected entry profile

A basic understanding of networking and some hands-on experience with an operating system will help you get started in this semester. However, curiosity and a willingness to learn are essential to develop practical expertise.

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1		

21.4. Outflow profile

After completing this semester, we recommend continuing with Advanced Networking & Cloud Technologies, where you will deepen your knowledge in you for various IT roles.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes					
Infrastructure	1	1	1	1	1
Software		1	1	1	
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

21.5. Examination and final grading

21.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount c

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape p aspect: throughout the entire semester, the assessors monitor your progress.

21.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You recor

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the deve

Developmental Feedback Scale

- *Undefined:* You have not yet undertaken any activities to demonstrate the learning outcome.
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This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't c

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using th gives you the opportunity to address and process the feedback received.

21.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discr

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achiev

21.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

21.6. Learning activities

During this semester, you and your team will analyze, design, and realize an IT solution for a real-world company. You will apply your knowledge of net

Throughout the project, you will manage and control your solution, ensuring it meets business requirements while maintaining security and performanc

Your learning journey will be supported by a mix of practical exercises, workshops, and challenges, helping you develop hands-on skills in perimeter se and coaches will guide you in refining your technical and professional skills.

By the end of the semester, you will have built a solid foundation in networking and cloud technologies, enabling you to design, implement, and manag

21.7. Learning materials

22. INFORMATION ABOUT [IN-OLB] OPEN LEARNING (30EC)

22.1. Content

ICT & Personalized Program is a route where you have a lot of freedom and flexibility to determine your personal focus and structure, within the bound Program. This gives great opportunities for starting multi-disciplinary projects. Projects should be IT related, based on relevant questions and developpr

Lots of freedom, but remember:

“Freedom comes with responsibility”

This large degree of freedom does not mean that there are no obligations. A large amount of autonomy and entrepreneurial behaviour is expected.

22.2. Learning outcomes

1. Learning outcome: Professional standard

You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory

2. Learning outcome: Personal leadership

You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professi

22.3. Expected entry profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1		

22.4. Outflow profile

The student builds a competence profile based upon the basic architectural layers (level 1) from OE1, this is taken to the next step at level 1. In additio
layer level grows with the phase of the students' study. Phase OE1, OE2 ~ level 1, phase OE3,4,5, ~ level 2 and phase OE6,7,8, ~ level 3.

A balanced competence profile does not have too many competencies (no focus) but also not too little (no widening). At least all 9 competences have t

Within Open Learning, students define their own dynamic competence profile per semester (exit level OEx) under supervision. In doing so, they draw u
successfully ending a phase. A learning outcome is a partial contribution to a competence step.

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

22.5. Examination and final grading

22.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount c

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape p
aspect: throughout the entire semester, the assessors monitor your progress.

22.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You recor

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the deve

Developmental Feedback Scale

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- *Orienting:* You have made a start and explored possibilities to demonstrate the learning outcome.

- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectation in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue to grow.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the feedback tool. This gives you the opportunity to address and process the feedback received.

22.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory'.

22.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

22.6. Learning activities

The student will determine the general lines of the content in this programme him or herself. Basically everything in IT should be possible. Because even the same speed, ... This means all workshops, presentations, etc., that will provide content, will be planned and scheduled on-demand or when the content is needed.

To facilitate this the student will create his/her own personal profile within the HBO-i Competence Framework, this framework gives a systematic overview of the competences. The student will be invited for an intake interview. This interview is about motivation and a possible context to use within open learning. In an intake document you will find the "open learning" definition. The intake interview will be an evaluation to see whether you really understand what open learning is about, so you know what you are getting into.

Because everyone has his own track, it is impossible to plan a curriculum that fits for everyone. Learning happens asynchronous, not everybody does the same things at the same time. The teachers notice it is necessary.

For example, workshops of more general applicability are offered to all students, like Scrum-101, Research, Concept validation, while more specific ones are offered to students who are interested in a specific area.

22.7. Learning materials

Apart from the Canvas LMS environment, anything which would be applicable can and will be used.

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23. INFORMATION ABOUT [SES] SIGNALS & EMBEDDED SYSTEMS (30EC)

HBO-ICT Minor

23.1. Content

The Minor **Signals and Embedded Systems** is aimed at students, who take an interest into the workings of embedded systems, how signals from the physical world are processed and how they can be used.

This minor explicitly invites students from various profiles to attend this minor but we mostly encourage students that have a background in ICT, Engineering or a related field.

The semester focuses on performing authentic, methodically structured, professional tasks, which contribute to new insights for the professional field. This includes setting up and monitoring their own learning path.

The research projects of this minor are based on real life problems from our Partners in Innovation and from the lectorates. You will work on one of these projects.

23.2. Learning outcomes

1. **Learning outcome:** Professional standard
You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory research, and the use of tools.
2. **Learning outcome:** Personal leadership
You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional development.
3. **Learning outcome:** You show insight into a combination of digital techniques for building embedded systems, versus the behavior of different kinds of signals.
Explanation: In the physical world all kinds of signals exist: audible, visual, electrical, mechanical, radio-frequent, and many more. You characterize and control yourself. This doesn't mean that you will have to know everything about every type of signal, but it means that you will know how to characterize and control yourself.
4. **Learning outcome:** Starting with a concrete application in mind, you provide a design for applying digital techniques, versus handling input signals.
Explanation: We aim at building well-behaved applications, and in this context it means that you create a design that can cope with the signals from the physical world, versus digital techniques.
5. **Learning outcome:** You set up and make use of a well-managed development environment. You provide a thorough transfer of knowledge and experience to other teams.
Explanation: Setting up a development environment is more than just having the right editor and compiler available. We expect that you will be able to transfer your knowledge and experience to other teams.

23.3. Expected entry profile

This minor explicitly invites students from various profiles to attend this minor but we mostly encourage students that have a background in ICT, Engine

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes					
Infrastructure					
Software	1	1	1	1	1
Hardware-interfacing	1	1	1	1	1

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

23.4. Outflow profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes					
Infrastructure					
Software	2	1	2	2	2
Hardware-interfacing	2	2	2	2	2

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

23.5. Examination and final grading

23.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount c

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape p aspect: throughout the entire semester, the assessors monitor your progress.

23.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You recor

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the dev

Developmental Feedback Scale

- *Undefined:* You have not yet undertaken any activities to demonstrate the learning outcome.
- *Orienting:* You have made a start and explored possibilities to demonstrate the learning outcome.
- *Beginning:* You have taken initial steps that contribute to demonstrating the learning outcome.
- *Proficient:* You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning o

- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectation in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue to grow.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the feedback tool. This gives you the opportunity to address and process the feedback received.

23.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory'.

23.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

23.6. Learning activities

With this semester, students choose a learning environment with a flexible learning pathway, and a high degree of self-management. Students make their own learning content is strongly dependent on the selected projects as well. Because everyone has his own track, it is impossible to plan a curriculum that fits everyone.

Learning happens asynchronously for these reasons: not every student does the same things, and the pace and entry level of students also differs. This is necessary.

More general workshops that address basics for all students, are mostly planned towards the beginning of the semester. Such workshops may be about the basics of programming.

Blended education

The semester works with blended learning, whereby, in addition to the activities on location, online education can take place that matches content and the pace of the semester is explained in the Canvas course. The decision to have parts of the education online is reserved for the degree programme.

23.7. Learning materials

This minor focuses on the interaction of application programs with the world around us, and how to accommodate for that in a controlled way. This means that you will use tools where needed. Such may be multimeters, oscilloscopes, soldering stations, all kinds of sensors and actuators, and more. It depends mostly on the project.

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24. INFORMATION ABOUT [O-IN-SDE] SOFTWARE DESIGN & ENGINEERING (30EC)

Software Life Cycle, Software Development, Design & Testing

24.1. Inhoud

In dit semester worden de beginselen van het Object Georiënteerd ontwerpen en programmeren behandeld. Er wordt voortgegaan op de stof van semester 1. We gaan object georiënteerd (OO) programmeren en daarbij allerlei OO-principes (SOLID) implementeren teneinde de kwaliteit van de code te verbeteren en de applicatie foutbestendig te maken en te houden.

24.2. Leeruitkomsten

1. Learning outcome: Professionele standaard

Je past individueel en in teamverband een professionele werkwijze toe op het gebied van projectorganisatie, communicatie met belanghebbenden en verslaggeving.

Explanation: * Project Organization: You collaborate functionally with your group and involved stakeholders (clients and teachers) in an iterative way. * Communication: You hold regular, goal-oriented meetings with your team. You proactively keep stakeholders informed about the process, progress and results. * Exploratory Research: You use reliable and relevant sources to support your decisions. You validate sources for reliability and relevance to the project. * Reporting: You provide stakeholders with 'artifacts'. Artifacts may include analysis documents, design documentation, code, and working software.

2. Learning outcome: Persoonlijk leiderschap

Je neemt initiatief om feedback te vragen en te reflecteren en benoemt jouw kernwaarden als vertrekpunt voor je studieloopbaan en je professionele ontwikkeling.

Explanation: * Feedback, Reflection, and Core Values: You regularly reflect on your progress by seeking feedback from team members and stakeholders.

3. Learning outcome: Learning Outcome 1 (Analyzing): You analyze and document validated user specifications based on stakeholder feedback and requirements.

Explanation: * User Specifications: You gather and validate the requirements stakeholders place on the application. These specifications are precise and measurable. * Documenting Specifications: You document these specifications in an analysis document and ensure they are validated by stakeholders, for example by using a review process.

4. Learning outcome: Learning Outcome 2 (Designing): You methodically translate validated user specifications into software and database designs.

Explanation: * Software and Database Designs: You create technical designs (such as architecture diagrams, domain models, and database designs). * Maintainable and Secure: The design is flexible, considering scalability, security, and future changes.

5. Learning outcome: Learning Outcome 3 (Implementation): You build and deliver multiple secure and maintainable applications (at least one of which is a web application). You use a relational database for persistence of the data relevant to your application.

Explanation: * Iterative Software Development Process: You develop applications by iteratively building, extending, and maintaining them, based on feedback and changing requirements.

- * Relevant Software Principles: You apply standard software development principles, such as OO principles and SOLID, in the design and implementation.
- * Database: You integrate a relational database into the application and ensure it communicates efficiently and securely with the software.

6. **Learning outcome:** Learning Outcome 4 (Managing): You use techniques such as version control, (automated) testing, and an iterative process

- Explanation:**
- * Version Control: You use a version control system (such as Git) to manage changes without disrupting existing functionality.
 - * Testing: You define and execute tests (such as unit tests, integration tests, and acceptance tests) to ensure the functionality and stability of the software.
 - * Iterative Process: You continuously improve code quality through small incremental changes and testing.

24.3. Verwacht ingangspatroon

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes					
Infrastructure					
Software		1	1	1	
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1		

24.4. Uitstroompatroon

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes					
Infrastructure					
Software	1	1	1	1	1
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

24.5. Toetsing en beoordeling

24.5.1. Toetsing: Holistisch en Longitudinaal

De beoordeling in dit semester is holistisch en longitudinaal. Holistisch betekent dat alles wat je als student doet meetelt voor de eindbeoordeling: de hele leeruitkomst.

De leeruitkomsten beschrijven het gewenste niveau, maar omdat deze onafhankelijk van het leertraject kunnen worden aangetoond, is er ruimte om een andere manier te kiezen. Dit is het longitudinale aspect: gedurende het hele semester volgen de beoordelaars je vooruitgang.

24.5.2. Feedback: Bevestiging en begeleiding

Docenten begeleiden en beoordelen je vooral door middel van feedback. Je hebt wekelijks gesprekken met een of meer docenten over je werk. Deze feedback is gericht op het verbeteren van je werk.

Daarnaast vinden op vaste momenten in het semester portfolio opleveringen plaats. Hierop krijg je feedback van je docenten op basis van de ontwikkelingsgerichte feedbackschaal

- **Onbepaald:** Je hebt nog geen activiteiten ondernomen voor het aantonen van de leeruitkomst.
- **Oriënterend:** Je hebt een begin gemaakt en de mogelijkheden verkend om de leeruitkomst aan te tonen.
- **Beginnend:** Je hebt eerste stappen gezet en deze uitgevoerd welke bijdragen aan het aantonen van de leeruitkomst.
- **Geoefend:** Je hebt meerdere keren laten zien dat je een basis gecreëerd hebt om de leeruitkomst aan te tonen. Je zal de leeruitkomst op een vol
- **Gevorderd:** Je hebt meerdere keren laten zien dat je met deze leeruitkomst bezig bent geweest met goed resultaat. Je hebt boven verwachting g blijft voortzetten.

Deze feedback is een momentopname van je huidige groei in de richting van het leerresultaat. Het is mogelijk dat je achteruit gaat in je ontwikkeling al

Binnen het semester zijn er minstens drie van deze formatieve evaluatiemomenten waarbij de beoordelaars feedback geven per leerresultaat aan de h beoordelaarsvergadering. Dit geeft je de gelegenheid om de ontvangen feedback te behandelen en te verwerken.

24.5.3. Assessment Richtlijnen

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discre

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achiev

24.5.4. Herkansingen en reparaties

Herkansingen of reparaties zijn niet van toepassing op dit semester. Er wordt van je verwacht dat je je werk een onbeperkt aantal keren opnieuw indier

24.6. Leeractiviteiten

We maken gebruik van *Blended Learning*, waarbij er wekelijks twee contactmomenten zijn. Studenten dienen zich wekelijks voor te bereiden alvorens bij deze activiteit niet om een beoordeling, maar om te oefenen met de leerstof en knelpunten/problemen onderkennen en deze te vertalen naar doelge

Een avond per week is er gedurende 4 lesuren een contactmoment op locatie en een andere avond is er gedurende 2 lesuren een online bijeenkomst, wegnemen van knelpunten bij het individuele voorbereiden. Het contactmoment op locatie is primair voor het behandelen van de opdrachten waarbij d

Semestercoach:

Iedere student krijgt aan het begin van ieder semester een betrokken vakdocent toegewezen als semestercoach. Gedurende het semester zal je meer je professionele leerdoelen. Daarnaast is de semestercoach jouw eerste aanspreekpunt en vertrouwenspersoon binnen het semester. Er wordt verwacht dat de student zelf het initiatief neemt om een gesprek met de semestercoach te voeren.

24.7. Leermaterialen

Deze cursus wordt ondersteund door een onderdeel in de online leeromgeving Canvas. Daarnaast zijn andere online bronnen, zoals forums, online cu

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25. INFORMATION ABOUT [IN-SDE] SOFTWARE DESIGN & ENGINEERING (30EC)

Software Life Cycle, Software Development, Design & Testing

25.1. Content

This semester you will search for what it means to be a good Software Engineer. To make your semester a success, we expect a **proactive** attitude fro or not. **Use feedback to learn and improve your work and attitude.** These are essential qualities for the software engineer.

Dit semester gaan we uitzoeken wat het betekent om een goede Software Engineer te zijn. Om van jouw semester een succes te maken verwachten w aan medestudenten, je peers, of het nou af is of niet. **Gebruik feedback om te leren en je werk en houding te verbeteren.** Dit zijn essentiële kwalite

25.2. Learning outcomes

1. **Learning outcome:** Learning Outcome 1 (Analyzing): You analyze and document validated user specifications based on stakeholder feedback a

Explanation: * User Specifications: You gather and validate the requirements stakeholders place on the application. These specifications are pric
* Documenting Specifications: You document these specifications in an analysis document and ensure they are validated by stakeholders, for ex

2. **Learning outcome:** Learning Outcome 2 (Designing): You methodically translate validated user specifications into software and database design

Explanation: * Software and Database Designs: You create technical designs (such as architecture diagrams, domain models, and database des
* Maintainable and Secure: The design is flexible, considering scalability, security, and future changes.

3. **Learning outcome:** Learning Outcome 3 (Implementation): You build and deliver multiple secure and maintainable applications (at least one of w
You use a relational database for persistence of the data relevant to your application.

Explanation: * Iterative Software Development Process: You develop applications by iteratively building, extending, and maintaining them, basec

- * Relevant Software Principles: You apply standard software development principles, such as OO principles and SOLID, in the design and implementation of the software.
- * Database: You integrate a relational database into the application and ensure it communicates efficiently and securely with the software.

4. **Learning outcome:** Learning Outcome 4 (Managing): You use techniques such as version control, (automated) testing, and an iterative process

- Explanation:**
- * Version Control: You use a version control system (such as Git) to manage changes without disrupting existing functionality.
 - * Testing: You define and execute tests (such as unit tests, integration tests, and acceptance tests) to ensure the functionality and stability of the software.
 - * Iterative Process: You continuously improve code quality through small incremental changes and testing.

5. **Learning outcome:** Professional standard

You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory research, and reporting.

- Explanation:**
- * Project Organization: You collaborate functionally with your group and involved stakeholders (clients and teachers) in an iterative process.
 - * Communication: You hold regular, goal-oriented meetings with your team. You proactively keep stakeholders informed about the process, progress, and challenges.
 - * Exploratory Research: You use reliable and relevant sources to support your decisions. You validate sources for reliability and relevance to the project.
 - * Reporting: You provide stakeholders with 'artifacts'. Artifacts may include analysis documents, design documentation, code, and working software.

6. **Learning outcome:** Personal leadership

You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional development.

Explanation: * Feedback, Reflection, and Core Values: You regularly reflect on your progress by seeking feedback from team members and stakeholders.

25.3. Expected entry profile

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1		

25.4. Outflow profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes					
Infrastructure					
Software	1	1	1	1	1
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

25.5. Examination and final grading

25.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work you do, the quality of your work, and the progress you make. The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning path. Throughout the entire semester, the assessors monitor your progress.

25.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record your progress and challenges in a portfolio. In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale.

Developmental Feedback Scale

- *Undefined*: You have not yet undertaken any activities to demonstrate the learning outcome.
- *Orienting*: You have made a start and explored possibilities to demonstrate the learning outcome.
- *Beginning*: You have taken initial steps that contribute to demonstrating the learning outcome.
- *Proficient*: You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome.
- *Advanced*: You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectation in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue to work on it.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the feedback tool. This gives you the opportunity to address and process the feedback received.

25.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory'.

25.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

25.6. Learning activities

Semester 2 is made up of many different learning activities. These activities are guided by the following teacher roles and stakeholders: Semester Coordinator, Lecturers, and Tutors.

1. Software Jumpstart: Circus Train Challenge

During the first two weeks you will be working on "Circus Train" project. This project will help you and your teachers to realize what you know well and what you need to learn.

- Create functional & non-functional requirements for a given case

Design

- Design a UI wireframe, based on the earlier created requirements
- Design a relational database, based on the earlier created requirements

Realise

- Implement a static webpage with HTML and CSS, based on the earlier created UI wireframe and functional requirements
- Create a database based on the earlier created database design
- Implement code, with an Object Oriented (OO) approach, no Object Relational Mapper (ORM), to send SQL queries to a database server

Manage & Control

- Debugging
- Unit testing & acceptance testing based on a test plan that is created by using the earlier created requirements

For professional development:

- Introduction to Learning outcomes
- Planning: intro, Research: intro, Portfolio: intro

2. Projects and Workshops

Semester 2 features two main projects: a Group Project (GP) and an Individual Project (IP), supplemented by various workshops.

2.1. Group Project (PiE)

The Group Project involves teams of 6 students, alternatively also group of 5-7 students can be created. Projects are provided by our Partners in Education. We would like to have one PiE for a group of 18 students (3 groups of 6 students per PiE). If we reflect on that on an entire OIL, there is need for about 18 PiEs. The students enter a cycle of iterations, and deliver project updates every three weeks to the PiE. In the end of the semester there is a show case where the students present their work.

2.2. Individual Project

Students undertake a multi-layered web application project based on object-oriented (OO) principles, crafted from a personal concept. This project is related to the learning outcomes. Direct references to Agile methodologies are reserved for Semester 3.

2.3. Workshops

Workshop sessions are designed around a continuation of tasks, either from previous workshops or new starting points for absentees or those preferring to start at the beginning of the semester.

25.7. Learning materials

All learning materials are available online, through the Learning Management System (LMS) Canvas.

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26. INFORMATION ABOUT [APA3A] STATISTICS (4EC)

26.1. Content

Statistics is about collecting and interpreting data using mathematical instruments, domain knowledge and common sense. Although mostly not explicit advanced topics like decision theory or data science.

In this introductory course, you will first learn how to summarize data using meaningful plots and numbers (*descriptive statistics*). Then you will learn how to test hypotheses (*inferential statistics*). Along the way, you will also train your ability to recognize statistical traps and not misuse the statistical tools.

26.2. Learning outcomes

1. Learning outcome: Analysis

Explanation: You learn to use correct statistical language (variables, datasets, distributions, conditions ...) to model relevant practical problems.

2. Learning outcome: Realization

Explanation: You will design and conduct a small statistical study of your own, from data collection to hypothesis testing and interpreting the results.

3. Learning outcome: Testing

Explanation: You reason about the correctness and trustworthiness of statistical models by using concepts like confidence intervals.

4. Learning outcome: Research

Explanation: Throughout the course, you adopt a curious attitude and take charge of your learning process. In the Statistics module, you will also

5. Learning outcome: Interaction

Explanation: There will be enough opportunity for interaction with your teachers and colleagues.

6. Learning outcome: Communication and Leadership

Explanation: There will be enough opportunity to describe and present your work, for instance in the research assignment.

26.3. Examination and final grading

26.3.1. Examination

The teachers will provide the students with formative feedback on deliverables and demonstrations. It is expected that this feedback will be processed

The student shows the work, alone or in a group context, but will be given feedback on an individual base.

The student work will be assessed on analysis, design and technical aspects, where the student will be informed whether the learning goals are met or not.

At these assessments the learning outcomes are determined based on state of the deliverables at that point. Ideally the learning outcomes are proficient assessments. It is expected that the student keeps on progressing on the learning outcomes, not doing so might lead to a lower grade of beginning or not yet

26.3.2. Repairs and resits

During the course, students are facilitated to process formative feedback on their learning goals in their deliverables and activities that are to be recorded

26.3.3. Tools allowed

Except for paper & pen, and a laptop, no additional equipment is needed. The regular rules about plagiarism apply.

26.3.4. Final grading

A student's growth is maintained in his portfolio. The student's growth on each learning outcome is registered according to one of the five levels of mastery

1. undefined,
2. orienting,
3. beginning,
4. proficient,
5. advanced.

After each formative assessment, the level of mastery for all learning outcomes is assessed for the student's deliverables.

At the end of the semester the student will be assessed on his performance of the semester. During the assessment meeting the teacher will use the student's growth. If the student is advanced, the grade will be at least Satisfactory. This does not mean that obtaining proficient is the minimum requirement to achieve a satisfactory grade.

26.4. Learning activities

Based on the characteristics of an academic pre-master and based on a highly knowledge-based scientific world, the course is structured according to the study program.

The education is designed according to the principles of the 4C/ID model (four-component instructional design model). Education is pre-structured in terms of

Knowledge, competences and professional skills are further gained and applied in practical assignments, which increase in size, abstraction and complexity.

Depending on the subjects, a variety of side-roads will be offered for in-depth knowledge and broader research.

26.5. Learning materials

All course content is in Canvas and Sharepoint. All software (compilers, IDEs, libraries, tools) is available for free.

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27. INFORMATION ABOUT [APA4B] SYNCHRONIZATION (2EC)

27.1. Content

Dive into the intricate world of synchronization and deadlock in this course on managing concurrent systems. Gain a deep understanding of semaphore programming, including race conditions and mutual exclusion. Learn techniques to detect, prevent, and resolve deadlocks, ensuring the smooth operation of concurrent applications.

One cannot board a plane before buying a ticket. Similarly, one cannot check e-mails before validating the appropriate credentials (for example, user name and password).

Daily life is full of synchronization, computer systems as well. Troubles are very subtle: solutions may seem to work in most of the cases, but we are often faced with situations where they do not.

27.2. Learning outcomes

1. **Learning outcome:** In addition to the general Academic Preparations learning outcomes, the student is able to: understand, recognize and solve synchronization problems.
2. **Learning outcome:** apply the common primitives like mutexes, semaphores, conditional variables
3. **Learning outcome:** apply common synchronization design patterns
4. **Learning outcome:** build a visual metaphor of a problem

27.3. Examination and final grading

27.3.1. Examination

The teachers will provide the students with formative feedback on deliverables and demonstrations. It is expected that this feedback will be processed and used to improve the student's work.

The student shows the work, alone or in a group context, but will be given feedback on an individual base.

The student work will be assessed on analysis, design and technical aspects, where the student will be informed whether the learning goals are met or not.

At these assessments the learning outcomes are determined based on state of the deliverables at that point. Ideally the learning outcomes are proficient assessments. It is expected that the student keeps on progressing on the learning outcomes, not doing so might lead to a lower grade of beginning or not yet.

27.3.2. Repairs and resits

During the course, students are facilitated to process formative feedback on their learning goals in their deliverables and activities that are to be recorded.

27.3.3. Tools allowed

Except for paper & pen, and a laptop, no additional equipment is needed. The regular rules about plagiarism apply.

27.3.4. Final grading

A student's growth is maintained in his portfolio. The student's growth on each learning outcome is registered according to one of the five levels of mastery.

1. undefined,
2. orienting,
3. beginning,
4. proficient,
5. advanced.

After each formative assessment, the level of mastery for all learning outcomes is assessed for the student's deliveries.

At the end of the semester the student will be assessed on his performance of the semester. During the assessment meeting the teacher will use the student's portfolio. If the student is advanced, the grade will be at least Satisfactory. This does not mean that obtaining proficient is the minimum requirement to achieve a satisfactory grade.

27.4. Learning activities

Based on the characteristics of an academic pre-master and based on a highly knowledge-based scientific world, the course is structured according to the study program.

The education is designed according to the principles of the 4C/ID model (four-component instructional design model). Education is pre-structured in terms of knowledge, competences and professional skills.

Knowledge, competences and professional skills are further gained and applied in practical assignments, which increase in size, abstraction and complexity.

Depending on the subjects, a variety of side-roads will be offered for in-depth knowledge and broader research.

27.5. Learning materials

28. INFORMATION ABOUT [MA-AMD] AI, MACHINE LEARNING & DATA (30EC)

Domain Analysis, Exploratory Data Analysis, Data Understanding, Prediction Modelling

28.1. Content

In this semester you will learn:

- to aggregate and prepare given datasets as well as other (open) datasets and use them in data analysis and identify opportunities for predictive a
- to use findings from data analysis to preprocess data, apply machine learning algorithms and evaluate the quality and usefulness of produced mc
- to deliver AI projects that follow the three 'Explainable AI' principles of transparency, interpretability, and explainability.

28.2. Learning outcomes

1. Learning outcome: Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve sta sustainable aspects.

2. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with

3. Learning outcome: Explainable AI

You deliver AI projects that follow the three 'Explainable AI' principles of transparency, interpretability, and explainability.

Explanation: Transparency means that the process by which the used input data results in prediction models is reproducible, reliably described & comparable to the domain knowledge and baselines. Explainability refers to the application of tools and methods that turn black-box models into g

4. Learning outcome: Data Preparation & Analysis

You are able to aggregate and prepare given datasets as well as other (open) datasets and use them in data analysis and identify opportunities fo

Explanation: Aggregate means acquiring data from a variety of different sources and in different formats and putting it together into a meaningfu those data is repeatable, transparent to others, and the results are suitable for data analysis. Data analysis implies amongst others: descriptive ai features, principle component analysis, summarization, anomaly detection, etc. and include an impact forecast.

5. Learning outcome: Model Engineering

You are able to use findings from data analysis to preprocess data, apply machine learning algorithms and evaluate the quality and usefulness of

Explanation: Findings from data analysis implies that your choice of data sources and feature selection is based on opportunities for predictive a formats that are more suitable for model training. Apply consists of training of different types of models like classification, regression, etc., as well over/underfitted etc. A defined domain refers to the fact that your evaluation must address the problem and impact definition as given by the dom

28.3. Expected entry profile

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

28.4. Outflow profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	1	1			
Organisational processes					
Infrastructure					
Software	2	2	2	2	2
Hardware-interfacing					

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

28.5. Examination and final grading

28.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount c
The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape p
aspect: throughout the entire semester, the assessors monitor your progress.

28.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You recor
In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the deve
Developmental Feedback Scale

- *Undefined:* You have not yet undertaken any activities to demonstrate the learning outcome.
- *Orienting:* You have made a start and explored possibilities to demonstrate the learning outcome.
- *Beginning:* You have taken initial steps that contribute to demonstrating the learning outcome.
- *Proficient:* You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning o
- *Advanced:* You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectat
development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't c
Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using th
gives you the opportunity to address and process the feedback received.

28.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discre

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achiev

28.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

28.6. Learning activities

During this semester you work on individual challenges, a group project and an open programme. Each one of these works has a different goal and dif
Exercises
These will be used to get acquainted with new theory regarding domain analysis, data provisioning and machine learning.

Individual challenge
All obtained knowledge will be used to develop an own AI model. Multiple iterations will be built on the AI model to improve the data quality, to apply dif
Group project
A project will be provided by Fontys partners for which will be investigated if and how predictive modelling can be an added value for there business pr
performance of the model.
However, this will never fully replace student's preparations and in all cases the responsibility for acquiring theoretical knowledge remains with the stud

28.7. Learning materials

A Canvas course including modules with (self) study materials, inspiration sessions and external sources is available.
[\[Return to table of content\]](#)

29. INFORMATION ABOUT [MA-AAI] APPLIED GENERATIVE AI (30EC)

Generative AI Solution Design, Responsible Implementation, Quality & Validation

29.1. Content

In this semester, you'll jump right into Generative AI with a T-shaped learning approach, specializing as a GenAI Architect, GenAI Engineer, or GenAI E

As a GenAI Architect, you'll figure out the best AI strategies for businesses, design integrated systems, and ensure everything runs ethically. You'll tackle

If you're more into building things from scratch, the GenAI Engineer track is your playground. You'll explore frameworks, tools, and integration methods

For those who love blending creativity with technology, the GenAI Experience Designer role is all about crafting user-friendly interfaces and smooth AI interactions

No matter your specialty, you'll focus on responsible AI—balancing innovation with transparency, accessibility, and privacy. Through hands-on projects, challenges of tomorrow with confidence and style.

29.2. Learning outcomes

1. **Learning outcome:** Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, and ensure sustainable aspects.

2. **Learning outcome:** Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your values.

3. **Learning outcome:** Analysing

You perform a detailed analysis of existing business processes and identify specific opportunities for the use of AI. You research which technologies are suitable for the organisation's needs.

Explanation: GenAI Architect

You translate business needs into concrete architectural options, with a strong focus on optimising processes and aligning with the organisation's strategy.

GenAI Engineer

You investigate the technical feasibility of integrating AI within existing infrastructures. You assess how AI models interact with current systems, identify bottlenecks, and propose solutions.

Experience Designer

You formulate design recommendations that ensure intuitive, inclusive, and trustworthy AI user experiences. You explore how interface choices influence user engagement and satisfaction.

4. **Learning outcome:** Advising

You advise stakeholders on technology choices, architecture and implementation strategies regarding GenAI, by placing various technical and strategic aspects in context.

Explanation: GenAI Architect

You map out current business processes and IT environments to identify where AI can add strategic value. You highlight inefficiencies and opportunities for improvement.

GenAI Engineer

You assess the technical feasibility of AI innovations within existing systems. You identify technical limitations, evaluate compatibility, and determine the required resources.

Experience Designer

You analyse user interaction patterns with a focus on conversation- and dialogue-driven interfaces. Through user observations and interviews, you gather insights to inform design decisions.

5. **Learning outcome:** Designing

You develop a design for an AI system in which the structure and coherence of the various components are clearly represented. You establish how the design aligns with business goals and user needs.

Explanation: GenAI Architect

You develop a detailed blueprint that illustrates how AI can be seamlessly integrated into business processes. The design visualises the interactive components and data flows.

GenAI Engineer

You translate the design into concrete technical specifications and develop an initial working model to test for robustness, scalability, and maintainability.

Experience Designer

You design sketches and interactive prototypes to visualise the user interface and AI-supported interactions. You focus on creating appealing, intuitive, and functional experiences.

6. **Learning outcome:** Realising

You develop, validate, and integrate AI solutions following a structured development process, ensuring that the solution functions according to requirements and user expectations.

Explanation: GenAI Architect

You ensure that the implementation of AI solutions aligns with the organisation's strategic vision and business processes. You monitor whether the solution meets the intended goals.

GenAI Engineer

You develop, validate, and integrate AI models into both existing and newly built systems using proven methods. You ensure technical quality, scalability, and security.

Experience Designer

You collaborate closely with the development team to ensure smooth functioning of user interface and interaction components. You create functional and user-centric experiences.

7. **Learning outcome:** Managing & Controlling

You manage and control the implemented AI solutions through systematic monitoring, evaluation, and risk management. You establish procedures for continuous improvement.

Explanation: GenAI Architect

You oversee the strategic and operational performance of AI solutions within business processes. You coordinate evaluations with stakeholders and implement improvements.

GenAI Engineer

You focus on technical monitoring by implementing systems that track critical performance indicators — such as response times and system errors — and address issues promptly.

Experience Designer

You continuously gather feedback on user experience and analyse interaction data. By organising usability tests and analysing user behaviour, you ensure the solution remains relevant and effective.

29.3. Expected entry profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

29.4. Outflow profile

Domain description HBO-i 2023

	Manage & control	Realise	Design	Advise	Analysis
User interaction	2	2	2	2	2
Organisational processes	2	2	2	2	2
Infrastructure					
Software	2	2	2	2	2
Hardware-interfacing					

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

29.5. Examination and final grading

29.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning aspect: throughout the entire semester, the assessors monitor your progress.

29.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the development

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the scale. This gives you the opportunity to address and process the feedback received.

29.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory' grade.

29.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

29.6. Learning activities

Your Three Learning Pillars



Industry Project

Real-world collaboration in action! Work in three-person teams where each member brings a different specialisation. Tackle authentic challenges from professional teamwork environments.

What you'll gain: Cross-disciplinary collaboration, industry experience, AI agent design, project management skills



Professional Research Project

Deep dive into your specialisation! Work within profile-specific groups alongside peers who share your role. Explore current industry challenges, etc.

What you'll gain: Deep expertise in your chosen field, research methodology, industry insights, ethical considerations



Knowledge Sharing Impact Labs

Become a T-shaped professional! Share your expertise whilst learning from peers' specialisations. Create valuable knowledge artefacts and apply them.

What you'll gain: T-shaped expertise, teaching skills, cross-disciplinary knowledge, collaborative leadership

29.7. Learning materials

A Canvas course featuring the necessary learning resources

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30. INFORMATION ABOUT [MA-BDA] BUSINESS IT & DATA ANALYSIS - MAIN (30EC)

Business Process Optimization, Dashboarding, Low Code Systems Development & Implementation

30.1. Content

The focus of this semester is on a medium to large sized organisation which is part of a supply chain. During this semester you show that you can **systematise informal insight** through an **Exploratory Data Analysis (EDA)**. Besides that, you will **create formal insight** in the **data** using **basic modelling**. Using **data** and **measure** and **monitor** the **usage** using given formats and/or **methods**.

Professional Task

The professional task, in which we work together with our Partners in Education (PiE's), offers you the opportunity to integrate and apply the above skill field of Personal Development (*professional skills*).

Personal Projects

Students carry out two personal projects on an individual basis: a Business personal project based on an authentic task to demonstrate their Business-come into play here.

30.2. Learning outcomes

1. Learning outcome: Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders and sustainable aspects.

Explanation: Methodological approach:

Identify the problem, determine the aim of a solution and pick an appropriate approach that is sound and reproducible.

Formulate project goals:

Define clear and specific objectives that the project aims to achieve.

Stakeholders:

Collaborate effectively with the various groups of interested parties including the client, own team members and teachers.

Applied research:

Apply systematic and practical investigations aimed at addressing specific problems, challenges, or questions with the goal of applying the findings.

Advice:

Offer one or more well-founded recommendations or suggestions with the intention of guiding your stakeholder's actions or decisions to help him/her.

Decisions:

Reach conclusions or resolutions after considering various alternatives.

Reports:

Structured documents that present information, analysis, findings, or recommendations on a particular subject.

Ethical, intercultural, and sustainable aspects:

Know the ethical standards and involve social ethical themes in the judgement process. Recognise your own boundaries and those of others and

2. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with

Explanation: Strengths and weaknesses:

Your internal characteristics that influence your abilities, behaviours, and overall performance. Recognizing and understanding these aspects and

Actions:

Make a well-considered choice of study or personal improvements, enhance your own learning capacity, recognise a learning need in yourself and

Personal growth:

Examine what type of professional you want to be in the long term, which field and type of positions you aspire to and how you can stand out from

Learning attitude:

Be aware, see opportunities and seize them, motivate yourself and others, be able to profile yourself, a team and others. Be aware of your devel

3. Learning outcome: Business analysis; You show that you can systematically analyse organisational processes and advise how these processes

Explanation: The organisation is medium to large size and part of a supply chain and therefore processes are assessed on governance, risk and

Analysis consists of the correlation of the bottlenecks and cause-effect relationships of the organisation within the supply chain. In your analysis y

4. Learning outcome: Exploratory Data Analysis; You create informal insight through an Exploratory Data Analysis (EDA).

Explanation: Creating insight, comprises the definition of the right sources, collecting usable data (subsets) from those sources, joining, transform (timeliness). Finally, you create visuals of the prepared data. The insight creation is in the visualisation of the data and the possible interpretation c

Exploratory Data Analysis (EDA) includes usage of SQL, tidyverse (with R Studio) and end user BI tools.

5. Learning outcome: Explanatory Data Analysis; You create formal insight in the data using basic statistical modelling.

Explanation: Explanatory Data Analysis continues where Exploratory Data Analysis stops: based on the data preparation and the insights gained

Use of models in Explanatory Data Analysis in LE3 Business will be restricted to linear modelling.

6. Learning outcome: Business optimisation; You realise the suggested IT system (or part thereof) based on the design, implement this in the orga

Explanation: Realise the implementation and acceptance of procedures in correlation with new or adapted information provision and control.

Educate and train end users in the renewed processes and use of a new IT.

Build and validate the Proof of Concept.

Structure a standard application, for example, using PowerAutomate.

30.3. Expected entry profile

Highly recommended to do before this semester: "Business IT & Data Analysis Introduction". Knowledge of relational databases and data analytical too "Business IT & Data Analysis Introduction".

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	1	1			
Organisational processes	1	1	1	1	1
Infrastructure	1	1			1
Software			1	1	
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

30.4. Outflow profile

Possible semesters after this: "AI, Machine Learning & Data", "Applied Generative AI" and "Industry 4.0 & Internet of Things" are popular options.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	1	1	1	1	
Organisational processes	2	2	2	2	2
Infrastructure					
Software	2	1	2	1	1
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

30.5. Examination and final grading

30.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning aspect: throughout the entire semester, the assessors monitor your progress.

30.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the development

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the scale. This gives you the opportunity to address and process the feedback received.

30.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory' grade.

30.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

30.6. Learning activities

Didactical Approach

Student-centred learning is the educational vision of Fontys. We provide you with opportunities to learn not only at school but also in the work field. As a result, you

The learning activities as well as the order of carrying them out are already set out for you in advance. This provides structure and the assurance of knowing what you are doing and applying it to practice.

You are in charge of your learning process. Based on the information from the lessons and workshops, you look for knowledge and sharpen your skills. The learning outcomes have been set and these form the basic starting points of this semester.

Learning Activities

Workshops by teachers as well as partners in education, lectures, presentations, demonstrations, videos, assignments, challenges, quizzes, interviews

Blended learning:

Keeping in line with the 21st century trends, blended learning is used to offer a rich and dynamic learning experience. Some instances are

- online courses (Datacamp, Microsoft learning paths, etc.) as well as reference materials and in-house videos to supplement the lessons in the class
- online quizzes following in-person lessons
- flipped classroom blended learning where students first investigate the professional task problem individually through online resources, then work together
- occasional hybrid meetings, for example, while working on the project (Teams+face-to-face)

30.7. Learning materials

Books

Theme	ISBN	Title	Author(s)	Publisher	Price
IT	1615470522	Super Charge Power BI: Power BI Is Better When You Learn to Write DAX	Matt Allington	Holy Macro! Books; 1st edition (June 1, 2018)	€ 29
IT	-	Super Charge Power BI: Power BI Is Better When You Learn to Write DAX (Kindle Edition)	Matt Allington	Holy Macro! Books; 1st edition (June 1, 2018)	€ 14,99
IT	1484251962	Expert T-SQL Window Functions in SQL Server 2019: The Hidden Secret to Fast Analytic and Reporting Queries	Kathi Kellenberger, Clayton Groom, Ed Pollack	Apress, 2nd edition (October 22, 2019)	€ 37,44

Software

Theme	Title	Url	Price
IT	Power BI Desktop	https://powerbi.microsoft.com/en-us/desktop/	Free
IT	PowerAutomate	https://www.linkedin.com/learning/search?entityType=LEARNING_PATH~COURSE&keywords=power%20automate	Free with LinkedIn Learning subscription from Student Square
Professional Development	Hogeschooltaal	https://www.hogeschooltaal.nl	± € 75
Business	LinkedIn Learning	https://www.linkedin.com/learning	Free

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31. INFORMATION ABOUT [MA-CSE] CYBER SECURITY ESSENTIALS (30EC)

Hacking & Defending Against Cyber Security Threats

31.1. Content

IT Security deals with protecting computer environments from security threats such as cyber criminals, hackers, malware, but also fraud. Multiple technical and considerations such as risk analysis, user awareness, security organisation, governance/compliance, privacy, security by design and business continuity

To determine if an IT-environment is secure enough, security tests can be performed. The environment is checked as if a real hacker would try to break the least network technology, web technology and system technology. Just having the knowledge is not enough; the large diversity of systems makes it necessary

Hacking is illegal, and therefore punishable by law. You are only allowed to apply hacking techniques to a client's environment if this client has given you permission. It is important to have some knowledge about law, legislation and responsible disclosure.

All above-mentioned technologies and knowledge areas will be part of this semester's project and learning activities.

31.2. Learning outcomes

1. Learning outcome: Offensive Security

You analyse the security of diverse IT environments, such as a business environment, a consumer product, or a technology, by ethical hacking and

client on security improvements of an IT environment on a physical, technical and organisational level.

Explanation: * Analyse the security: A pen-test or penetration test is a vulnerability assessment or security analysis. In a security analysis all layers.

* Diverse IT environments: A business environment can be analysed by means of a pen-test or red teaming. Security of a consumer product or a to account for a responsible and ethical approach. Technology can be IT, IoT or OT (industrial).

* Ethical hacking according to a methodical approach: Each student must show for at least 2 different types of security analysis: pentesting and v the type and scope a pen-test, confidentiality contract, pen-test plan, pen-test report, and the presentation and delivery of findings and an advice

* Security threats come from threat actors such as cyber criminals, script kiddies, frauds and can be caused by user errors or calamities

Business risks indicate the impact of the threat to the business continuity of a customer in the case the threat really happens

* Common risk analysis method has to be at least a qualitative method of risk analysis that groups the risk in categories low - medium – high

* Advice on security improvements to mitigate the threats and risks must be based on standards, best-practices, and security principles. Security environment, can be physical, technical (on network, system, application and software level) or organisational (procedural, human, legal).

2. Learning outcome: Defensive Security

You realise a secure IT environment, considering functional requirements as well as the following non-functional requirements: security, monitoring

Explanation: * Secure IT environment: IT components, interacting and communicating over a network, with connection to the Internet

* Functional & non-functional requirements: Determine (analyse) the need with respect to security, monitoring, ethics, compliance, usability

* Realise: Translate the requirements into a secure solution design and implement and build this design

* Procedural response: Incident response is a structured methodology for handling security incidents, breaches, and cyber threats. A well-defined prevent future attacks.

* Security incidents: A computer security incident is a violation or imminent threat of violation of computer security policies, acceptable use policies

* Analyse in an efficient and methodical way: It should address the response phases as defined by a standardised framework like NIST

3. Learning outcome: Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, and sustainable aspects.

Explanation: * Research: You collect requirements and approach a challenge from multiple perspectives.

* Advice: You show the skills to inform and advise stakeholders within your project about your projects' deliverables.

* Methodologies: Use frameworks and methodologies from the Cyber Security domain: OWASP top 10, OSINT, the Cyber Kill Chain, CVE and C

4. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with

Explanation: * Goals and Actions: You use this reflection and feedback to look for opportunities that further your development and act on those opportunities

* Adjusting: You investigate and try out strategies to adapt to changes and setbacks. You align your professional ambitions with personal values. `

31.3. Expected entry profile

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

31.4. Outflow profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes	1	1	1		1
Infrastructure	2	2	2	2	2
Software		2			
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

31.5. Examination and final grading

31.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work you do, the quality of your work, and the progress you make throughout the semester.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning path. Throughout the entire semester, the assessors monitor your progress.

31.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record these conversations in your portfolio.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome with good results.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue to work on it.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale, giving you the opportunity to address and process the feedback received.

31.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory' or better.

31.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

31.6. Learning activities

The following learning activities, and corresponding learning tasks, help the student in showing the learning outcomes:

- The **group project** will function as the overall authentic task for the semester. In the project, students will be looking for vulnerabilities in ICT environments.
- For **individual learning**, the student has to acquire and train basic knowledge and skills with regard to the needed techniques and relevant knowledge activities considering research competences and internship preparation.

31.7. Learning materials

The Canvas course contains the relevant study materials and references to, and ideas about, where to look for materials. For some subjects, you are also expected to look for materials on your own.

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32. INFORMATION ABOUT [MA-FED] FRONT END DEVELOPMENT (30EC)

32.1. Content

Embark on a human-centered journey through modern web development with mentors who guide you as senior colleagues. Develop expertise in Node.js and Express.js.

Let's face it – AI can churn out functioning code faster than you can say "npm install," but it still can't tell when a user is about to rage-quit your app because of a poor user experience.

Build secure applications with testing strategies that ensure quality experiences. Learn APIs, databases, and modern deployment practices – focused on building real-world applications.

All skills are taught through collaborative problem-solving rather than abstract exercises. Your mentors will challenge you with authentic scenarios while you learn to think like a professional.

This experience values technical rigor and personal growth equally. You'll become a full-stack professional who builds with cutting-edge tech while understanding the human impact of your code.

Ready to become technically brilliant and emotionally intelligent?

32.2. Learning outcomes

1. Learning outcome: User Interaction

You design and implement accessible user interfaces based on user research, test them with representative users, and justify design decisions using user-centered design principles.

Explanation: You apply user-centered design methods to research needs, create prototypes, and implement interfaces that follow accessibility guidelines and user feedback.

2. Learning outcome: Development

You develop user-friendly web applications using modern frameworks, implement appropriate testing and version control practices, and ensure scalability and performance.

Explanation: You write clean, structured code for frontend and backend components, use version control effectively, and implement testing strategies and their rationale.

3. Learning outcome: Iterative Process

You apply an iterative design and development approach with clearly defined goals for each cycle, systematically document your process, and document your progress.

Explanation: You work through multiple design and development cycles with specific objectives, collect and analyze feedback systematically, and document your progress.

4. Learning outcome: Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, and ensure sustainable aspects.

5. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your values and goals.

32.3. Expected entry profile

Students should have a solid foundation in web development fundamentals before enrolling in this topic. This means we expect you to have worked with HTML, CSS, JavaScript, and PHP.

We encourage engineers to join this more web-focused topic to broaden their knowledge and learn more about user interaction. Students that have a background in software development are also welcome.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	1	1	1	1	1
Organisational processes					
Infrastructure					
Software	1	1		1	
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

32.4. Outflow profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	2	2	2	2	2
Organisational processes					
Infrastructure					
Software	2	2	1	2	1
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

32.5. Examination and final grading

32.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning experience. Throughout the entire semester, the assessors monitor your progress.

32.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record your progress. In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectation in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue to

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the scale. This gives you the opportunity to address and process the feedback received.

32.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a

32.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

32.6. Learning activities

You are expected to demonstrate the competencies in these learning outcomes multiple times using different forms of evidence. This can be accomplished through the following activities:

1. **Client Project:** Demonstrate your user interaction and web development skills through a real-world project with an external client, where you add value to the client.
2. **Showcase Portfolio:** Present a collection of your best work that demonstrates your growth and proficiency in both UI/UX design and web development.
3. **Personal Project:** Apply user interaction principles and web development practices in a self-initiated project that reflects your personal interests and goals.

32.7. Learning materials

- Canvas
- Onsite Workshops

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33. INFORMATION ABOUT [MA-FSD] FULL STACK DEVELOPMENT (30EC)

Agile Development, Quality Assurance & Software Process Automation

33.1. Content

You will learn how to create a full-stack web application in which the front and back-end layers are separated. You will learn how to do this in a modern way. You will work with a dedicated frontend framework of your choice and will pay explicit attention to achieve a good user experience of your product. For the back-end, you will use a dedicated framework.

You will also learn how to automate your Agile software development and deployment process, and how Software quality plays a central role in it. You will learn the role of IT systems for business.

33.2. Learning outcomes

1. **Learning outcome:** 1. Full-stack Application
Both individually and in teams, you develop full-stack web applications. You design user-friendly applications by following a user-centered design process.

Explanation: Full-stack web application: Developing separate front and backend.

Architectural choices: Choices about data persistency, communication protocols (request/response message exchanges, simultaneous two-way communication).

2. **Learning outcome:** 2. Agile Development
You analyze and design the business context of the group-project case and the stakeholders' needs. You choose and apply the most suitable agile methodology.

Explanation: Agile software development methods: The underlying 12 agile principles and 4 values of agile methods.

Stakeholders' needs: Functional and non-functional

Business context: The software must support the processes of the business ensuring alignment with the customer's operational needs.

3. **Learning outcome:** 3. Software Development Process Automation

You analyze, design, and realize a continuous integration (CI) and continuous deployment (CD) development environment for a full-stack web ap

Explanation: Software Development Process: The process should address phases such as Analysis, Design, Realization, Testing, and Deployme

Best Practices: Utilize industry-standard practices, possibly including use of generative AI during your development work.

4. **Learning outcome:** 4. Software Quality Assurance

You justify the usage of tools and techniques and apply them to visualize the software quality. You make use of different test techniques against a

Explanation: Software Quality: Quality is assured on different levels of the process and product and agreed upon with the customer of your proje

Software Quality Criteria: You address criteria like maintainability (e.g. static code analysis, code reviews, refactoring), security and performance.

5. **Learning outcome:** Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve sta sustainable aspects.

Explanation: Involve stakeholders: You actively ask and apply feedback from project stakeholders and advise them on solutions.

Conduct applied research: You conduct and document research activities related to your projects to make and justify your decisions using resear

Intercultural, ethical, and sustainable: You analyze and manage the cultural, ethical, and sustainable aspects in your projects applying theoretical

6. **Learning outcome:** Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with

Explanation: Aware of: You use tools to identify and understand your strengths and weaknesses.

Personal growth: You undertake activities to improve your weaknesses and seize opportunities using your strengths. You choose and prepare for

Learning attitude: You are proactive in planning and executing your learning activities, in formulating feedback questions, requesting and process

33.3. Expected entry profile

Students need to have completed the Software Design & Engineering semester from the introduction phase or have knowledge covered in this semest follow a small self-paced module on software analysis & advice.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes					
Infrastructure					
Software	1	1	1	1	1
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

33.4. Outflow profile

This semester prepares you for an internship in software and the advanced-level semester Complex Software Systems. Additionally, it provides you wit

Domain description HBO-i 2023

Analysis	Advise	Design	Realise	Manage & control
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User interaction			1	1	
Organisational processes	1		1	1	
Infrastructure	1		1	1	
Software	2	2	2	2	2
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

33.5. Examination and final grading

33.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning aspect: throughout the entire semester, the assessors monitor your progress.

33.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the development

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectation in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the scale. This gives you the opportunity to address and process the feedback received.

33.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory' grade.

33.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

33.6. Learning activities

The education is a mix of short presentations/workshops, project work, guest lectures from companies. Frequent interactions between students and coaches

You will work on two projects. Both projects will be for developing a web application. The first project is your individual project which will be based on your own idea

Next to it, you will work also on a group project which reflects a case presented by one of our partners in education (external company). In the group project, you will focus on usability and security, a business context expecting from you professional behavior. The group and business context may introduce for example cultural differences

33.7. Learning materials

You are free in choosing your learning materials and learning activities. We do provide starting (and for some new topics even extensive) learning materials. All materials are provided in the Canvas courses.

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34. INFORMATION ABOUT [MA-GD] GAME DESIGN (30EC)

34.1. Content

Games often appeal to us because of one simple fact: they're fun! But why is that? In order to find that out, *Game Design* aims to transform you from a

To do this, we will give you the toolkit you'll need to identify the feelings you want to evoke in the player, the message you want to convey, and what kind of game to create something that's more than the sum of its parts.

You won't just be designing, of course: you'll also be developing your game. The goal is to turn your design into a playable demo using one of the industry design practices, storytelling basics, sound design, and more. We will host various open playtesting events to validate your design.

You'll be creating worlds; players will traverse the great plains and climb the majestic mountains that you envisioned.

34.2. Learning outcomes

1. Learning outcome: Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, and address sustainable aspects.

Explanation: Methodological approach: You identify relevant challenges of the client and use a variety of research strategies, methods, and activities to achieve project goals. You can formulate a design challenge that is appropriate to the assignment.

Involve stakeholders: You collect different perspectives such as target-user needs and stakeholder requirements and involve those parties in the design process.

Conduct applied research: You can logically chain research activities to create value towards the project goals.

Advice: You have the communication skills to inform and advise stakeholders within your project about your project's insights.

Decisions: You can substantiate the choices in your project based on analyses and documented in designs.

Reports: You can document your applied research steps effectively in concise reports (such as devlogs).

Ethical, intercultural, and sustainable aspects: When applicable, you consider and act according to ethical standards, sustainable practices, and values.

2. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your values and goals.

Explanation: Strengths and weaknesses: You reflect on your own learning ability and actions, using feedback that you asked and received.

Actions: You use this reflection and feedback to look for opportunities that further your development and act on those opportunities.

Personal growth: You investigate and try out strategies to adapt to changes and setbacks. You align your professional ambitions with personal values and goals.

Learning attitude: You adopt a pro-active, open-minded approach to learning, you embrace feedback, and you are committed to self-improvement.

3. Learning outcome: Analysis

You are able to obtain desired results(1) by researching game design related subjects methodically(2) and thoroughly(3).

Explanation: 1. With desired results we mean that you can clearly define the relevant research question at hand to scope the problem at hand to means you show that you use justified research methods such as the DOT research framework. 3. Thorough means that you use different methods to research the same subject.

4. Learning outcome: Advice

You are able to make recommendations(1) about important decisions in the game design process based on the conclusions and results from the analysis.

Explanation: 1. With recommendations we mean suggestions that you make that will help achieve desired outcomes. 2. With clearly communicated advice we mean that you can explain your advice in a clear and concise manner.

5. Learning outcome: Design

You show that you can design a game in an analytical(1) way using existing frameworks, theories, and models(2). You can validate(3) game systems and prototypes.

Explanation: 1. Analytical means that your work is based on prior analysis insights. You support your design choices with sources such as: Target market analysis, user research, and competitive analysis. 2. Existing frameworks such as: UML, ERD, Design Patterns, MDA framework and Self Determination Theory as well as concept development methods such as paper prototyping, roleplaying and wizard of Oz and use the results to refine your design. 4. Common industry standard methods and prototypes such as paper prototypes, mock-ups, and user tests.

6. Learning outcome: Realization

You show that you can develop(1) and validate(2) game prototypes, based on a design(3) where you create, find or adapt(4) audio-visual and narrative assets.

Explanation: 1. With develop we mean that you are able to create technical prototypes based on prior work or from scratch. 2. You are able to validate your prototypes by testing them with users. 3. With design we mean that you can create a design document that describes the game's mechanics, story, and assets. 4. With create, find or adapt we mean that you can create basic assets, but more realistically will find existing assets. With audio-visual and narrative assets, we mean all 'art' assets for your games such as: 2d art, 3d art, synthesized and/or recorded sound effects, music, and voice acting. 7. A complex context means that the scope and complexity of both the situation and the product are high.

7. Learning outcome: Manage and Control

You are able to work in a structured and methodical way both individually as well as in a group context(1) where you use relevant processes and tools.

Explanation: 1. With both individually as well as in a group context we mean that you show you can adjust the process accordingly. 2. With relevant tools we mean that you use tools that are relevant to the game design process. 3. With iterative process we mean that you show a process of cyclic repeating steps to improve quality.

34.3. Expected entry profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	1	1	1	1	1
Organisational processes					
Infrastructure					
Software			1	1	1
Hardware-interfacing					

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	

34.4. Outflow profile

Domain description HBO-i 2023

	Manage & control	Realise	Design	Advise	Analysis
User interaction	2	2	2	2	2
Organisational processes					
Infrastructure					
Software		2	2	2	2
Hardware-interfacing					

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

34.5. Examination and final grading

34.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning aspect: throughout the entire semester, the assessors monitor your progress.

34.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using this scale, giving you the opportunity to address and process the feedback received.

34.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve

34.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

34.6. Learning activities

During the semester, lectures and practicals will be offered in the form of workshops. These workshops will deal with subjects related to technology and progress on a regular basis.

Teachers will provide students with feedback on their work. Students are expected to record and process this feedback. For the Industry Project Assignment discussed. The student is expected to demonstrate Industry Project progress to the client at set intervals.

Blended education

The semester works with blended learning, whereby, in addition to the activities on location, online education can take place that matches content and semester is explained in the Canvas course. The decision to have parts of the education online is reserved for the degree programme.

34.7. Learning materials

Canvas course including all linked information.

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35. INFORMATION ABOUT [MA-II] INDUSTRY 4.0 & INTERNET OF THINGS (30EC)

Machine to Machine Interaction, Data Analytics & Machine Learning, Business Data Processing

35.1. Content

The domain of "Smart Industry" (also known as Industry 4.0) is booming. It is the further automation of the traditional manufacturing and process industry. Computing, data analysis and artificial intelligence are being integrated. These new technologies allow for further automation, better communication, and

The fourth industrial revolution is about the "smart factory", with a modular structure in which cyber physical systems (CPS) monitor physical processes. "cyber physical systems" communicate and cooperate with each other and with people in synchronous timing, both internally and across organizational

The industry and business community currently have a very large and growing demand for professionals who not only understand the context, but also is so large that professionals within this domain must not only be able to operate as a generalist, but also must be able to specialize. This means that it is necessary because the projects and systems are multidisciplinary and so these professionals need to thrive within multidisciplinary programs and projects.

35.2. Learning outcomes

1. Learning outcome: Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, and sustainable aspects.

2. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with

3. Learning outcome: Analysis: You analyse problems, processes and/or challenges of an (existing) Smart Industry system or parts thereof. You are

Explanation: You perform an analysis on a problem, process and/or system in which you conduct a research within the domain and come up with improvement, data analysis, transformation of data into information, human-machine interaction, process control, machine-to-machine communication. Solutions are met. You consider KPI's and added (business) value of your solutions to the stakeholders.

4. Learning outcome: Design: You design possible new solutions, improvements and extensions of existing solutions based on the given advice in

Explanation: You design solutions according to the requirements of the stakeholders. These solutions consist of a mixture of software and/or hardware. They enable production, distribution, analysis and visualization of (real-time) data and processes, and as such serve for further optimization of human factors. You show that quality aspects, like usability, safety, security, robustness, testability, extendibility and maintainability are met.

5. Learning outcome: Implementation: You implement possible new solutions, improvements and extensions of existing solutions based on the given

Explanation: You implement parts of a Smart Industry system, such as process controllers, communication, IoT protocols, and/or you set up an implementation for the target user group. You integrate the separate components in a working system and/or in an existing system. You draft and execute a

35.3. Expected entry profile

The domain of Smart industry is very broad and, as suggested by the topics listed above, it is suitable and accessible for all students with any prior knowledge or experience on new topics. This is achieved by the given freedom to select your own program within the semester.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	1			1	
Organisational processes	1	1			
Infrastructure	1			1	
Software	1		1	1	
Hardware-interfacing	1		1	1	

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

35.4. Outflow profile

The semester Industry 4.0 & IoT prepares for:

- a possible internship in the direction of Smart Industry
- advanced semester Industry 5.0 and Study program Smart Industry & IoT
- study programme Robotics

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	2			2	
Organisational processes	2	2			
Infrastructure	2			2	
Software	2		2	2	
Hardware-interfacing	2	2	2	2	2

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

35.5. Examination and final grading

35.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of learning outcomes achieved.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning path. Throughout the entire semester, the assessors monitor your progress.

35.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record these conversations in your portfolio.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue to work on it.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. This gives you the opportunity to address and process the feedback received.

35.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory' grade.

35.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

35.6. Learning activities

Students participate in a mixed learning environment that is both project based and task-based. This means that an authentic question, challenge or problem guidance gradually change from teacher-driven to a more student-driven approach during the semester and will be offered whenever needed and relevant speed and level is possible. It is expected that students prepared themselves for a session with teachers or experts by studying the provided study materials.

Blended education

The semester works with blended learning, whereby, in addition to the activities on location, online education can take place that matches content and the semester is explained in the Canvas course. The decision to have parts of the education online is reserved for the degree programme.

35.7. Learning materials

A Canvas course including modules with (self) study materials, inspiration sessions and external sources is available.

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36. INFORMATION ABOUT [MA-ID] INTELLIGENT DEVICES (30EC)

Distributed Embedded Software, Object Oriented Design, Edge AI, Internet of Things

36.1. Content

The theme of this semester is “Intelligent Devices.”

In today's world, embedded systems are at the heart of our increasingly connected, data-driven society. This semester focuses on how these intelligent

Key subtopics we will explore include:

- **Embedded Systems Programming**

Develop low-level programming skills that enable efficient use of hardware resources — a must for building high-performance, power-conscious embedded systems.

- **Distributed Systems and Connectivity**

Discover how modern embedded systems communicate within large-scale networks — including the Internet of Things — to form the digital backbone.

- **Professional Software Design**

Learn to design robust, maintainable software using C++ and SOLID principles while following industry standards and best practices.

- **Real-Time Operating Systems (RTOS)**

Dive into the principles of RTOS, a cornerstone for applications requiring precise timing, responsiveness, and reliability, such as in edge computing.

- **Future Technologies and Trends**

Explore the frontiers of embedded innovation, including **Edge AI**, which brings intelligent decision-making closer to the source of data; **embedded ML**, which enables intelligent systems to learn from data; and **quantum computing**, which promises to revolutionize embedded systems.

This semester combines theory with hands-on experience to help you build the skills needed to design and implement intelligent embedded systems for a variety of applications.

36.2. Learning outcomes

1. **Learning outcome:** Programming Techniques

You successfully implement maintainable, clear, and documented object-oriented software by applying relevant Software Development Life Cycle (SDLC) programming tasks, utilising skills in build processes, mocking, and simulation, while adhering to industry standards.

2. **Learning outcome:** Software Design Techniques

You design robust, maintainable object-oriented systems by applying SOLID principles and following the phases of the Software Development Life Cycle (SDLC). You use Object-Oriented Analysis (OOA) and Object-Oriented Design (OOD) using UML diagrams (class, state, sequence) relevant to embedded systems.

3. **Learning outcome:** Embedded Systems

You design and implement low-level embedded systems, including configuring and controlling peripherals (e.g., timers, sensors, actuators) and real-time edge AI/tinyML, extended low-level programming or any other relevant embedded topic. You focus on the Design, Testing, and Implementation phases.

4. **Learning outcome:** Communication

You design and implement scalable and extendable communication solutions for interconnecting components or subsystems in a distributed system. You analyze system state behavior. The focus is on selecting the most suitable communication solution for the specific project context, ensuring the system's scalability.

5. **Learning outcome:** Real-time Operating Systems

You apply basic real-time operating system (RTOS) concepts, including task/thread management, synchronization, and inter-task communication structuring, while following the Software Development Life Cycle (SDLC). Emphasis is placed on understanding the role of RTOS in optimizing system performance.

6. **Learning outcome:** Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, and ensure sustainable aspects.

7. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with

36.3. Expected entry profile

To maximize success in this semester, prior knowledge from the following semesters is recommended:

- Intelligent Technologies
- Software Design & Engineering (with some affinity for and enthusiasm toward embedded world)

However, you are welcome to join from other semesters if you enjoy a challenge! Embedded systems can be a valuable addition to your knowledge in :

Domain description HBO-i 2023

	Manage & control	Realise	Design	Advise	Analysis
User interaction					
Organisational processes					
Infrastructure		1	1		
Software		1	1	1	1
Hardware-interfacing	1	1	1	1	1

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

36.4. Outflow profile

After completing this semester, students are encouraged to further develop their expertise in:

- Industry 4.0 & the Internet of Things
- Industry 5.0
- High-Tech Embedded Software Engineering

Domain description HBO-i 2023

	Manage & control	Realise	Design	Advise	Analysis
User interaction					
Organisational processes					
Infrastructure		1	1		
Software	1	1	2	2	2
Hardware-interfacing	2	2	2	2	2

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

36.5. Examination and final grading

36.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount c

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape it. Throughout the entire semester, the assessors monitor your progress.

36.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the development

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectation in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the scale, giving you the opportunity to address and process the feedback received.

36.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory'.

36.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

36.6. Learning activities

You will participate in workshops where you gain and apply new knowledge. This knowledge is then put into practice through hands-on challenges and bring in their own ideas to demonstrate proficiency in the learning outcomes.

36.7. Learning materials

A Canvas course including workshops with links to online sources, hardware for inspiration projects and challenges.

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37. INFORMATION ABOUT [MA-MC] MEDIA CREATION (30EC)

37.1. Content

In the Media Creation Main semester, students found their own simulated media agencies as a group. This way, students further develop their professional skills as junior professionals. Key aspects include:

1. Preparing for internships, aiming for proficiency in UX and media creation, suitable for small media agencies.
2. Gaining a comprehensive understanding of the study program's main semesters, internship opportunities, and professional possibilities.
3. Guidance from lecturers to prepare for successful internship completion and advice on achieving professional goals in ICT.

The media creation semester serves as a building block to fill a student's "toolbox" with creative and technical skills, tool knowledge, empathy, design principles, and shares some activities and materials.

37.2. Learning outcomes

1. **Learning outcome:** Conceptualize, design, and develop professional media products
You create engaging concepts and translate them into professional validated media products by applying user-centered design principles, visual communication, and emerging technologies.

Explanation: Engaging & validated media products = You find this out through testing, is the target audience attracted, triggered, satisfied or otherwise. You create engaging concepts and translate them into professional validated media products by applying user-centered design principles, visual communication, and emerging technologies.

User-centered design principles = Your digital products are designed and build in accordance with guidelines from the media-field, like the usability principles.

Emerging technologies = You explore technologies that are relevant for the current age of ICT, like Artificial Intelligence, Mixed reality, internet of things, etc.

2. **Learning outcome:** Transferable production
You create professional documentation and use version control for your products in a personal and team context. You communicate design decisions and recommendations.

Explanation: Professional documentation = You create and validate logical structures as ways of documenting your process for the stakeholders.

Version control in personal and team context = Employing fitting version control systems like Git or others to manage and track different versions of your work.

Design decisions and recommendations = Communicating with stakeholders using methods like storyboarding and brand guides to facilitate (re)production.

3. Learning outcome: Creative iterations

You present the successive iterations of your creative process, and the connections between them, of your methodically substantiated, iterative d

Explanation: Successive iterations = Validated and explained improvements done on a previous version based on feedback, new insights or (usi

Creative process = Every step you take for creating products in your project guided by a design model like Double Diamond, Design thinking and

Connections = The logical relationships or linkages between successive iterations.

4. Learning outcome: Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve sta sustainable aspects.

Explanation: Stakeholder = For example a company, an artist, a professional, ... from the industry with a media creation task that you work on w

Methodological approach = i.e. DOT Framework, Double Diamond, Agile / Scrum, ...

Ethical, intercultural and sustainable aspects = You keep in mind the numerous ways of impact of your creations (also for example the energy use

5. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with

Explanation: Promote your personal growth = You produce your own personal portfolio to prove this and all the other learning outcomes. The for a self-crafted website, a collection of showreels (with documentation), ...

37.3. Expected entry profile

Students who have finished their first year successfully in one of the introduction semesters or students coming from a different main semester.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	1	1	1	1	1
Organisational processes					
Infrastructure					
Software					
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

37.4. Outflow profile

The semester media creation prepares the student for an internship and / or is an ideal fit for a following semester in Game Design, Front End Develop

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	2	2	2	2	2
Organisational processes	2				
Infrastructure					
Software	1	1			
Hardware-interfacing	1	1			

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

37.5. Examination and final grading

37.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning aspect: throughout the entire semester, the assessors monitor your progress.

37.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectation in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the scale. This gives you the opportunity to address and process the feedback received.

37.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory' grade.

37.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

37.6. Learning activities

Project based work

The semester revolves around a company or stakeholder's project assignment (e.g., Partner in Education, relevant industry professional). Students' educational block can span multiple projects. The projects must cover and facilitate the semester's learning outcomes.

Technical workshops

Technical workshops are optional educational blocks where teachers organize interactive teaching moments centered around a specific topic / category and the project challenge. Besides a set of predetermined (and published on canvas) activities, teachers can organize their teaching materials and content around progress in projects and their personal learning journey, but they can also be adapted to emerging topics as needed.

Showcase Events

There will be three showcase events for the project groups ("Media agencies") throughout the semester. These events are arranged in week 7, 12, and 17. Progress solutions and minimal viable prototypes can be tested by stakeholders. The third event is a public conference-style event where the final solutions are presented.

Feedback sessions - Official formative feedback / Portfolio review (at least 3x)

Students submit their portfolio in Canvas before the deadline set by the teacher team. Each portfolio review has its own 'hand-in assignment' with Rubric and appointment: these are always on-site physical meetings. Teachers indicate grading per LO (based on scale) in Canvas. Teachers put feedback and feedforward in the portfolio. Teachers are used to present a quick rundown of the student's products aligned with the LOs. Afterwards the teacher duo is available for questions and oral explanation. Teacher duos are encouraged to use part of their allocated time to prepare the feedback on site (not in the evenings or during weekends). The do's and don'ts for students on canvas to make the entire process transparent. Every teacher (assessor) duo should complement each other in skills/LOs. For the second

Individual activities

This optional educational block is for individual student work. Some students may choose not to attend technical workshops and develop their learning activities must contribute to students' growth, learning outcomes, and project-based work.

37.7. Learning materials

Learning materials are published on canvas.

38. INFORMATION ABOUT [MA-MAD] MOBILE APPS DEVELOPMENT (30EC)

38.1. Content

In the SM semester you will work on a wide variety of developing techniques to make (native) apps for mobile platforms. You will work on both the conc independently creating a design based on your own idea, to turn it into a testable prototype with a (natively) programmed proof-of-concept with a minim

The activities of app developers are often the following:

- Mapping the wishes of the client and/or user.
- Conducting consultations with, for example, functional designers, UX designers, UI designers and programmers.
- Design, develop and / or implement new apps and further develop existing apps (based on consultation with client and development team).
- Monitoring the relationship between the functional and technical side of the app development (in collaboration with the rest of the development te
- Informing back-end developers and ensuring a good link between the front-end of the app and back-end development.
- Preparation and management of (process) documentation.
- Thinking about new possibilities and applications for existing apps and opportunities for new apps.

38.2. Learning outcomes

1. Learning outcome: Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve sta sustainable aspects.

2. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with

3. Learning outcome: Analysis: you show you can convert a problem or opportunity into multiple smart mobile concepts based on thorough analysi

Explanation: You use common research methods (like the DOT-framework) to analyse a problem, trends in the market and relevant stakeholder mobile challenges. Mobile Concept: A crystallised idea based on a problem description with a distinguishable research component.

4. Learning outcome: Advice: Based on relevant analysis, you advice on suitable user interaction and technological design, improvement of design

Explanation: You consider the following aspects in your advice: iterative way of working, outcomes of user tests, create POC's. You process the

5. Learning outcome: Design: You translate concepts into user friendly designs and validate these designs through user tested iterations.

Explanation: Translate: Using design theory and methodologies, you are capable of designing engaging solutions. Engaging: the outcome from i quality standards. Designs: Products that describe the problem, target audience, interaction narrative, user interface and comply with software en You incorporate the output of previous design and analysis steps into every new design iteration in a meaningful way.

6. Learning outcome: Management & Control: You continuously apply techniques to manage a development process and to assure its quality in di

Explanation: Stakeholders are all clients, (future) developers, all users. You process the feedback given in your work.

38.3. Expected entry profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	1	1	1	1	1
Organisational processes					
Infrastructure					
Software			1	1	1
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

38.4. Outflow profile

Domain description HBO-i 2023

							Manage & control
							Realise
							Design
							Advise
							Analysis
User interaction	2	2	2	2	2	2	
Organisational processes							
Infrastructure							
Software	1	1	2	2	2	2	
Hardware-interfacing							

Professional Development BSc 2020

Investigative problem solving	2
Targeted interaction	2
Personal leadership	2
Future oriented organisation	2
Professional development	2

38.5. Examination and final grading

38.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount c

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning. In this aspect: throughout the entire semester, the assessors monitor your progress.

38.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the deve

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't c

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the form below. This form gives you the opportunity to address and process the feedback received.

38.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessor.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve

38.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

38.6. Learning activities

A new development, following demands and suggestions from the Mobile Development community, allows for either broadening and/or deepening of the choice. The platform for which we develop is chosen based upon a student's own preference and on the requirements of the prototype, however it is a programmed app running on a mobile device, accompanied by a design portfolio. The student should involve the teachers in the learning process via WhatsApp.

For the Industry Project you will work on an app for an external commissioning party, preferably one of our PIEs (dCentralize, Peercode etc), together with a client to convince this client of its feasibility. Of course the idea is to come up with an app as impressive as possible. PIERs of all this means you make optimal use of the concept may be grand in its setup, with an App as the proof of concept. The knowledge and skills acquired in the workshops should be clearly applied in the assignment in duos.

There will be a focus on either:

- Descriptive ethics and target group analysis, or,
- A proof of concept of an innovative feature that contributes to the group project.

Blended education

The semester works with blended learning, whereby, in addition to the activities on location, online education can take place that matches content and semester is explained in the Canvas course. The decision to have parts of the education online is reserved for the degree programme.

38.7. Learning materials

Theory and practice

Lectures and practicals will be offered in the form of workshops. These workshops will deal with subjects related to techniques and design and student: Progress is shown and discussed on a weekly basis with the teachers, at the students' initiative and recorded in FeedPulse by the students. There will processes are also discussed.

With regard to the theoretical part of the course, the students will be able to attend (online) workshops in which they can work on their assignments tog

The result of these workshops will be handed in at the LMS Canvas the same day and students will be able to view their own progress. Furthermore th project management environment.

Individual working sessions

Students are expected to work on their mobile products all days of the work week, even when nothing is rostered or scheduled.

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39. INFORMATION ABOUT [MA-NCA] NETWORK & CLOUD AUTOMATION (30EC)

Deploying & Automating Intelligent IT Services, DevSecOps

39.1. Content

During this semester, you will explore network and cloud automation by designing and implementing automated IT infrastructure solutions. You will acq

Throughout the semester, you will develop and manage automated solutions, ensuring they align with business requirements while maintaining security. You will enhance security through Security Orchestration Automation and Response (SOAR), ensuring proactive threat detection and response. Additic performance and system health.

39.2. Learning outcomes

- 1. Learning outcome:** Professional standard
Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve sta sustainable aspects.
- 2. Learning outcome:** Personal leadership
You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with
- 3. Learning outcome:** Enhanced Interoperability
After conducting methodical research, you are able to analyze the requirements, provide advice, create a design, and implement the rollout for a protocols, communication, and applications.
- 4. Learning outcome:** Enhanced Securing
After conducting methodical research, you are able to analyze the requirements, provide advice, create a design, and implement modern and trac requirements of a large SME.
- 5. Learning outcome:** Enhanced Automation
After conducting methodical research, you are, based on the applicable standards, able to analyze the requirements of a large SME and orchestr
- 6. Learning outcome:** Manage & Control enhanced
After conducting methodical research, you are able to advise and design an architecture based on the analyzed requirements of a large SME tha

39.3. Expected entry profile

We recommend to finish Network & Cloud Basics semester which is a foundation in networking and cloud technologies, where you learn core concepts

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes					
Infrastructure	1	1	1	1	
Software		1	1	1	
Hardware-interfacing					

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

39.4. Outflow profile

By the end of the semester, you will have a strong foundation in network and cloud automation, enabling you to design, implement, and manage modern

Domain description HBO-i 2023

	Manage & control	Realise	Design	Advise	Analysis
User interaction					
Organisational processes	2	2			
Infrastructure	2	2	2	2	2
Software	2	1	1	1	
Hardware-interfacing					

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

39.5. Examination and final grading

39.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning aspect: throughout the entire semester, the assessors monitor your progress.

39.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using this scale, giving you the opportunity to address and process the feedback received.

39.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a

39.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

39.6. Learning activities

Your learning will be supported by hands-on exercises, workshops, and real-world challenges, helping you build practical skills in infrastructure automation and professional competencies.

39.7. Learning materials

A Canvas course featuring the necessary learning resources

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40. INFORMATION ABOUT [O-MA-OLC] OPEN LEARNING (30EC)

40.1. Content

We have Open Education for PRO from the second year of your study. This means that you will define your own personal learning route. Do you want to learn in a particular technology? It's up to you!

You will work as much as possible in a context that is most supportive of your personal learning trajectory. This means it is rather impossible to determine the learning process determined at the beginning of the process.

With this in mind, Education for professionals can be defined as follows:

Educating responsive professionals for all possible careers in IT. This is done by focusing on talent development and personalized competence development.

Students who have successfully concluded OE2 can start in OE3. Starting level would be the learning outcomes output of the OE2. The output level is based on a student is pursuing.

The output level is based on the achieved growth on the set competence indicators of the student's personal competence profile.

The student will determine the general lines of the content in this program him or herself. Basically, everything in IT should be possible. Because everyone learns at the same speed, in the same manner, etc. This means all workshops, presentations, etc., that will provide content, will be planned and scheduled on-demand.

To facilitate this, the student will create his/her own personal profile within the HBO-i Competence Framework, this framework gives a systematic overview of the competences.

A complete explanation of the structure of the framework and how it functions can be found in the Canvas course.

Blended Learning

At Education for Professionals, we apply the design principles of Blended Learning. This means that we learn with each other online and face-to-face. We learn in groups or individually.

- Face to face contact (18:00-21:00)

Every Tuesday evening from 18:00 to 21:00, we learn in groups or individually at FICT. The following activities could be scheduled:

1. Status updates: Give your group members and coaches insight in your learning process. What have you done last week? Show your plan for the week.
2. Discuss about a topic:

You will zoom in into a topic which is introduced by a student (or coach).

1. Self-work/Collaboration: Students are given the time to work together on a challenge or a smaller task which is related to the discussion topic talk.
2. Inspiration sessions / Workshops: Inspiration sessions and workshops could be scheduled.

- Online contact (18:00-19:30)

Of course, students can do their own study activities online in all sorts of ways. For example, the student asks fellow students to discuss deliverables or assignments.

40.2. Learning outcomes

1. **Learning outcome:** Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, and sustainable aspects.

Explanation: Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, and sustainable aspects.

2. **Learning outcome:** Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your strengths and weaknesses.

Explanation: You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your strengths and weaknesses.

3. **Learning outcome:** Analysis:

You can obtain desired results by analyzing the processes, products and information flows in a methodical and thorough way.

Explanation: You know what you are researching and why, taking into account the client, target audience & market.

Using several justified research methods based on the DOT-framework

Applying multiple different methods of analysis or focusing on several different aspects. For example: client needs, audience needs, other related factors.

4. **Learning outcome:** Advice:

You can give advice about the organization, processes and/or information important for decisions related to your challenge.

Explanation: Your advice is based on the conclusions and results from the analysis.

Your advice can be based on cost aspects and quality properties such as: availability, performance, security and scalability.

5. **Learning outcome:** Design:

You are able to design (parts of) systems, communicate and validate about it for both functional/technical and aesthetic requirements using industry standards.

Explanation: With design we mean that you can use a structured and methodical approach in your design. You use a set of concept development. The design covers both the functional/technical and aesthetic design of your products. You communicate your design in a well presented and structured way. You validate your designed systems in an early stage using efficient and quick prototyping methods such as paper prototyping and use the results.

6. Learning outcome: Realization:

You can implement and validate a product based on a design in a structured way using innovative and/or relevant technology adhering to quality.

Explanation: We expect that you validate your product and the quality of the product in a structured way using methods such as: User testing, etc. Structured way means using industry standard development processes such as Scrum/Agile and Kanban.

With innovative and/or relevant technology we mean that you experiment and use upcoming and promising technology that suits your needs. Your product adheres to previously defined quality criteria in accordance with stakeholders. Many non-functional requirements such as stability, performance, etc.

7. Learning outcome: Manage and Control:

You can perform activities that focus on control, monitoring and optimization of the development, commissioning and use of ICT-systems. These activities include:

Explanation: We expect that you build your controlling, monitoring and optimisation of the development in a structured way.

We expect that you can communicate with stakeholders about your activities who are related to manage and control.

40.3. Expected entry profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

40.4. Outflow profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

40.5. Examination and final grading

40.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work you do, the quality of your work, etc.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning. In this aspect: throughout the entire semester, the assessors monitor your progress.

40.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record these conversations in your portfolio.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome in a structured way.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue to work on it.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using this scale. This gives you the opportunity to address and process the feedback received.

40.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory' or better.

40.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

40.6. Learning activities

The teachers will provide you with formative feedback on deliverables and demonstrations. You show the work, alone or in a group context, but will be (Manage and Control, Future oriented organization, Investigative Problem Solving, Personal Leadership and Targeted Interaction). Besides the continuous

At these assessments the competences are evaluated based on state of the deliverables and your personal development at that point. This will lead to you keep on progressing on the competences, not doing so might lead to a lower grade at the next formative assessment such as beginning or orientation

The first and leading collection to show progress is your professional portfolio. The portfolio is built from the project deliverables you work on in each cluster

The second type of deliverables is proof. All documents and artifacts that you create as a means to design, compare or research can serve as proof. It

Blended education

The semester works with blended learning, whereby, in addition to the activities on location, online education can take place that matches content and semester is explained in the Canvas course.

40.7. Learning materials

All information about the education program Education for Professionals OE3 could be found in the CANVAS course. Select your own study materials y

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41. INFORMATION ABOUT [O-MA-OLD] OPEN LEARNING (30EC)

41.1. Content

We have Open Education for 'Education for professionals' from the second year of your study. This means that you will define your own personal learning you want to specialize in a particular technology? It's up to you!

You will work as much as possible in a context that is most supportive of your personal learning trajectory. This means it is rather impossible to determine determined at the beginning of the process.

With this in mind, Education for professionals can be defined as follows:

Educating responsive professionals for all possible careers in IT. This is done by focusing on talent development and personalized competence

Students who have successfully concluded OE3 can start in OE4. Starting level would be the learning outcomes output of the OE3. The output level is that a student is pursuing.

The output level is based on the achieved growth on the set competence indicators of the student's personal competence profile.

The Open Learning program will run as a part time course with at least two contact moments a week:

- Tuesday evening: 18u-21u: Fontys Location Eindhoven: TQ
- Thursday evening: 18u-19.30u: Online contact moment

The student will determine the general lines of the content in this program him or herself. Basically, everything in IT should be possible. Because everyone the same speed, in the same manner, etc. This means all workshops, presentations, etc., that will provide content, will be planned and scheduled on-demand

To facilitate this, the student will create his/her own personal profile within the HBO-i Competence Framework, this framework gives a systematic overview

A complete explanation of the structure of the framework and how it functions can be found in the Canvas course.

Blended Learning

At Education for Professionals, we apply the design principles of Blended Learning. This means that we learn with each other online and face-to-face. 1

- Face to face contact (18:00-21:00)

Every Tuesday evening from 18:00 to 21:00, we learn in groups or individually at FICT. The following activities could be scheduled:

1. Status updates: Give your group members and coaches insight in your learning process. What have you done last week? Show your plan for the
2. Discuss about a topic:

You will zoom in into a topic which is introduced by a student (or coach).

1. Self-work/Collaboration: Students are given the time to work together on a challenge or a smaller task which is related to the discussion topic talk
2. Inspiration sessions / Workshops: Inspiration sessions and workshops could be scheduled.

- Online contact (18:00-19:30)

Of course, students can do their own study activities online in all sorts of ways. For example, the student asks fellow students to discuss deliverables or

41.2. Learning outcomes

1. **Learning outcome:** Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders and sustainable aspects.

Explanation: Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals

intercultural, and sustainable aspects.

2. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with

Explanation: You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions

3. Learning outcome: Analysis:

You can obtain desired results by analyzing the processes, products and information flows in a methodical and thorough way.

Explanation: You know what you are researching and why, taking into account the client, target audience & market.

Using several justified research methods based on the DOT-framework

Applying multiple different methods of analysis or focusing on several different aspects. For example: client needs, audience needs, other related

4. Learning outcome: Advice:

You can give advice about the organization, processes and/or information important for decisions related to your challenge.

Explanation: Your advice is based on the conclusions and results from the analysis.

Your advice can be based on cost aspects and quality properties such as: availability, performance, security and scalability.

5. Learning outcome: Analysis

You can obtain desired results by analyzing the processes, products and information flows in a methodical and thorough way.

You know what you are researching and why, taking into account the client, target audience & market.

Using several justified research methods based on the DOT-framework

Applying multiple different methods of analysis or focusing on several different aspects. For example: client needs, audience needs, other related

Advice

You can give advice about the organization, processes and/or information important for decisions related to your challenge.

Your advice is based on the conclusions and results from the analysis.

Your advice can be based on cost aspects and quality properties such as: availability, performance, security and scalability.

Design:

You are able to design (parts of) systems, communicate and validate about it for both functional/technical and aesthetic requirements using industry

Explanation: With design we mean that you can use a structured and methodical approach in your design. You use a set of concept development

The design covers both the functional/technical and aesthetic design of your products. You communicate your design in a well presented and structured

You validate your designed systems in an early stage using efficient and quick prototyping methods such as paper prototyping and use the results

6. Learning outcome: Realization:

You can implement and validate a product based on a design in a structured way using innovative and/or relevant technology adhering to quality

Explanation: We expect that you validate your product and the quality of the product in a structured way using methods such as: User testing, etc.

Structured way means using industry standard development processes such as Scrum/Agile and Kanban.

With innovative and/or relevant technology we mean that you experiment and use upcoming and promising technology that suits your needs.

Your product adheres to previously defined quality criteria in accordance with stakeholders. Many non-functional requirements such as stability, performance

7. Learning outcome: Manage and Control:

You can perform activities that focus on control, monitoring and optimization of the development, commissioning and use of ICT-systems. These activities

Explanation: We expect that you build your controlling, monitoring and optimisation of the development in a structured way.

We expect that you can communicate with stakeholders about your activities who are related to manage and control.

41.3. Expected entry profile

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

41.4. Outflow profile

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

41.5. Examination and final grading

41.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work you do, the quality of your work, and the learning outcomes you achieve. The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning path. Throughout the entire semester, the assessors monitor your progress.

41.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record your progress in the portfolio. In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale.

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome with good results.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectation in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue to work on it. Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale, giving you the opportunity to address and process the feedback received.

41.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory' grade.

41.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

41.6. Learning activities

The teachers will provide you with formative feedback on deliverables and demonstrations. You are expected to record the feedback in FeedPulse and the learning outcomes (Analysis, Advice, Design, Realisation, Manage and Control, Future oriented organization, Investigative Problem Solving, Personal Leadership).

At these demo's the competences are evaluated based on state of the deliverables and your personal development at that point. This will lead to a grade. Keep on progressing on the competences, not doing so might lead to a lower grade at the next formative assessment such as beginning or orienting.

41.7. Learning materials

All information about the education program Education for Professionals OE4 could be found in the CANVAS course. Select your own study materials. [\[Return to table of content\]](#)

42. INFORMATION ABOUT [O-MA-OLE] OPEN LEARNING (30EC)

42.1. Content

We have Open Education for PRO from the second year of your study. This means that you will define your own personal learning route. Do you want to learn a particular technology? It's up to you!

You will work as much as possible in a context that is most supportive of your personal learning trajectory. This means it is rather impossible to determine the content of the program at the beginning of the process.

With this in mind, Education for professionals can be defined as follows:

Educating responsive professionals for all possible careers in IT. This is done by focusing on talent development and personalized competence development.

Students who have successfully concluded OE4 can start in OE5. Starting level would be the learning outcomes output of the OE4. The output level is the level a student is pursuing.

The Open Learning program will run as a part time course with at least two contact moments a week:

- Tuesday evening: 18u-21u: Fontys Location Eindhoven: TQ
- Thursday evening: 18u-19.30u: Online contact moment

The student will determine the general lines of the content in this program him or herself. Basically, everything in IT should be possible. Because everyone works at the same speed, in the same manner, etc. This means all workshops, presentations, etc., that will provide content, will be planned and scheduled on-demand.

To facilitate this, the student will create his/her own personal profile within the HBO-i Competence Framework, this framework gives a systematic overview of the competences required for the profession.

A complete explanation of the structure of the framework and how it functions can be found in the Canvas course.

Blended Learning

At Education for Professionals, we apply the design principles of Blended Learning. This means that we learn with each other online and face-to-face. \

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Every Tuesday evening from 18:00 to 21:00, we learn in groups or individually at FHICT. The following activities could be scheduled:

1. Status updates: Give your group members and coaches insight in your learning process. What have you done last week? Show your plan for the
2. Discuss about a topic:

You will zoom in into a topic which is introduced by a student (or coach).

1. Self-work/Collaboration: Students are given the time to work together on a challenge or a smaller task which is related to the discussion topic talk
2. Inspiration sessions / Workshops: Inspiration sessions and workshops could be scheduled.

- Online contact (18:00-19:30)

Of course, students can do their own study activities online in all sorts of ways. For example, the student asks fellow students to discuss deliverables o

42.2. Learning outcomes

1. Learning outcome: Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve sta sustainable aspects.

Explanation: Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goe intercultural, and sustainable aspects.

2. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with

Explanation: You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actio

3. Learning outcome: Analysis:

You can obtain desired results by analyzing the processes, products and information flows in a methodical and thorough way.

Explanation: You know what you are researching and why, taking into account the client, target audience & market.

Using several justified research methods based on the DOT-framework

Applying multiple different methods of analysis or focusing on several different aspects. For example: client needs, audience needs, other related

4. Learning outcome: Advice:

You can give advice about the organization, processes and/or information important for decisions related to your challenge.

Explanation: Your advice is based on the conclusions and results from the analysis.

Your advice can be based on cost aspects and quality properties such as: availability, performance, security and scalability.

5. Learning outcome: Design:

You are able to design (parts of) systems, communicate and validate about it for both functional/technical and aesthetic requirements using indus

Explanation: With design we mean that you can use a structured and methodical approach in your design. You use a set of concept developmer

The design covers both the functional/technical and aesthetic design of your products. You communicate your design in a well presented and stru

You validate your designed systems in an early stage using efficient and quick prototyping methods such as paper prototyping and use the results

6. Learning outcome: Realization:

You can implement and validate a product based on a design in a structured way using innovative and/or relevant technology adhering to quality

Explanation: We expect that you validate your product and the quality of the product in a structured way using methods such as: User testing, ex

Structured way means using industry standard development processes such as Scrum/Agile and Kanban.

With innovative and/or relevant technology we mean that you experiment and use upcoming and promising technology that suits your needs.

Your product adheres to previously defined quality criteria in accordance with stakeholders. Many non-functional requirements such as stability, p

7. Learning outcome: Manage and Control:

You can perform activities that focus on control, monitoring and optimization of the development, commissioning and use of ICT-systems. These :

Explanation: We expect that you build your controlling, monitoring and optimisation of the development in a structured way.

We expect that you can communicate with stakeholders about your activities who are related to manage and control.

42.3. Expected entry profile

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

42.4. Outflow profile

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

42.5. Examination and final grading

42.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning aspect: throughout the entire semester, the assessors monitor your progress.

42.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the scale, which gives you the opportunity to address and process the feedback received.

42.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a

42.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

42.6. Learning activities

The teachers will provide you with formative feedback on deliverables and demonstrations. You are expected to record the feedback in FeedPulse and the learning outcomes (Analysis, Advice, Design, Realisation, Manage and Control, Future oriented organization, Investigative Problem Solving, Personal Leadership)

At these demo's the competences are evaluated based on state of the deliverables and your personal development at that point. This will lead to a grade. Keep on progressing on the competences, not doing so might lead to a lower grade at the next formative assessment such as beginning or orienting.

42.7. Learning materials

All information about the education program Education for Professionals OE5 could be found in the CANVAS course. Select your own study materials and

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43. INFORMATION ABOUT [MA-OLC] OPEN LEARNING (30EC)

43.1. Content

ICT & Personalized Program is a route where you have a lot of freedom and flexibility to determine your personal focus and structure, within the boundaries of the Program. This gives great opportunities for starting multi-disciplinary projects. Projects should be IT related, based on relevant questions and developments

Lots of freedom, but remember:

“Freedom comes with responsibility”

This large degree of freedom does not mean that there are no obligations. A large amount of autonomy and entrepreneurial behaviour is expected.

43.2. Learning outcomes

1. **Learning outcome:** Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders and consider sustainable aspects.

2. **Learning outcome:** Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your values and goals.

43.3. Expected entry profile

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

43.4. Outflow profile

The student builds a competence profile based upon the basic architectural layers (level 1) from OE2, this is taken to the next step at level 2. In addition, the layer level grows with the phase of the students' study. Phase OE1, OE2 ~ level 1, phase OE3,4,5, ~ level 2 and phase OE6,7,8, ~ level 3.

A balanced competence profile does not have too many competencies (no focus) but also not too little (no widening). At least all 9 activities (including personal leadership) are covered.

Within Open Learning, students define their own dynamic competence profile per semester (exit level OEx) under supervision. In doing so, they draw up a competence profile successfully ending a phase. A learning outcome is a partial contribution to a competence step.

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

43.5. Examination and final grading

43.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work you do and the quality of your work.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning path. In addition, throughout the entire semester, the assessors monitor your progress.

43.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record these conversations in your portfolio.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome in the final assessment.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue to work on it.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. This gives you the opportunity to address and process the feedback received.

43.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory' grade.

43.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

43.6. Learning activities

The student will determine the general lines of the content in this programme him or herself. Basically everything in IT should be possible. Because even the same speed, ... This means all workshops, presentations, etc., that will provide content, will be planned and scheduled on-demand or when the course is running.

To facilitate this the student will create his/her own personal profile within the HBO-i Competence Framework, this framework gives a systematic overview of the competences. The student will be invited for an intake interview. This interview is about motivation and a possible context to use within open learning. In an intake document the student will describe an "extreme interaction". The intake interview will be an evaluation to see whether you really understand what open learning is about, so you know what you are capable of.

Because everyone has his own track, it is impossible to plan a curriculum that fits for everyone. Learning happens asynchronous, not everybody does the same things at the same time. The teachers notice it is necessary.

For example, workshops of more general applicability are offered to all students, like Scrum-101, Research, Concept validation, while more specific ones are offered to students who are interested in a specific field.

43.7. Learning materials

Apart from the Canvas LMS environment, anything which would be applicable can and will be used.

[Return to table of content]

44. INFORMATION ABOUT [MA-OLD] OPEN LEARNING (30EC)

44.1. Content

ICT & Personalized Program is a route where you have a lot of freedom and flexibility to determine your personal focus and structure, within the boundaries of the Program. This gives great opportunities for starting multi-disciplinary projects. Projects should be IT related, based on relevant questions and developments in the field.

Lots of freedom, but remember:

"Freedom comes with responsibility"

This large degree of freedom does not mean that there are no obligations. A large amount of autonomy and entrepreneurial behaviour is expected.

44.2. Learning outcomes

1. Learning outcome: Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders and consider sustainable aspects.

2. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your own goals.

44.3. Expected entry profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

44.4. Outflow profile

The student builds a competence profile based upon the basic architectural layers (level 1) from OE2, this is taken to the next step at level 2. In addition, the layer level grows with the phase of the students' study. Phase OE1, OE2 ~ level 1, phase OE3,4,5, ~ level 2 and phase OE6,7,8, ~ level 3.

A balanced competence profile does not have too many competencies (no focus) but also not too little (no widening). At least all 9 competencies have to be reached.

Within Open Learning, students define their own dynamic competence profile per semester (exit level OEx) under supervision. In doing so, they draw up a plan of action successfully ending a phase. A learning outcome is a partial contribution to a competence step.

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

44.5. Examination and final grading

44.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning aspect: throughout the entire semester, the assessors monitor your progress.

44.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the development

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the scale, which gives you the opportunity to address and process the feedback received.

44.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a

44.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

44.6. Learning activities

The student will determine the general lines of the content in this programme him or herself. Basically, everything in IT should be possible. Because everyone learns

To facilitate this the student will create his/her own personal profile within the HBO-i Competence Framework, this framework gives a systematic overview of the competences. The student will be invited for an intake interview. This interview is about motivation and a possible context to use within open learning. In an intake document you will be asked to describe your "extreme interaction". The intake interview will be an evaluation to see whether you really understand what open learning is about, so you know what you

Because everyone has his own track, it is impossible to plan a curriculum that fits for everyone. Learning happens asynchronous, not everybody does it at the same time. The teachers notice it is necessary.

For example, workshops of more general applicability are offered to all students, like Scrum-101, Research, Concept validation, while more specific subjects are offered to

44.7. Learning materials

Apart from the Canvas LMS environment, anything which would be applicable can and will be used.

[\[Return to table of content\]](#)

45. INFORMATION ABOUT [AV-BDA] BUSINESS IT & DATA ANALYSIS - ADVANCED (30 ECTS)

IT Governance, IT Architecture, Predictive Modelling

45.1. Content

The Advanced Semester Business IT & Data Analysis is focused on developing professional skills together with relevant technical skills in Information Management, integrating appropriate technology components and solutions.

The work of the semester is designed for learning how to work effectively and reliably on real-world projects in collaborative groups. Each student will contribute to the success of the group, and that they have succeeded in gaining those learning outcomes.

At the start of the semester, projects will be presented by Fontys Partners in Innovation. There may also be projects from other Fontys partners and private enterprise. The students will be assigned a selection of these projects to work on in their project groups.

The projects will require the students to first formulate research questions and then use modern research methodologies to develop answers to those questions.

After completion of this semester, students should be able to manage their own work together with a group, and to work together to turn problems into solutions.

The course will take place in Strijp TQ, with students being required to be present at specified times during the week. A substantial amount of time will be spent on

In addition to the core project work of the semester, participation in additional events and activities, such as Career Day and Innovation Insight, will be encouraged. Attendance at all scheduled or notified events is mandatory unless they are explicitly marked as optional.

45.2. Learning outcomes

1. **Learning outcome:** Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide advice and sustainable arguments.

Explanation: Responsibility: You have the obligation to fulfil tasks, duties, or expectations associated with your particular role or position. You take and deliver the expected results.

Relevant selected methodologies: You select a methodology depending on the specific goals, context, and nature of the research or project. Select

Advice to your stakeholders: You consider the context, the nature of the relationship, the type of cooperative partner and the specific needs or characteristics

Complex and uncertain contexts: Identify the hallmarks and roles of the environment of the assignment and know the business legitimization.

Future-oriented choices: You make decisions and take actions that are focused on long-term goals, outcomes, and sustainability while considering task division.

2. **Learning outcome:** Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show

Explanation: Formulate goals and actions: You define specific objectives and determine the steps or actions needed to achieve them. You take initiative

Leadership: Refers to the ability to take initiative and responsibility for your own actions.

Long term development as an ICT professional: You focus on continuous learning, skill enhancement, and adaptation to the evolving landscape of

Carrying out these actions: Execute and implement specific tasks or activities to achieve a goal or desired (learning) outcome.

Achieving your goals: Successfully reach the desired (learning) outcomes or milestones that have been set intentionally.

3. **Learning outcome:** IT Governance and Strategic Alignment

You analyse the consequences of a (strategic) change of direction on organisational processes and their information provision and provide advice and values.

Explanation: You show understanding of the, possibly unstructured, strategic and operational issues of IT governance in the enterprise. You analyse in a manner based on known best practices. You determine the aim of an evidence-based solution, and choose an appropriate approach. You drive the

You are able to apply IT within a given strategy, but also to develop a strategy for business development based on new technological possibilities in a pragmatic, critical and based on sources.

In a sustainable manner means that your solution is solid and adaptable in the future. You practice continuous awareness of the potential consequences of your own line of reasoning alongside others.

For implementation, you apply known models for change management from the IT perspective, as well as from the organisational and human perspective.

4. **Learning outcome:** Business and IT Architecture You design the architecture of the organisation and its activities in several complementary view processes, based on an implementation plan that is tailored to the enterprise and its technology and market space.

Explanation: Enterprise architecture refers to dual meanings: both the architecture of the enterprise itself (the business architecture), and the type of choices in IT architecture in relation to the (type of) enterprise.

Design means that you analyse and design new or existing organisational processes in coherence with new or existing information systems, while taking into account process and ask questions from various perspectives, you match these questions with a fitting approach that is pragmatic, critical and based on sources

Frameworks, rules and principles are derived from business and IT standards and best practices, including IT audit, and Change management.

5. **Learning outcome:** Develop business processes and information systems You analyse, design and modify processes and systems that support acceptance criteria based on quality properties, including risk analysis carried out with attention to security aspects, among others. You provide a

Explanation: You design and develop - applying modern technologies and methods - tools and applications that add true value for the stakeholders to make. You use (process) data to improve the business operations and validate the outputs and outcomes by thorough testing, and you apply new knowledge aware of your role in the context of the IT assignment, exploring and tackling the tasks involved, addressing others, searching for opportunities for

6. **Learning outcome:** Implement and improve business processes and information systems You are able to implement, prototype, and/or integrate you take into account the use of existing components and libraries, apply design-quality criteria, and safeguard the integrity, security and performance

Explanation: You realise working prototypes of information systems, applications or tools, or parts of them. Throughout all phases of the development You can manage small and large data sets and process these into information and useful business intelligence. You follow industry standards and

45.3. Expected entry profile

You are able to analyse business processes, execute a gap analysis and gather and document (user) requirements and use cases.

You are able to extract data, do an EDA, clean and transform the data and create insightful visualisations.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	1	1	1	1	1
Organisational processes	2	2	2	2	2
Infrastructure	1	1			1
Software	1	2	2	1	1
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

45.4. Outflow profile

You are capable of working on multiple projects simultaneously. You can advise a company on strategic level. You can research to find the best solution data driven decisions.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	2	1	1	1	1
Organisational processes	3	3	3	3	3
Infrastructure	2	1	1		
Software	2	2	1	2	2
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	3	3	3	3

45.5. Examination and final grading

45.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount c

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape p aspect: throughout the entire semester, the assessors monitor your progress.

45.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You recor

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the deve

Developmental Feedback Scale

- *Undefined:* You have not yet undertaken any activities to demonstrate the learning outcome.
- *Orienting:* You have made a start and explored possibilities to demonstrate the learning outcome.
- *Beginning:* You have taken initial steps that contribute to demonstrating the learning outcome.
- *Proficient:* You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning o
- *Advanced:* You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectat development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't c

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using this gives you the opportunity to address and process the feedback received.

45.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory'.

45.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

45.6. Learning activities

Workshops and lectures, seminars, self-study and research, presentations, quizzes, interviews, presentations to the customer, support for professional development.

There are certain regular events and activities that are part of the FICT semester calendar, such as Career Day and Innovation Insight, in which all students participate.

The semester includes an International Week which is an opportunity for cultural interchange and exchange of ideas. During this week the students of FICT have the opportunity for students of FICT to pay a return visit to the partner institutions involved.

45.7. Learning materials

Course materials and links will be made available through Canvas, and portflow will be the sole site for submission and assessment of work and assignments.

There will also be a dedicated team in MS Teams for the course, which is to be used for the everyday processes of the course, including project team communication.

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46. INFORMATION ABOUT [AV-CSS] COMPLEX SOFTWARE SYSTEMS (30EC)

Development of Complex Software Environments, Non-Functional & Data Requirements

46.1. Content

Enterprise software can be described as large-scale distributed software aimed at an organization rather than end-users. In this software the *quality of service* is typically, this load is not distributed evenly over time. Services provided by enterprise software have to be available around the clock. Such software should be designed and require following of the industry best practices to resolve them.

In this semester we will learn how to design, implement and validate a high quality enterprise software according to the industry best practices. The main goal is to develop a software that meets the requirements of the industry.

In this semester, we will use a strategy of research based learning. Students will be challenged to solve a real world problem with real world stakeholders based on industry best practices.

46.2. Learning outcomes

1. Learning outcome: Scalable Architectures

Besides functionality, you develop architecture of an enterprise software based on explicitly stated software quality requirements. You explicitly focus on scalability. You design your system with future adaptation in mind. You assess the extent in which the quality requirements are met by your software.

Explanation: Software is a composition of the source code, data, infrastructure and runtime. Relevant quality requirements are expressed as non-functional requirements and be supportive for the system architecture realisation.

You investigate architectural patterns that work well in the context of your projects. Your chosen architecture must support verifiably multiple relevant quality requirements.

2. Learning outcome: Development and Operations (DevOps)

You set up environments, tools and processes which support your continuous software development process. Your deployment environment supports automated deployments.

Explanation: Your software is available to developers and other stakeholders during development and in production environment. Your software is deployed to a cloud environment.

You define environments which will be used by developers in the development and creation of software. These infrastructure environments are designed to be reusable.

You choose a deployment strategy to deploy reliably and repeatedly while aiming for a zero downtime in a non-local environment. You support automated deployments and investigate which software performance indicators you will need to collect and monitor those while the software is running. You validate that your software is running in a cloud environment using CI/CD principles).

3. Learning outcome: Cloud native

You develop your software according to the best practices of cloud native development. You deploy (parts of) your application to a cloud platform. You use cloud services for your software quality.

Explanation: You develop software that is completely or partially composed of cloud services. You investigate what tools, architecture design and cost of ownership. You estimate the total cost of software ownership in the cloud and take precautions to reduce it. In your architectural decisions you discuss the impact of cloud services.

4. Learning outcome: Security by Design

You investigate how to minimize security risks for your application, and you incorporate best practices in your whole software development process.

Explanation: You investigate which security risks are most common (for instance OWASP top 10) and you investigate which best practices are used. You use common techniques (for instance misuse cases, trust boundaries) in analysis and design of your architecture. You implement common techniques when they still occur.

5. Learning outcome: Distributed Data

You apply best practices for handling and storing large amount of various data types in your software. You use the non-functional requirements of compromising other software qualities.

Explanation: You create non-functional and functional requirements focused on data storage and handling. You thoroughly investigate which data volume, access pattern and data variety. You identify the sensitive data in your software and take precautions to protect it. You apply legal requirements.

6. Learning outcome: Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide adequate sustainable arguments.

Explanation: You develop and deliver professional products of high quality within a complex context, both individually and as a team, that fits the context may contain multiple stakeholders with different interests, is research heavy, has social impact, and may have legal, ethical, intercultural aspects.

You apply critical thinking in your day-to-day work and while doing research. Each of the research results can be validated and is relevant and valuable advice stakeholders.

Your final solution is designed with the possibility for future further development and is transferrable. You deliver professional products according to people's work. You use appropriate communication considering your role in a team, your audience and the medium to convey your message and

7. Learning outcome: Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show

Explanation: Leadership means that you can proactively define your own personal and technical goals and follow-up actions. You reflect on your received feedback and act upon it to improve your work.

A professional attitude is the way you conduct yourself in a professional setting. Common traits associated with a professional attitude are respect

For your long-term development you are aware of multiple career paths and can reflect which ones fit best, considering your (potential) skills and you adjust your personal development plans. The preparing for graduation or finding a minor should be part of your plan (for instance, by finding a

46.3. Expected entry profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction			1	1	
Organisational processes	1		1	1	
Infrastructure	1		1	1	
Software	2	2	2	2	2
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

46.4. Outflow profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction			2	1	
Organisational processes			1	2	
Infrastructure	1	1	2	2	1
Software	3	3	3	3	3

Hardware-interfacing				
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Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	3	3	3	3

46.5. Examination and final grading

46.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your progress throughout the entire semester, the assessors monitor your progress.

46.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental

Developmental Feedback Scale

- *Undefined:* You have not yet undertaken any activities to demonstrate the learning outcome.
- *Orienting:* You have made a start and explored possibilities to demonstrate the learning outcome.
- *Beginning:* You have taken initial steps that contribute to demonstrating the learning outcome.
- *Proficient:* You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome
- *Advanced:* You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the scale. This gives you the opportunity to address and process the feedback received.

46.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a

46.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

46.6. Learning activities

Advanced Software is a research-based semester. By doing research, you will learn methods and technology that are common in a daily practice of the preferred project methodology and planning. There is a flexible learning path in which focus on learning outcomes is determined and you can choose from

You will spend half of your time on your Group Project (including knowledge exchange among students) and half of your time on your Individual Project

Typically, the group project is part of a long-lasting research project initiated by a Partner in Innovation or professorship (dutch: Lectoraat), named here trigger you into doing research, learning, and applying your knowledge in the context of an authentic project. Examples of projects that students have worked on: a problem that the client ask you to solve is authentic: the client doesn't have an answer to this problem themselves and very interested in further development in the context of a large software project. It may also be possible to create a multidisciplinary environment by joining projects with students from different

The role of the individual project is two-fold. First, it serves as a playground for the technical skills required in the group projects – you learn technology through your individual group project.

The semester will be divided in sprints of 3 weeks. You learn by acquiring knowledge and skills through workshops and educational materials, exchanging ideas, and further study and define your own learning path.

During this semester, you will be part of a group of 5 or 6 students. You will learn by means of an applied research in a group project, and an individual graduation preparation, societal impact, and the research framework.

You will be coached and judged by three teachers. One of them is your tutor for the group project and your semester coach. The other two will guide you

HackIT

The semester starts with a kick-off on Monday, followed by a four-day hackathon on Monday through Thursday. On Monday you will be assigned to the teams. You will present your results on Thursday afternoon.

46.7. Learning materials

47. INFORMATION ABOUT [AV-CSP] CYBER SECURITY PROFESSIONAL (30EC)

Security Innovation & Red, Blue or Yellow Specialisation

47.1. Content

The **Cyber Security Professional semester** prepares students to take on a specialised role in cybersecurity by working on real-world security challenges (**compliance**)—and build a portfolio that reflects their expertise in that role. For example, a Red Teamer may conduct a penetration test and report the assessment or develop an incident response plan based on relevant frameworks (e.g., ISO 27001 or NIST).

Throughout the semester, students are expected to define and carry out their own research or project work using structured methodologies—such as in small teams or individually, often in open-ended or changing scenarios, where they must analyse and respond to both technical and organisational security.

In addition to technical development, students plan and reflect on their personal growth as ICT professionals. This includes setting learning goals, keep but also consider broader ethical, sustainable, and intercultural impacts of their work, ensuring their solutions are relevant in a real-world professional context.

47.2. Learning outcomes

1. Learning outcome: Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide adequate sustainable arguments.

2. Learning outcome: Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show

3. Learning outcome: The Security Specialist

The student demonstrates expertise and competence in a self-chosen professional security role. This is evidenced through a portfolio containing

Explanation: Expertise and competence refer to the student's ability to perform, explain, and evaluate typical tasks and deliverables associated with

- * Applying technical skills using relevant technologies and tools,
- * Selecting and following appropriate standards and procedures,
- * Producing professional reports,
- * Collaborating within and outside the organization,
- * Addressing ethical considerations,
- * Sharing knowledge and advising peers.

The chosen professional role must reflect an existing cybersecurity profession with defined responsibilities and tasks. Pre-defined options include

4. Learning outcome: The Researcher and Developer

The student demonstrates the ability to research and develop (R&D) innovative solutions within an open and complex context. This is shown through reflection.

Explanation: Research involves:

- * Defining a structured research approach with clear research questions, using a combination and triangulation of diverse research strategies, and
- * Justifying choices and considering alternatives during research or project execution;
- * Planning and carrying out the transfer of R&D outcomes to clients, stakeholders, target groups, the broader community, and society.

Development includes:

- * Applying technical development skills;
- * Learning and implementing new technologies or technological concepts;
- * Demonstrating project management skills in your role as a developer and innovator, in collaboration with peers, stakeholders, and experts.

Innovative solutions are those that clearly contribute in a new way to the needs or challenges of stakeholders or target groups. This includes adding An open and complex setting refers to the student's ability to handle situations that begin with an undefined scope or uncertain outcomes, and to

47.3. Expected entry profile

The standard preparation semester is main Cyber Security Essentials.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes					
Infrastructure	2	2	2	2	2
Software					
Hardware-interfacing					

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

47.4. Outflow profile

The outflow is level 3 competences in analysis, design, realization, advice, and manage and control. The semester prepares you for a graduation internship.

Domain description HBO-i 2023

	Manage & control	Realise	Design	Advise	Analysis
User interaction					
Organisational processes	2	2	2	2	
Infrastructure	3	3	3	3	3
Software					
Hardware-interfacing					

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	3	3	3	3

47.5. Examination and final grading

47.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work you do and the quality of your work.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning path. Throughout the entire semester, the assessors monitor your progress.

47.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record these conversations in your portfolio.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome in your final project.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue to work on it.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using this scale. This gives you the opportunity to address and process the feedback received.

47.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory' grade.

47.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

47.6. Learning activities

The global learning activities are:

- a security specialisation, in which a student learns and show to be able to perform the typical tasks and responsibilities of a self-chosen security p
- a group project, in which the student learns and shows research and development skills, from innovation and working with stakeholders to technic
- a personal project, in which the student has maximum freedom to choose personal learning goals.

47.7. Learning materials

The Canvas course contains the relevant study materials and references to and ideas about where to look for materials. The students are expected to :

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48. INFORMATION ABOUT [AV-EAI] ENHANCED AI TECHNIQUES (30EC)

Artificial & Convolutional Neural Networks, Deep Learning, Reinforcement Learning & NLP

48.1. Content

This document describes the semester Enhanced AI Techniques (which was until summer 2025 termed semester "AI-Advanced"). Because the field of semester, a profound basic knowledge of AI techniques and principles is therefore expected at the start.

48.2. Learning outcomes

1. **Learning outcome:** (1) Data preparation; You are able to prepare and store a given dataset in such a way that it can be used in your data analys

Explanation: Preparing a dataset consists of determining the right sources, collecting the useful data (subsets) from those sources, merging ther transparent to others, and the results are suitable for data analysis and/or modelling. Storing a dataset includes investigating cloud solutions and

2. **Learning outcome:** (2) Data analysis & model engineering; You are able to reliably apply machine (including deep) learning algorithms and othe

Explanation: Reliably refers to the process being reproducible, results being explainable to humans and based on decision making that is consid Learning (RL) and Natural Language Processing (NLP). Applying consists of the following iterative steps: 1) exploratory data analysis; 2) feature using grid search and cross-validation.

3. **Learning outcome:** Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide ad sustainable arguments.

Explanation: Appropriate communication means reporting and/or presenting the approach, process or outcomes of a data analysis in a methodo role and on the projected results. You effectively manage cultural differences in your interactions with all involved. Interaction about data analysis communicate results, including appropriate data visualizations (such as data stories, infographics, or (a set of) static or interactive plot(s)).

Assess the context means the organizational and societal environment in which the data analysis project takes place. You show that you can ider organization and society. Multiple perspectives include social and ethical considerations, law compliance, organizational data maturity, alignment based automated decision-making will be an important and integral part of your learning process. A viable business case adds value for the stake foresight). Sustainable manner means you take long-term effects (people, planet, profit, purpose) of your solution into account.

Identifying the problem means you can formulate research questions in order to determine the aim of your solution in an unstructured and multidis phases/components: 1) project proposal 2) data preparation 3) data analysis/modelling and evaluation 4) results and deployment. Coming up with verifiable sources. You are able to methodically and creatively find answers to applied research questions, considering alternatives and critically a

4. **Learning outcome:** Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show

Explanation: Entrepreneurial mindset includes being aware, seeing opportunities and seizing them, motivating oneself and others, being able to enhancing ones' own learning capacity, demanding and giving active feedback, all with respect to the learning outcomes. Professional ambitions from others in field of AI/data.

5. **Learning outcome:** (5) Graduation preparation; You create chances to acquire and define an internship assignment based on a match between :

48.3. Expected entry profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction			1	1	
Organisational processes				1	
Infrastructure				1	
Software	2	2	2	2	2
Hardware-interfacing					

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

48.4. Outflow profile

Domain description HBO-i 2023

	Manage & control	Realise	Design	Advise	Analysis
User interaction		2	2		
Organisational processes					2
Infrastructure					2
Software	3	3	3	3	3
Hardware-interfacing					

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	3	3	3	3

48.5. Examination and final grading

48.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning path. Throughout the entire semester, the assessors monitor your progress.

48.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
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- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the scale. This gives you the opportunity to address and process the feedback received.

48.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a

48.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

48.6. Learning activities

This (S6/7) semester "Enhanced AI Techniques" focusses strongly on more advanced machine learning (ML) algorithms and techniques than were pro

The focus in this semester lies on such deep learning techniques as:

- Artificial Neural Networks (ANN)
- Convolutional Neural Networks (CNN)
- Reinforcement Learning (RL)
- Natural Language Processing (NLP)

Within this course, the terms 'machine learning' and 'deep learning' are understood as "a set of modern computer algorithms with which a structured da be provided where needed. We consider mathematics as a useful means, which can provide a deeper insight in the underlying principles of an algorithm

Students are expected to clean data in such a way that the process of cleaning is structured, explainable and repeatable. In order to explore and disco semesters. The focus of this Enhanced AI Techniques semester is on unstructured data in the format of images, video, speech, and or written text.

Moreover, the 'data practitioner' in this semester should be able to reflect on their personal professional development, including but not limited to: apply effectively verbally and in writing about research ideas and findings, and skills that challenge the student in these fields as to ensure an environment in

Students are stongly encouraged to use the data science methodology, as created by IBM Analytics and to use the steps and deliverables as describer

The semester is roughly divided in 3 parts that are intertwined through the course of the semester:

First, the **Core program** of this semester is composed of a recap of the basic machine learning algorithms from the second year of the study, workshop

- Data Visualisation
- Ethics & Law

Second, around this Core programme is a (group)**Project** in which students approach a relevant real world case that requires the application of the the opportunities to work on their learning outcomes.

Third, starting about halfway the semester, is a smaller **Data Challenge**, in which students work either alone or in duo's on an AI topic of their own choi Processing), this Data Challenge provides ample opportunity to show the learning outcomes at the appropriate level. The choice for the challenge topic course provides some starting points with references to study materials in other interesting AI topics and the goal is to expand that 'library' as we go.

With regard to support from the tutors:

For the group project, each student group will get a **technical tutor** and a **process tutor**, which they will meet by default every week. The latter also fu **challenge tutors**, guiding their work. Before starting the Data Challenge, the student formulates a plan and discussed this plan with their tutors. As me include the possibility for the student to become proficient in them before the end of the semester.

The didactic form used in the Enhanced AI Techniques semester is **Research-Based Learning** (see Canvas). The basic rule is to have optimal use of

48.7. Learning materials

A Canvas course including modules with slides, assignments, and references to external sources and free online study materials (for example, MOOC)

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49. INFORMATION ABOUT [AV-GP] GAME PRODUCTION (30EC)

49.1. Content

Game Production builds upon your *Game Development* and *Game Design* experience, elevating it to a more professional level. You learn to work toget experts. This demands a more professional, structured approach and project scoping to produce a *vertical slice*—a polished yet limited segment of you the continuation of your game's development to an alpha release at the end of the semester.

The semester also offers an opportunity for an in-depth exploration of a chosen topic, allowing you to specialize and contemplate your focus for the res

Ambitious students also get the chance to start a real indie game studio, assisted by a network of game industry professionals (restrictions apply).

49.2. Learning outcomes

1. Learning outcome: Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide ad sustainable arguments.

Explanation: Responsibility: You manage planning and monitoring of your project effectively, adapting to the stakeholders' way of working and cc solutions.

Relevant methodologies: You identify relevant challenges of the client and use a variety of research strategies, methods, and activities (such as th Advice: You describe and document your work effectively, so that your work is reproducible and transparent, and your outcomes can be used by c

Future-oriented choices: When applicable, you consider and act according to ethical standards, sustainable practices, and work towards respons

2. Learning outcome: Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show

Explanation: Independently formulate goals and actions: You reflect on your own learning ability and actions, using feedback that you asked and Leadership: You can take the lead in planning and setting goals for your own project and development. You take responsibility for your role in a gr

ICT professional: You know which role you envision in game development or the wider IT-landscape and what role you play in a team. You can ju Achieving your goals: You can prepare for your graduation internship or choice of minor that suits your personal development goals. This means i

3. Learning outcome: Analysis

You are able to obtain desired results(1) by researching game related subjects (design and technical) methodically(2) and thorough(3) using the t

Explanation: 1. With desired results we mean that you can clearly define the relevant research question at hand to scope the problem at hand to

means you show that you use justified research methods such as the DOT research framework. 3. Thorough means that you triangulate using different industry such as: software algorithms, engines, game genres, new hardware, business models. You know when and how to use these in your project.

4. Learning outcome: Advice

You are able to make recommendations(1) about the game design process and important decisions therein based on the conclusions and results of your research.

Explanation: 1. With recommendations we mean suggestions that you make that will help achieve desired outcomes and long-term goals. 2. With presentations. Contribution to competence growth: UI.AD.3, SE-AD.2

5. Learning outcome: Design

You show that you can design a complex game in an analytical(1) way using existing frameworks, theories, and models(2). You can validate(3) your design.

Explanation: 1. Analytical means that your work is based on prior analysis insights. You support your design choices with sources such as: Targeted frameworks such as: UML, ERD, Design Patterns, MDA framework and Self Determination Theory as well as concept development methods such as paper prototyping, roleplaying and wizard of Oz and use the results to refine your design. 4. Common industry standard method and prototypes such as paper prototypes, mock-ups, and user tests. Contribution to competence growth: UI.DE.3, SE.DE.2

6. Learning outcome: Realisation

You show that you can develop(1) and validate(2) game prototypes, based on a design(3) where you create, find or adapt(4) audio-visual and narrative assets.

Explanation: 1. With develop we mean that you are able to create technical prototypes based on prior work or from scratch. 2. You are able to validate your prototypes based on prior designs. 4. With create, find or adapt we mean that you can create basic assets, but more realistically will find existing assets. With audio-visual and narrative assets, we mean all 'art' assets for your games such as: 2d art, 3d art, synthesized and/or recorded sound effects, Controllers, Industry Standard Game Engines. Contribution to competence growth: UI.RE.3, SE.RE.3

7. Learning outcome: Manage and Control

You are able to work in a structured and methodical way both individually as well as in a group context(1) where you use relevant processes(2) and tools(3) to manage and control your project.

Explanation: 1. With both individually as well as in a group context we mean that you show you can adjust the process accordingly. 2. With relevant control tools and build management tools. 3. With quality we mean that you are able to define a level of quality to adhere to in accordance with standard automated tests, code reviews and continuous integration and building setups. Contribution to competence growth: UI.MC.3, SE.MC.3

49.3. Expected entry profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	2	2	2	2	2
Organisational processes					
Infrastructure					
Software			2	2	2
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

49.4. Outflow profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	3	3	3	3	3
Organisational processes					
Infrastructure					
Software	2	2	2	3	3
Hardware-interfacing					

Professional Development BSc 2020

	Investigative problem solving	3	3	3	3
	Targeted interaction	3	3	3	3
	Personal leadership	3	3	3	3
	Future oriented organisation	3	3	3	3
Professional development		3	3	3	3

49.5. Examination and final grading

49.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work you do and the quality of your work.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning. Throughout the entire semester, the assessors monitor your progress.

49.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record these conversations in your portfolio.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome with good results.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue to work on it.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. This gives you the opportunity to address and process the feedback received.

49.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory' or better.

49.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

49.6. Learning activities

Game Studio Project (with industry partner)

Students form their own group with the goal to start a multidisciplinary game design team, in essence their own game company (including company name and design, build, and validate a transferable prototype).

In addition, the group can define, design, build, and publish an entertainment game based on their own ideas. This structure allows students to prepare and critically review the quality of the project.

Deep Dive (individual)

In order to allow students to dive into a topic that falls outside their role or scope of the Game Studio project, they work on a two-week project called the Deep Dive.

Workshops

During the semester a range of workshops on both game design and game technology as well as professional development skills are offered in the form of workshops on their portfolio.

Way of working

The entire semester is organized as a project using scrum. In daily standups each project is discussed on process level. In bi-weekly sprint demo's the progress is shown.

Every scheduled day there will be experts and tutors available. There will be regular community meetings to learn from others and exchange solutions.

Blended education

The semester works with blended learning, whereby, in addition to the activities on location, online education can take place that matches content and the semester is explained in the Canvas course. The decision to have parts of the education online is reserved for the degree programme.

49.7. Learning materials

Canvas course including all linked information.

50. INFORMATION ABOUT [GRAA] GRADUATION FALL (30EC)

50.1. Content

Your last semester of your bachelor study is the graduation internship. During that period, you demonstrate in (professional) practice that you have intentions of graduating at, it is also the ideal opportunity to know at which company to quickly start your professional career.

Internships are about working and learning in a professional and authentic practice. It is an assignment/work that:

- you perform at an institution or company;
- is in an IT field that you already have knowledge about.
- is acquired by you and fulfilled by you in an independent manner, you are the project leader.
- results in a final portfolio including a presentation and an IT product.

Duration internship

The internship lasts a whole semester: regularly from Fontys week 1 to Fontys week 18. The assessment takes place at the end of the semester (Fontys week 18). Off that you accumulate according to your internship agreement can be taken in consultation with the company, including company closures and mandatory

Sick leave

Let's hope not, but it is of course possible that you become ill during your internship or miss days on your internship for some other reason. Because you have a problem to achieve the goals of your graduation internship.

Internship permission

In order to start the graduation internship, you must meet the admission criteria. This means you must have obtained all study points preceding the graduation internship of your choice.

It is recommended to start searching for a graduation project in time. This would normally be at the beginning of the semester prior to your graduation semester.

50.2. Learning outcomes

1. Learning outcome: Professional duties

You carry out professional duties at a bachelor level, producing professional-grade work aligned with the IT subdomain (IT-area) you are specialised in.

2. Learning outcome: Situation-orientation

You apply your previously acquired knowledge and skills in a new, authentic context to deliver valuable and relevant results for both the project and the organisation.

3. Learning outcome: Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide adequate and sustainable arguments.

4. Learning outcome: Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show initiative and motivation.

50.3. Expected entry profile

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	3	3	3	3

50.4. Outflow profile

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	3	3	3	3

50.5. Examination and final grading

50.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work you do, the quality of your work, and the progress you make.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning path. Throughout the entire semester, the assessors monitor your progress.

50.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record these conversations in your portfolio.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome with good results.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue to work on it.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale, giving you the opportunity to address and process the feedback received.

50.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory' or better.

50.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

50.6. Learning activities

Key moments of examination

Week 3-5	Project plan/project document and company visit by first assessor (onsite or online). The first assessor provides feedback during the first company visit in week 3-5, and the assessors give feedback on the project plan/project document. (Formative assessment)
Week 9/10	Intervision moment at school, sharing results, status and plan so far and receiving and reflecting on feedback. Company supervisor and assessors give feedback and feedforward, also based upon judgment from the student on status and progress. (Formative assessment)
Week 10-17	Hand in portfolio products with a request for feedback; formative assessment.
Week 17	Hand in portfolio. (Deadline Tuesday in week 17).
Week 19-20	Final presentation and/or demonstration of the final product, combined with accountability about the execution and evaluation of the process. Summative assessment by first and second assessor after advice from the other jury members (external expert and company supervisor). The session is public. Interested public do not have to register in advance. In case the student prefers a private session due to personal circumstances or company confidentiality reasons, the student must contact the first assessor no later than week 10 of the school's schedule to register the request in Canvas.

The internship is largely independently acquired by you as a student and carried out at an external client. A Fontys lectrate, staff-department or other internal support and a qualified external company supervisor. The full details on the criteria for the assignment, the company and company supervisor can be found on the website of the Fontys University of Applied Sciences.

During your internship you will be supervised by someone from the company; your company supervisor. In addition, you will receive guidance from school. This is the chairperson during the criteria based interview and also co-assessor. During your internship, your internship teacher is your first point of contact.

50.7. Learning materials

There is no standard teaching material available for the graduation internship. You can, however, use the documents that you can find on Canvas. Here

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51. INFORMATION ABOUT [AV-HIT] HI-TECH EMBEDDED SOFTWARE ENGINEERING (HTESE)

Hi-Tech Embedded Software Design & Solutions

51.1. Content

The unit of learning outcomes Hi-Tech Embedded Software Engineering, is the final preparation for graduation.

The semester focuses on performing authentic, methodically structured, professional tasks, which contribute to new insights for the professional field. They learn to set up and monitor their own learning path.

The research projects of this semester are based on practical problems from our Partners in Innovation and from the lectorates and fit the theme of this semester.

During the semester you will work on a research project from a Partner or Lectorate, and a personal project.

The semester is concluded by the Innovations Insight Fair in week 18, at which the results of all research projects are presented.

51.2. Learning outcomes

1. Learning outcome: Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide adequate and sustainable arguments.

2. Learning outcome: Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show initiative and responsibility.

3. Learning outcome: Analysis

You specify a distributed computer system including timing, resource use and performance, taking into account safety aspects. And you compile a detailed design.

Explanation: Based on a practical problem, you identify the direction of the solution and an appropriate approach.

A distributed computer system means: a system that consists of multiple hardware and software components, including networks, that work together to achieve a common goal.

Acceptance test plan means that you define a plan and methods in advance for validating your solution strategy and implementation (for sub-project).

Integration test plan means that you define in advance a plan and methods that ensure that subsystems connect seamlessly during design and development.

4. Learning outcome: Advise

Based on your analysis, you provide technical advisory on the (distributed) computer system that is to be realised, including hardware and software requirements.

Explanation: You can justify your advice for the direction of the solution towards your stakeholders. Together with a plan of action to come to solution.

Technical advice means: a substantiated advice for solving the problems identified during the analysis and design.

5. Learning outcome: Design

You design a distributed computer system including determining actuators, sensors, timing, resource usage and performance.

Explanation: You design a system consisting of hardware and software components based on your advice, and substantiate this design. You use decomposition and abstraction (for example, design by contract) with the aim of making reusable components, and a correct interaction between components. A distributed computer system consists of multiple hardware and software components, including networks, that work together in a coordinated way for a common goal.

6. Learning outcome: Realisation

You realise a complete computer system (or parts thereof) including network, hardware and system software based on your own design. And you document the implementation.

Explanation: You translate your advice and your design into an implementation. You take care of the quality of the implementation. You create code and make use of standards, methods and techniques relevant to the work field. And you ensure proper integration with existing code.

7. Learning outcome: Manage & Control

You set up and make use of a system for version management, release management, teamwork support and automated testing for hard- and software.

Explanation: Version control means: managing various versions of system components and changing components in a controlled manner using version control systems.

Release management means: the controlled and planned preparation of software for a release. And tracking component and system versions relevant to the release.

Automated testing means: the application of tools in a software development process for continuous and automated software testing and reporting.

51.3. Expected entry profile

To prepare you for this semester, you need to have finished a main semester and an internship, that both cover a good balance of hardware interfacing

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes					
Infrastructure			1	1	
Software	1	1	2	2	2
Hardware-interfacing	2	2	2	2	2

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

51.4. Outflow profile

This semester prepares you for the graduation semester.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes					
Infrastructure			1	1	
Software	2	2	3	3	3
Hardware-interfacing	3	3	3	3	3

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	3	3	3	3

51.5. Examination and final grading

51.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount c

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape p aspect: throughout the entire semester, the assessors monitor your progress.

51.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You recor

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the deve

Developmental Feedback Scale

- *Undefined:* You have not yet undertaken any activities to demonstrate the learning outcome.
- *Orienting:* You have made a start and explored possibilities to demonstrate the learning outcome.
- *Beginning:* You have taken initial steps that contribute to demonstrating the learning outcome.

- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectation in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue to grow.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the feedback tool. This gives you the opportunity to address and process the feedback received.

51.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory'.

51.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

51.6. Learning activities

With this semester, students choose a learning environment with a flexible learning pathway, and a high degree of self-management and learner agency in a research project. The learning content is strongly dependent on the selected projects as well. Because everyone has his own track, it is impossible to provide a fixed learning path.

Learning happens asynchronously for these reasons: not every student does the same things, and the pace of students also differs. The result of this is that learning is personalized.

Workshops of more general applicability are offered to all students, such as workshops about the student journey, the scrum process, making substantial progress, and the importance of feedback.

Blended education

The semester works with blended learning, whereby, in addition to the activities on location, online education can take place that matches content and the pace of the semester is explained in the Canvas course. The decision to have parts of the education online is reserved for the degree programme.

51.7. Learning materials

Apart from the Canvas LMS environment, the InnovationLab's GitLab and MS Teams anything which would be applicable can and will be used.

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52. INFORMATION ABOUT [AV-HC] HUMAN CENTERED DESIGN (30EC)

52.1. Content

In the Human Centered Design semester, you learn to solve complex issues in an investigative and applied way where, searching for the best solution, you use the latest edge technologies, assisted by artificial intelligence. This includes front end programming languages and frameworks to create prototypes, MVP's and user interfaces.

The emphasis is on performing tasks independently (even when working together in a group), showing leadership and taking responsibility. In all these activities, you are encouraged to take initiative and to be creative.

You choose your own projects and bear responsibility for its success!

52.2. Learning outcomes

1. Learning outcome: User interaction (analyse & advise):

You analyse the user, the user interaction and the user experience, also taking state of the art interactive technologies into account. You select a user interface design approach and justify your choice.

Explanation: By analyse, we mean that you carry out research and record the conclusions. By state of the art interactive technologies, we mean the latest edge technologies. By design process and describe the different phases of this process for your project. By advise we mean that you communicate and document your advice.

2. Learning outcome: User interaction (execution & validation):

You execute and evaluate the user experience of an interactive product. You document the development process for the stakeholders.

Explanation: By execute, we mean that you produce a range of prototype(s) and/or MVP(s) you can present to stakeholders and users. By evaluate, we mean that you can substantiate and validate the choices you make in relation to the design process and the user experience (including its interactions).

3. Learning outcome: Realisation of technical products:

You create & design technical products with existing components or libraries using predetermined quality criteria and version control.

Explanation: By you create, we mean that during the semester you will independently build your (showcase) portfolio and realise custom-made technical products. By existing components or libraries, we mean that you have investigated which program language, framework, or library is suitable for the project. By example, scalability, user-friendliness, security aspects and different types of devices. By version control we mean that you use a system to store and manage your code.

4. Learning outcome: Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide adequate and sustainable arguments.

5. Learning outcome: Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show initiative and take responsibility for your own development.

52.3. Expected entry profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	2	2	2	2	2
Organisational processes					
Infrastructure					
Software	2	2	1	2	1
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

52.4. Outflow profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	3	3	3	3	3
Organisational processes					
Infrastructure					
Software	2	2	2	2	1
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	3	3	3	3

52.5. Examination and final grading

52.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning aspect: throughout the entire semester, the assessors monitor your progress.

52.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the development

Developmental Feedback Scale

- *Undefined:* You have not yet undertaken any activities to demonstrate the learning outcome.
- *Orienting:* You have made a start and explored possibilities to demonstrate the learning outcome.
- *Beginning:* You have taken initial steps that contribute to demonstrating the learning outcome.
- *Proficient:* You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome.
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This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't c

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using th gives you the opportunity to address and process the feedback received.

52.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discr

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achiev

52.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

52.6. Learning activities

On-site teaching: The semester is organised so that students can work at least 2.5 half-day sessions on location. On the other days, all scheduled act

Blended education: Blended learning can be used during the semester where, in addition to on-site teaching activities, online teaching can take place offer parts of the education online is reserved for the course.

Individual contribution: Students work with fellow students, stakeholders, and lecturers throughout the semester, but are ultimately assessed individu chosen study cases (which may be worked on in pairs) but also to the group project.

Group project(s): Projects for a group of students with a client/ stakeholder. The project runs for an average of 6-10 weeks (this may vary). Students p lectorate may be the client or fulfil the role of expert and/or consultant. In these projects students usually work with by 4 to 6 students but also this may

International collaboration: During the semester, there are several opportunities for international and intercultural collaboration. This contributes to th but also during the semester.

Workshops and inspiration sessions: A series of optional, interactive sessions on topics from the different "focus areas" and professional skills in wh teachers or initiated by you

52.7. Learning materials

CANVAS contains all the teaching materials and is also where you can submit your assignments. In addition to verbal feedback, you will also receive w have submitted, your semester report and the feedback you have received.

CANVAS also contains exploratory resources. These resources may include information sites, online courses, videos, articles, books, etc. These resou

In addition to CANVAS, we also use MS Teams as an online work environment for communication and announcements. If necessary, communication c

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53. INFORMATION ABOUT [AV-IE] IMMERSIVE EXPERIENCES (30EC)

53.1. Content

During this semester, you'll explore how to create powerful experiences through a blend of storytelling, (digital) craftsmanship and emotional engagemer technologies, photography, lasercutting, 3D-printing, etc. You'll learn how to transform ideas into meaningful experiences, whether through a gesture-c

The program emphasizes both the visible and hidden aspects of immersive experience design. You'll learn how to:

- Create emotional resonance
- Develop concepts and narratives
- Design and/or develop meaningful interactions
- Integrate physical and digital elements
- Work with immersive technology
- Use data to enhance and personalize experiences

This unique combination allows you to create experiences that deeply connect with people, whether through high-tech immersive environments, tactile

53.2. Learning outcomes

1. Learning outcome: LO1: Analysing the Experencer –User

You analyze and apply insights into how people perceive, feel, and remember experiences in order to design emotionally engaging and meaningf

Explanation: • User-Centered Analysis — You analyze user behavior by considering their physical, psychological, emotional, and perceptual per

- Contextual Awareness — You Integrate user insights into broader social and ethical contexts ensuring the experience is relevant, inclusive, and
- Design with behavioral and sensory insight — You understand and apply knowledge of human behavior, sensory perception, memory, and emot
- Iteratively evaluate and adapt — You continuously reflect and adapt based on user feedback and emotional response and translating these exp

2. Learning outcome: LO2: Creating the Interaction – Design

You create value through innovative concept development and interaction design, by translating user insights and contextual understanding into i

Explanation: • Conceptual Innovation. You develop original, relevant, and experience-driven concepts that respond to user needs and project co

- Interaction Design. You design immersive interactions that are technically sound, emotionally resonant, and appropriate for the project's goals —
- Strategic Design Process. You apply a structured and reasoned design process, supported by user research and context analysis, to translate a
- Iterative Development and Value Creation. You prototype, test, and refine your design to deliver measurable value and impact.

3. Learning outcome: LO3: Realising the Experience –Technology

You realize and evaluate immersive experiences by applying appropriate tools and technologies, ensuring the solution aligns with user needs, prc

Explanation: • Technical Proficiency. You apply immersive technologies creatively (e.g. XR, AI, spatial computing) to build impactful prototypes o
• Tool-Based Creativity. You integrate both digital and physical tools in creative and meaningful ways to develop hybrid and multisensory experien
• Phased Realization. You realizes experiences using phase-appropriate methods —from early mockups to functional prototypes — ensuring align
• Professional Delivery. You deliver immersive products with appropriate documentation, usability, and transferability—reflecting professional stan

4. Learning outcome: Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show

Explanation: You demonstrate personal leadership by

- Taking initiative. You'll identify opportunities, set realistic goals, and pursue them, demonstrating enterprise and a forward-thinking mindset
- Reflection and growth. You'll reflect on your own progress, gather feedback, and make well-informed decisions about your future direction and p
- Profiling yourself. You choose at least one field of expertise within the domain of Immersive Experience and clarify your ambitions, your unique

5. Learning outcome: Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide ad sustainable arguments.

Explanation: You demonstrate professional standard by:

- Taking responsibility and solve problems. You'll identify, analyze, and solve complex problems within your project's context
- Considering ethics and context. You'll account for social, ethical, and sustainable aspects, reflecting this in your decisions and designs
- Collaboration and expertise sharing. You collaborate with stakeholders, validate your choices, and proactively share your expertise to foster stro
- Organising and execution. You'll manage your project's phases (timing, budget, delivery) while honoring ethical standards and ensuring sustaina
- Applying methodological research approach. You use structured design processes grounded in methodical research.

53.3. Expected entry profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	2	2	2	2	2
Organisational processes	1				
Infrastructure					
Software	1		1		
Hardware-interfacing	1				

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

53.4. Outflow profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	3	3	3	3	3
Organisational processes	2				
Infrastructure					
Software	2		2		
Hardware-interfacing	2				

Professional Development BSc 2020

	Investigative problem solving	3	3	3	3
	Targeted interaction	3	3	3	3
	Personal leadership	3	3	3	3
	Future oriented organisation	3	3	3	3
Professional development		3	3	3	3

53.5. Examination and final grading

53.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning path. Throughout the entire semester, the assessors monitor your progress.

53.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record these conversations in your portfolio.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome in your portfolio.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue to work on it.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. This gives you the opportunity to address and process the feedback received.

53.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory' or better.

53.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

53.6. Learning activities

On-site teaching: The semester is organised so that students can work at least 2.5 half-day sessions on location. On the other days, all scheduled activities take place online.

Blended education: Blended learning can be used during the semester where, in addition to on-site teaching activities, online teaching can take place in Canvas. The part of the education online is reserved for the course.

Individual contribution: Students work with fellow students, stakeholders, and lecturers throughout the semester, but are ultimately assessed individually on their chosen study cases (which may be worked on in pairs) but also to the group project, which is usually worked on by 4 to 6 students.

Group project(s): Projects for a group of students with a client/ stakeholder. The project runs for an average of 8-12 weeks (this may vary). Students from the pre-graduate may be the client or fulfil the role of expert and/or consultant.

International collaboration: During the semester, there are several opportunities for international and intercultural collaboration. This contributes to the learning process but also during the semester.

Workshops and inspiration sessions: A series of optional, interactive sessions on topics from the different "focus areas" and professional skills in which teachers or initiated by you.

53.7. Learning materials

CANVAS contains all the teaching material and this is also the hand-in place for the elaboration of problems. In addition to oral feedback, written feedback assignments, the semester report and the feedback received.

There are also orientating sources in CANVAS. The exploratory sources can be information sites, online courses, videos, articles, books, etc. These sources are listed in the Canvas.

For communication and announcements we will also use MS Teams as an online working environment in addition to CANVAS. Should it be necessary, we will use MS Teams.

Portflow is an extension of Canvas and will be used to assess your portfolio. In Portflow, students can visualise and self-account for their development. For an explanation on how to use Portflow: <https://beleidswiki.fhict.nl/doku.php?id=en:beleid:portflow>

Feedpulse is an extension of Canvas and is used within this semester to support the feedback process (capturing development-oriented feedback on student work). For an explanation on how to use Feedpulse: <https://beleidswiki.fhict.nl/doku.php?id=en:beleid:feedpulse>

54. INFORMATION ABOUT [AV-IV] INDUSTRY 5.0 (30EC)

AI & Robotics, Digital Twins, Swarm Computing, Smart Cities, Supply Chain Ecosystems

54.1. Content

In this semester students will learn by setting out relevant challenges and address these in a structured and methodical manner. Challenges should allow understanding of processes while setting these skills in the perspective of what they mean in terms of solving the challenge (creating or adding value).

For the 'safe' Single and Dual projects the challenges are set by the combination of context and the Body of Knowledge for Industry 5.0 (Advanced Semester) is based on the diagnosis of the cases of external stakeholders.

Portfolio

As a result of his/her methodical working and attitude, during the semester the student will create a set of professional (half)products that show the

() rigor of the processes,

() the compliance to professionalism and professional standards, () the relevance to the context,

() the quality of outcomes based on SMART challenge attributes and () student's created or added value.

The set of products with a reading guide is the portfolio. The portfolio is what the student uses to show how he/she has reached the learning outcomes

54.2. Learning outcomes

1. Learning outcome: Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide adequate sustainable arguments.

2. Learning outcome: Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show

3. Learning outcome: Analysis and Advise - You are able to find, interpret and utilize information from the (smart industry) domain and the context

Explanation: - This means that, with the collected information, you are able to identify what is relevant and useful for your challenge(s) and project. You understand your added value. NOTE – summarize and present (advise) can be applied to other states (design, realisation and management as well)

4. Learning outcome: Design - Based on the results of rigor analysis you are able to create, present and defend a plan that holds a promise to solve

5. Learning outcome: Realization - You are able to bring your plan to execution while proactively monitoring and preserving the compliance ('to what') through validation and testing.

6. Learning outcome: Manage & Control: You set up and make use of a system for version management, release management, teamwork support further on the project.

Explanation: Version control means: managing various versions of system components and changing components in a controlled manner using and planned preparation of software for a release. And tracking component and system versions released for customer use. Automated testing means

54.3. Expected entry profile

The standard preparation semester for this advanced semester is Industry 4.0 & IoT. If you have not previously completed this semester but are interested

Upon completion of Industry 5.0 semester, you will achieve Level 3 competencies in analysis, design, realization, advise and management & control.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	2			2	
Organisational processes	2	2			
Infrastructure	2			2	
Software	2		2	2	
Hardware-interfacing	2	2	2	2	2

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

54.4. Outflow profile

This semester prepares you for:

- A graduation in the area of Smart Industry & IoT.
- Further specialisation in the area of Robotics.
- Other ICT advanced programs where the gained knowledge and skills can be applied, such as, embedded software engineering, AI, software engineering.

Domain description HBO-i 2023

	Manage & control	Realise	Design	Advise	Analysis
User interaction	2	2			
Organisational processes	2	3			
Infrastructure	2			2	
Software	2	2	2		
Hardware-interfacing	3	3	3	3	3

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	3	3	3	3

54.5. Examination and final grading

54.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work you do throughout the semester.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning. In other words, you can choose your own path to reach the learning outcomes. Throughout the entire semester, the assessors monitor your progress.

54.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record these conversations in your portfolio.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome with good results.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue to work on it.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using this scale. This gives you the opportunity to address and process the feedback received.

54.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory' grade.

54.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

54.6. Learning activities

Students build their own individual portfolio. Within this semester, students choose a learning environment with a flexible learning path, and act on a high Triple Iteration Project (TriP) and the individual project. They are challenged to explore new knowledge in a methodical and structured manner based on oriented and professional development oriented support all of the projects.

During the Industry project students apply and extend their gained knowledge and insights in the more complex context of a project with an external project. Next to the project's focus on quality and applied knowledge the transferability is an explicit aspect of each project.

Blended education

The semester works with blended learning, whereby, in addition to the activities on location, online education can take place that matches content and semester is explained in the Canvas course. The decision to have parts of the education online is reserved for the degree programme.

54.7. Learning materials

A Canvas course including modules with (self) study materials, inspiration sessions and external sources is available.

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55. INFORMATION ABOUT [AV-NCI] NETWORK & CLOUD ARCHITECTURE INNOVATION

Interoperability, Agility & Innovation of Enterprise Architecture, DevSecOps

55.1. Content

The Network and Cloud Architecture Innovation semester helps you deepen your knowledge of designing, managing, and improving IT infrastructures, implementation, and management, preparing you for complex IT environments.

You'll apply structured methodologies to create secure and efficient infrastructures while aligning with industry best practices. Frameworks like TOGAF Throughout the semester, you'll receive continuous feedback from coaches and peers. Two coaches will regularly check in to support your progress and

55.2. Learning outcomes

- 1. Learning outcome:** Professional standard
You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide adequate sustainable arguments.
- 2. Learning outcome:** Personal leadership
You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show
- 3. Learning outcome:** Architecture of an enterprise infrastructure
You can methodically translate the requirements for an enterprise infrastructure into different alternatives for an infrastructure design and justify them.
Explanation: When drawing up the various alternatives for the infrastructure architecture, the applications, data storage and performance such as
- 4. Learning outcome:** Realization of enterprise infrastructure solutions
You can realize an enterprise infrastructure in a methodical way and demonstrate with an extensive test report and demos that the infrastructure is
Explanation: When realizing the infrastructure, the student justifies the choices made and demonstrates how the infrastructure works with tests (
- 5. Learning outcome:** Analytics for enterprise infrastructure
You can collect data from various stakeholders / interest groups in a methodical way to arrive at requirements for the design of an enterprise infrastructure
Explanation: When preparing requirements for the infrastructure, the application landscape, data storage, accessibility and performance such as
- 6. Learning outcome:** Setting up management for an enterprise Infrastructure
You can in a methodical way the proactive management of an enterprise designing infrastructure, designing, testing and show that with monitoring
Explanation: When realizing proactive management, you continuously have the status of the infrastructure in view (monitoring & analyses) and you
- 7. Learning outcome:** Innovations and Trends
You have insight into innovations/trends for the infrastructure and/or the management of the infrastructure and you can research these innovations
Explanation: There are constant new developments within the infrastructure. By extensively researching an innovation and/or trends, you show that

55.3. Expected entry profile

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes	2	1	1	1	2
Infrastructure	2	2	2	2	2
Software	1	1		1	1
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

55.4. Outflow profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes	2	2	1	2	2
Infrastructure	3	3	3	3	3
Software	1	1		1	1
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	3	3	3	3

55.5. Examination and final grading

55.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning. In this aspect: throughout the entire semester, the assessors monitor your progress.

55.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the scale. This gives you the opportunity to address and process the feedback received.

55.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory' or better.

55.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

55.6. Learning activities

This semester is structured around agile, self-directed learning, with multiple iterations. You are responsible for planning your projects while receiving support from the lecturers.

- Group Project – Work with industry partners through Partners in Innovation on real network and cloud architecture challenges.
- Individual Project – Choose your own research topic and develop a project that fits your professional interests and learning goals.

You'll conduct research, ensuring your findings are valuable for both stakeholders and industry. You'll also collaborate with classmates, industry professionals and lecturers to achieve your goals.

55.7. Learning materials

The Canvas course contains the relevant study materials and references to and ideas about where to look for materials. The students are expected to use these materials to guide their learning.

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56. INFORMATION ABOUT [AV-OLF] OPEN LEARNING (30EC)

56.1. Content

ICT & Personalized Program is a route where you have a lot of freedom and flexibility to determine your personal focus and structure, within the boundaries of the Program. This gives great opportunities for starting multi-disciplinary projects. Projects should be IT related, based on relevant questions and developments in the field.

Lots of freedom, but remember:

“Freedom comes with responsibility”

This large degree of freedom does not mean that there are no obligations. A large amount of autonomy and entrepreneurial behaviour is expected.

56.2. Learning outcomes

1. **Learning outcome:** Professional standard
You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide adequate and sustainable arguments.
2. **Learning outcome:** Personal leadership
You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show initiative and responsibility in your own development.

56.3. Expected entry profile

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

56.4. Outflow profile

The output level is based on the achieved growth on the set competence indicators of the students personal competence profile. This should connect a student's personal development to the program's goals.

A balanced competence profile does not have too many competencies (no focus) but also not too little (no widening). At least all 9 competences (including level 2).

Within Open Learning, students define their own dynamic competence profile per semester (exit level OEx) under supervision. In doing so, they draw up a plan for their personal development, successfully ending a phase. A learning outcome is a partial contribution to a competence step.

Professional Development BSc 2020

	Investigative problem solving	3	3	3	3
	Targeted interaction	3	3	3	3
	Personal leadership	3	3	3	3
	Future oriented organisation	3	3	3	3
Professional development		3	3	3	3

56.5. Examination and final grading

56.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning path. Throughout the entire semester, the assessors monitor your progress.

56.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record these conversations.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
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- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue to work on it.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale, giving you the opportunity to address and process the feedback received.

56.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory' grade.

56.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

56.6. Learning activities

The student will determine the general lines of the content in this programme him or herself. Basically everything in IT should be possible. Because everyone has their own pace, ... This means all workshops, presentations, etc., that will provide content, will be planned and scheduled on-demand or when the content is relevant.

To facilitate this the student will create his/her own personal profile within the HBO-i Competence Framework, this framework gives a systematic overview of the competences. A student will be invited for an intake interview. This interview is about motivation and a possible context to use within open learning. In an intake document you will describe your 'extreme interaction'. The intake interview will be an evaluation to see whether you really understand what open learning is about, so you know what you want to learn.

Because everyone has his own track, it is impossible to plan a curriculum that fits for everyone. Learning happens asynchronous, not everybody does the same things at the same time, but the teachers notice it is necessary.

For example, workshops of more general applicability are offered to all students, like Scrum-101, Research, Concept validation, while more specific subjects are offered to students who are interested in that field.

56.7. Learning materials

Apart from the Canvas LMS environment, anything which would be applicable can and will be used.

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57. INFORMATION ABOUT [AV-OLG] OPEN LEARNING (30EC)

57.1. Content

ICT & Personalized Program or Open Learning is a route where you have a lot of freedom and flexibility to determine your personal focus and structure your Personalized Program. This gives great opportunities for starting multi-disciplinary projects. Projects should be IT related, based on relevant questions or challenges.

Lots of freedom, but remember:

“Freedom comes with responsibility”

This large degree of freedom does not mean that there are no obligations. A large amount of autonomy and entrepreneurial behaviour is expected.

57.2. Learning outcomes

1. Learning outcome: Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide ad sustainable arguments.

2. Learning outcome: Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show

57.3. Expected entry profile

Any finished combination in the Main phase will suffice.

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

57.4. Outflow profile

The output level is based on the achieved growth on the set competence indicators of the students personal competence profile. This should connect a

A balanced competence profile does not have too many competencies (no focus) but also not too little (no widening). At least all 9 competences (includ least level 2.

Within Open Learning, students define their own dynamic competence profile per semester (exit level OEx) under supervision. In doing so, they draw u successfully ending a phase. A learning outcome is a partial contribution to a competence step.

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	3	3	3	3

57.5. Examination and final grading

57.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount c

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape p aspect: throughout the entire semester, the assessors monitor your progress.

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57.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

57.6. Learning activities

The student will determine the general lines of the content in this programme him or herself. Basically everything in IT should be possible. Because everyone has the same speed, ... This means all workshops, presentations, etc., that will provide content, will be planned and scheduled on-demand or when the course

To facilitate this the student will create his/her own personal profile within the HBO-i Competence Framework, this framework gives a systematic overview. The student will be invited for an intake interview. This interview is about motivation and a possible context to use within open learning. In an intake document "extreme interaction". The intake interview will be an evaluation to see whether you really understand what open learning is about, so you know what you

Because everyone has his own track, it is impossible to plan a curriculum that fits for everyone. Learning happens asynchronous, not everybody does the same, so the teachers notice it is necessary.

For example, workshops of more general applicability are offered to all students, like Scrum-101, Research, Concept validation, while more specific ones are offered to students

57.7. Learning materials

Apart of the Canvas LMS environment, anything which would be applicable can and will be used.

[Return to table of content]

58. INFORMATION ABOUT [O-AV-OLF] OPEN LEARNING (30EC)

58.1. Content

We have Open Education for PRO from the second year of your study. This means that you will define your own personal learning route. Do you want to focus on a particular technology? It's up to you!

You will work as much as possible in a context that is most supportive of your personal learning trajectory. This means it is rather impossible to determine the content at the beginning of the process.

With this in mind, Education for professionals can be defined as follows:

Educating responsive professionals for all possible careers in IT. This is done by focusing on talent development and personalized competence development.

Students who have successfully concluded OE5 can start in OE6. Starting level would be the learning outcomes output of the OE5. The output level is the level a student is pursuing.

The student will determine the general lines of the content in this program him or herself. Basically, everything in IT should be possible. Because everyone has the same speed, in the same manner, etc. This means all workshops, presentations, etc., that will provide content, will be planned and scheduled on-demand or when the course

To facilitate this, the student will create his/her own personal profile within the HBO-i Competence Framework, this framework gives a systematic overview of the framework and how it functions can be found in the Canvas course.

A complete explanation of the structure of the framework and how it functions can be found in the Canvas course.

Blended Learning

At Education for Professionals, we apply the design principles of Blended Learning. This means that we learn with each other online and face-to-face. We

- Face to face contact (18:00-21:00)

Every Tuesday evening from 18:00 to 21:00, we learn in groups or individually at FHICT. The following activities could be scheduled:

1. Status updates: Give your group members and coaches insight in your learning process. What have you done last week? Show your plan for the week.
2. Discuss about a topic:

You will zoom in into a topic which is introduced by a student (or coach).

1. Self-work/Collaboration: Students are given the time to work together on a challenge or a smaller task which is related to the discussion topic talk
2. Inspiration sessions / Workshops: Inspiration sessions and workshops could be scheduled.

- Online contact (18:00-19:30)

Of course, students can do their own study activities online in all sorts of ways. For example, the student asks fellow students to discuss deliverables or

58.2. Learning outcomes

1. **Learning outcome:** Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide adequate sustainable arguments.

Explanation: You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide adequate sustainable arguments.

intercultural, and sustainable arguments.

2. Learning outcome: Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show

Explanation: You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional

3. Learning outcome: Analysis:

You can obtain desired results by analyzing the processes, products and information flows in a methodical and thorough way.

Explanation: You know what you are researching and why, taking into account the client, target audience & market.

Using several justified research methods based on the DOT-framework

Applying multiple different methods of analysis or focusing on several different aspects. For example: client needs, audience needs, other related

4. Learning outcome: Advice:

You can give advice about the organization, processes and/or information important for decisions related to your challenge.

Explanation: Your advice is based on the conclusions and results from the analysis.

Your advice can be based on cost aspects and quality properties such as: availability, performance, security and scalability.

5. Learning outcome: Design:

You are able to design (parts of) systems, communicate and validate about it for both functional/technical and aesthetic requirements using industry

Explanation: With design we mean that you can use a structured and methodical approach in your design. You use a set of concept development

The design covers both the functional/technical and aesthetic design of your products. You communicate your design in a well presented and structured

You validate your designed systems in an early stage using efficient and quick prototyping methods such as paper prototyping and use the results

6. Learning outcome: Realization:

You can implement and validate a product based on a design in a structured way using innovative and/or relevant technology adhering to quality

Explanation: We expect that you validate your product and the quality of the product in a structured way using methods such as: User testing, experiments

Structured way means using industry standard development processes such as Scrum/Agile and Kanban.

With innovative and/or relevant technology we mean that you experiment and use upcoming and promising technology that suits your needs.

Your product adheres to previously defined quality criteria in accordance with stakeholders. Many non-functional requirements such as stability, performance

7. Learning outcome: Manage and Control:

You can perform activities that focus on control, monitoring and optimization of the development, commissioning and use of ICT-systems. These activities

Explanation: We expect that you build your controlling, monitoring and optimisation of the development in a structured way.

We expect that you can communicate with stakeholders about your activities who are related to manage and control.

58.3. Expected entry profile

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

58.4. Outflow profile

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	3	3	3	3

58.5. Examination and final grading

58.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your

58.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the development

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue to grow.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the feedback scale. This gives you the opportunity to address and process the feedback received.

58.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory'.

58.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

58.6. Learning activities

The teachers will provide you with formative feedback on deliverables and demonstrations. You are expected to record the feedback in FeedPulse and the learning outcomes (Analysis, Advice, Design, Realisation, Manage and Control, Future oriented organization, Investigative Problem Solving, Personal Leadership).

At these demo's the competences are evaluated based on state of the deliverables and your personal development at that point. This will lead to a grade. Keep on progressing on the competences, not doing so might lead to a lower grade at the next formative assessment such as beginning or orienting.

58.7. Learning materials

All information about the education program Education for Professionals OE6 could be found in the CANVAS course. Select your own study materials.

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59. INFORMATION ABOUT [O-AV-OLG] OPEN LEARNING (30EC)

59.1. Content

We have Open Education for PRO from the second year of your study. This means that you will define your own personal learning route. Do you want to learn in a particular technology? It's up to you!

You will work as much as possible in a context that is most supportive of your personal learning trajectory. This means it is rather impossible to determine the content at the beginning of the process.

With this in mind, Education for professionals can be defined as follows:

Educating responsive professionals for all possible careers in IT. This is done by focusing on talent development and personalized competence development.

Students who have successfully concluded OE6 can start in OE7. Starting level would be the learning outcomes output of the OE6. The output level is based on the achieved growth on the set competence indicators of the student's personal competence profile. At the end of the semester, the output level is determined.

The output level is based on the achieved growth on the set competence indicators of the student's personal competence profile. At the end of the semester, the output level is determined.

- For all competences (Advising - Analysing - Designing - Realising and Manage & Control) and the PO-skills proficiency level 3 has been demonstrated.

The student will determine the general lines of the content in this program him or herself. Basically, everything in IT should be possible. Because everyone learns at the same speed, in the same manner, etc. This means all workshops, presentations, etc., that will provide content, will be planned and scheduled on-demand.

To facilitate this, the student will create his/her own personal profile within the HBO-i Competence Framework, this framework gives a systematic overview of the competences.

A complete explanation of the structure of the framework and how it functions can be found in the Canvas course.

Blended Learning

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Every Tuesday evening from 18:00 to 21:00, we learn in groups or individually at FHICT. The following activities could be scheduled:

1. Status updates: Give your group members and coaches insight in your learning process. What have you done last week? Show your plan for the next week.
2. Discuss about a topic: ...

You will zoom in into a topic which is introduced by a student (or coach).

1. Self-work/Collaboration: Students are given the time to work together on a challenge or a smaller task which is related to the discussion topic talk
2. Inspiration sessions / Workshops: Inspiration sessions and workshops could be scheduled.

- Online contact (18:00-19:30)

Of course, students can do their own study activities online in all sorts of ways. For example, the student asks fellow students to discuss deliverables o

59.2. Learning outcomes

1. **Learning outcome:** Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide ad sustainable arguments.

Explanation: You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, a intercultural, and sustainable arguments..

2. **Learning outcome:** Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show

Explanation: You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professi

3. **Learning outcome:** Analysis:

You can obtain desired results by analyzing the processes, products and information flows in a methodical and thorough way.

Explanation: You know what you are researching and why, taking into account the client, target audience & market.

Using several justified research methods based on the DOT-framework

Applying multiple different methods of analysis or focusing on several different aspects. For example: client needs, audience needs, other related

4. **Learning outcome:** Advice:

You can give advice about the organization, processes and/or information important for decisions related to your challenge.

Explanation: Your advice is based on the conclusions and results from the analysis.

Your advice can be based on cost aspects and quality properties such as: availability, performance, security and scalability.

5. **Learning outcome:** Design:

You are able to design (parts of) systems, communicate and validate about it for both functional/technical and aesthetic requirements using indus

Explanation: With design we mean that you can use a structured and methodical approach in your design. You use a set of concept developmer

The design covers both the functional/technical and aesthetic design of your products. You communicate your design in a well presented and stru

You validate your designed systems in an early stage using efficient and quick prototyping methods such as paper prototyping and use the results

6. **Learning outcome:** Realization:

You can implement and validate a product based on a design in a structured way using innovative and/or relevant technology adhering to quality

Explanation: We expect that you validate your product and the quality of the product in a structured way using methods such as: User testing, ex

Structured way means using industry standard development processes such as Scrum/Agile and Kanban.

With innovative and/or relevant technology we mean that you experiment and use upcoming and promising technology that suits your needs.

Your product adheres to previously defined quality criteria in accordance with stakeholders. Many non-functional requirements such as stability, p

7. **Learning outcome:** Manage and Control:

You can perform activities that focus on control, monitoring and optimization of the development, commissioning and use of ICT-systems. These :

Explanation: We expect that you build your controlling, monitoring and optimisation of the development in a structured way.

We expect that you can communicate with stakeholders about your activities who are related to manage and control.

59.3. Expected entry profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

59.4. Outflow profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	3	3	3	3

59.5. Examination and final grading

59.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work you do, the quality of your work, and the progress you make.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning. In addition, throughout the entire semester, the assessors monitor your progress.

59.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record your progress in the portfolio.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome with good results.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue to work on it.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale, giving you the opportunity to address and process the feedback received.

59.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory' grade.

59.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

59.6. Learning activities

The teachers will provide you with formative feedback on deliverables and demonstrations. You are expected to record the feedback in FeedPulse and the learning outcomes (Analysis, Advice, Design, Realisation, Manage and Control, Future oriented organization, Investigative Problem Solving, Personal Leadership).

At these demo's the competences are evaluated based on state of the deliverables and your personal development at that point. This will lead to a grade. Keep on progressing on the competences, not doing so might lead to a lower grade at the next formative assessment such as beginning or orienting.

59.7. Learning materials

All information about the education program Education for Professionals OE7 could be found in the CANVAS course. Select your own study materials.

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60. INFORMATION ABOUT [GA] GLOBAL ACTING IN IT (30EC)

60.1. Content

Globalization and digitalization kickstarted a revolution, changing everything from how we live to how we work. Students are not just witnessing this history but are part of it. In a semester long project with real world impact. Guided by the UN Sustainable Development Goals, they will explore how future technology can shape our world.

The three keystones of the program are:

- Intercultural awareness and practice
- International collaboration and experience
- Examining IT from a global perspective, with focus on having a positive impact on society, the environment and beyond

The objectives of the minor are:

- Become a responsive ICT professional: adaptive, resilient, innovative and responsible
- Global citizenship & intercultural awareness
- Communication and collaboration across borders
- You can study and work on real life projects in an international environment

The IT topics taught by the 5 partner institutes during this minor are:

- Data Science
- UX Design
- Internet of Things
- Artificial Intelligence
- Immersive Technology

60.2. Learning outcomes

1. Learning outcome: Professional duties in international context

You execute professional IT duties within an international, multicultural environment, delivering professional products that address global challenges for an international project. Your professional products demonstrate technical competency while considering international perspectives,

2. Learning outcome: Situation orientation

You apply your previously acquired IT knowledge and skills in an authentic international context, working collaboratively with students from international universities. Your contributions remain relevant to international stakeholder(s) while navigating different educational systems, technological infrastructure,

3. Learning outcome: Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, and ensure sustainable aspects.

4. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your values and goals.

Explanation: You actively develop intercultural competencies using theoretical frameworks and reflecting on your growth through methodological approaches and vision for becoming an internationally minded IT professional capable of working in global, cross cultural environments.

60.3. Expected entry profile

Students entering the Global Acting in IT minor should possess HBO-i Level 2 competencies within their chosen IT discipline (software development, cybersecurity, and tools within their field and collaborate in team settings. They should be comfortable with fundamental technical concepts, able to analyze requirements, and communicate effectively.

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

60.4. Outflow profile

Through the international minor experience, students will advance to HBO-i Level 3 within their own IT discipline, where they can handle complex, multidisciplinary tasks, demonstrate the ability to work autonomously in uncertain, cross cultural contexts, lead international project initiatives, conduct independent research and innovation, and communicate effectively.

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

60.5. Examination and final grading

60.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work you do, the quality of your work, and the progress you make.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape your learning path. Throughout the entire semester, the assessors monitor your progress.

60.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record these conversations in your portfolio.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome with good results.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't continue to work on it.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. This gives you the opportunity to address and process the feedback received.

60.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory' or better.

60.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

60.6. Learning activities

This minor consists of the following activities:

- A start-up week with introductions and start-up on:
 - Intercultural development & their professional development portfolio (PDP)
 - The overarching project. This project is related to the UN sustainable development goals. Students with work on this project throughout the semester.
- Five 3-week modules organised by the 5 partner institutes. The students learn about the following IT topics, apply this to their project and keep track of their progress:
 - Data Science (Belgium Campus iTversity, South-Africa)
 - Ux Design (Universitat de Lleida, Spain)
 - Internet of Things (University of applied sciences Upper Austria)
 - Artificial Intelligence (PXL, Belgium)
 - Immersive Technology (FICT, Netherlands)
- Two weeks for a repair assignment if applicable.

60.7. Learning materials

Canvas is used as the central location for all learning materials. Each institute can use this to share any relevant learning materials with the students.

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61. INFORMATION ABOUT [SIT] SMART INDUSTRY TECHNOLOGIES (30EC)

HBO-ICT Minor

61.1. Content

The overall goal of Smart Industry is to provide solutions so that the production, products and service systems become smart and interconnected, flexible and adaptable (IoT) with network technologies (5G), cloud/edge computing, data analytics and artificial intelligence (AI), smart supply chains, with manufacturing and adjustable on the fly to meet highly specific end-user demands. It is not surprising that manufacturers and organisations are making radical changes to their production processes.

The semester focuses on performing authentic, methodically structured, professional tasks, which contribute to new insights for the professional field. The students take responsibility for their own learning path. The learning process will be supported by workshops, contacts with experts and coaching in various forms and methods.

The research projects of this minor are based on real life problems from our company partners and from the lectorates. Students will work on one of the projects, broadening their knowledge and learning how new technologies are integrated into Smart Industry, or as **specialists**, by deepening their expertise and knowledge in a specific area.

The **Smart Industry Technologies Minor** aims at students from Fontys ICT and Fontys Engineering who are interested in enriching own knowledge, skills and competences. The minor is also open for students from Mechatronics at Engineering. But besides those, this minor explicitly invites students from other profiles at the Engineering department to join. The learning environment makes this minor unique among other Smart Industry study programs in the Netherlands.

The minor emphasizes hands-on learning through real-world projects developed in collaboration with industry partners. These projects involve vertical integration, providing a complete understanding of how technologies come together in real-world industrial environments.

To support these projects, students will have access to a range of resources, including specialized workshops, expert sessions, practical exercises, and other ICT/Engineering units, such as: Smart Industry and IoT Specialisation (ICT), Minor Adaptive Robotics (Engineering), and Technology Advancement (ICT). The minor aims to apply theoretical knowledge to practical research challenges.

The support includes but is not limited to:

1. Dedicated workshops to cover the basics of Network, Security, Data Analytics, System Engineering, Industrial Automation.
2. Dedicated Advanced workshops on IoT, Infrastructure, Vision, AI, Digital Twin, Industrial Automation, Security.
3. Workshops/ expert sessions on demand from students in the range of the above topics, and broader on Robotics, Additive manufacturing, Embedded Systems, etc.

61.2. Learning outcomes

1. Learning outcome: Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, and address sustainable aspects.

2. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your own strengths and weaknesses.

3. Learning outcome: Smart Industry proficiency

The student is able to define and explain the core concepts, principles, and technologies underlying smart industry. The student is able to analyse applications, and reflect on the impact of these solutions on the productivity and efficiency.

4. Learning outcome: Smart Industrial Integrated solutions

The student is able to design and implement solutions in which he/she integrates various technologies relevant in Smart Industry, such as:

- o Monitoring and controlling smart industrial processes and systems
- o Configuring and programming sensors, actuators, and controllers, as well as network and infrastructure, to enable communication and interope
- o Collecting relevant data and utilizing them for decision making regarding improvements and optimizations
- o Addressing safety and security risks, and implementing best practices to safeguard data networks, and devices against threats and unauthorize

61.3. Expected entry profile

The admission and outflow profiles are somewhat loosened, due to the diverse admission group we are targeting. Students from the Institute of Engine

We expect students to have competences from the HBO-i domain description on level 1 (Analyse, Advice, Design, Realisation, Control & Manage) as 'Solving) on HBO-i level 1.

In the HBO-Engineering domain description there are equivalent competences: Analyse, Design, Realize, Control, and Manage, Advise, Research and given HBO-I competences and we expect students coming from the engineering domain to have these competences on level 1 of the HBO Engineering

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

61.4. Outflow profile

Domain description HBO-i 2023

	Manage & control	Realise	Design	Advise	Analysis
User interaction	1				
Organisational processes	1	1			
Infrastructure	2				
Software	1		1		
Hardware-interfacing	2	2	2	2	2

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

61.5. Examination and final grading

61.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount c

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape p aspect: throughout the entire semester, the assessors monitor your progress.

61.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You recor

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Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the feedback tool. This gives you the opportunity to address and process the feedback received.

61.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a 'Satisfactory' or better.

61.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

61.6. Learning activities

Authentic questions, challenges and problems from professional practice are the basis for all learning activities, and different smaller steps learning tasks during the semester. Coaching and guidance will be offered whenever needed and relevant. Learning content is offered when required or requested in themselves for instruction sessions with teachers or experts by studying the provided study material. During these sessions focus will be on clarifying the content.

Learning happens asynchronously for these reasons: not every student does the same things, and the pace and entry level of students also differs. The support is necessary. The support includes but is not limited to:

1. Dedicated workshops, mostly planned towards the beginning of the semester, that cover the basics of Network, Security, Data Analytics, Systems.
2. Dedicated Advanced workshops on IoT, Infrastructure, Vision, AI, Digital Twin, Industrial Automation, Security.
3. Workshops/ expert sessions on demand from students in the range of the above topics, and broader on Robotics, Additive manufacturing, Embedded systems.

The support will be tailored to meet the needs of each project, ensuring students receive the guidance necessary to apply theoretical knowledge to practice.

Blended education

The semester works with blended learning, whereby, in addition to the activities on location, online education can take place that matches content and the semester is explained in the Canvas course. The decision to have parts of the education online is reserved for the degree programme.

61.7. Learning materials

A Canvas course including modules with (self) study materials, inspiration sessions and external sources is available.

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FONTYS ICT PROGRAMME GUIDE

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31. [MA-AMD]	AI, Machine Learning & Data
32. [MA-AAI]	Applied Generative AI
33. [MA-CSE]	Cyber Security Attack and Defend
34. [MA-FED]	Front End Development
35. [MA-FSD]	Full Stack Development
36. [MA-GD]	Game Design
37. [MA-II]	Industry 4.0 & Internet of Things
38. [MA-ID]	Intelligent Devices
39. [MA-ISP]	Internship
40. [MA-BDA]	IT & Process Transformations
41. [MA-MC]	Media Creation
42. [MA-MAD]	Mobile Apps Development
43. [MA-NCA]	Network & Cloud Automation
44. [MA-OLD]	Open Learning

45. [MA-OLC] Open Learning
 46. [O-MA-OLD] Open Learning
 47. [O-MA-OLE] Open Learning
 48. [O-MA-OLC] Open Learning
 49. [AV-CSS] Complex Software Systems
 50. [AV-CSP] Cyber Security Professional
 51. [AV-EAI] Enhanced AI Techniques
 52. [AV-GP] Game Production
 53. [GRAa] Graduation Fall
 54. [GRAb] Graduation Spring
 55. [AV-HIT] Hi-Tech Embedded Software Engineering
 56. [AV-HC] Human Centered Design
 57. [AV-IE] Immersive Experiences
 58. [AV-IV] Industry 5.0
 59. [AV-NCI] Network & Cloud Architecture Innovation
 60. [AV-OLF] Open Learning
 61. [AV-OLG] Open Learning
 62. [O-AV-OLF] Open Learning
 63. [O-AV-OLG] Open Learning
 64. [AV-BDA] Strategic Information & Data Architectures
 65. [GA] Global Acting in IT
 66. [SIT] Smart Industry Technologies
-

1. INFORMATION ABOUT [O-IN-AIB] AI BASICS (30EC)

Data Preparation & Analysis, Machine Learning, Large Language Models & AI Ethics

1.1. Content

In this semester you'll dive headfirst into AI's exciting world. You'll master fundamental techniques like classification algorithms, bias detection, and model evaluation while seeing how they solve actual problems.

You explore concepts including training data balance, decision boundaries, and prediction accuracy—the building blocks of effective AI systems. You'll learn how to review model performance, detect overfitting, and implement improvements through hyperparameter tuning.

The course covers practical applications: from clustering customer data and visualizing complex relationships to using sentiment analysis for content reliability across digital platforms. You'll discover how recommendation systems work and explore the creative side of AI through generative models.

Through hands-on projects like analyzing satellite imagery, optimizing delivery routes, and predicting customer behavior, you'll gain experience with real-world scenarios. You'll also examine how AI can enhance accessibility and sustainability in urban environments.

By semester's end, you'll be equipped to translate client requests into AI solutions and make smart, responsible decisions in your future career. Let the adventure begin!

1.2. Learning outcomes

1. **Learning outcome:** Professional standard

You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory research, and reporting.

2. **Learning outcome:** Personal leadership

You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional development.

3. **Learning outcome:** Analysing

You can analyze data, prepare and use it for AI applications within a specific domain, where you assess the data

quality, use visualizations to present insights, and evaluate the impact of your choices.

Explanation: In real-world AI, raw data is rarely ready to use. You'll act as both detective—questioning biases, origins, and flaws—and storyteller, turning complex data into actionable insights. Each domain brings unique ethical considerations and specialised knowledge, affecting how data is analysed. The outcome stresses that all data decisions have consequences, making impact evaluation critical to building reliable AI foundations.

4. Learning outcome: Advising

You can make recommendations based on your analysis of AI applications, analyze their societal and ethical impact, and provide advice on this to stakeholders, where you make use of existing frameworks for responsible and transparent AI.

Explanation: This outcome highlights your role as both technical interpreter and ethical guide. Beyond what AI can do, you'll explore what it should do, assessing impacts on people, organisations, and society. You'll identify potential harms and benefits, apply established guidelines, and tailor advice to diverse audiences, maintaining accuracy and ethical rigour throughout.

5. Learning outcome: Designing

You can design AI applications that meet the needs of both stakeholders and users, where you take into consideration model choice, feasibility, quality, and scalability.

Explanation: This is about crafting AI solutions that are not just clever but useful. You'll balance stakeholder goals with user needs, choosing models that suit the context, data, and deployment constraints. Effective design also involves making sound technical design decisions—from selecting appropriate algorithms to structuring system architecture—that align with resource limits, long-term performance, and integration needs. Designing becomes a blend of human-centred thinking and technical planning.

6. Learning outcome: Realising

You can implement existing AI models and services in practical applications, where you can recognize and justify their possibilities and limitations. You have insight into the functioning of various AI models, can implement simple techniques yourself, and validate the achieved results.

Explanation: Here, theory becomes action. You'll bring AI systems to life, making informed decisions about which models to use and why. It's about understanding core principles, being hands-on with basic techniques, and ensuring your implementations work reliably in the real world, not just in theory.

7. Learning outcome: Managing & Controlling

You can monitor and manage the performance and quality of AI systems, define appropriate indicators, manage model versions, and take into account the operational requirements of stakeholders.

Explanation: Managing AI means continuous attention to its reliability and relevance. You'll monitor key metrics, define what success looks like for each use case, and handle versioning as models evolve. Operational context matters—meeting expectations around availability, integration, and transparency ensures that AI systems deliver lasting value.

1.3. Expected entry profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development			1	1

1.4. Outflow profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	1	1	1	1	1
Organisational processes	1	1	1	1	1
Infrastructure					
Software	1	1	1	1	1
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

1.5. Examination and final grading

1.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

1.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- *Undefined:* You have not yet undertaken any activities to demonstrate the learning outcome.
- *Orienting:* You have made a start and explored possibilities to demonstrate the learning outcome.
- *Beginning:* You have taken initial steps that contribute to demonstrating the learning outcome.
- *Proficient:* You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome at a satisfactory level if you continue your development in this way.
- *Advanced:* You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations and are focused on continuous improvement. You will demonstrate the learning outcome at a more than satisfactory level if you continue your development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

1.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

1.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

1.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

AI is not a replacement or threat, but a collaborative actor in your learning. We use AI to strengthen creativity, deepen understanding, and expand educational possibilities. You are expected to learn to work with AI effectively and productively: a core professional competency for ICT professionals.

The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available, including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

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1.7. Learning activities

1.8. Learning materials

[Return to table of content]

2. INFORMATION ABOUT [IN-AIB] AI BASICS (30EC)

Data Preparation & Analysis, Machine Learning, Large Language Models & AI Ethics

2.1. Content

In this semester you'll dive headfirst into AI's exciting world. You'll master fundamental techniques like classification algorithms, bias detection, and model evaluation while seeing how they solve actual problems.

You explore concepts including training data balance, decision boundaries, and prediction accuracy—the building blocks of effective AI systems. You'll learn how to review model performance, detect overfitting, and implement improvements through hyperparameter tuning.

The course covers practical applications: from clustering customer data and visualizing complex relationships to using sentiment analysis for content reliability across digital platforms. You'll discover how recommendation systems work and explore the creative side of AI through generative models.

Through hands-on projects like analyzing satellite imagery, optimizing delivery routes, and predicting customer behavior, you'll gain experience with real-world scenarios. You'll also examine how AI can enhance accessibility and sustainability in urban environments.

By semester's end, you'll be equipped to translate client requests into AI solutions and make smart, responsible decisions in your future career. Let the adventure begin!

2.2. Learning outcomes

1. **Learning outcome:** Professional standard

You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory research, and reporting.

2. **Learning outcome:** Personal leadership

You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional development.

3. **Learning outcome:** Analysing

You can analyze data, prepare and use it for AI applications within a specific domain, where you assess the data quality, use visualizations to present insights, and evaluate the impact of your choices.

Explanation: In real-world AI, raw data is rarely ready to use. You'll act as both detective—questioning biases, origins, and flaws—and storyteller, turning complex data into actionable insights. Each domain brings unique ethical considerations and specialised knowledge, affecting how data is analysed. The outcome stresses that all data decisions have consequences, making impact evaluation critical to building reliable AI foundations.

4. **Learning outcome:** Advising

You can make recommendations based on your analysis of AI applications, analyze their societal and ethical impact, and provide advice on this to stakeholders, where you make use of existing frameworks for responsible and transparent AI.

Explanation: This outcome highlights your role as both technical interpreter and ethical guide. Beyond what AI can do, you'll explore what it should do, assessing impacts on people, organisations, and society. You'll identify potential harms and benefits, apply established guidelines, and tailor advice to diverse audiences, maintaining accuracy and ethical rigour throughout.

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Explanation: Managing AI means continuous attention to its reliability and relevance. You'll monitor key metrics, define what success looks like for each use case, and handle versioning as models evolve. Operational context matters—meeting expectations around availability, integration, and transparency ensures that AI systems deliver lasting value.

2.3. Expected entry profile

				Investigative problem solving	
			Targeted interaction		
		Personal leadership			
	Future oriented organisation				
Professional development	1	1			

2.4. Outflow profile

Domain description HBO-i 2023

					Manage & control
					Realise
					Design
					Advise
					Analysis
User interaction	1	1	1	1	1
Organisational processes	1	1	1	1	1
Infrastructure					
Software	1	1	1	1	1
Hardware-interfacing					

Professional Development BSc 2020

					Investigative problem solving
					Targeted interaction
					Personal leadership
					Future oriented organisation
Professional development	1	1	1	1	1

2.5. Examination and final grading

2.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

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2.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

2.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

2.6. AI Usage Guidelines

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AI as a Collaborative Partner

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Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

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2.7. Learning activities

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3. INFORMATION ABOUT [APA2B] AUTOMATA (3EC)

3.1. Content

Algorithms are at the heart of any computer application solving non-trivial problems in transportation, healthcare, education, social media etc. Writing efficient algorithms (because nobody likes slow apps and stack overflows.) requires

patience, analytical talent, abstract thinking and creativity!

In this course you learn the basics of formal languages and automata: regular languages, regular expressions, deterministic finite automata, non-deterministic finite automata, powerset construction.

Compilers systematically translate high-level languages designed for humans into a program written in low-level assembly more suited to machines or to another language. In this module you learn about the essential components of each compiler: lexical analysis and parsing. You learn how to describe a language (programming language or a domain specific language) by grammars and how to use them in combination with a parser generator, to parse a character sequence (program) into a parse tree.

3.2. Learning outcomes

3.3. Examination and final grading

3.3.1. Examination

The teachers will provide the students with formative feedback on deliverables and demonstrations. It is expected that this feedback will be processed in further deliverables, within reason.

The student shows the work, alone or in a group context, but will be given feedback on an individual base.

The student work will be assessed on analysis, design and technical aspects, where the student will be informed whether the learning goals are met concerning the criteria.

At these assessments the learning outcomes are determined based on state of the deliverables at that point. Ideally the learning outcomes are proficient at these assessments. This grade is a formative indicator of mastery of learning goals at that moment and don't give any guarantee for further assessments. It is expected that the student keeps on progressing on the learning outcomes, not doing so might lead to a lower grade of beginning or orienting.

3.3.2. Repairs and resits

During the course, students are facilitated to process formative feedback on their learning goals in their deliverables and activities that are to be recorded in their portfolio (Canvas). All learning activities are related to the professional assignment.

3.3.3. Tools allowed

Except for paper & pen, and a laptop, no additional equipment is needed. The regular rules about plagiarism apply.

3.3.4. Final grading

A student's growth is maintained in his portfolio. The student's growth on each learning outcome is registered according to one of the five levels of mastery.

1. undefined,
2. orienting,
3. beginning,
4. proficient,
5. advanced.

After each formative assessment, the level of mastery for all learning outcomes is assessed for the student's deliveries.

At the end of the semester the student will be assessed on his performance of the semester. During the assessment meeting the teacher will use the student's portfolio. The overall grade is based on the shown performance and will be graded by the examiners. If all learning outcomes are proficient or advanced, the grade will be at least Satisfactory. This does not mean that obtaining proficient is the minimum requirement to achieve a satisfactory grade.

3.4. Learning activities

Based on the characteristics of an academic premaster and based on a highly knowledge-based scientific world, the course is structured according the research-based learning method. It is a structured and predictable learning path, in which the learning outcomes and assessment criteria are determined by the study program.

The education is designed according to the principles of the 4C/ID model (four-component instructional design model). Education is pre-structured in terms of content, working methods and guidance.

Knowledge, competences and professional skills are further gained and applied in practical assignments, which increase in size, abstraction and complexity during the study program. The student gets more and more choice in a pre-sorted set

of learning tasks that are designed according to 4C/ID.

Depending on the subjects, a variety of side-roads will be offered for in-depth knowledge and broader research.

3.5. Learning materials

All course content is in Canvas and Sharepoint. All software (compilers, IDEs, libraries, tools) is available for free.

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4. INFORMATION ABOUT [IN-CST] CYBER SECURITY PRINCIPLES & TECHNIQUES (30EC)

4.1. Content

This semester introduces you to the fundamentals of cyber security and is suitable both for aspiring security specialists and for ICT students who want to become more security-aware professionals. You will explore different security roles and domains through hands-on challenges, gaining insight into how universal security principles apply across areas such as enterprise IT, IoT and cyber-physical systems.

You will build a solid technical foundation for advanced cyber security semesters, supported by curated learning resources and practical workshops. The content is developed with security experts and tailored to beginners, with room for customization if you already have prior experience.

4.2. Learning outcomes

1. Learning outcome: Professional standard

You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory research, and reporting.

Explanation: Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

Success criteria:

- You explain how you applied appropriate Security and Privacy Engineering Principles (as mentioned in NIST SP 800-53r5).
- You have explicitly applied appropriate parts of professional standards (Mitre ATT&CK Matrices, OWASP guidelines, CIS benchmarks).
- You explain how security law and regulation (Cyber Resilience Act, Network and Information Security Directive 2, General Data Protection Regulation) are relevant to organizations in the professional field.
- You have performed applied security research in pairs using the DOT research framework.
- You explain where you explicitly considered ethical aspects in a security context.

2. Learning outcome: Personal leadership

You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional development.

Explanation: You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

Success criteria:

- You take initiative in your learning process (planning challenges, keeping your semester coach up to date, asking/documenting feedback) at least once per week and at least twice from each expert teacher with at least one formative indication moment in between.
- You take responsibility in group work (being present, contribute your share of the work).
- You explore the security community outside of FICT, e.g. by participating in a security related event (hackerspace, CTF, conference).
- You reflect on your learning and think about what this means for your future study and career path.

3. Learning outcome: Applied Cyber Security Techniques

Apply relevant security techniques to- solve realistic security challenges in professional ICT environments (IT, OT, and IoT). These challenges are aligned with ENISA roles: Legal/Policy/Compliance Officer, Digital Forensics Investigator, Cyber Security Implementer, Penetration Tester, and Cyber Incident Responder.

Explanation: Success criteria:

- Individually, you demonstrate the application of appropriate security techniques to address practical challenges associated with each ENISA role.
- Individually, you can explain what these security techniques are and in what way these improve security of an entity. To be able to do this, you should be able to explain the entity that is being secured (e.g. OS, network).
- In pairs, you apply appropriate security techniques in practical research.
- As a project team, you collaboratively implement appropriate security techniques in a project and present results.

4.3. Expected entry profile

There are no prerequisites for this semester, other than a finished start semester.

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development			1	1

4.4. Outflow profile

After completing this semester, we recommend continuing with the Main semester **Cyber Security: Attack & Defend**. This follow-up semester allows you to further strengthen your technical and analytical skills and deepen your hands-on experience with real-world security scenarios. It is a logical next step for students who have the ambition to develop into a cyber security professional.

Domain description HBO-i 2023

	Manage & control	Realise	Design	Advise	Analysis
User interaction					
Organisational processes					
Infrastructure		1		1	1
Software		1		1	1
Hardware-interfacing					1

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

4.5. Examination and final grading

4.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

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Ethical Boundaries

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Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

4.7. Learning activities

During this semester you will explore different security roles and domains by working on hands-on challenges. This allows you to get a good insight in the world of security and experience how universal security principles can be applied in different domains such as enterprise IT, IoT and cyberphysical systems.

The security roles you will explore are closely related to these ENISA roles:

- Legal/Policy/Compliance Officer
- Digital Forensics Investigator
- Cyber Security Implementer
- Penetration Tester
- Cyber Incident Responder

In addition you will gain solid, foundational technical skills that prepare you for the technically challenging main and advanced cyber security semesters.

To support your learning we have curated a selection of recommended (online) security resources and on top of that we offer workshops.

The semester content has been selected in collaboration with security experts and is tailored to suit beginners. However, if you have prior experience in the field, we can discuss customized adjustments to better match your level.

4.8. Learning materials

A Canvas course featuring the necessary learning resources

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5. INFORMATION ABOUT [O-IN-CST] CYBER SECURITY PRINCIPLES & TECHNIQUES (30EC)

5.1. Content

This semester introduces you to the fundamentals of cyber security and is suitable both for aspiring security specialists and for ICT students who want to become more security-aware professionals. You will explore different security roles and domains through hands-on challenges, gaining insight into how universal security principles apply across areas such as enterprise IT, IoT and cyber-physical systems.

You will build a solid technical foundation for advanced cyber security semesters, supported by curated learning resources and practical workshops. The content is developed with security experts and tailored to beginners, with room for customization if you already have prior experience.

5.2. Learning outcomes

1. Learning outcome: Professional standard

You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory research, and reporting.

Explanation: Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

Success criteria:

- You explain how you applied appropriate Security and Privacy Engineering Principles (as mentioned in NIST SP 800-53r5).
- You have explicitly applied appropriate parts of professional standards (Mitre ATT&CK Matrices, OWASP

guidelines, CIS benchmarks).

- You explain how security law and regulation (Cyber Resilience Act, Network and Information Security Directive 2, General Data Protection Regulation) are relevant to organizations in the professional field.
- You have performed applied security research in pairs using the DOT research framework.
- You explain where you explicitly considered ethical aspects in a security context.

2. Learning outcome: Personal leadership

You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional development.

Explanation: You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

Success criteria:

- You take initiative in your learning process (planning challenges, keeping your semester coach up to date, asking/documenting feedback) at least once per week and at least twice from each expert teacher with at least one formative indication moment in between.
- You take responsibility in group work (being present, contribute your share of the work).
- You explore the security community outside of FICT, e.g. by participating in a security related event (hackerspace, CTF, conference).
- You reflect on your learning and think about what this means for your future study and career path.

3. Learning outcome: Applied Cyber Security Techniques

Apply relevant security techniques to- solve realistic security challenges in professional ICT environments (IT, OT, and IoT). These challenges are aligned with ENISA roles: Legal/Policy/Compliance Officer, Digital Forensics Investigator, Cyber Security Implementer, Penetration Tester, and Cyber Incident Responder.

Explanation: Success criteria:

- Individually, you demonstrate the application of appropriate security techniques to address practical challenges associated with each ENISA role.
- Individually, you can explain what these security techniques are and in what way these improve security of an entity. To be able to do this, you should be able to explain the entity that is being secured (e.g. OS, network).
- In pairs, you apply appropriate security techniques in practical research.
- As a project team, you collaboratively implement appropriate security techniques in a project and present results.

5.3. Expected entry profile

There are no prerequisites for this semester, other than a finished start semester.

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

5.4. Outflow profile

After completing this semester, we recommend continuing with the Main semester **Cyber Security: Attack & Defend**. This follow-up semester allows you to further strengthen your technical and analytical skills and deepen your hands-on experience with real-world security scenarios. It is a logical next step for students who have the ambition to develop into a cyber security professional.

Domain description HBO-i 2023

	Manage & control	Realise	Design	Advise	Analysis
User interaction					

Organisational processes				
Infrastructure	1	1		1
Software	1	1		1
Hardware-interfacing	1			

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

5.5. Examination and final grading

5.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

5.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome at a satisfactory level if you continue your development in this way.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations and are focused on continuous improvement. You will demonstrate the learning outcome at a more than satisfactory level if you continue your development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

5.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

5.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

5.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

AI is not a replacement or threat, but a collaborative actor in your learning. We use AI to strengthen creativity, deepen understanding, and expand educational possibilities. You are expected to learn to work with AI effectively and productively: a core professional competency for ICT professionals.

The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available, including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

5.7. Learning activities

During this semester you will explore different security roles and domains by working on hands-on challenges. This allows you to get a good insight in the world of security and experience how universal security principles can be applied in different domains such as enterprise IT, IoT and cyberphysical systems.

The security roles you will explore are closely related to these ENISA roles:

- Legal/Policy/Compliance Officer
- Digital Forensics Investigator
- Cyber Security Implementer
- Penetration Tester
- Cyber Incident Responder

In addition you will gain solid, foundational technical skills that prepare you for the technically challenging main and advanced cyber security semesters.

To support your learning we have curated a selection of recommended (online) security resources and on top of that we offer workshops.

The semester content has been selected in collaboration with security experts and is tailored to suit beginners. However, if you have prior experience in the field, we can discuss customized adjustments to better match your level.

5.8. Learning materials

A Canvas course featuring the necessary learning resources

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6. INFORMATION ABOUT [APA1B] DATA SCIENCE MATHEMATICS (4EC)

6.1. Content

Data Science, responsible for transforming (big) data into useful information, has become an unmissable link in almost every industry, from entertainment to healthcare. Personalized recommendations, identifying disease in medical scans, detecting fraud, comprehensive covid19 visualizations.. are just some examples of data science applications.

The subjects of the course are:

- vector and matrix operations
- linear regression and gradient descent
- bias & variance
- classification
- K-nearest neighbour
- perceptrons
- forward & backward propagation
- (deep) neural networks

6.2. Learning outcomes

6.3. Examination and final grading

6.3.1. Examination

The teachers will provide the students with formative feedback on deliverables and demonstrations. It is expected that this feedback will be processed in further deliverables, within reason.

The student shows the work, alone or in a group context, but will be given feedback on an individual base.

The student work will be assessed on analysis, design and technical aspects, where the student will be informed whether the learning goals are met concerning the criteria.

At these assessments the learning outcomes are determined based on state of the deliverables at that point. Ideally the learning outcomes are proficient at these assessments. This grade is a formative indicator of mastery of learning goals at that moment and don't give any guarantee for further assessments. It is expected that the student keeps on progressing on the learning outcomes, not doing so might lead to a lower grade of beginning or orienting.

6.3.2. Repairs and resits

During the course, students are facilitated to process formative feedback on their learning goals in their deliverables and activities that are to be recorded in their portfolio (Canvas). All learning activities are related to the professional assignment.

6.3.3. Tools allowed

Except for paper & pen, and a laptop, no additional equipment is needed. The regular rules about plagiarism apply.

6.3.4. Final grading

A student's growth is maintained in his portfolio. The student's growth on each learning outcome is registered according to one of the five levels of mastery.

1. undefined,
2. orienting,
3. beginning,
4. proficient,
5. advanced.

After each formative assessment, the level of mastery for all learning outcomes is assessed for the student's deliveries.

At the end of the semester the student will be assessed on his performance of the semester. During the assessment meeting the teacher will use the student's portfolio. The overall grade is based on the shown performance and will be graded by the examiners. If all learning outcomes are proficient or advanced, the grade will be at least Satisfactory. This does not mean that obtaining proficient is the minimum requirement to achieve a satisfactory grade.

6.4. Learning activities

Based on the characteristics of an academic premaster and based on a highly knowledge-based scientific world, the course is structured according the course-based learning method. It is a structured and predictable learning path, in

which the learning outcomes and assessment criteria are determined by the study program.

The education is designed according to the principles of the 4C/ID model (four-component instructional design model). Education is pre-structured in terms of content, working methods and guidance.

Knowledge, competences and professional skills are further gained and applied in practical assignments, which increase in size, abstraction and complexity during the study program. The student gets more and more choice in a pre-sorted set of learning tasks that are designed according to 4C/ID.

Depending on the subjects, a variety of side-roads will be offered for in-depth knowledge and broader research.

6.5. Learning materials

All course content is in Canvas and Sharepoint. All software (compilers, IDEs, libraries, tools) is available for free.

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7. INFORMATION ABOUT [APA2A] DATA STRUCTURES & ALGORITHMS 1 (4EC)

7.1. Content

Algorithms are at the heart of any computer application solving non-trivial problems in transportation, healthcare, education, social media etc. Writing efficient algorithms (because nobody likes slow apps and stack overflows..) requires patience, analytical talent, abstract thinking and creativity!

This course provides a comprehensive introduction to fundamental data structures and algorithms used in computer science. You'll learn how to design, implement, and analyze efficient algorithms to solve various computational problems.

This course will help you train these skills by examining several classic algorithms for sorting, searching and optimization problems. You will learn to reason about algorithm complexity with measures like the big-Oh notation, and to use efficient data structures like hashtables, trees etc. Examples of optimization strategies discussed are greedy, backtracking, dynamic programming.

It explores algorithmic techniques including searching, sorting, graph traversal, dynamic programming, and divide-and-conquer. Topics also include algorithm analysis, complexity theory, and understanding trade-offs between time and space efficiency. Through lectures, problem-solving exercises, and programming assignments, you'll develop a solid foundation in data structures and algorithms that are essential for solving real-world computational challenges.

7.2. Learning outcomes

1. Learning outcome: Analysis

Explanation: You learn to estimate algorithm complexity, to recognize appropriate algorithmic strategies for solving real problems etc.

2. Learning outcome: Design

Explanation: You design efficient algorithms and reason about their correctness and complexity.

3. Learning outcome: Realization

Explanation: You implement your algorithmic designs into actual working solutions.

4. Learning outcome: Testing

Explanation: You design and apply unit tests to guarantee correctness of the solutions, and apply performance tests as well.

5. Learning outcome: Research

Explanation: Throughout the course, you adopt a curious attitude and take charge of your learning process. You are invited to investigate in more depth algorithms that you find interesting.

6. **Learning outcome:** Interaction

Explanation: There will be enough opportunity for interaction with your teachers and colleagues.

7. **Learning outcome:** Communication & leadership

Explanation: You describe and justify, mostly in writing, the correctness and complexity of your algorithms.

7.3. Examination and final grading

7.3.1. Examination

The teachers will provide the students with formative feedback on deliverables and demonstrations. It is expected that this feedback will be processed in further deliverables, within reason.

The student shows the work, alone or in a group context, but will be given feedback on an individual base.

The student work will be assessed on analysis, design and technical aspects, where the student will be informed whether the learning goals are met concerning the criteria.

At these assessments the learning outcomes are determined based on state of the deliverables at that point. Ideally the learning outcomes are proficient at these assessments. This grade is a formative indicator of mastery of learning goals at that moment and don't give any guarantee for further assessments. It is expected that the student keeps on progressing on the learning outcomes, not doing so might lead to a lower grade of beginning or orienting.

7.3.2. Repairs and resits

During the course, students are facilitated to process formative feedback on their learning goals in their deliverables and activities that are to be recorded in their portfolio (Canvas). All learning activities are related to the professional assignment.

7.3.3. Tools allowed

Except for paper & pen, and a laptop, no additional equipment is needed. The regular rules about plagiarism apply.

7.3.4. Final grading

A student's growth is maintained in his portfolio. The student's growth on each learning outcome is registered according to one of the five levels of mastery.

1. undefined,
2. orienting,
3. beginning,
4. proficient,
5. advanced.

After each formative assessment, the level of mastery for all learning outcomes is assessed for the student's deliveries.

At the end of the semester the student will be assessed on his performance of the semester. During the assessment meeting the teacher will use the student's portfolio. The overall grade is based on the shown performance and will be graded by the examiners. If all learning outcomes are proficient or advanced, the grade will be at least Satisfactory. This does not mean that obtaining proficient is the minimum requirement to achieve a satisfactory grade.

7.4. Learning activities

Based on the characteristics of an academic premaster and based on a highly knowledge-based scientific world, the course is structured according the research-based learning method. It is a structured and predictable learning path, in which the learning outcomes and assessment criteria are determined by the study program.

The education is designed according to the principles of the 4C/ID model (four-component instructional design model). Education is pre-structured in terms of content, working methods and guidance.

Knowledge, competences and professional skills are further gained and applied in practical assignments, which increase in size, abstraction and complexity during the study program. The student gets more and more choice in a pre-sorted set of learning tasks that are designed according to 4C/ID.

Depending on the subjects, a variety of side-roads will be offered for in-depth knowledge and broader research.

7.5. Learning materials

All course content is in Canvas and Sharepoint. All software (compilers, IDEs, libraries, tools) is available for free.

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8. INFORMATION ABOUT [APA4C] DATA STRUCTURES & ALGORITHMS 2 (3EC)

8.1. Content

Explore the fundamental question of P versus NP in this course on computational complexity. Delve into the distinction between the complexity classes P and NP, studying reductions, NP-completeness, and the famous Cook-Levin theorem. Investigate the implications of solving P versus NP, including its impact on cryptography, optimization, and practical computing. Gain insights into current developments and research directions in the field.

Through lectures, discussions, and problem-solving exercises, students will develop a solid foundation in computational complexity theory and gain the ability to analyze and reason about the complexities of computational problems.

8.2. Learning outcomes

1. **Learning outcome:** In addition to the general Academic Preparations learning outcomes, the student is able to: • don't focus only on your current problem, but be aware that differently looking problems might give you the solution • the core of theoretical computer science

8.3. Examination and final grading

8.3.1. Examination

The teachers will provide the students with formative feedback on deliverables and demonstrations. It is expected that this feedback will be processed in further deliverables, within reason.

The student shows the work, alone or in a group context, but will be given feedback on an individual base.

The student work will be assessed on analysis, design and technical aspects, where the student will be informed whether the learning goals are met concerning the criteria.

At these assessments the learning outcomes are determined based on state of the deliverables at that point. Ideally the learning outcomes are proficient at these assessments. This grade is a formative indicator of mastery of learning goals at that moment and don't give any guarantee for further assessments. It is expected that the student keeps on progressing on the learning outcomes, not doing so might lead to a lower grade of beginning or orienting.

8.3.2. Repairs and resits

During the course, students are facilitated to process formative feedback on their learning goals in their deliverables and activities that are to be recorded in their portfolio (Canvas). All learning activities are related to the professional assignment.

8.3.3. Tools allowed

Except for paper & pen, and a laptop, no additional equipment is needed. The regular rules about plagiarism apply.

8.3.4. Final grading

A student's growth is maintained in his portfolio. The student's growth on each learning outcome is registered according to one of the five levels of mastery.

1. undefined,
2. orienting,
3. beginning,

4. proficient,
5. advanced.

After each formative assessment, the level of mastery for all learning outcomes is assessed for the student's deliveries.

At the end of the semester the student will be assessed on his performance of the semester. During the assessment meeting the teacher will use the student's portfolio. The overall grade is based on the shown performance and will be graded by the examiners. If all learning outcomes are proficient or advanced, the grade will be at least Satisfactory. This does not mean that obtaining proficient is the minimum requirement to achieve a satisfactory grade.

8.4. Learning activities

Based on the characteristics of an academic premaster and based on a highly knowledge-based scientific world, the course is structured according to the research-based learning method. It is a structured and predictable learning path, in which the learning outcomes and assessment criteria are determined by the study program.

The education is designed according to the principles of the 4C/ID model (four-component instructional design model). Education is pre-structured in terms of content, working methods and guidance.

Knowledge, competences and professional skills are further gained and applied in practical assignments, which increase in size, abstraction and complexity during the study program. The student gets more and more choice in a pre-sorted set of learning tasks that are designed according to 4C/ID.

Depending on the subjects, a variety of side-roads will be offered for in-depth knowledge and broader research.

8.5. Learning materials

All course content is in Canvas and Sharepoint. All software (compilers, IDEs, libraries, tools) is available for free.

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9. INFORMATION ABOUT [APA3B] DECISION THEORY (4EC)

9.1. Content

This course focuses on studying the theoretical foundations and practical applications of decision-making under uncertainty.

It covers Reinforcement Learning (RL) to solve MDPs where an agent learns through interactions with its environment. You'll explore various RL algorithms, such as Q-learning and Monte Carlo methods, to train agents to make optimal decisions in uncertain environments.

Throughout the course, you can expect a mix of theoretical concepts, algorithmic techniques, and practical implementations. Assignments and projects involve coding exercises, problem-solving, and analysis of MDP models and algorithms.

9.2. Learning outcomes

1. **Learning outcome:** Students will be able to identify and formulate basic decision problems as Markov decision processes (mini project).
2. **Learning outcome:** Students will be able to implement basic automated decision-making algorithms to solve decision problems (mini project).

9.3. Examination and final grading

9.3.1. Examination

The teachers will provide the students with formative feedback on deliverables and demonstrations. It is expected that this feedback will be processed in further deliverables, within reason.

The student shows the work, alone or in a group context, but will be given feedback on an individual base.

The student work will be assessed on analysis, design and technical aspects, where the student will be informed whether the learning goals are met concerning the criteria.

At these assessments the learning outcomes are determined based on state of the deliverables at that point. Ideally the learning outcomes are proficient at these assessments. This grade is a formative indicator of mastery of learning goals at that moment and don't give any guarantee for further assessments. It is expected that the student keeps on progressing on the learning outcomes, not doing so might lead to a lower grade of beginning or orienting.

9.3.2. Repairs and resits

During the course, students are facilitated to process formative feedback on their learning goals in their deliverables and activities that are to be recorded in their portfolio (Canvas). All learning activities are related to the professional assignment.

9.3.3. Tools allowed

Except for paper & pen, and a laptop, no additional equipment is needed. The regular rules about plagiarism apply.

9.3.4. Final grading

A student's growth is maintained in his portfolio. The student's growth on each learning outcome is registered according to one of the five levels of mastery.

1. undefined,
2. orienting,
3. beginning,
4. proficient,
5. advanced.

After each formative assessment, the level of mastery for all learning outcomes is assessed for the student's deliveries.

At the end of the semester the student will be assessed on his performance of the semester. During the assessment meeting the teacher will use the student's portfolio. The overall grade is based on the shown performance and will be graded by the examiners. If all learning outcomes are proficient or advanced, the grade will be at least Satisfactory. This does not mean that obtaining proficient is the minimum requirement to achieve a satisfactory grade.

9.4. Learning activities

Based on the characteristics of an academic premaster and based on a highly knowledge-based scientific world, the course is structured according the research-based learning method. It is a structured and predictable learning path, in which the learning outcomes and assessment criteria are determined by the study program.

The education is designed according to the principles of the 4C/ID model (four-component instructional design model). Education is pre-structured in terms of content, working methods and guidance.

Knowledge, competences and professional skills are further gained and applied in practical assignments, which increase in size, abstraction and complexity during the study program. The student gets more and more choice in a pre-sorted set of learning tasks that are designed according to 4C/ID.

Depending on the subjects, a variety of side-roads will be offered for in-depth knowledge and broader research.

9.5. Learning materials

All course content is in Canvas and Sharepoint. All software (compilers, IDEs, libraries, tools) is available for free.

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10. INFORMATION ABOUT [APA4A] FUNCTIONAL PROGRAMMING (2EC)

10.1. Content

Functional Programming is a programming style where mathematical functions (that is, expressions with input parameters and an output result) are fundamental and running a program basically means evaluating a complex function. When

compared to traditional imperative and OO programming styles, FP is much faster to develop and easier to test/debug, and is also very suitable for parallel applications. Lisp, Haskell, Scala, Elm are examples of purely functional programming languages very much used today.

This course aims at exposing you to the functional way of thinking and introducing FP concepts and techniques that have been proven useful in practice, like list comprehensions, generators, map, filter, reduce. To facilitate the comparison between programming paradigms and to encourage assimilation of the FP style in your programming practice, “mixed” languages like F# and Python can be investigated.

10.2. Learning outcomes

1. Learning outcome: Analysis

Explanation: You learn to use the functional programming style confidently, understand the pros and cons and recognize, in a programming situation, whether employing it is desirable or not.

2. Learning outcome: Design

Explanation: You design programs using functional programming paradigms (modular, reusable, without side-effects).

3. Learning outcome: Realization

Explanation: You program with basic functional concepts like filter, map, reduce, recursion.

4. Learning outcome: Testing

Explanation: You reason about the correctness and time/space complexity of your programs, and measure their practical performance.

5. Learning outcome: Research

Explanation: Throughout the course, you adopt a curious attitude and take charge of your learning process. In the FP module, you will also work on an explicit research assignment.

6. Learning outcome: Interaction

Explanation: There will be enough opportunity for interaction with your teachers and colleagues.

7. Learning outcome: Communication and leadership

Explanation: There will be enough opportunity to describe and present your work, for instance in the research assignment.

10.3. Examination and final grading

10.3.1. Examination

The teachers will provide the students with formative feedback on deliverables and demonstrations. It is expected that this feedback will be processed in further deliverables, within reason.

The student shows the work, alone or in a group context, but will be given feedback on an individual base.

The student work will be assessed on analysis, design and technical aspects, where the student will be informed whether the learning goals are met concerning the criteria.

At these assessments the learning outcomes are determined based on state of the deliverables at that point. Ideally the learning outcomes are proficient at these assessments. This grade is a formative indicator of mastery of learning goals at that moment and don't give any guarantee for further assessments. It is expected that the student keeps on progressing on the learning outcomes, not doing so might lead to a lower grade of beginning or orienting.

10.3.2. Repairs and resits

During the course, students are facilitated to process formative feedback on their learning goals in their deliverables and activities that are to be recorded in their portfolio (Canvas). All learning activities are related to the professional assignment.

10.3.3. Tools allowed

Except for paper & pen, and a laptop, no additional equipment is needed. The regular rules about plagiarism apply.

10.3.4. Final grading

A student's growth is maintained in his portfolio. The student's growth on each learning outcome is registered according to one of the five levels of mastery.

1. undefined,
2. orienting,
3. beginning,
4. proficient,
5. advanced.

After each formative assessment, the level of mastery for all learning outcomes is assessed for the student's deliveries.

At the end of the semester the student will be assessed on his performance of the semester. During the assessment meeting the teacher will use the student's portfolio. The overall grade is based on the shown performance and will be graded by the examiners. If all learning outcomes are proficient or advanced, the grade will be at least Satisfactory. This does not mean that obtaining proficient is the minimum requirement to achieve a satisfactory grade.

10.4. Learning activities

Based on the characteristics of an academic premaster and based on a highly knowledge-based scientific world, the course is structured according to the research-based learning method. It is a structured and predictable learning path, in which the learning outcomes and assessment criteria are determined by the study program.

The education is designed according to the principles of the 4C/ID model (four-component instructional design model). Education is pre-structured in terms of content, working methods and guidance.

Knowledge, competences and professional skills are further gained and applied in practical assignments, which increase in size, abstraction and complexity during the study program. The student gets more and more choice in a pre-sorted set of learning tasks that are designed according to 4C/ID.

Depending on the subjects, a variety of side-roads will be offered for in-depth knowledge and broader research.

10.5. Learning materials

All course content is in Canvas and Sharepoint. All software (compilers, IDEs, libraries, tools) is available for free.

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11. INFORMATION ABOUT [IN-GV] GAME DEVELOPMENT (30EC)

Game Engine Exploration, Playful Interaction & Scripting

11.1. Content

Game Development lets you apply your ICT skills in the most fun and interactive field of all: games!

This semester introduces you to software development using industry-standard game engines. You build confidence in core programming principles while building game mechanics and experiencing playful interactions.

After starting with the best game development practices and patterns, you'll then get to choose between many of the facets of game development, such as: input systems, shaders, 2d and 3d assets handling and creation, animation, VR integration, and simple game AI.

11.2. Learning outcomes

1. Learning outcome: Analysis

You can research basic principles of game development in a methodical fashion and analyze technical and design-related issues to make well-founded choices.

Explanation: Basic principles: game engines and techniques that fit your assignment.

Methodical: use appropriate research methods such as the DOT framework.

Well-founded choices: substantiate your choices with context (DOT framework), examples and link your analysis to

your design.

2. Learning outcome: Advice

You can make substantiated recommendations about game development choices, based on research and/or analysis.

Explanation: Substantiated: make sure to base your advice on your knowledge and analyses.

3. Learning outcome: Design

You can design and structure the applied techniques using existing game development frameworks, theories, and models, and substantiate your design choices.

Explanation: Techniques: this includes techniques that you use for game mechanisms, but also, for example, audiovisual or aesthetic components.

Frameworks, theories and models: demonstrate that you know how to use for example, UML, ERD, and Design Patterns.

Substantiated: justify your design choices based on your knowledge and analyses.

4. Learning outcome: Realization

You can develop a functional game prototype with the use of a game engine by applying your technical skills based on, and linked to, your designs. You demonstrate this both individually and in groups.

Explanation: Functional game prototype: a working version of your game that is ready to be tested or played.

Individually and in groups: during the semester you are expected to realize both individual products and to make a substantial contribution to the group product.

5. Learning outcome: Management & Control

You can collaborate effectively on a game development project by using, for example, an iterative process, version control, and basic project management methods.

Explanation: Collaborate effectively: you make a meaningful contribution to the group process and actively participate in group meetings.

Iterative process: work systematically using a cycle of design, development, testing and collecting and analyzing feedback, so that the design can be further adjusted if necessary.

Version control and project management methods: make visible use of tools such as Git, and methods such as Agile/Scrum.

6. Learning outcome: Professional standard

You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory research, and reporting.

Explanation: • Professional practice: You show effort in being professional in your work and communication. You use validated professional tools and frameworks in your daily work.

• Reporting: You fill your portfolio with evidence of your work, create multiple Devlogs and add a reading guide to bring it all together.

7. Learning outcome: Personal leadership

You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional development.

Explanation: • Take the initiative: You approach your teachers to show your work and request feedback on specific elements.

• Professional development: You reflect on your own strengths and weaknesses and formulate specific personal goals to work on during the semester.

11.3. Expected entry profile

No previous knowledge or skills levels are required for entry into this semester. Therefore, all semester 1 profiles are welcome, although a keen interest in or knowledge of games will of course make for a smoother experience.

Professional Development BSc 2020

					Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1						

11.4. Outflow profile

This semester is part of a three-semester track on Game-related topics. We suggest directly following up with the main-level semester on Game Design to dive deeper into the craft of designing and developing games from scratch. Then you can take up the advanced semester Game Production towards the end of your studies.

Game Development is both technically challenging and reliant on good insights into the user experience. Therefore, any user experience of software-oriented topic could also be logical next steps.

Below is an explanation of the the expected outflow levels in relation to the HBO-i framework.

	Analysis	Advise	Design	Realise	Manage & control
User interaction	Level: 1 Explanation: students analyse the group assignments to identify the core elements by analysing the wishes of the external client and translating them into a fitting, interactive game design.	Level: 1 Explanation: students advise their project groups on their area of expertise and the client about the best fitting game engine and other technical options.	Level: 1 Explanation: students use an iterative design process, including prototyping, to design the necessary technical building blocks to realize the desired interaction in the game.	Level: 1 Explanation: students realise and user-test at least two game prototypes on their effectiveness in terms of user interaction (i.e., motivation)	Level: 1 Explanation: students demonstrate their project management skills by displaying their products and results, as well as the process leading to them through devlogs, reflections, and presentations in an online portfolio.
Software	Level: 1 Explanation: students link the 'Pizza' assignment to the appropriate game engine and analyse the best techniques to realize the wishes of the external client.	Level: 1 Explanation: students formulate and discuss specific requirements for the desired game, based on the available game engines, which they use to formulate technical their advise to the group and their stakeholders.	Level: 1 Explanation: students design the appropriate game mechanics to go with the overall game, including the technology on how to make and test them.	Level: 1 Explanation: students combine the game mechanics they realise into a full game prototype, applying multiple advanced software development techniques, which they test with potential end users.	Level: 1 Explanation: students manage their projects by making use of industry standard version control systems appropriate to the assignment.

Domain description HBO-i 2023

						Manage & control
						Realise
						Design
						Advise
						Analysis
User interaction	1	1	1	1	1	1

Organisational processes					
Infrastructure					
Software	1	1	1	1	1
Hardware-interfacing					

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

11.5. Examination and final grading

11.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

11.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome at a satisfactory level if you continue your development in this way.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations and are focused on continuous improvement. You will demonstrate the learning outcome at a more than satisfactory level if you continue your development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

11.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

11.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

11.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

AI is not a replacement or threat, but a collaborative actor in your learning. We use AI to strengthen creativity, deepen understanding, and expand educational possibilities. You are expected to learn to work with AI effectively and productively: a core professional competency for ICT professionals.

The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available, including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

11.7. Learning activities

During the semester, lectures and practicals will be offered in the form of workshops. These workshops will deal with subjects related to various game technologies and students will practice using each of the three most (currently) used game engines in the field: Unity, Godot, and Unreal.

Students will report to a dedicated coach/teacher to present and discuss progress on a regular basis, and teachers will provide students with regular feedback and feed forward on their work. Students are expected to record and process this feedback.

The first phase of this semester will consist of 4 weeks of getting to know each other, the semester, as well as the game engines and other commonly used technologies, accompanied by a number of small, individual assignments.

The second phase consists of a 7-week group project, where you will create your first game, based on an industry case and feedbacked on by a number of industry experts.

The final phase consists of another 7-week project where you will tackle a new, but similar group challenge.

Each phase end with a showcase (or playtest) event, and throughout the semester, multiple portfolio reviews will be conducted by the tutors.

11.8. Learning materials

Canvas course including all linked information.

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12. INFORMATION ABOUT [IN-IT] INTELLIGENT TECHNOLOGIES (30EC)

Embedded software, Microcontrollers and Hardware interfacing

12.1. Content

This semester, you'll learn to design, analyze, build, and test intelligent solutions that seamlessly integrate hardware and software. You'll develop problem-solving, communication, and research skills while gaining insights into electrical concepts, microcontrollers, and sensors and actuators such as motion detectors and environmental sensors and various types of motors. You'll also explore the possibilities and limitations of embedded systems. Beyond that, you'll learn to write technically sound software and design larger systems using industry standard tools where separate modules communicate and collaborate to achieve a common goal.

Equally important, you'll refine your interpersonal skills in Agile teams, working on a solution for a partner company. You'll apply structured development processes, and using tools like Scrum, version control, and work management systems.

Semester Coach

The semester coach provides individual guidance and group process guidance. The semester coach is partly responsible for the guidance and assessment of the professional skills learning outcomes (i.e. professional standard and personal leadership). The semester coach supports you through various discussions, acting situationally by choosing the right role and approach at the appropriate moment.

Points of Contact

The lecturers of this semester are your first point of contact for the topics they are involved with. The semester project leader is available for matters within the semester, and the curriculum owner is available for matters that transcend semesters.

The semester coach is often a point of contact for personal conversations, such as private situations that impact your studies.

12.2. Learning outcomes

1. Learning outcome: Applied Procedural Programming for Embedded Systems:

The student applies procedural programming techniques by developing software applications across embedded systems that are fully tested, function according to given requirements and design and meet specified quality standards.

2. Learning outcome: Structured Software Design and Documentation:

The student demonstrates the ability to describe a system and develop a software design using a specified development methodology. This includes the steps of analysis, advice, design for a given set of requirements and using standard tools to document the design. The student produces a design document and a test plan that adhere to applicable quality standards.

3. Learning outcome: Embedded Systems with Actuators and Sensors:

The student demonstrates the ability to build their own embedded system based on given requirements. The system has multiple actuators and sensors, connected by the student, that are controlled via custom software. It must meet provided timing criteria. The student can locate errors using measurement tools, report on them, and deliver a functioning and tested system. The student is also able to describe the architecture and operation of the microcontroller and perform an analysis of actuators and sensors.

4. Learning outcome: System Communication using Transport Protocols:

The student demonstrates that different systems can communicate with each other based on existing transport layers such as TCP/IP and a custom message exchange agreement. The student is capable of making a well-founded choice for the implementation based on an analysis of these transport layers. The student delivers a functioning and tested system along with the corresponding documentation.

5. Learning outcome: Personal leadership

You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional development.

6. Learning outcome: Professional standard

You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory research, and reporting.

12.3. Expected entry profile

Basic programming skills at semester 1 level are advised. Experience with the Arduino platform is an advantage. Example projects include small embedded systems that you built by integrating various libraries and components to make a proof-of-concept. Examples include a hotel safe deposit box, turnstile pool access based on a card reader, a simple thermostat for controlling an electric heater, etc.

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development			1	1

12.4. Outflow profile

After completing this semester students are encouraged to further develop their expertise in:

- Intelligent devices
- Industry 4.0 & the Internet of Things

Domain description HBO-i 2023

	Manage & control	Realise	Design	Advise	Analysis
User interaction					
Organisational processes					
Infrastructure		1	1		
Software		1	1	1	
Hardware-interfacing	1	1	1	1	1

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

12.5. Examination and final grading

12.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

12.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
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This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

12.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

12.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

12.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

AI is not a replacement or threat, but a collaborative actor in your learning. We use AI to strengthen creativity, deepen understanding, and expand educational possibilities. You are expected to learn to work with AI effectively and productively: a core professional competency for ICT professionals.

The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available, including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

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12.7. Learning activities

There are four main lines.

1. Individual project: You work on a project individually. You can choose from a predefined project or define your own. Each sprint you'll add functionality or improve the existing functionality.
2. Industry project: You work in a group on a project in cooperation with a company partner. The partner company sets the domain context of the project and you set a problem to solve and define a project within that.
3. Personal Leadership: You work on personal leadership individually and put it in practice during the semester. You do this in repeating cycles by setting goals and actions, executing them and reviewing your outcomes of those actions and goals. Finally you reflect and adjust the goals.
4. Workshops: Teachers will give various kinds of workshops such as inspirational, explanations, demo's or practice.

12.8. Learning materials

Required:

1. Canvas course. The course contains various learning materials such as task descriptions, workshops and other resources.
2. Laptop. You need a laptop on which to do your work. Linux, Mac or Windows.
3. C Compiler, Git, Code editor, Documentation authoring tools. The required tools are either free/open source or supplied by Fontys

Recommend:

1. An Arduino R3 or R4, or another microcontroller
2. A multimeter (e.g. UNI-T UT132D).
3. Logic Analyzer (e.g. IKALogic SQ series (SQ25,SQ50) or Analog Devices ADALM2000 or Analog Discovery version 2 or 3).
4. Portable power supply or +5v wall plug

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13. INFORMATION ABOUT [O-IN-IT] INTELLIGENT TECHNOLOGIES (30EC)

Embedded software, Microcontrollers and Hardware interfacing

13.1. Content

This semester, you'll learn to design, analyze, build, and test intelligent solutions that seamlessly integrate hardware and software. You'll develop problem-solving, communication, and research skills while gaining insights into electrical concepts, microcontrollers, and sensors and actuators such as motion detectors and environmental sensors and various types of motors. You'll also explore the possibilities and limitations of embedded systems. Beyond that, you'll learn to write technically sound software and design larger systems using industry standard tools where separate modules communicate and collaborate to achieve a common goal.

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semesters.

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13.2. Learning outcomes

1. Learning outcome: Professional standard

You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory research, and reporting.

2. Learning outcome: Personal leadership

You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional development.

3. Learning outcome: Applied Procedural Programming for Embedded Systems:

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13.3. Expected entry profile

Basic programming skills at semester 1 level are advised. Experience with the Arduino platform is an advantage. Example projects include small embedded systems that you built by integrating various libraries and components to make a proof-of-concept. Examples include a hotel safe deposit box, turnstile pool access based on a card reader, a simple thermostat for controlling an electric heater, etc.

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1		

13.4. Outflow profile

After completing this semester students are encouraged to further develop their expertise in:

- Intelligent devices
- Industry 4.0 & the Internet of Things

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes					
Infrastructure			1	1	
Software			1	1	1
Hardware-interfacing	1	1	1	1	1

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

13.5. Examination and final grading

13.5.1. Assessment: Holistic and Longitudinal

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13.7. Learning activities

We use a Blended Learning approach, with two contact sessions each week. Students must prepare every week before attending these sessions. The preparation is usually theoretical and includes an activity that needs to be submitted. This activity is not graded, but it helps students to practice the learning material, identify problems or difficulties, and turn these into clear, focused questions.

One evening per week, there is an in-person session of 4 lesson hours at a physical location. On another evening, there is an online session of 2 lesson hours. One lesson hour lasts 45 minutes. The online session is mainly for answering students' questions and helping with any problems they face while preparing on their own. The in-person session is mainly for working on assignments. During this session, the teacher gives feedback (written and/or spoken) on the submitted work and explains new topics.

13.8. Learning materials

Required:

1. Canvas course. The course contains various learning materials such as task descriptions, workshops and other resources.
2. Laptop. You need a laptop on which to do your work. Linux, Mac or Windows.
3. C Compiler, Git, Code editor, Documentation authoring tools. The required tools are either free/open source or supplied by Fontys

Recommend:

1. An Arduino R3 or R4, or another microcontroller
2. A multimeter (e.g. UNI-T UT132D).

3. Logic Analyzer (e.g. IKALogic SQ series (SQ25,SQ50) or Analog Devices ADALM2000 or Analog Discovery version 2 or 3).
4. Portable power supply or +5v wall plug

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14. INFORMATION ABOUT [IN-IM] INTERACTIVE MEDIA (30EC)

14.1. Content

The theme of the semester is Interactive Media. The semester requires students to explore the future field of Media Design in its full scope. Several projects are given to students that all focus on a different aspect of Media Design. Students document the results of various group projects, an individual project, documenting all the material from the project in portfolio forms the basis of the assessment.

The semester offers students the freedom to make their own choices and therefore requires the student to act as a professional-to-be. This means fulfilling obligations, actively participating in the programme from nine to four every day, taking the initiative to seek out discussions with lecturers and collecting feedback, and complying with agreements in the student group.

14.2. Learning outcomes

1. **Learning outcome:** Interactive Media: You iteratively create interactive media products that are based on exploratory research that leads to a good user experience.

Explanation: Iteratively

This means you don't make your product all at once and hope it works. Instead, you:

- Get to know the context of the project
- Create a first version (a prototype).
- Show it to clients, teachers and end users.
- Get feedback from them.
- Use that feedback to improve your product in the next version.
- Repeat this process several times—each version is called an iteration.

The goal is to improve your product step-by-step, based on what you learn along the way. Show these steps in your portfolio.

Create

In this context create not just making things but also applying the theoretical knowledge you will acquire during the semester and using the insight you get from the stakeholders.

Interactive media products

These are not final, fully working products but interactive prototypes. They can vary in how detailed they are:

- Low-fidelity prototypes: simple wireframes or sketches.
- Mid-fidelity prototypes: clickable digital mockups.
- High-fidelity prototypes: nearly complete versions that look and feel close to the final product.

Each version helps you and others understand what the product might look like and how it will work.

Exploratory user research

This means your design choices are based on research, not just on guesses or personal opinions. That includes:

- Starting with a clear research question.
- Choosing the right research methods (like interviews, surveys, usability tests, etc.).
- Collecting real insights from users and/or stakeholders.
- Applying those insights in your project to make better decisions in your iterations.

By doing this, you make sure the product you're creating actually fits the needs of real users.

2. **Learning outcome:** Front-end development: You develop several interactive media products using front-end development languages and commit it to a version control environment.

Explanation: Develop

You write code that results in products that can be used by others. The code itself is well structured.

Front-end development languages

This means you're learning how to build the visual and interactive part of a website — the part users actually see and use. You'll be using:

- HTML – to structure the content (like headings, paragraphs, images)
- CSS – to style your website (like colors, fonts, layout)
- JavaScript – to make your site interactive (like buttons that do things, forms that respond)

Commit code in a version control environment

This is about keeping your code organized, understandable, and trackable using Git. Instead of just throwing all your code online at once, you:

- Create a README file to explain your project and how to use it
- Write comments in your code to explain what it does
- Use commits to save your progress step-by-step, with clear messages (like "added homepage", "fixed form bug")
- You start using version control in group projects

3. **Learning outcome:** Visual Design:

You design products that are substantiated and validated by making use of the visual design process and through design research.

Explanation: Visual design process

The visual design process consists of the following steps

- Learn from work that others made by looking at it and formulate what makes the work interesting/good (inspiration).
- Start from uncertainty then go from quantity to quality, by making use of design research. Which will lead to the best visual, dynamic and interactive solution
- From analog sketches on paper, to applications created with software like adobe creative suite and Figma.

Design research

- Many variations and iterations based on design elements are needed in all phases of the project
- Get feedback from clients, teachers and end-users
- Select your work
- Present your work to teachers and clients
- Reflect on your work and the process.

Components of visual design

The following design elements must be considered in the design process

- Composition
- Layout
- Typography
- Animation
- Form
- Color

4. **Learning outcome:** Professional standard

You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory research, and reporting.

5. **Learning outcome:** Personal leadership

You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional development.

14.3. Expected entry profile

Students who have finished their first semester successfully. An interest in several or at least one of the following topics is of course helpful namely, front-end development, User Experience and/or Visual Design.

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1		

14.4. Outflow profile

By the end of the semester, you worked on many different topics and products. So you have a background in the topics of visual design, user experience and front-end development. This gives you a good preparation for several upcoming semesters like media production, front end development, smart mobile, game design and open learning.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	1	1	1	1	1
Organisational processes					
Infrastructure					
Software	1	1		1	
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

14.5. Examination and final grading

14.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

14.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- *Undefined:* You have not yet undertaken any activities to demonstrate the learning outcome.
- *Orienting:* You have made a start and explored possibilities to demonstrate the learning outcome.
- *Beginning:* You have taken initial steps that contribute to demonstrating the learning outcome.
- *Proficient:* You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome at a satisfactory level if you continue your development in this way.
- *Advanced:* You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations and are focused on continuous improvement. You will demonstrate the learning outcome at a more than satisfactory level if you continue your development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is

no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

14.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

14.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

14.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

AI is not a replacement or threat, but a collaborative actor in your learning. We use AI to strengthen creativity, deepen understanding, and expand educational possibilities. You are expected to learn to work with AI effectively and productively: a core professional competency for ICT professionals.

The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available, including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

14.7. Learning activities

1. The semester includes several group projects with external clients. The clients can be a Partner in Education or a collaborating Fontys Institute. The projects have different points of interest.
2. The semester includes an individual project. The student chooses his or her own subject and actively approaches one or more lecturers for individual guidance. The planning of the individual project can be found in Canvas.
3. Throughout the semester, students work on their portfolio in Portflow. The material for this comes from the group project, the individual project and the lessons.

The implementation of educational activities may differ in terms of planning from the implementations in Tilburg and Eindhoven. The information in Canvas is leading in this.

14.8. Learning materials

Canvas course including all linked information.

15. INFORMATION ABOUT [O-IN-IM] INTERACTIVE MEDIA (30EC)

15.1. Content

The theme of the semester is Interactive Media. The semester requires students to explore the future field of Media Design in its full scope. Several projects are given to students that all focus on a different aspect of Media Design. Students document the results of various group projects, an individual project, documenting all the material from the project in portfolio forms the basis of the assessment.

The semester offers students the freedom to make their own choices and therefore requires the student to act as a professional-to-be. This means fulfilling obligations, actively participating in the programme from nine to four every day, taking the initiative to seek out discussions with lecturers and collecting feedback, and complying with agreements in the student group.

15.2. Learning outcomes

1. **Learning outcome:** Professional standard

You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory research, and reporting.

2. **Learning outcome:** Personal leadership

You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional development.

3. **Learning outcome:** Learning outcome: Interactive Media: You iteratively create interactive media products that are based on exploratory research that leads to a good user experience.

Explanation: Explanation: Iteratively

This means you don't make your product all at once and hope it works. Instead, you:

- Get to know the context of the project
- Create a first version (a prototype).
- Show it to clients, teachers and end users.
- Get feedback from them.
- Use that feedback to improve your product in the next version.
- Repeat this process several times—each version is called an iteration.

The goal is to improve your product step-by-step, based on what you learn along the way. Show these steps in your portfolio.

Create

In this context create not just making things but also applying the theoretical knowledge you will acquire during the semester and using the insight you get from the stakeholders.

Interactive media products

These are not final, fully working products but interactive prototypes. They can vary in how detailed they are:

- Low-fidelity prototypes: simple wireframes or sketches.
- Mid-fidelity prototypes: clickable digital mockups.
- High-fidelity prototypes: nearly complete versions that look and feel close to the final product.

Each version helps you and others understand what the product might look like and how it will work.

Exploratory user research

This means your design choices are based on research, not just on guesses or personal opinions. That includes:

- Starting with a clear research question.
- Choosing the right research methods (like interviews, surveys, usability tests, etc.).
- Collecting real insights from users and/or stakeholders.
- Applying those insights in your project to make better decisions in your iterations.

By doing this, you make sure the product you're creating actually fits the needs of real users.

4. **Learning outcome:** Learning outcome: Front-end development: You develop several interactive media products using front-end development languages and commit it to a version control environment.

Explanation: Explanation: Develop

You write code that results in products that can be used by others. The code itself is well structured.

Front-end development languages

— This means you're learning how to build the visual and interactive part of a website — the part users actually see

and use. You'll be using:

- HTML – to structure the content (like headings, paragraphs, images)
- CSS – to style your website (like colors, fonts, layout)
- JavaScript – to make your site interactive (like buttons that do things, forms that respond)

Commit code in a version control environment

This is about keeping your code organized, understandable, and trackable using Git. Instead of just throwing all your code online at once, you:

- Create a README file to explain your project and how to use it
- Write comments in your code to explain what it does
- Use commits to save your progress step-by-step, with clear messages (like "added homepage", "fixed form bug")
- You start using version control in group projects

5. **Learning outcome:** Learning outcome: Visual Design:

You design products that are substantiated and validated by making use of the visual design process and through design research.

Explanation: Explanation: Visual design process

The visual design process consists of the following steps

- Learn from work that others made by looking at it and formulate what makes the work interesting/good (inspiration).
- Start from uncertainty then go from quantity to quality, by making use of design research. Which will lead to the best visual, dynamic and interactive solution
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Design research

- Many variations and iterations based on design elements are needed in all phases of the project
- Get feedback from clients, teachers and end-users
- Select your work
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- Reflect on your work and the process.

Components of visual design

The following design elements must be considered in the design process

- Composition
- Layout
- Typography
- Animation
- Form
- Color

15.3. Expected entry profile

Students who have finished their first semester successfully. An interest in several or at least one of the following topics is of course helpful namely, front-end development, User Experience and/or Visual Design.

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	

15.4. Outflow profile

By the end of the semester, you worked on many different topics and products. So you have a background in the topics of visual design, user experience and front-end development. This gives you a good preparation for several upcoming semesters like media production, front end development, smart mobile, game design and open learning.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	1	1	1	1	1
Organisational processes					
Infrastructure					
Software	1	1		1	
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	

15.5. Examination and final grading

15.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

15.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

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15.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

15.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

15.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

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Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

15.7. Learning activities

1. The semester includes several group projects with external clients. The clients can be a Partner in Education or a collaborating Fontys Institute. The projects have different points of interest.
2. The semester includes an individual project. The student chooses his or her own subject and actively approaches one or more lecturers for individual guidance. The planning of the individual project can be found in Canvas.
3. Throughout the semester, students work on their portfolio in Portflow. The material for this comes from the group project, the individual project and the lessons.

The implementation of educational activities may differ in terms of planning from the implementations in Tilburg and Eindhoven. The information in Canvas is leading in this.

15.8. Learning materials

Canvas course including all linked information.

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16. INFORMATION ABOUT [ICT5A] INTERNSHIP ICT FALL (30EC)

16.1. Content

NOTE: This description is only for student that leave on internship in the FALL semester. There is a separate description for the spring semester.

You will do an internship in the 5th semester of your studies. During that period, you demonstrate in (professional) practice that you have integrated the required professional and technical development and that you can also apply it independently. You also get the chance to get a taste of professional practice and thus get a better idea of what it has to offer.

Internships are about working and learning in professional practice. It is an assignment/work that:

- you perform at an institution or company;
- matches the choice of study and what you can do with it later;
- asks you to make the action very clear, so that it can be understood by others, why and for what purpose you did or did not do something;
- results in a portfolio with professional products.

Duration internship

The internship lasts a whole semester: from Fontys week 1 to Fontys week 18. The assessment takes place at the end of the semester (Fontys week 19). Holidays and public holidays that fall within the semester count as internship weeks. So you have no school holidays this semester. Days off that you accumulate (see internship agreement) can be taken in consultation with the company. Days off and/or company closures, which are mandatory within the company, are of course not counted as days off.

An internship must have a minimum of 90 days. You perform these 90 days between Monday Fontys week 1 and Friday Fontys week 18.

Sick leave

Let's hope not, but it is of course possible that you become ill during a (prolonged) period of your internship or miss days on your internship for some other reason. Because you basically continue to work on holidays, you have some leeway. Being ill or taking leave for another reason does not have to be a problem to achieve the minimum number of days.

What are internship days?

Internship days are all days on which you work on your internship and/or on your internship assignment. You can include visits that you make in the context of your internship or for your education, such as counseling moments at school or visiting a minor market.

Internship permission

In order to start arranging an internship, you must have internship permission from the examination board. Please note: before you can actually start the internship, the internship coordinator for your study route must approve the internship location you have found based on the project proposal that you have completed and handed in yourself.

Permission to go on an internship is given when you have completed one of the semesters of the second year of study. You will then first follow the other semester of the second year (which you have not yet completed) and do an internship immediately after that (regardless of whether you pass this other semester). Hence, you will always have one semester to find a suitable internship. It is recommended to start searching in time. After your internship, you will have to retake any semester from the second year that you have not yet passed until it is completed.

16.2. Learning outcomes

Learning outcome 1:

[Professional duties] You carry out the professional duties on a junior bachelor level resulting in professional products in line with the IT-area you are working in.

Learning outcome 2:

[Situation-orientation] You apply your previously acquired knowledge and skills in an authentic context to deliver relevant results for the project and company.

Learning outcome 3:

[Professional Standard] You take responsibility for solving an IT issue. You conduct research using a relevant method you select and advise your stakeholders in a complex and uncertain context. You substantiate future-oriented choices with ethical and sustainable arguments.

Learning outcome 4:

[Personal Leadership] You are entrepreneurial around your projects and personal development, you pay attention to your own learning ability and keep in mind what kind of IT professional and/or what type of positions you aspire to.

16.3. Examination and final grading

16.3.1. Examination

Key moments of examination

Week 3-5	Projectplan/projectdocument. Formative assessment; the first assessor provides feedback during the first company visit in week 3-5. The second assessor provides online feedback on assignment only.
Week 9/10	Midterm review; formative assessment.
Week 10-17	Hand in concept portfolio with a request for feedback; formative assessment.
Week 17	Portfolio including proces description and achievements. Hand in at the beginning of week 17.
Week 18	Formative assessment during the second company visit in week 18. Final presentation and/or demonstration of the final product, combined with accountability about the execution and evaluation of the process. Feedback and advised grading from internship coach and company coach.
Week 19/20	Summative assessment bij first and second assessor. Criterion-based Interview on learning outcomes taking into account all aspects of your internship.

16.3.2. Repairs and resits

According to the TER this is a practice based semester. Hence there are no options for repair or retake. If you fail your internship, you should either redo a semester from year 2 before restarting an internship or, if you have passed both of these semesters, start a new internship.

16.3.3. Tools allowed

Not applicable. The regular rules about plagiarism apply.

16.3.4. Final grading

The assessment will take place in week 19. The assessment is taken by two assessors: the first assessor, which is the teacher that guided you during your internship and the second assessor, which is another examiner from school. Assessment are portfolio-based and are conducted in the form of a criterion-based interview (CGI) to determine if all learning outcomes have been sufficiently achieved.

If a student has at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessor meeting.

16.4. Learning activities

The internship is largely independently acquired by you as a student and carried out at an external client. This may be both at home and abroad. This requires entrepreneurship on your part; we expect your own initiative when formulating an assignment and executing it. It is your own responsibility to find a suitable assignment and internship position. Failing to do so may force you to put your studies on hold until you found a suitable position, since you will not be allowed to continue to later semesters without completing an internship.

During your internship you will be supervised by someone from the company; your company supervisor. In addition, you will receive guidance from school by an internship teacher (the 1st assessor). He mainly guides you process-wise and is co-evaluator at the end. You will also be assigned a 2nd assessor. This is a co-assessor at the final assessment. During your internship, your internship teacher is your first point of contact from school and also acts as a semester coach.

16.5. Learning materials

There is no standard teaching material available for the internship. You can, however, use the documents that you can find on Canvas. Here you will find, among other things, the research framework and support for creating your portfolio.

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17. INFORMATION ABOUT [ICT5B] INTERNSHIP ICT SPRING (30EC)

17.1. Content

NOTE: This description is only for student that leave on internship in the SPRING semester. There is a separate description for the fall semester.

You will do an internship in the 5th semester of your studies. During that period, you demonstrate in (professional) practice that you have integrated the required professional and technical development and that you can also apply it independently. You also get the chance to get a taste of professional practice and thus get a better idea of what it has to offer.

Internships are about working and learning in professional practice. It is an assignment/work that:

- you perform at an institution or company;
- matches the choice of study and what you can do with it later;
- asks you to make the action very clear, so that it can be understood by others, why and for what purpose you did or did not do something;
- results in a portfolio with professional products.

Duration internship

The internship lasts a whole semester: from Fontys week 1 to Fontys week 18. The assessment takes place at the end of the semester (Fontys week 19). Holidays and public holidays that fall within the semester count as internship weeks. So you have no school holidays this semester. Days off that you accumulate (see internship agreement) can be taken in consultation with the company. Days off and/or company closures, which are mandatory within the company, are of course not counted as days off.

An internship must have a minimum of 90 days. You perform these 90 days between Monday Fontys week 1 and Friday Fontys week 18.

Sick leave

Let's hope not, but it is of course possible that you become ill during a (prolonged) period of your internship or miss days on your internship for some other reason. Because you basically continue to work on holidays, you have some leeway. Being ill or taking leave for another reason does not have to be a problem to achieve the minimum number of days.

What are internship days?

Internship days are all days on which you work on your internship and/or on your internship assignment. You can include visits that you make in the context of your internship or for your education, such as counseling moments at school or visiting a minor market.

Internship permission

In order to start arranging an internship, you must have internship permission from the examination board. Please note: before you can actually start the internship, the internship coordinator for your study route must approve the internship location you have found based on the project proposal that you have completed and handed in yourself.

Permission to go on an internship is given when you have completed one of the semesters of the second year of study. You will then first follow the other semester of the second year (which you have not yet completed) and do an internship immediately after that (regardless of whether you pass this other semester). Hence, you will always have one semester to find a suitable internship. It is recommended to start searching in time (i.e. as soon as you know you have permission). After your internship, you will have to retake any semester from the second year that you have not yet passed until it is completed.

17.2. Learning outcomes

Learning outcome 1:

[Professional duties] You carry out professional duties at a junior bachelor level, producing professional-grade work aligned with the IT subdomain (IT-area) you are working in.

Learning outcome 2:

[Situation-orientation] You apply your previously acquired knowledge and skills in a new, authentic context to deliver valuable and relevant results for both the project and the company.

Learning outcome 3:

[Professional Standard] Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

Learning outcome 4:

[Personal Leadership] You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

17.3. Examination and final grading

17.3.1. Examination

Key moments of examination

Week 3-5	Projectplan/projectdocument. Formative assessment; the first assessor provides feedback during the first company visit in week 3-5.
Week 9/10	Additional feedback moment at school; formative assessment.
Week 10-17	Hand in concept portfolio with a request for feedback; formative assessment.
Week 17	Portfolio including proces description and achievements. Hand in at the beginning of week 17.
Week 18	Formative assessment during the second company visit in week 18. Final presentation and/or demonstration of the final product, combined with accountability about the execution and evaluation of the process. Feedback and advised grading from internship coach and company coach.
Week 19/20	Summative assessment bij first and second assessor. Criterion-based Interview on learning outcomes taking into account all aspects of your internship.

17.3.2. Repairs and resits

According to the TER this is a practice based semester. Hence there are no options for repair or retake. If you fail your internship, you should either redo a semester from year 2 before restarting an internship or, if you have passed both of these semesters, start a new internship.

17.3.3. Tools allowed

Not applicable. The regular rules about plagiarism apply.

17.3.4. Final grading

The assessment will take place in week 19. The assessment is taken by two assessors: the first assessor, which is the teacher that guided you during your internship and the second assessor, which is another examiner from school. Assessment are portfolio-based and are conducted in the form of a criterion-based interview (CGI) to determine if all learning outcomes have been sufficiently achieved.

If a student has at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors.

17.4. Learning activities

The internship is largely independently acquired by you as a student and carried out at an external client. This may be both at home and abroad. This requires entrepreneurship on your part; we expect your own initiative when formulating an assignment and executing it. It is your own responsibility to find a suitable assignment and internship position. Failing to do so may force you to put your studies on hold until you found a suitable position, since you will not be allowed to continue to later semesters without completing an internship.

During your internship you will be supervised by someone from the company; your company supervisor. In addition, you will receive guidance from school by an internship teacher (the 1st assessor). He mainly guides you process-wise and is co-evaluator at the end. You will also be assigned a 2nd assessor. This is a co-assessor at the final assessment. During your internship, your internship teacher is your first point of contact from school and also acts as your semester coach.

17.5. Learning materials

There is no standard teaching material available for the internship. You can, however, use the documents that you can find on Canvas. Here you will find, among other things, the research framework and support for creating your portfolio.

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18. INFORMATION ABOUT [O-STA] INTRODUCTION TO IT (30EC)

18.1. Inhoud

Introductie

Welkom bij het Startsemester van Fontys ICT PRO.

Je staat aan het begin van een nieuwe stap in je carrière. Dit is het begin van een innoverende studie aan Fontys ICT. Een hogeschool waarbij je vanaf de eerste dag aan de slag gaat met het opdoen van kennis, het aanleren van vaardigheden en het verder uitbouwen van een juiste beroepshouding.

Voor de één is het wellicht al even geleden dat je met studeren bezig bent geweest, voor de ander is er wellicht een directe overgang van MBO naar HBO. De één heeft al een redelijke ICT-basis, de ander heeft ervoor gekozen zich om te scholen. In alle gevallen zal dit een aanzienlijke verandering teweegbrengen. Je combineert je studie met je werk- en privéleven. Die privé-omgeving kan er voor iedereen anders uitzien. Thuiswonend, samenwonend, stappen, kinderen, mantelzorg, ziekte, verhuizingen/verbouwingen, enz. Dit alles zal van invloed zijn op je studie, en zowel jij als je omgeving zal rekening moeten houden met het feit dat je hier aan begonnen bent.

Dit betekent dat je in het begin op zoek moet gaan naar een goed studieritme. Wat zijn geschikte momenten om te studeren? Hoe zorg je ervoor dat je hier voldoende tijd voor plant? Mensen om je heen moeten wennen aan het idee dat je bezig bent met je studie en ervaren dat ze je niet meer zo maar kunnen storen met vragen of verhalen.

Ons onderwijs is gebaseerd op de visie “student nabij onderwijs en toetsen”. Hierbij staat ontwikkelingsgerichte feedback voorop, en die feedback dien je zelf te verkrijgen. Dat betekent regelmatig uitwerkingen bij docenten voorleggen of het bespreken van vraagstukken met medestudenten om te kijken of je de goede kant op gaat. Wanneer je moet samenwerken in groepen zal iedereen rekening moeten houden met elkaar. Mocht het niet soepel lopen, ga hierover met elkaar in gesprek en kijk of je elkaar hierbij kunt helpen. Dit is één van de aandachtspunten binnen dit eerste semester.

Je leert veelal in de context van levensechte projecten (challenges). Tijdens het semester moet je eigenaarschap nemen van je eigen leerproces. Je leert de basis van ICT binnen de context van voorbeelden en taken die je in je latere werk als ICT-professional zult tegenkomen. Vanuit deze basis word je gestimuleerd om gefundeerde keuzes te maken voor je eigen leerpad.

De focus van dit semester ligt op oriëntatie op het gebied van ICT en wat ons onderwijs te bieden heeft, en het maken van een keuze voor je tweede semester. Daarbij bouw je ook je beroepsvaardigheden verder uit die voor alle ICT-professionals relevant zijn. Tijdens dit semester kies je voor welk semester 2 onderwerp je je wilt inschrijven en bereid je je daarop ook voor.

Daarnaast oriënteer je je op de juiste opleiding: Associate degree of Bachelor. Aan het einde van dit semester kan hier ook nog tussen overgestapt worden.

De leeromgeving is zo georganiseerd dat je nauw contact hebt met je docenten en medestudenten. De fysieke leeromgeving, ook wel Fontys Innovation Lab genoemd, bestaat met name uit tribunes voorgeplande workshops, maar ook uit meer algemene ruimtes waarbij gewerkt wordt aan oefeningen en challenges, leren en ontmoeten van medestudenten en docenten.

18.2. Leeruitkomsten

1. Learning outcome: Professionele standaard

Je past individueel en in teamverband een professionele werkwijze toe op het gebied van projectorganisatie, communicatie met belanghebbenden, verkennend onderzoek en rapportage.

Explanation: Professionele standaard

Je past de beroepspraktijk toe, zowel individueel als in teamverband, op het gebied van projectorganisatie, communicatie met belanghebbenden, verkennend onderzoek en verslaglegging.

Projectorganisatie:

Voor een bepaalde opdracht maak je -met begeleiding- een planning (timing & deeltaken) met je teamleden. Je maakt afspraken over de verwachte kwaliteit.

Communicatie en rapportage:

Je houdt rekening met de directe belanghebbende wanneer je aan een opdracht werkt. Je let op wat je wilt communiceren en in welke vorm je dit doet. Je herkent taken binnen het groepswork en neemt je rol binnen de groep op je. Je bent je bewust van vooroordelen en de gevolgen die vooroordelen kunnen hebben.

Verkennend onderzoek:

Je onderzoekt en stelt vragen tijdens het oplossingsproces. Je beantwoordt vragen door een passende aanpak te gebruiken: pragmatisch, kritisch en gebaseerd op meerdere bronnen.

2. Learning outcome: Persoonlijk leiderschap

Je neemt initiatief om feedback te vragen en te reflecteren en benoemt jouw kernwaarden als vertrekpunt voor je studieloopbaan en je professionele ontwikkeling.

Explanation: Persoonlijk leiderschap

Je neemt initiatief in het vragen om en reflecteren op feedback. Je identificeert je eigen kernwaarden als basis voor je studieloopbaan en professionele ontwikkeling.

Feedback en reflectie:

Je zoekt actief feedback (van collega's en coaches) op je leeractiviteiten die gericht zijn op zowel technische als professionele leeruitkomsten.

Je denkt na over de ontvangen feedback om leermogelijkheden te identificeren en je inspanningen te richten op professionele, persoonlijke en academische groei.

Je herkent je eigen leerbehoeften en staat open voor feedback. Je gebruikt feedback als een hulpmiddel om je vaardigheden en kennis te verbeteren.

Kernwaarden:

Door zelfevaluatie en reflectie identificeer je wat je drijft en motiveert en herken je je persoonlijke kwaliteiten en talenten in de context van je persoonlijke en professionele leven.

Je maakt weloverwogen keuzes voor je ontwikkeling en studieprogramma, waarbij je ervoor zorgt dat deze aansluiten bij de uitkomst van je reflectie op drijfveren en kwaliteiten.

Je beschrijft je motieven en kwaliteiten in de context van je toekomstige rol als ICT-professional, waarbij je begrijpt hoe ze je carrièrepad en professionele gedrag beïnvloeden.

3. Learning outcome: Oriëntatie

Je oriënteert je op het brede IT werkveld door je te verdiepen in verschillende onderdelen van het leerplan. Je kiest een vervolgstap die bij je past en onderbouwt deze keuze.

Explanation: Verdiepen:

Door leeractiviteiten in verschillende contexten uit te voeren, maak je kennis met verschillende onderdelen van ons curriculum.

Onderbouwen:

Op basis van deze activiteiten leg je uit waarom je voor een bepaalde volgende stap kiest.

4. Learning outcome: Maken van professionele IT-producten

Je creëert één of meer IT beroepsproducten. Hiervoor analyseer je de probleemcontext, adviseer je over te maken keuzes, ontwerp je een passende oplossing, realiseer je deze en valideer je de werking.

Explanation: Analyseren:

Je onderzoekt de opdracht en begrijpt de randvoorwaarden van je oplossing. Je legt de verzamelde requirements en kwaliteitscriteria vast op een gestandaardiseerde manier.

Voorbeelden van professionele producten: analysedocument, testplan, use cases, user stories, product backlog, procesanalyse.

Advies:

Je legt uit waarom de professionele producten en de keuzes die je hebt gemaakt bijdragen aan het oplossen van het probleem .

Voorbeelden van professionele producten: technische documentatie, productvergelijkingen, adviesrapportage, verbetervoorstellen.

Ontwerp:

Je bedenkt een oplossing op basis van de eisen en kwaliteitscriteria en legt je keuzes vast. Je gebruikt hiervoor standaardmethoden.

Voorbeelden van beroepsproducten: klassendiagram, stroomdiagram, databaseschema, architectuurontwerp, systeemontwerp, hardwareontwerp, pseudocode, interactieontwerp, wireframes, schetsen, ontwerpverklaringen, rapid-, low- en medium fidelity prototypes, organisatieprocessen, gegevensstromen, communicatieprotocol .

Realiseren:

Je maakt je oplossing op basis van je ontwerp, met aandacht voor kwaliteitscriteria.

Voorbeelden van professionele producten: broncode, geïnstalleerde software, screenshots of opnames van werkende software, high fidelity interactieve prototypes, UX-ontwerpen, implementatieplan, cloudinfrastructuur, procesbeschrijving, embedded software inclusief hardware .

Valideren:

Je test je producten met standaardtechnieken en volgens een standaardwerkwijze en bepaalt of de oplossing voldoet aan de vereisten en kwaliteitscriteria. Je legt de resultaten vast en verbetert je product(en) waar nodig .

Voorbeelden van professionele producten: feedback van belanghebbenden, testrapportage, opname van demosessie, gebruikerstests, gebruikstests, gevalideerd professioneel product.

18.3. Verwacht ingangprofiel

18.4. Uitstroomprofiel

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development			1	1

18.5. Toetsing en beoordeling

18.5.1. Toetsing: Holistisch en Longitudinaal

De beoordeling in dit semester is holistisch en longitudinaal. Holistisch betekent dat alles wat je als student doet meetelt voor de eindbeoordeling: de hoeveelheid werk, de kwaliteit van het werk, je professionaliteit en de aangetoonde competenties.

De leeruitkomsten beschrijven het gewenste niveau, maar omdat deze onafhankelijk van het leertraject kunnen worden aangetoond, is er ruimte om een deel van je eigen leerproces vorm te geven. Daarom is het belangrijk om je docenten regelmatig te raadplegen om er zeker van te zijn dat je de juiste dingen doet. Dit is het longitudinale aspect: gedurende het hele semester volgen de beoordelaars je vooruitgang.

18.5.2. Feedback: Bevestiging en begeleiding

Docenten begeleiden en beoordelen je vooral door middel van feedback. Je hebt wekelijks gesprekken met een of meer docenten over je werk. Deze feedback leg je zelf vast in FeedPulse, waarna docenten reageren met informatie over je voortgang.

Daarnaast vinden op vaste momenten in het semester portfolio opleveringen plaats. Hierop krijg je feedback van je docenten op basis van de developmental feedback scale. Deze schaal heeft vijf niveaus.

Ontwikkelingsgerichte feedbackschaal

- **Onbepaald:** Je hebt nog geen activiteiten ondernomen voor het aantonen van de leeruitkomst.
- **Oriënterend:** Je hebt een begin gemaakt en de mogelijkheden verkend om de leeruitkomst aan te tonen.
- **Beginnend:** Je hebt eerste stappen gezet en deze uitgevoerd welke bijdragen aan het aantonen van de leeruitkomst.
- **Ge oefend:** Je hebt meerdere keren laten zien dat je een basis gecreëerd hebt om de leeruitkomst aan te tonen. Je zal de leeruitkomst op een voldoende niveau aantonen, als je jouw ontwikkeling op deze manier blijft voortzetten
- **Gevorderd:** Je hebt meerdere keren laten zien dat je met deze leeruitkomst bezig bent geweest met goed resultaat. Je hebt boven verwachting gepresteerd en bent gericht op continue verbetering. Je zal de leeruitkomst op een ruim voldoende niveau aantonen, als je jouw ontwikkeling op deze manier blijft voortzetten.

Deze feedback is een momentopname van je huidige groei in de richting van het leerresultaat. Het is mogelijk dat je achteruit gaat in je ontwikkeling als je bepaald gedrag niet consequent laat zien in de praktijk.

Binnen het semester zijn er minstens drie van deze formatieve evaluatiemomenten waarbij de beoordelaars feedback geven per leerresultaat aan de hand van de ontwikkelingsgerichte feedbackschaal. Het laatste van deze drie formatieve evaluatiemomenten is uiterlijk twee weken voor de beoordelaarsvergadering. Dit geeft je de gelegenheid om de ontvangen feedback te behandelen en te verwerken.

18.5.3. Assessment Richtlijnen

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

18.5.4. Herkansingen en reparaties

Herkansingen of reparaties zijn niet van toepassing op dit semester. Er wordt van je verwacht dat je je werk een onbeperkt aantal keren opnieuw indient voor feedback.

18.6. Richtlijnen AI gebruik

Bij Fontys ICT hebben we een visie op het gebruik van Gen AI (AI), die te vinden is in het AI-manifest. Hierin stellen we verwachtingen vast voor hoe je tijdens je leerproces met AI omgaat.

AI als samenwerkingspartner

AI is geen vervanging of bedreiging, maar een samenwerkingspartner in je leerproces. We gebruiken AI om creativiteit te versterken, begrip te verdiepen en onderwijsmogelijkheden uit te breiden. Er wordt van je verwacht dat je leert om effectief en productief met AI te werken: een kerncompetentie voor ICT-professionals.

Het kernprincipe: jij bent de architect, niet het gereedschap. AI versnelt je werk, maar jij blijft verantwoordelijk voor alles wat je levert. Pas bij het gebruik van AI het transparantieprincipe toe: wees duidelijk en eerlijk over wat AI heeft bijgedragen en wat je eigen analyse, oordeel of creatie is.

Validatie en kritische betrokkenheid

Werken met AI betekent dat je de output voortdurend kritisch en nieuwsgierig evalueert. AI kan feiten hallucineren, onveilige code produceren of vooroordelen inbouwen. Het is essentieel om het vermogen te ontwikkelen om te evalueren, vragen te stellen en te verifiëren. Uw beslissingen zijn belangrijk, dus zorg ervoor dat u ze kunt uitleggen. Documenteer uw validatieproces voor belangrijke AI-bijdragen.

Ethische grenzen

Verantwoord gebruik van AI vereist het respecteren van duidelijke grenzen. Voer nooit persoonlijke gegevens, bedrijfsgeheimen of vertrouwelijke informatie in openbare AI-tools in. De basisrichtlijnen van de AVG/GDPR zijn altijd van toepassing. Gebruik institutionele AI-tools wanneer deze beschikbaar zijn, waaronder de faciliteiten die Fontys ICT biedt. Respecteer intellectueel eigendom: begrijp welke licenties van toepassing zijn op door AI gegenereerde content en zorg ervoor dat je je hieraan houdt.

Je AI-praktijken hebben niet alleen invloed op jou, maar ook op Fontys ICT en eventuele professionele partners waarmee je samenwerkt.

18.7. Leeractiviteiten

Je leert eerst de basis van ICT aan de hand van voorbeelden en opdrachten die je later als ICT-professional tegenkomt. Van daaruit word je gestimuleerd om weloverwogen keuzes te maken voor je eigen leerroute. De focus van dit semester ligt op oriëntatie op het vakgebied ICT en wat onze opleiding te bieden heeft, en het maken van een keuze voor je tweede semester. Daarbij ontwikkel je ook professionele vaardigheden die relevant zijn als ICT-professional. Tijdens dit semester kies je voor welk onderwerp je in semester 2 wilt inschrijven en bereid je je daar ook op voor.

Je kunt ook nog kiezen om te switchen tussen Associate degree of Bachelor.

De leeromgeving is zo ingericht dat je nauw samenwerkt met je docenten en medestudenten.

Gedurende het semester wordt aan drie challenges gewerkt:

- Eerste challenge:
Gedurende de eerste vier weken ga je aan de slag met de eerste challenge van dit semester. Dit voer je individueel uit en heeft een vaste context.
- Tweede challenge:
Gedurende vijf weken ga je individueel aan de slag met een challenge waarbij je enkele keuzemogelijkheden hebt.
- Derde challenge:
In deze laatste challenge van negen weken ga je aan de slag met een grotere challenge waarbij je met een groep van vier tot zes studenten aan de slag gaat met een multidisciplinair project. Dit betekent dat groepsleden verschillende keuzes hebben gemaakt en vanuit deze keuze aan de challenge zullen bijdragen. Ook hierbij kan gekozen uit een lijst met challenges.

18.8. Leermaterialen

Canvas bevat al het lesmateriaal.

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19. INFORMATION ABOUT [STA] INTRODUCTION TO IT (30EC)

19.1. Content

Welcome to Startsemester at Fontys ICT.

Our education is based on the vision "student centered education". You will learn in the context of real-life projects. During the semester you will need to take ownership of your own learning process. You will first learn the basics of ICT within the context of examples and tasks that you will encounter in your later work as an ICT professional. From this base, you will be stimulated to make well-founded choices for your own learning path.

The focus of this semester is on orientation on the field of ICT and what our education has to offer and making a choice for your second semester. In doing so you will also develop professional skills that are relevant to all ICT professionals. During this semester you will choose in which semester 2 topic you would like to enroll and prepare for it as well:

- Semester 2 topics to choose from
 - AI Basics
 - Business IT & Data Analysis
 - Game Development
 - Smart Industry
 - Cyber Security Principles & Techniques
 - Intelligent Technologies
 - Interactive Media
 - Network & Cloud Basics
 - Software Design & Engineering
 - Open Learning

You can also choose to switch to

- Associate Degree
- Accelerated program

The learning environment is organised in such a way that you interact closely with your teachers and fellow students. The physical learning environment, also called Open ICT Lab (OIL), consists of classrooms or tribunes for planned workshops, but also of more general areas for supported self-study, working on exercises & projects, learning and meeting up with fellow-students and teachers.

19.2. Learning outcomes

1. Learning outcome: Professional standard

You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory research, and reporting.

Explanation: Project organisation

For a given assignment you - with guidance - make a planning (timing & part-tasks) with your team members. You make agreements about the expected quality.

Communication and reporting

You take the direct stakeholder into account when working on an assignment. You pay attention to what you want to communicate and in which shape you do this. You recognize tasks within the group work, and take on your role within the group. You are aware of bias and the consequences bias could have.

Exploratory research

You explore and ask questions throughout the solution process. You answer questions by using an appropriate approach: pragmatic, critical and based on multiple sources.

2. Learning outcome: Personal leadership

You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional development.

Explanation: Feedback and reflection

You plan your assignments efficiently and realistically. You work result-oriented on your assignments. You recognize learning needs in yourself, are open to feedback, and reflect on this.

Core values

You identify your own core values. You make considered choices for your development and your study program.

3. Learning outcome: Orientation

You orient yourself to the broad IT work field by immersing yourself in various parts of the curriculum. You choose a next step that suits you and substantiate this choice.

Explanation: Immersing

By carrying out learning activities in different contexts, you get acquainted with various parts of our curriculum. Substantiate

You explain, on the basis of these activities, why you choose a particular next step.

4. Learning outcome: Creating professional IT products

You create one or more IT professional products. To do so, you analyse the problem context, advise on choices to be made, design a suitable solution, realise it and validate its operation.

Explanation: Analyse

You research the assignment and understand the preconditions of your solution. You record the collected requirements and quality criteria in a standardised way.

Examples of professional products: analysis document, test plan, use cases, user stories, product backlog, process analysis.

Advise

You explain why the professional products and the choices you made contribute to solving the problem.

Examples of professional products: technical documentation, product comparisons, advisory reporting, improvement proposals.

Design

You devise a solution based on the requirements and quality criteria and record your choices. You use standard ways to do so.

Examples of professional products: class diagram, flow chart, database diagram, architecture design, system design, hardware design, pseudo-code, interaction design, wireframes, sketches, design statements, rapid-, low and medium fidelity prototypes, organisational processes, data flows, communication protocol.

Realise

You make your solution based on your design, paying attention to quality criteria.

Examples of professional products: source code, installed software, screenshots or recording of working software, high fidelity interactive prototypes, UX designs, implementation plan, cloud infrastructure, process description, embedded software including hardware.

Validate

You test your products using standard techniques and according to a standard operating procedure and determine whether the solution meets the requirements and quality criteria. You record the results and improve your product(s) where necessary.

Examples of professional products: stakeholder feedback, test reporting, recording of demo session, user tests, usability tests, validated professional product.

19.3. Expected entry profile

19.4. Outflow profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development			1	1

19.5. Examination and final grading

19.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

19.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome at a satisfactory level if you continue your development in this way.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations and are focused on continuous improvement. You will demonstrate the learning

outcome at a more than satisfactory level if you continue your development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

19.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

19.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

19.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

AI is not a replacement or threat, but a collaborative actor in your learning. We use AI to strengthen creativity, deepen understanding, and expand educational possibilities. You are expected to learn to work with AI effectively and productively: a core professional competency for ICT professionals.

The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available, including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

19.7. Learning activities

Project one:	During the first four weeks you will start working on the first project of this semester. The first project is an individual project and has a fixed context.
Project two:	The next four weeks you will be executing a second individual project. Within this Canvas module three possible projects are being displayed. You are free to chose one of these projects.

Project three and four	For the next five weeks you will work on project three and four simultaneously. The third project is an individual one. The fourth and fifth projects will be a group project. A group exists out of 6 students. You are free to choose any project out of the project list, as long as you take a different one for each project phase.
Project five	During the last five weeks of this semester you will execute the fifth project. Project five is a group project and will be the last project of this semester.

19.8. Learning materials

Canvas contains all the learning materials.

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20. INFORMATION ABOUT [APA1A] LOGIC & SET THEORY (4EC)

20.1. Content

Explore the fundamental principles of logic and set theory in this course. Learn propositional and predicate logic, proof techniques. Study set theory, including operations, relations, functions, and infinite sets. Gain a understanding of set theory and its implications. Develop essential skills in logical reasoning and mathematical foundations.

Furthermore, it introduces constraint programming and Satisfiability Modulo Theories (SMT) solvers. You'll learn how to model real-world problems using constraints and use SMT solvers to efficiently solve them. Topics include constraint modeling, SMT solving algorithms, advanced topics like quantified constraints and program verification, and practical applications in areas such as scheduling and resource allocation. The course combines lectures, problem-solving exercises, and programming assignments to develop your skills in constraint programming and SMT solving.

20.2. Learning outcomes

20.3. Examination and final grading

20.3.1. Examination

The teachers will provide the students with formative feedback on deliverables and demonstrations. It is expected that this feedback will be processed in further deliverables, within reason.

The student shows the work, alone or in a group context, but will be given feedback on an individual base.

The student work will be assessed on analysis, design and technical aspects, where the student will be informed whether the learning goals are met concerning the criteria.

At these assessments the learning outcomes are determined based on state of the deliverables at that point. Ideally the learning outcomes are proficient at these assessments. This grade is a formative indicator of mastery of learning goals at that moment and don't give any guarantee for further assessments. It is expected that the student keeps on progressing on the learning outcomes, not doing so might lead to a lower grade of beginning or orienting.

20.3.2. Repairs and resits

During the course, students are facilitated to process formative feedback on their learning goals in their deliverables and activities that are to be recorded in their portfolio (Canvas). All learning activities are related to the professional assignment.

20.3.3. Tools allowed

Except for paper & pen, and a laptop, no additional equipment is needed. The regular rules about plagiarism apply.

20.3.4. Final grading

A student's growth is maintained in his portfolio. The student's growth on each learning outcome is registered according to one of the five levels of mastery.

1. undefined,
2. orienting,
3. beginning,
4. proficient,
5. advanced.

After each formative assessment, the level of mastery for all learning outcomes is assessed for the student's deliveries.

At the end of the semester the student will be assessed on his performance of the semester. During the assessment meeting the teacher will use the student's portfolio. The overall grade is based on the shown performance and will be graded by the examiners. If all learning outcomes are proficient or advanced, the grade will be at least Satisfactory. This does not mean that obtaining proficient is the minimum requirement to achieve a satisfactory grade.

20.4. Learning activities

Based on the characteristics of an academic premaster and based on a highly knowledge-based scientific world, the course is structured according to the research-based learning method. It is a structured and predictable learning path, in which the learning outcomes and assessment criteria are determined by the study program.

The education is designed according to the principles of the 4C/ID model (four-component instructional design model). Education is pre-structured in terms of content, working methods and guidance.

Knowledge, competences and professional skills are further gained and applied in practical assignments, which increase in size, abstraction and complexity during the study program. The student gets more and more choice in a pre-sorted set of learning tasks that are designed according to 4C/ID.

Depending on the subjects, a variety of side-roads will be offered for in-depth knowledge and broader research.

20.5. Learning materials

All course content is in Canvas and Sharepoint. All software (compilers, IDEs, libraries, tools) is available for free.

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21. INFORMATION ABOUT [O-IN-NCB] NETWORK & CLOUD BASICS (30EC)

Connecting, Provisioning & Managing IT Services, DevOps

21.1. Content

In today's digital world, IT infrastructure plays a crucial role in business operations. In this semester, you will explore the fundamentals of networking, cloud platforms, and DevOps, gaining the technical knowledge and hands-on skills needed to manage modern IT environments.

You will learn how networks are designed, deployed, and secured, covering key concepts such as perimeter defense security and device management. Additionally, you will explore different cloud platforms, understanding how they support business agility and scalability.

DevOps practices will be introduced to help you automate and streamline IT processes, ensuring efficient collaboration between development and operations teams. You will also work with operating systems, understanding their role in networked environments and enterprise IT infrastructure.

Throughout the semester, you will not only develop technical expertise but also gain insights into real-world IT challenges, preparing you for a career in networking, cloud computing, and IT security.

21.2. Learning outcomes

1. **Learning outcome:** Professional standard

— You apply professional practice, both individually and in teams, in the areas of project organisation, communication

with stakeholders, exploratory research, and reporting.

2. Learning outcome: Personal leadership

You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional development.

3. Learning outcome: Basic Interoperability

After conducting research, you are able to design and implement a sustainable infrastructure based on the specified requirements of a small SME, taking into account the different types of data, connections, protocols, communication, and applications. After conducting research, you are able to design and implement a sustainable infrastructure based on the specified requirements of a small SME, taking into account the different types of data, connections, protocols, communication, and applications.

4. Learning outcome: Basic Securing

After conducting research, you are able to design and implement modern and traditional basic security measures across all infrastructure domains based on the specified requirements of a small SME.

5. Learning outcome: Basic Automation

After conducting research, you are able to automate the deployment of an infrastructure, application, and business process by programming scripts, based on the specified requirements of a small SME.

6. Learning outcome: Manage & Control basics

After conducting research, you are able to design an infrastructure based on the specified requirements of a small SME that considers certain management aspects and processes and complies with local laws and regulations.

21.3. Expected entry profile

A basic understanding of networking and some hands-on experience with an operating system will help you get started in this semester. However, curiosity and a problem-solving mindset are just as important. No advanced technical skills are required—this semester is designed to build on foundational knowledge and develop practical expertise.

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1		

21.4. Outflow profile

Domain description HBO-i 2023

	Manage & control	Realise	Design	Advise	Analysis
User interaction					
Organisational processes					
Infrastructure	1	1	1	1	1
Software		1	1	1	
Hardware-interfacing					

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

21.5. Examination and final grading

21.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

21.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome at a satisfactory level if you continue your development in this way.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations and are focused on continuous improvement. You will demonstrate the learning outcome at a more than satisfactory level if you continue your development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

21.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

21.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

21.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

AI is not a replacement or threat, but a collaborative actor in your learning. We use AI to strengthen creativity, deepen understanding, and expand educational possibilities. You are expected to learn to work with AI effectively and productively: a core professional competency for ICT professionals.

The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available, including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

21.7. Learning activities

During this semester, you and your team will analyze, design, and realize an IT solution for a real-world company. You will apply your knowledge of networking, cloud platforms, and DevOps to create a secure and efficient IT environment.

Throughout the project, you will manage and control your solution, ensuring it meets business requirements while maintaining security and performance. You will learn to advise stakeholders by assessing IT needs, proposing improvements, and justifying your design choices.

Your learning journey will be supported by a mix of practical exercises, workshops, and challenges, helping you develop hands-on skills in perimeter security, device management, and operating systems. Collaboration with external stakeholders will give you insight into real-world IT challenges, while teachers and coaches will guide you in refining your technical and professional skills.

By the end of the semester, you will have built a solid foundation in networking and cloud technologies, enabling you to design, implement, and manage IT solutions in a business context.

21.8. Learning materials

A Canvas course featuring the necessary learning resources

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22. INFORMATION ABOUT [IN-NCB] NETWORK & CLOUD BASICS (30EC)

Connecting, Provisioning & Managing IT Services, DevOps

22.1. Content

In today's digital world, IT infrastructure plays a crucial role in business operations. In this semester, you will explore the fundamentals of networking, cloud platforms, and DevOps, gaining the technical knowledge and hands-on skills needed to manage modern IT environments.

You will learn how networks are designed, deployed, and secured, covering key concepts such as perimeter defense security and device management. Additionally, you will explore different cloud platforms, understanding how they support business agility and scalability.

DevOps practices will be introduced to help you automate and streamline IT processes, ensuring efficient collaboration between development and operations teams. You will also work with operating systems, understanding their role in

networked environments and enterprise IT infrastructure.

Throughout the semester, you will not only develop technical expertise but also gain insights into real-world IT challenges, preparing you for a career in networking, cloud computing, and IT security.

22.2. Learning outcomes

- 1. Learning outcome:** Professional standard
You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory research, and reporting.
- 2. Learning outcome:** Personal leadership
You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional development.
- 3. Learning outcome:** Basic Interoperability
After conducting research, you are able to design and implement a sustainable infrastructure based on the specified requirements of a small SME, taking into account the different types of data, connections, protocols, communication, and applications. After conducting research, you are able to design and implement a sustainable infrastructure based on the specified requirements of a small SME, taking into account the different types of data, connections, protocols, communication, and applications.
- 4. Learning outcome:** Basic Securing
After conducting research, you are able to design and implement modern and traditional basic security measures across all infrastructure domains based on the specified requirements of a small SME.
- 5. Learning outcome:** Basic Automation
After conducting research, you are able to automate the deployment of an infrastructure, application, and business process by programming scripts, based on the specified requirements of a small SME.
- 6. Learning outcome:** Manage & Control basics
After conducting research, you are able to design an infrastructure based on the specified requirements of a small SME that considers certain management aspects and processes and complies with local laws and regulations.

22.3. Expected entry profile

A basic understanding of networking and some hands-on experience with an operating system will help you get started in this semester. However, curiosity and a problem-solving mindset are just as important. No advanced technical skills are required—this semester is designed to build on foundational knowledge and develop practical expertise.

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1		

22.4. Outflow profile

After completing this semester, we recommend continuing with Advanced Networking & Cloud Technologies, where you will deepen your knowledge in security, automation, and cloud management. The skills acquired here will also be valuable in DevOps, IT security, and infrastructure management, preparing you for various IT roles.

Domain description HBO-i 2023

	Manage & control	Realise	Design	Advise	Analysis
User interaction					

Organisational processes					
Infrastructure	1	1	1	1	1
Software		1	1	1	
Hardware-interfacing					

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

22.5. Examination and final grading

22.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

22.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome at a satisfactory level if you continue your development in this way.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations and are focused on continuous improvement. You will demonstrate the learning outcome at a more than satisfactory level if you continue your development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

22.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

22.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

22.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

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The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available, including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

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22.7. Learning activities

During this semester, you and your team will analyze, design, and realize an IT solution for a real-world company. You will apply your knowledge of networking, cloud platforms, and DevOps to create a secure and efficient IT environment.

Throughout the project, you will manage and control your solution, ensuring it meets business requirements while maintaining security and performance. You will learn to advise stakeholders by assessing IT needs, proposing improvements, and justifying your design choices.

Your learning journey will be supported by a mix of practical exercises, workshops, and challenges, helping you develop hands-on skills in perimeter security, device management, and operating systems. Collaboration with external stakeholders will give you insight into real-world IT challenges, while teachers and coaches will guide you in refining your technical and professional skills.

By the end of the semester, you will have built a solid foundation in networking and cloud technologies, enabling you to design, implement, and manage IT solutions in a business context.

22.8. Learning materials

A Canvas course featuring the necessary learning resources

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23. INFORMATION ABOUT [IN-OLB] OPEN LEARNING (30EC)

23.1. Content

ICT & Personalized Program is a route where you have a lot of freedom and flexibility to determine your personal focus and structure, within the boundaries of the HBO-I competence framework. In practice, this means that anything within these boundaries potentially can be used in ICT & Personalized Program. This gives great opportunities for starting multi-disciplinary projects. Projects should be IT related, based on relevant questions and developments from the professional field and where possible externally validated.

Lots of freedom, but remember:

“Freedom comes with responsibility”

This large degree of freedom does not mean that there are no obligations. A large amount of autonomy and entrepreneurial behaviour is expected.

23.2. Learning outcomes

1. Learning outcome: Professional standard

You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory research, and reporting.

2. Learning outcome: Personal leadership

You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional development.

23.3. Expected entry profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development			1	1

23.4. Outflow profile

The student builds a competence profile based upon the basic architectural layers (level 1) from OE1, this is taken to the next step at level 1. In addition, a student can deepen or broaden and will touch other architectural layers. These will initially be linked at level 1. Generally speaking the core architectural layer level grows with the phase of the students' study. Phase OE1, OE2 ~ level 1, phase OE3,4,5, ~ level 2 and phase OE6,7,8, ~ level 3.

A balanced competence profile does not have too many competencies (no focus) but also not too little (no widening). At least all 9 competences have to be covered on level 1.

Within Open Learning, students define their own dynamic competence profile per semester (exit level OEx) under supervision. In doing so, they draw up their own criteria for challenges (projects) that contribute to growth within the HBO-i competence framework. A complete competence level is reached by successfully ending a phase. A learning outcome is a partial contribution to a competence step.

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

23.5. Examination and final grading

23.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

23.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

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23.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

23.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

23.6. AI Usage Guidelines

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Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

23.7. Learning activities

The student will determine the general lines of the content in this programme him or herself. Basically everything in IT should be possible. Because everyone has his own track, it is impossible to plan a curriculum that fits for everyone. Learning happens asynchrone, not everybody does the same things, with the same speed, ... This means all workshops, presentations, etc., that will provide content, will be planned and scheduled on-demand or when the coaches notice it is necessary.

To facilitate this the student will create his/her own personal profile within the HBO-i Competence Framework, this framework gives a systematic overview of the whole IT landscape including their professional skills. This profile will be created shortly after the start of the semester. Before enrollment, the student will be invited for an intake interview. This interview is about motivation and a possible context to use within open learning. In an intake document, it is outlined which topics the student would like to engage. This usually is a very rough outline. For example, "Artificial intelligence", "Encryption" or extreme interaction". The intake interview will be an evaluation to see whether you really understand what open learning is about, so you know what you're up to.

Because everyone has his own track, it is impossible to plan a curriculum that fits for everyone. Learning happens asynchronous, not everybody does the same things, with the same speed, ... This means all workshops, lessons, ... that will provide content, will be planned and scheduled on-demand or when the teachers notice it is necessary.

For example, workshops of more general applicability are offered to all students, like Scrum-101, Research, Concept validation, while more specific subjects are offered to a smaller group of students.

23.8. Learning materials

Apart of the Canvas LMS environment, anything which would be applicable can and will be used.

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24. INFORMATION ABOUT [O-IN-BDA] PROCESS DATA IN ORGANISATIONS (30EC)

Business Analysis, Process Modelling & Low Code Development, Data Handling & Statistics

24.1. Inhoud

In semester 2 Business ga je werken aan veel verschillende thema's. Je bekwaamt je in bedrijfskundige thema's, statistische basisvaardigheden, data-analyse en overkoepelende vaardigheden als professional.

In semester 1 heb je al kennis gemaakt met de verschillende thema's bij Business. In semester twee bouwen we hiërarchisch voort.

Leerinhoud:

OE2 PRO Business		
Organisatie	ICT	PO
<i>Organisatie en informatie</i> Strategische analyse: - Balanced scorekaart - BMC, Porter Informatiebeheer: - Requirements	<i>Programmeren R/SQL of Python/Power-BI</i> Datamodelleren en SQL - ERD-modellering - Query op basis van datamodel: - Volledige SQL selecteren - Subquery's voldoen aan Common Table	Professionele standaard: Je past individueel en in teamverband een professionele werkwijze toe op het gebied van projectorganisatie, communicatie met

• MoSCoW-methode • KPI's • Dataset genereren	Expressions R/tidyverse en SQL • Query's op basis van het omgekeerde omgekeerde • Valideren van opgeloste query's met SQL en vervalsing • Visualiseer van queryresultaten met ggplot	belanghebbenden, verkennend onderzoek en rapportage.
Organisaties begrijpen: • Processen beschrijven, ontwerpen, verbeteren, ist / soll / knelpuntanalyse, BPMN 2.0 • Interne controle en beheersingsmaatregelen met functiescheiding	Statistiek • Variabelen en meetniveaus • Gegevens presenteren • Centrummaten • Spreidingsmaten • Normaalverdeling	Persoonlijk Leiderschap: Je neemt initiatief om feedback te vragen en te reflecteren en benoemt jouw kernwaarden als vertrekpunt voor je studieloopbaan en je professionele ontwikkeling.
Doel = voorwaarden opgesteld om IT-organisaties te ondersteunen.	Doel = relatie zien tussen data en informatie (basis). Focus op visualiseren.	

Professionele vaardigheden

In semester 2 komen veel verschillende onderwerpen aan bod die allemaal essentieel zijn bij het begrijpen, verbeteren en ontwikkelen van bedrijfsprocessen met behulp van ICT-middelen. Omdat geen enkel onderwerp op zichzelf staat en juist het begrijpen en samenvoegen van de verbindende verbindende kracht is van een goede ICT & Business professional, staat toepassing van op toegevoegde kennis en kunde centraal in dit semester.

In het startsemester heb je je kunnen oriënteren op ICT én jezelf. Ook heb je ervaren en geleerd wat er allemaal komt kijken bij het samenwerken in een project: elkaar goede feedback geven, plannen, vergaderen en het maken van goede onderlinge afspraken. Hierop pakken we bij ICT & Business in semester 2 deur. Je toont niet alleen wat je kunt, maar ook hoe je dat bereikt hebt. Je werkt op een prettige manier samen, je neemt niet alles klakkeloos aan en reflecteert op wat je nog wilt leren. Dit soort vaardigheden komen overeen met jouw professionele ontwikkeling (PO).

24.2. Leeruitkomsten

1. Learning outcome: Professionele standaard

Je past individueel en in teamverband een professionele werkwijze toe op het gebied van projectorganisatie, communicatie met belanghebbenden, verkennend onderzoek en rapportage.

Explanation: Je herkent ethische en maatschappelijk kwesties. Je herkent eigen en andermans grenzen en handelt daarnaar. Je faseert -onder begeleiding- een gegeven opdracht in tijd en deeltaken waarin alle teamleden zich kunnen vinden. Je maakt afspraken over de verwachte kwaliteit. Je denkt in de uitwerking na over het gevolg van de oplossing voor betrokken mensen en organisaties. Je bent je bewust van de impact die culturele verschillen hebben op je werk en leven.

Je houdt rekening met directe belanghebbenden bij de opdracht. Je hebt aandacht voor wat je wil communiceren en in welke vorm. Je neemt je eigen rol in de groep. Je herkent taken in het groepswerk. Je spreekt anderen aan op hun rol. Je bent je bewust van de impact die culturele verschillen hebben op jou en op andere mensen met wie je omgaat.

Je blijft gedurende het hele oplosproces nieuwsgierig en vragen stellen. Je beantwoordt vragen met een passende aanpak: pragmatisch, kritisch en gebaseerd op bronnen. Je bent je ervan bewust dat culturele verschillen van invloed zijn op de keuzes die je maakt voor je oplossingen.

2. Learning outcome: Persoonlijk leiderschap

Je neemt initiatief om feedback te vragen en te reflecteren en benoemt jouw kernwaarden als vertrekpunt voor je studieloopbaan en je professionele ontwikkeling.

Explanation: Je ziet en grijpt kansen. Je motiveert jezelf. Je neemt verantwoordelijkheid voor jouw handelen. Je werkt resultaatgericht aan je opdracht of taak. Je maakt overwogen keuzes in je studieprogramma. Je herkent bij jezelf leerbehoeften. Je staat open voor feedback en reflecteert daarop. Je begrijpt wat cultuur is. Je herkent je eigen cultuur en een aantal dimensies van cultuurverschillen.

3. Learning outcome: (3) Bedrijfsanalyse; Je laat zien dat je een organisatie in kaart kan brengen. Je stelt een advies op voor de optimalisatie daarvan. In je advies neem je naast extra opbrengst en kostenvoordeel, ook niet-financiële

opbrengsten zoals verbetering van werkplezier en klanttevredenheid mee.

Explanation: De organisatie is een kleine of middelgrote organisatie. Organisaties bestaan met een reden. Dit bepaalt uiteindelijk inrichting van de organisatie en de toepassing van IT. Het analyseren van een organisatie en haar omgeving is een basisvaardigheid voor iedere IT-professional.

4. **Learning outcome:** (4) Data prepareren; Je modelleert, ontwerpt, bewaart en bevraagt een dataset met behulp van correcte en zinvolle ontwerpmethodes.

Explanation: Modelleren en ontwerpen van een dataset bestaat uit het creëren van een datamodel binnen de context. Gebaseerd op dit datamodel ontwerp je en implementeer je een database om de gegevens van de opdrachtgever op te slaan. Gebaseerd op de KPI's van de opdrachtgever implementeer je management rapportages in de database.

5. **Learning outcome:** (5) Dataset analyseren; Je bereidt gegevens voor en analyseert deze data statistisch zodat je zinvolle informatie creëert op basis van data-analyse en rapportage tools

Explanation: Prepareren van data het gebruiken van opschoon- en transformatie technieken om de data om te zetten naar een gestructureerde, samenhangende dataset die bruikbaar is voor exploratory data analysis. De analyse bestaat uit het creëren van statistieken, grafieken en conclusies die resulteren in een rapport dat past bij de opdrachtgever.

24.3. Verwacht ingangsprofiel

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes	1	1	1		
Infrastructure					
Software					
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1		

24.4. Uitstroomprofiel

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	1	1	1	1	1
Organisational processes	1	1	1	1	1
Infrastructure					
Software	1	1	1	1	1

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

24.5. Toetsing en beoordeling

24.5.1. Toetsing: Holistisch en Longitudinaal

De beoordeling in dit semester is holistisch en longitudinaal. Holistisch betekent dat alles wat je als student doet meetelt voor de eindbeoordeling: de hoeveelheid werk, de kwaliteit van het werk, je professionaliteit en de aangetoonde competenties.

De leeruitkomsten beschrijven het gewenste niveau, maar omdat deze onafhankelijk van het leertraject kunnen worden aangetoond, is er ruimte om een deel van je eigen leerproces vorm te geven. Daarom is het belangrijk om je docenten regelmatig te raadplegen om er zeker van te zijn dat je de juiste dingen doet. Dit is het longitudinale aspect: gedurende het hele semester volgen de beoordelaars je vooruitgang.

24.5.2. Feedback: Bevestiging en begeleiding

Docenten begeleiden en beoordelen je vooral door middel van feedback. Je hebt wekelijks gesprekken met een of meer docenten over je werk. Deze feedback leg je zelf vast in FeedPulse, waarna docenten reageren met informatie over je voortgang.

Daarnaast vinden op vaste momenten in het semester portfolio opleveringen plaats. Hierop krijg je feedback van je docenten op basis van de developmental feedback scale. Deze schaal heeft vijf niveaus.

Ontwikkelingsgerichte feedbackschaal

- *Onbepaald:* Je hebt nog geen activiteiten ondernomen voor het aantonen van de leeruitkomst.
- *Oriënterend:* Je hebt een begin gemaakt en de mogelijkheden verkend om de leeruitkomst aan te tonen.
- *Beginnend:* Je hebt eerste stappen gezet en deze uitgevoerd welke bijdragen aan het aantonen van de leeruitkomst.
- *Geoevend:* Je hebt meerdere keren laten zien dat je een basis gecreëerd hebt om de leeruitkomst aan te tonen. Je zal de leeruitkomst op een voldoende niveau aantonen, als je jouw ontwikkeling op deze manier blijft voortzetten
- *Gevorderd:* Je hebt meerdere keren laten zien dat je met deze leeruitkomst bezig bent geweest met goed resultaat. Je hebt boven verwachting gepresteerd en bent gericht op continue verbetering. Je zal de leeruitkomst op een ruim voldoende niveau aantonen, als je jouw ontwikkeling op deze manier blijft voortzetten.

Deze feedback is een momentopname van je huidige groei in de richting van het leerresultaat. Het is mogelijk dat je achteruit gaat in je ontwikkeling als je bepaald gedrag niet consequent laat zien in de praktijk.

Binnen het semester zijn er minstens drie van deze formatieve evaluatiemomenten waarbij de beoordelaars feedback geven per leerresultaat aan de hand van de ontwikkelingsgerichte feedbackschaal. Het laatste van deze drie formatieve evaluatiemomenten is uiterlijk twee weken voor de beoordelaarsvergadering. Dit geeft je de gelegenheid om de ontvangen feedback te behandelen en te verwerken.

24.5.3. Assessment Richtlijnen

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

24.5.4. Herkansingen en reparaties

Herkansingen of reparaties zijn niet van toepassing op dit semester. Er wordt van je verwacht dat je je werk een onbeperkt aantal keren opnieuw indient voor feedback.

24.6. Richtlijnen AI gebruik

Bij Fontys ICT hebben we een visie op het gebruik van Gen AI (AI), die te vinden is in het AI-manifest. Hierin stellen we verwachtingen vast voor hoe je tijdens je leerproces met AI omgaat.

AI als samenwerkingspartner

AI is geen vervanging of bedreiging, maar een samenwerkingspartner in je leerproces. We gebruiken AI om creativiteit te versterken, begrip te verdiepen en onderwijsmogelijkheden uit te breiden. Er wordt van je verwacht dat je leert om effectief en productief met AI te werken: een kerncompetentie voor ICT-professionals.

Het kernprincipe: jij bent de architect, niet het gereedschap. AI versnelt je werk, maar jij blijft verantwoordelijk voor alles wat je levert. Pas bij het gebruik van AI het transparantieprincipe toe: wees duidelijk en eerlijk over wat AI heeft bijgedragen en wat je eigen analyse, oordeel of creatie is.

Validatie en kritische betrokkenheid

Werken met AI betekent dat je de output voortdurend kritisch en nieuwsgierig evalueert. AI kan feiten hallucineren, onveilige code produceren of vooroordelen inbouwen. Het is essentieel om het vermogen te ontwikkelen om te evalueren, vragen te stellen en te verifiëren. Uw beslissingen zijn belangrijk, dus zorg ervoor dat u ze kunt uitleggen. Documenteer uw validatieproces voor belangrijke AI-bijdragen.

Ethische grenzen

Verantwoord gebruik van AI vereist het respecteren van duidelijke grenzen. Voer nooit persoonlijke gegevens, bedrijfsgeheimen of vertrouwelijke informatie in openbare AI-tools in. De basisrichtlijnen van de AVG/GDPR zijn altijd van toepassing. Gebruik institutionele AI-tools wanneer deze beschikbaar zijn, waaronder de faciliteiten die Fontys ICT biedt. Respecteer intellectueel eigendom: begrijp welke licenties van toepassing zijn op door AI gegenereerde content en zorg ervoor dat je je hieraan houdt.

Je AI-praktijken hebben niet alleen invloed op jou, maar ook op Fontys ICT en eventuele professionele partners waarmee je samenwerkt.

24.7. Leeractiviteiten

Inspiratiecolleges, werkcolleges, gastcolleges, workshops, groepsgewijs werken aan een casusopdracht, onderzoeksopdracht bij (of in samenwerking met) een extern bedrijf, projectwerk (proftaak), werkgroep besprekingen en zelfstudie.

24.8. Leermaterialen

(Status: X = verplicht, O = aanschaffen na overleg)

ISBN	Titel	Druk	Auteur	Uitgever	Prijs	Status
9789001831592	Basisvaardigheden toegepaste statistiek HO	2e	Buuren, H. en Reus, G.J.	Noordhoff Uitgevers	€24,95	X
9781615473021	Slaying Excel Dragons (e-book)		Mike Girvin	Holy Macro Books	€18,00	O
9781491910399	R for Data Science	2e	Hadley Wickham & Garrett Grolemund	O'Reilly	€34,99	O

Diverse bronnen worden beschikbaar gesteld via de Canvas course.

- Rapportageteknik; Bas Andeweg, Sjaak Baars, Jaap de Jong, Rien Elling, ISBN 9789001841744 (gedurende de hele opleiding)
- Onderzoeksvaardigheden, aan de hand van triangulatieframework, zie site: ictresearchmethods.nl
- Licentie "Hogeschooltaal", met Fontys account inloggen, kosten €70,= voor 5 studiejaren.

Software (alle zijn freeware):

- Voor Excel: MS professional
- Voor R: R Studio, gratis
- Datacamp course (inlogaccount startsemester)

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25. INFORMATION ABOUT [IN-BDA] PROCESS DATA IN ORGANISATIONS (30EC)

Business Analysis, Process Modelling & Low Code Development, Data Handling & Statistics

25.1. Content

In the introduction semester Process Data in Organisations, you will work on a variety of topics. You will become proficient in business themes, data analysis, and overarching professional skills. Because the themes cannot exist in isolation, in this semester you will work on three professional products that bring these themes together. You will work on these in the context of a professional task which will be performed for an external stakeholder.

25.2. Learning outcomes

1. **Learning outcome:** Professional standard
You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory research, and reporting.
2. **Learning outcome:** Personal leadership
You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional development.
3. **Learning outcome:** Analysis
You investigate and describe information needs and process flows within a simple business context, selecting relevant data and assessing its usability and relevance based on agreed criteria.
4. **Learning outcome:** Advising
You formulate a concise, substantiated advice on improving data-driven processes or enhancing information provision, taking into account technical, ethical, and organisational aspects.
5. **Learning outcome:** Designing
You design an initial concept of an information system that meets stakeholder needs, feasibility, and scalability, and you explain the design choices made.
6. **Learning outcome:** Realising
You implement and test, on a small scale, a simple information system, validating the solution with end users based on agreed acceptance criteria.
7. **Learning outcome:** Manage & Control
You pay structured attention to maintenance, technical, ethical, and privacy aspects of the solution, and document this in line with recognised guidelines.

25.3. Expected entry profile

There are no prerequisites for this semester so all are welcome. Students with natural sense curiosity as well as a willingness to interact with and help people will have the smoothest experience.

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

25.4. Outflow profile

Domain description HBO-i 2023

	Manage & control	Realise	Design	Advise	Analysis
User interaction	1	1	1	1	1
Organisational processes	1	1	1	1	1
Infrastructure					
Software	1	1	1	1	1
Hardware-interfacing					

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

25.5. Examination and final grading

25.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

25.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- *Undefined*: You have not yet undertaken any activities to demonstrate the learning outcome.
- *Orienting*: You have made a start and explored possibilities to demonstrate the learning outcome.
- *Beginning*: You have taken initial steps that contribute to demonstrating the learning outcome.

- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome at a satisfactory level if you continue your development in this way.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations and are focused on continuous improvement. You will demonstrate the learning outcome at a more than satisfactory level if you continue your development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

25.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

25.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

25.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

AI is not a replacement or threat, but a collaborative actor in your learning. We use AI to strengthen creativity, deepen understanding, and expand educational possibilities. You are expected to learn to work with AI effectively and productively: a core professional competency for ICT professionals.

The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available, including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

25.7. Learning activities

Your daily activities will consist of working on a variety of assignments and projects. This includes exploring relevant topics, applying your knowledge and skills in practice, and developing solutions individually as well as collaboratively. You will contribute your work to team-based projects, where individual inputs are combined into shared deliverables that are presented to a client or stakeholder.

25.8. Learning materials

All necessary materials will be provided. No extra materials are needed for this semester.

26. INFORMATION ABOUT [SES] SIGNALS & EMBEDDED SYSTEMS (30EC)

HBO-ICT Minor

26.1. Content

The Minor **Signals and Embedded Systems** is aimed at students, who take an interest into the workings of embedded systems, how signals from the world around us can be measured and used in applications, and how real-world signals can be generated by these applications.

This minor explicitly invites students from various profiles to attend this minor but we mostly encourage students that have a background in ICT, Engineering or Mechatronics or students having affinity with technology, software engineering, electrical engineering when they come from other majors.

The semester focuses on performing authentic, methodically structured, professional tasks, which contribute to new insights for the professional field. The focus is on learning, by carrying out research and development activities, within practice-oriented research. Students are (mainly) responsible for setting up and monitoring their own learning path.

The research projects of this minor are based on real life problems from our Partners in Innovation and from the lectorates. You will work on one of these research projects, and on a project of your own.

26.2. Learning outcomes

1. **Learning outcome:** Professional standard
You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory research, and reporting.
2. **Learning outcome:** Personal leadership
You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional development.
3. **Learning outcome:** You show insight into a combination of digital techniques for building embedded systems, versus the behavior of different kinds of signals from the physical world. You give recommendations how they can be used in an application (Analysis and Advice).

Explanation: In the physical world all kinds of signals exist: audible, visual, electrical, mechanical, radio-frequent, and many more. You characterize such signals with respect to their static and dynamic characteristics. And you do this for both signals that are meant to be 'sensed' and for signals that you control yourself. This doesn't mean that you will have to know everything about every type of signal, but it means that you will know how to characterize 'any' type of unknown signal. The same counts for digital techniques: you don't need to know everything, but you do know how to apply them.

4. **Learning outcome:** Starting with a concrete application in mind, you provide a design for applying digital techniques, versus handling input signals and/or applying output signals. You implement such a design in an actual proof-of-concept (Design and Realization).

Explanation: We aim at building well-behaved applications, and in this context it means that you create a design that can cope with the signals from the physical world on one side, and with an intended use of these signals within the application on the other side. You may select whether you put more focus on signals, versus digital techniques.

5. **Learning outcome:** You set up and make use of a well-managed development environment. You provide a thorough transfer of knowledge and work-products (Manage and Control).

Explanation: Setting up a development environment is more than just having the right editor and compiler available. We expect that you will be able to arrange revision control of source code and documents, that you do so in the context of a team, and that you archive and transfer your work products properly to other teams.

26.3. Expected entry profile

This minor explicitly invites students from various profiles to attend this minor but we mostly encourage students that have a background in ICT, Engineering or Mechatronics or students having affinity with technology, software engineering, electrical engineering or related when they come from other majors.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes					
Infrastructure					
Software	1	1	1	1	1
Hardware-interfacing	1	1	1	1	1

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

26.4. Outflow profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes					
Infrastructure					
Software	2	1	2	2	2
Hardware-interfacing	2	2	2	2	2

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

26.5. Examination and final grading

26.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

26.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

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26.5.3. Assessment Guidelines

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This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

26.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

26.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

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The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available, including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

26.7. Learning activities

With this semester, students choose a learning environment with a flexible learning pathway, and a high degree of self-management. Students make choices for their own learning path and choose research activities, development activities and study materials which fit the chosen research project. The learning content is strongly dependent on the selected projects as well. Because everyone has his own track, it is impossible to plan a curriculum that fully accommodates for all possible outcomes.

Learning happens asynchronously for these reasons: not every student does the same things, and the pace and entry level of students also differs. The result of this is that a significant number of workshops, lectures, and so, that offer in-depth content are scheduled on- demand or when coaches detect it is necessary.

More general workshops that address basics for all students, are mostly planned towards the beginning of the semester. Such workshops may be about the scrum process, structured problem solving, tooling, working with hardware, how to perform good measurements, and such.

Blended education

The semester works with blended learning, whereby, in addition to the activities on location, online education can take place that matches content and didactics, or when there are planning and scheduling issues. The basic rule is to have optimal use of teaching time on location. The blended set-up for the semester is explained in the Canvas course. The decision to have parts of the education online is reserved for the degree programme.

26.8. Learning materials

This minor focuses on the interaction of application programs with the world around us, and how to accommodate for that in a controlled way. This means that, apart from the generally obvious learning materials that mostly live inside a computer or laptop, the student must be provided with hardware tools where needed. Such may be multimeters, oscilloscopes, soldering stations, all kinds of sensors and actuators, and more. It depends mostly on the projects what is needed, and thus what must be provided.

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27. INFORMATION ABOUT [IN-SDE] SOFTWARE DESIGN & ENGINEERING (30EC)

Software Life Cycle, Software Development, Design & Testing

27.1. Content

This semester you will search for what it means to be a good Software Engineer. To make your semester a success, we expect a **proactive** attitude from you. Be curious and actively seek feedback. Show your work often not only to your teachers, but also to fellow students, your peers, whether it is finished or not. **Use feedback to learn and improve your work and attitude.** These are essential qualities for the software engineer.

Dit semester gaan we uitzoeken wat het betekent om een goede Software Engineer te zijn. Om van jouw semester een succes te maken verwachten we een **proactieve** houding van je. Wees nieuwsgierig en ga actief op zoek naar feedback. Laat je werk vaak zien aan zoveel mogelijk docenten, maar ook aan medestudenten, je peers, of het nou af is of niet. **Gebruik feedback om te leren en je werk en houding te verbeteren.** Dit zijn essentiële kwaliteiten van een software engineer.

27.2. Learning outcomes

1. **Learning outcome:** Learning Outcome 1 (Analyzing): You analyze and document validated user specifications based on stakeholder feedback and according to a standard method for simple, non-distributed applications.

Explanation: * User Specifications: You gather and validate the requirements stakeholders place on the application. These specifications are prioritized based on their value to the stakeholders.

* Documenting Specifications: You document these specifications in an analysis document and ensure they are validated by stakeholders, for example through acceptance tests.

2. **Learning outcome:** Learning Outcome 2 (Designing): You methodically translate validated user specifications into software and database designs, considering maintainability and security. You are able to justify your choices.

Explanation: * Software and Database Designs: You create technical designs (such as architecture diagrams, domain models, and database designs) using appropriate standard notation methods, accurately translating the specifications into technical solutions.

* Maintainable and Secure: The design is flexible, considering scalability, security, and future changes.

3. **Learning outcome:** Learning Outcome 3 (Implementation): You build and deliver multiple secure and maintainable applications (at least one of which is web-based) in an iterative software development process. In your code, you apply relevant software principles to ensure maintainability and security. You use a relational database for persistence of the data relevant to your application.

Explanation: * Iterative Software Development Process: You develop applications by iteratively building, extending, and maintaining them, based on the technical design (LO 4) and validated stakeholder specifications (LO 3).

* Relevant Software Principles: You apply standard software development principles, such as OO principles and SOLID, in the design and implementation of the application. You can explain how the chosen principles contribute to a maintainable and secure application.

* Database: You integrate a relational database into the application and ensure it communicates efficiently and securely with the software.

4. **Learning outcome:** Learning Outcome 4 (Managing): You use techniques such as version control, (automated) testing, and an iterative process to improve the quality of your code. You can explain how the chosen techniques contribute to this improvement.

Explanation: * Version Control: You use a version control system (such as Git) to manage changes without disrupting existing functionality.

* Testing: You define and execute tests (such as unit tests, integration tests, and acceptance tests) to ensure the functionality and stability of the software.

* Iterative Process: You continuously improve code quality through small incremental changes and testing.

5. **Learning outcome:** Professional standard

You apply professional practice, both individually and in teams, in the areas of project organisation, communication with stakeholders, exploratory research, and reporting.

Explanation: * Project Organization: You collaborate functionally with your group and involved stakeholders (clients and teachers) in an iterative process to reach a common end result. You take initiative to continuously improve the collaboration process.

* Communication: You hold regular, goal-oriented meetings with your team. You proactively keep stakeholders informed about the process, progress, and decisions made. You ensure communication is appropriate to the relationship with the involved parties.

* Exploratory Research: You use reliable and relevant sources to support your decisions. You validate sources for reliability and relevance to the project using a standard method.

* Reporting: You provide stakeholders with 'artifacts'. Artifacts may include analysis documents, design documentation, code, and working software. Documentation is complete, compact, concrete, spell-checked, and tailored to the involved stakeholders.

6. **Learning outcome:** Personal leadership

You take the initiative in asking for, and reflecting on, feedback. You identify your own core values as the basis for your study career and professional development.

Explanation: * Feedback, Reflection, and Core Values: You regularly reflect on your progress by seeking feedback from team members and stakeholders (such as clients and teachers). You improve your work based on received feedback and your own critical reflection.

27.3. Expected entry profile

Professional Development BSc 2020

				Investigative problem solving	
			Targeted interaction		
		Personal leadership			
	Future oriented organisation				
Professional development	1	1			

27.4. Outflow profile

Domain description HBO-i 2023

					Manage & control
					Realise
					Design
					Advise
					Analysis
User interaction					
Organisational processes					
Infrastructure					
Software	1	1	1	1	1
Hardware-interfacing					

Professional Development BSc 2020

					Investigative problem solving
			Targeted interaction		
		Personal leadership			
	Future oriented organisation				
Professional development	1	1	1	1	1

27.5. Examination and final grading

27.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

27.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- *Undefined*: You have not yet undertaken any activities to demonstrate the learning outcome.
- *Orienting*: You have made a start and explored possibilities to demonstrate the learning outcome.
- *Beginning*: You have taken initial steps that contribute to demonstrating the learning outcome.
- *Proficient*: You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome at a satisfactory level if you continue your development in this way.

- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations and are focused on continuous improvement. You will demonstrate the learning outcome at a more than satisfactory level if you continue your development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

27.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

27.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

27.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

AI is not a replacement or threat, but a collaborative actor in your learning. We use AI to strengthen creativity, deepen understanding, and expand educational possibilities. You are expected to learn to work with AI effectively and productively: a core professional competency for ICT professionals.

The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available, including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

27.7. Learning activities

Semester 2 is made up of many different learning activities. These activities are guided by the following teacher roles and stakeholders: Semester Coach (SEC), two Technical Coaches (TECs) and Partner in Education (PiE).

1. Software Jumpstart: Circus Train Challenge

During the first two weeks you will be working on "Circus Train" project. This project will help you and your teachers to realize what you know well and what the focus needs to be in the first part of the semester. The project will involve the following software development activities:

Analysis

- Create functional & non-function requirements for a given case

Design

- Design a UI wireframe, based on the earlier created requirements
- Design a relational database, based on the earlier created requirements

Realise

- Implement a static webpage with HTML and CSS, based on the earlier created UI wireframe and functional requirements
- Create a database based on the earlier created database design
- Implement code, with an Object Oriented (OO) approach, no Object Relational Mapper (ORM), to send SQL queries to a database server

Manage & Control

- Debugging
- Unit testing & acceptance testing based on a test plan that is created by using the earlier created requirements

For professional development:

- Introduction to Learning outcomes
- Planning: intro, Research: intro, Portfolio: intro

2. Projects and Workshops

Semester 2 features two main projects: a Group Project (GP) and an Individual Project (IP), supplemented by various workshops.

2.1. Group Project (PiE)

The Group Project involves teams of 6 students, alternative also group of 5-7 students can be created. Projects are provided by our Partners in Education (PiEs).

The students enter a cycle of iterations, and deliver project updates every three weeks to the PiE. In the end of the semester there is a show case where all groups and PiEs are present.

2.2. Individual Project

Students undertake a multi-layered web application project based on object-oriented (OO) principles, crafted from a personal concept. This project is requiring students to engage in the full software development lifecycle, from analysis to design, implementation and testing.

2.3. Workshops

Workshop sessions are designed around a continuation of tasks, either from previous workshops or new starting points for absentees or those preferring a fresh start. Workshops are not mandatory. The workshops are guided by the Technical Coach (TEC) role. The schedule will be shared at the start of the semester.

27.8. Learning materials

All learning materials are available online, through the Learning Management System (LMS) Canvas.

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28. INFORMATION ABOUT [O-IN-SDE] SOFTWARE DESIGN & ENGINEERING (30EC)

Software Life Cycle, Software Development, Design & Testing

28.1. Inhoud

Dit semester gaan we uitzoeken wat het betekent om een goede Software Engineer te zijn. Om van jouw semester een succes te maken verwachten we een **proactieve** houding van je. Wees nieuwsgierig en ga actief op zoek naar feedback. Laat je werk vaak zien aan zoveel mogelijk docenten, maar ook aan medestudenten, je peers, of het nou af is of

niet. **Gebruik feedback om te leren en je werk en houding te verbeteren.** Dit zijn essentiële kwaliteiten van een software engineer.

28.2. Leerkomsten

1. **Learning outcome:** Learning Outcome 1 (Analyzing): You analyze and document validated user specifications based on stakeholder feedback and according to a standard method for simple, non-distributed applications.

Explanation: * User Specifications: You gather and validate the requirements stakeholders place on the application. These specifications are prioritized based on their value to the stakeholders.

* Documenting Specifications: You document these specifications in an analysis document and ensure they are validated by stakeholders, for example through acceptance tests.

2. **Learning outcome:** Learning Outcome 2 (Designing): You methodically translate validated user specifications into software and database designs, considering maintainability and security. You are able to justify your choices.

Explanation: * Software and Database Designs: You create technical designs (such as architecture diagrams, domain models, and database designs) using appropriate standard notation methods, accurately translating the specifications into technical solutions.

* Maintainable and Secure: The design is flexible, considering scalability, security, and future changes.

3. **Learning outcome:** Learning Outcome 3 (Implementation): You build and deliver multiple secure and maintainable applications (at least one of which is web-based) in an iterative software development process. In your code, you apply relevant software principles to ensure maintainability and security. You use a relational database for persistence of the data relevant to your application.

Explanation: * Iterative Software Development Process: You develop applications by iteratively building, extending, and maintaining them, based on the technical design (LO 4) and validated stakeholder specifications (LO 3).

* Relevant Software Principles: You apply standard software development principles, such as OO principles and SOLID, in the design and implementation of the application. You can explain how the chosen principles contribute to a maintainable and secure application.

* Database: You integrate a relational database into the application and ensure it communicates efficiently and securely with the software.

4. **Learning outcome:** Learning Outcome 4 (Managing): You use techniques such as version control, (automated) testing, and an iterative process to improve the quality of your code. You can explain how the chosen techniques contribute to this improvement.

Explanation: * Version Control: You use a version control system (such as Git) to manage changes without disrupting existing functionality.

* Testing: You define and execute tests (such as unit tests, integration tests, and acceptance tests) to ensure the functionality and stability of the software.

* Iterative Process: You continuously improve code quality through small incremental changes and testing.

5. **Learning outcome:** Professionele standaard

Je past individueel en in teamverband een professionele werkwijze toe op het gebied van projectorganisatie, communicatie met belanghebbenden, verkennend onderzoek en rapportage.

Explanation: * Project Organization: You collaborate functionally with your group and involved stakeholders (clients and teachers) in an iterative process to reach a common end result. You take initiative to continuously improve the collaboration process.

* Communication: You hold regular, goal-oriented meetings with your team. You proactively keep stakeholders informed about the process, progress, and decisions made. You ensure communication is appropriate to the relationship with the involved parties.

* Exploratory Research: You use reliable and relevant sources to support your decisions. You validate sources for reliability and relevance to the project using a standard method.

* Reporting: You provide stakeholders with 'artifacts'. Artifacts may include analysis documents, design documentation, code, and working software. Documentation is complete, compact, concrete, spell-checked, and tailored to the involved stakeholders.

6. **Learning outcome:** Persoonlijk leiderschap

Je neemt initiatief om feedback te vragen en te reflecteren en benoemt jouw kernwaarden als vertrekpunt voor je studieloopbaan en je professionele ontwikkeling.

Explanation: * Feedback, Reflection, and Core Values: You regularly reflect on your progress by seeking feedback from team members and stakeholders (such as clients and teachers). You improve your work based on received feedback and your own critical reflection.

28.3. Verwacht ingangspatrofiel

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development			1	1

28.4. Uitstroomprofiel

Domain description HBO-i 2023

	Manage & control	Realise	Design	Advise	Analysis
User interaction					
Organisational processes					
Infrastructure					
Software	1	1	1	1	1
Hardware-interfacing					

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

28.5. Toetsing en beoordeling

28.5.1. Toetsing: Holistisch en Longitudinaal

De beoordeling in dit semester is holistisch en longitudinaal. Holistisch betekent dat alles wat je als student doet meetelt voor de eindbeoordeling: de hoeveelheid werk, de kwaliteit van het werk, je professionaliteit en de aangetoonde competenties.

De leeruitkomsten beschrijven het gewenste niveau, maar omdat deze onafhankelijk van het leertraject kunnen worden aangetoond, is er ruimte om een deel van je eigen leerproces vorm te geven. Daarom is het belangrijk om je docenten regelmatig te raadplegen om er zeker van te zijn dat je de juiste dingen doet. Dit is het longitudinale aspect: gedurende het hele semester volgen de beoordelaars je vooruitgang.

28.5.2. Feedback: Bevestiging en begeleiding

Docenten begeleiden en beoordelen je vooral door middel van feedback. Je hebt wekelijks gesprekken met een of meer docenten over je werk. Deze feedback leg je zelf vast in FeedPulse, waarna docenten reageren met informatie over je voortgang.

Daarnaast vinden op vaste momenten in het semester portfolio opleveringen plaats. Hierop krijg je feedback van je docenten op basis van de developmentaal feedback scale. Deze schaal heeft vijf niveaus.

Ontwikkelingsgerichte feedbackschaal

- **Onbepaald:** Je hebt nog geen activiteiten ondernomen voor het aantonen van de leeruitkomst.
- **Oriënterend:** Je hebt een begin gemaakt en de mogelijkheden verkend om de leeruitkomst aan te tonen.
- **Beginnend:** Je hebt eerste stappen gezet en deze uitgevoerd welke bijdragen aan het aantonen van de leeruitkomst.
- **Geoefend:** Je hebt meerdere keren laten zien dat je een basis gecreëerd hebt om de leeruitkomst aan te tonen. Je zal de leeruitkomst op een voldoende niveau aantonen, als je jouw ontwikkeling op deze manier blijft voortzetten
- **Gevorderd:** Je hebt meerdere keren laten zien dat je met deze leeruitkomst bezig bent geweest met goed resultaat. Je hebt boven verwachting gepresteerd en bent gericht op continue verbetering. Je zal de leeruitkomst op een ruim voldoende niveau aantonen, als je jouw ontwikkeling op deze manier blijft voortzetten.

Deze feedback is een momentopname van je huidige groei in de richting van het leerresultaat. Het is mogelijk dat je achteruit gaat in je ontwikkeling als je bepaald gedrag niet consequent laat zien in de praktijk.

Binnen het semester zijn er minstens drie van deze formatieve evaluatiemomenten waarbij de beoordelaars feedback geven per leerresultaat aan de hand van de ontwikkelingsgerichte feedbackschaal. Het laatste van deze drie formatieve evaluatiemomenten is uiterlijk twee weken voor de beoordelaarsvergadering. Dit geeft je de gelegenheid om de ontvangen feedback te behandelen en te verwerken.

28.5.3. Assessment Richtlijnen

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

28.5.4. Herkansingen en reparaties

Herkansingen of reparaties zijn niet van toepassing op dit semester. Er wordt van je verwacht dat je je werk een onbeperkt aantal keren opnieuw indient voor feedback.

28.6. Richtlijnen AI gebruik

Bij Fontys ICT hebben we een visie op het gebruik van Gen AI (AI), die te vinden is in het AI-manifest. Hierin stellen we verwachtingen vast voor hoe je tijdens je leerproces met AI omgaat.

AI als samenwerkingspartner

AI is geen vervanging of bedreiging, maar een samenwerkingspartner in je leerproces. We gebruiken AI om creativiteit te versterken, begrip te verdiepen en onderwijsmogelijkheden uit te breiden. Er wordt van je verwacht dat je leert om effectief en productief met AI te werken: een kerncompetentie voor ICT-professionals.

Het kernprincipe: jij bent de architect, niet het gereedschap. AI versnelt je werk, maar jij blijft verantwoordelijk voor alles wat je levert. Pas bij het gebruik van AI het transparantieprincipe toe: wees duidelijk en eerlijk over wat AI heeft bijgedragen en wat je eigen analyse, oordeel of creatie is.

Validatie en kritische betrokkenheid

Werken met AI betekent dat je de output voortdurend kritisch en nieuwsgierig evalueert. AI kan feiten hallucineren, onveilige code produceren of vooroordelen inbouwen. Het is essentieel om het vermogen te ontwikkelen om te evalueren, vragen te stellen en te verifiëren. Uw beslissingen zijn belangrijk, dus zorg ervoor dat u ze kunt uitleggen. Documenteer uw validatieproces voor belangrijke AI-bijdragen.

Ethische grenzen

Verantwoord gebruik van AI vereist het respecteren van duidelijke grenzen. Voer nooit persoonlijke gegevens, bedrijfsgeheimen of vertrouwelijke informatie in openbare AI-tools in. De basisrichtlijnen van de AVG/GDPR zijn altijd van toepassing. Gebruik institutionele AI-tools wanneer deze beschikbaar zijn, waaronder de faciliteiten die Fontys ICT biedt. Respecteer intellectueel eigendom: begrijp welke licenties van toepassing zijn op door AI gegenereerde content en zorg ervoor dat je je hieraan houdt.

Je AI-praktijken hebben niet alleen invloed op jou, maar ook op Fontys ICT en eventuele professionele partners waarmee je samenwerkt.

28.7. Leeractiviteiten

We maken gebruik van *Blended Learning*, waarbij er wekelijks twee contactmomenten zijn. Studenten dienen zich wekelijks voor te bereiden alvorens zij aan het contactmoment deelnemen. Een voorbereiding bevat veelal een theoretisch karakter en een activiteit welke ingeleverd dient te worden. Het gaat bij deze activiteit niet om een beoordeling, maar om te oefenen met de leerstof en knelpunten/problemen onderkennen en deze te vertalen naar doelgerichte vragen.

Een avond per week is er gedurende 4 lesuren een contactmoment op locatie en een andere avond is er gedurende 2 lesuren een online bijeenkomst, waarbij 1 lesuur 45 minuten duurt. Het online contactmoment in de week is primair voor het beantwoorden van vragen door studenten en voor het wegnemen van knelpunten bij het individuele voorbereiden. Het contactmoment op locatie is primair voor het behandelen van de opdrachten waarbij de docent feedback (schriftelijk en/of mondeling) geeft op het gemaakte werk en voor het bespreken van nieuwe onderwerpen.

Semestercoach:

Iedere student krijgt aan het begin van ieder semester een betrokken vakdocent toegewezen als semestercoach. Gedurende het semester zal je meerdere malen contact hebben met deze semestercoach over de voortgang van je studie, de mogelijkheid om je leeruitkomsten te koppelen aan je werkplek en je professionele leerdoelen. Daarnaast is de semestercoach jouw eerste aanspreekpunt en vertrouwenspersoon binnen het semester. Er wordt verwacht dat de student zelf het initiatief neemt om een gesprek met de semestercoach te voeren.

28.8. Leermaterialen

Deze cursus wordt ondersteund door een onderdeel in de online leeromgeving Canvas. Daarnaast zijn andere online bronnen, zoals forums, online cursussen (bijv. Udemy) of films/screencasts op youtube, ook toegestaan.

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29. INFORMATION ABOUT [APA3A] STATISTICS (4EC)

29.1. Content

Statistics is about collecting and interpreting data using mathematical instruments, domain knowledge and common sense. Although mostly not explicitly stated, statistics plays a role in a wide variety of fields: news, sports, medicine, politics, scientific research etc. It is also a key building block for more advanced topics like decision theory or data science.

In this introductory course, you will first learn how to summarize data using meaningful plots and numbers (*descriptive statistics*). Then you will learn how to setup simple statistical studies and use techniques like hypothesis testing to draw general conclusions from a limited amount of data (*inferential statistics*). Along the way, you will also train your ability to recognize statistical traps and not misuse the statistical tools.

29.2. Learning outcomes

1. Learning outcome: Analysis

Explanation: You learn to use correct statistical language (variables, datasets, distributions, conditions ...) to model relevant practical problems.

2. Learning outcome: Realization

Explanation: You will design and conduct a small statistical study of your own, from data collection to hypothesis testing and interpreting the results.

3. Learning outcome: Testing

Explanation: You reason about the correctness and trustworthiness of statistical models by using concepts like confidence intervals.

4. Learning outcome: Research

Explanation: Throughout the course, you adopt a curious attitude and take charge of your learning process. In the Statistics module, you will also work on an explicit research assignment.

5. Learning outcome: Interaction

Explanation: There will be enough opportunity for interaction with your teachers and colleagues.

6. Learning outcome: Communication and Leadership

Explanation: There will be enough opportunity to describe and present your work, for instance in the research assignment.

29.3. Examination and final grading

29.3.1. Examination

The teachers will provide the students with formative feedback on deliverables and demonstrations. It is expected that this feedback will be processed in further deliverables, within reason.

The student shows the work, alone or in a group context, but will be given feedback on an individual base.

The student work will be assessed on analysis, design and technical aspects, where the student will be informed whether the learning goals are met concerning the criteria.

At these assessments the learning outcomes are determined based on state of the deliverables at that point. Ideally the learning outcomes are proficient at these assessments. This grade is a formative indicator of mastery of learning goals at that moment and don't give any guarantee for further assessments. It is expected that the student keeps on progressing on the learning outcomes, not doing so might lead to a lower grade of beginning or orienting.

29.3.2. Repairs and resits

During the course, students are facilitated to process formative feedback on their learning goals in their deliverables and activities that are to be recorded in their portfolio (Canvas). All learning activities are related to the professional assignment.

29.3.3. Tools allowed

Except for paper & pen, and a laptop, no additional equipment is needed. The regular rules about plagiarism apply.

29.3.4. Final grading

A student's growth is maintained in his portfolio. The student's growth on each learning outcome is registered according to one of the five levels of mastery.

1. undefined,
2. orienting,
3. beginning,
4. proficient,
5. advanced.

After each formative assessment, the level of mastery for all learning outcomes is assessed for the student's deliveries.

At the end of the semester the student will be assessed on his performance of the semester. During the assessment meeting the teacher will use the student's portfolio. The overall grade is based on the shown performance and will be graded by the examiners. If all learning outcomes are proficient or advanced, the grade will be at least Satisfactory. This does not mean that obtaining proficient is the minimum requirement to achieve a satisfactory grade.

29.4. Learning activities

Based on the characteristics of an academic premaster and based on a highly knowledge-based scientific world, the course is structured according the research-based learning method. It is a structured and predictable learning path, in which the learning outcomes and assessment criteria are determined by the study program.

The education is designed according to the principles of the 4C/ID model (four-component instructional design model). Education is pre-structured in terms of content, working methods and guidance.

Knowledge, competences and professional skills are further gained and applied in practical assignments, which increase in size, abstraction and complexity during the study program. The student gets more and more choice in a pre-sorted set of learning tasks that are designed according to 4C/ID.

Depending on the subjects, a variety of side-roads will be offered for in-depth knowledge and broader research.

29.5. Learning materials

All course content is in Canvas and Sharepoint. All software (compilers, IDEs, libraries, tools) is available for free.

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30. INFORMATION ABOUT [APA4B] SYNCHRONIZATION (2EC)

30.1. Content

Dive into the intricate world of synchronization and deadlock in this course on managing concurrent systems. Gain a deep understanding of semaphores, their role in coordinating shared resources, and how they enable synchronization in multi-threaded environments. Explore the challenges of concurrent programming, including race conditions and mutual exclusion. Learn techniques to detect, prevent, and resolve deadlocks, ensuring the smooth operation of complex systems. Through practical examples and hands-on exercises, develop the skills necessary to design and implement robust, deadlock-free concurrent applications.

One cannot board a plane before buying a ticket. Similarly, one cannot check e-mails before validating the appropriate credentials (for example, user name and password). In the same way, an ATM will not provide any service until it receives a correct PIN.

Daily life is full of synchronization, computer systems as well. Troubles are very subtle: solutions may seem to work in most of the cases, but we are only interested when they work properly in all cases. Remember Murphy's Law: "Anything that can go wrong will go wrong".

30.2. Learning outcomes

1. **Learning outcome:** In addition to the general Academic Preparations learning outcomes, the student is able to: understand, recognize and solve the problems around synchronization between threads like mutual exclusion and deadlock
2. **Learning outcome:** apply the common primitives like mutexes, semaphores, conditional variables
3. **Learning outcome:** apply common synchronization design patterns
4. **Learning outcome:** build a visual metaphor of a problem

30.3. Examination and final grading

30.3.1. Examination

The teachers will provide the students with formative feedback on deliverables and demonstrations. It is expected that this feedback will be processed in further deliverables, within reason.

The student shows the work, alone or in a group context, but will be given feedback on an individual base.

The student work will be assessed on analysis, design and technical aspects, where the student will be informed whether the learning goals are met concerning the criteria.

At these assessments the learning outcomes are determined based on state of the deliverables at that point. Ideally the learning outcomes are proficient at these assessments. This grade is a formative indicator of mastery of learning goals at that moment and don't give any guarantee for further assessments. It is expected that the student keeps on progressing on the learning outcomes, not doing so might lead to a lower grade of beginning or orienting.

30.3.2. Repairs and resits

During the course, students are facilitated to process formative feedback on their learning goals in their deliverables and activities that are to be recorded in their portfolio (Canvas). All learning activities are related to the professional assignment.

30.3.3. Tools allowed

Except for paper & pen, and a laptop, no additional equipment is needed. The regular rules about plagiarism apply.

30.3.4. Final grading

A student's growth is maintained in his portfolio. The student's growth on each learning outcome is registered according to one of the five levels of mastery.

1. undefined,
2. orienting,
3. beginning,
4. proficient,
5. advanced.

After each formative assessment, the level of mastery for all learning outcomes is assessed for the student's deliveries.

At the end of the semester the student will be assessed on his performance of the semester. During the assessment meeting the teacher will use the student's portfolio. The overall grade is based on the shown performance and will be graded by the examiners. If all learning outcomes are proficient or advanced, the grade will be at least Satisfactory. This does not mean that obtaining proficient is the minimum requirement to achieve a satisfactory grade.

30.4. Learning activities

Based on the characteristics of an academic premaster and based on a highly knowledge-based scientific world, the course is structured according to the research-based learning method. It is a structured and predictable learning path, in which the learning outcomes and assessment criteria are determined by the study program.

The education is designed according to the principles of the 4C/ID model (four-component instructional design model). Education is pre-structured in terms of content, working methods and guidance.

Knowledge, competences and professional skills are further gained and applied in practical assignments, which increase in size, abstraction and complexity during the study program. The student gets more and more choice in a pre-sorted set of learning tasks that are designed according to 4C/ID.

Depending on the subjects, a variety of side-roads will be offered for in-depth knowledge and broader research.

30.5. Learning materials

All course content is in Canvas and Sharepoint. All software (compilers, IDEs, libraries, tools) is available for free.

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31. INFORMATION ABOUT [MA-AMD] AI, MACHINE LEARNING & DATA (30EC)

Domain Analysis, Exploratory Data Analysis, Data Understanding, Prediction Modelling

31.1. Content

In this semester you will learn:

- to prepare structured and unstructured datasets on theoretical data quality standards by means of domain analysis and exploratory data analysis;
- to use findings from your data design to preprocess data, train and validate machine learning algorithms and evaluate the quality and usefulness of produced models for a defined domain;
- to deliver AI solutions that follow the three Explainable AI principles of transparency, interpretability, and explainability.

31.2. Learning outcomes

1. Learning outcome: Analysing

You analyse the possibilities for AI solutions by conducting exploratory investigation of the domain and context while aligning stakeholder objectives and aggregating and exploring data systematically.

Explanation: Exploratory research means research that enhances your own understanding of the context, by

means of expert interviews, widely acknowledged standard work, and other viable sources.

Aggregating means acquiring data from a variety of different sources and in different formats and putting it together into a meaningful larger total dataset.

Exploring the data includes (amongst others): descriptive analytics, unbalanced data, inherent bias, statistical overviews, detecting outliers and inconsistencies, etc.

2. **Learning outcome:** Advising

You advise stakeholders on AI solutions by recommending an analytical approach for a given problem and context, founded on analytical insights and sustainable AI frameworks.

Explanation: Analytical approach consists of defining the target variable of the prediction, the nature of the target variable and to find out which of the features from the data are good indicators for the chosen target variable.

Analytical insights can be identified by finding correlations between features, principal component analysis, summarization, anomaly detection, etc.

Sustainable AI frameworks can include the Fairness, Accountability and Transparency (FAT) or the EU's Ethics Guidelines for Trustworthy AI.

3. **Learning outcome:** Designing

You design an iterative process to create an interpretable AI solution for a defined domain, based on findings from data analysis.

Explanation: Iterative process means that modelling is a cyclic process aiming for continuous improvement, and the alterations are compared systematically.

Interpretable AI solution addresses the possibility for humans to comprehend the project cohesion and results by making them comparable to the domain knowledge and baselines.

Defined domain refers to the idea that your evaluation must address the problem and impact definition as given by the domain stakeholders, and evaluation metrics must be translated to be meaningful to them.

Findings from data analysis implies that your choice of data quality requirements is based on opportunities for predictive analytics that you previously identified.

4. **Learning outcome:** Realising

You realise AI solutions by preparing and preprocessing data according to quality criteria, applying appropriate machine learning techniques, and validating models against industry-standard evaluation metrics.

Explanation: Preparing the data consists of cleaning the data according to theories of data quality, in such a way that the process of cleaning and preparing those data is repeatable, transparent to others, and the results are suitable for data analysis.

Preprocessing refers to applying systematic ways like feature selection, encoding, scaling, etc. of turning raw datasets into formats that are more suitable for model training.

Validating means judging the results of machine learning against industry standard evaluation metrics.

5. **Learning outcome:** Manage & Control

You manage AI solutions by following the three 'Explainable AI' principles (transparency, interpretability, and explainability) and make it transferable within its context.

Explanation: Transparency means that the process by which the used input data results in prediction models is reproducible, reliably described and its decisions are motivated.

Explainability refers to the application of tools and methods that turn black-box models into grey/white-box models by having the model draw out its decision-making process and/or describe its feature importance.

Transferable means that all products related to the development and maintenance of the AI solution are documented such that the process is reproducible for continuation of usage or further research.

6. **Learning outcome:** Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research*, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

Explanation: Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research*, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

*Associate degree (graduation level): action-oriented research.

7. **Learning outcome:** Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

Explanation: You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your

learning attitude.

31.3. Expected entry profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes			1	1	
Infrastructure					
Software	1	1	1	1	1
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

31.4. Outflow profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	1	1			
Organisational processes	1	1	2	2	1
Infrastructure					
Software	2	2	2	1	2
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

31.5. Examination and final grading

31.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

31.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
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- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations and are focused on continuous improvement. You will demonstrate the learning outcome at a more than satisfactory level if you continue your development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

31.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

31.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

31.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

AI is not a replacement or threat, but a collaborative actor in your learning. We use AI to strengthen creativity, deepen understanding, and expand educational possibilities. You are expected to learn to work with AI effectively and productively: a core professional competency for ICT professionals.

The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available, including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

31.7. Learning activities

During this semester you work on individual challenges and a group project. Each one of these works has a different goal and different context, however, they all follow the same process in terms of domain analysis, data design, predictive modelling and deployment.

Lab assignment

These will be used to get acquainted with new theory regarding domain analysis, data provisioning and machine learning.

Individual challenges

All obtained knowledge will be used to develop an own AI model by means of machine learning. Multiple iterations will be built on the AI model to improve the data quality, to apply different machine learning techniques and finetune them, and to apply and to align the requirements of the model with the stakeholder.

Industry Project

A project will be provided by Fontys partners for which will be investigated if and how predictive modelling can be an added value for there business process. Like the individual challenge, multiple iterations of the AI model will be developed to match the expectations of the Fontys partner and to improve the performance of the model.

However, this will never fully replace student's preparations and in all cases the responsibility for acquiring theoretical knowledge remains with the student.

31.8. Learning materials

A Canvas course including modules with (self) study materials, inspiration sessions and external sources is available.

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32. INFORMATION ABOUT [MA-AAI] APPLIED GENERATIVE AI (30EC)

Generative AI Solution Design, Responsible Implementation, Quality & Validation

32.1. Content

In this semester, you'll develop into a T-shaped GenAI professional by choosing your specialization—GenAI Architect, GenAI Engineer, or GenAI Experience Designer—while building collaborative capability across all three domains. This isn't about becoming a generalist who knows a little about everything. The course is built on a fundamental principle: specialists who collaborate effectively create better solutions than generalists working alone. Your vertical depth comes from focused expertise in one role. Your horizontal breadth comes from working on cross-functional teams where an Architect, Engineer, and Designer build real applications together.

As a GenAI Architect, you'll design AI strategies and integrated systems, mastering deployment patterns, scalability trade-offs, and governance frameworks—while understanding enough about implementation and experience to make architecture decisions that your team can actually build and users can actually use.

As a GenAI Engineer, you'll build robust prototypes and optimize performance, mastering frameworks, debugging approaches, and integration techniques—while understanding enough about architecture and experience to implement systems that fit within larger designs and create intuitive user interactions.

As a GenAI Experience Designer, you'll craft intuitive interfaces and smooth AI interactions, mastering user research, conversation design, and usability principles—while understanding enough about architecture and engineering to design experiences that are both user-centered and feasible to build.

You'll work on two substantial projects: an Industry Project with real partners where mixed-role teams tackle authentic challenges, and a Professional Research Project where you develop deep expertise in your specialization through systematic investigation.

Throughout the semester, Morning Momentum Sessions alternate between Role Guild meetings (specialists with

specialists, diving deep) and Project Stand-ups (cross-functional coordination and integration). This rhythm builds both depth in your specialization and breadth across GenAI work.

No matter your role, you'll focus on responsible AI—balancing innovation with transparency, accessibility, and privacy. Through systematic Decision Logs, real-world problem solving, and evidence-based reasoning, you'll build both the specialized expertise and collaborative capability that make you valuable on professional teams. By semester's end, you won't just be "ready to tackle AI challenges"—you'll be a T-shaped professional who can lead in your domain while working effectively across the full GenAI application lifecycle.

32.2. Learning outcomes

1. **Learning outcome:** Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research*, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

2. **Learning outcome:** Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

3. **Learning outcome:** Analysing

Given a realistic organizational context with incomplete information, you investigate the current GenAI landscape and analyse existing challenges and workflows to discover where GenAI can add value. You explore different solution options hands-on and explain why certain solutions fit this context better than others, given the available resources, potential risks, and expected benefits — with documented reasoning, that can be defended when challenged, that shows your investigation process, the criteria you used, and how you weighed trade-offs.

Explanation: GenAI Engineer

Your analysis focuses on technical feasibility: models, APIs, integration patterns, performance. Hands-on means working code, benchmarks, and experiments — not just reading about options.

Experience Designer

Your analysis focuses on users and interaction: needs, behaviours, expectations. Hands-on means talking to real users and building low-fidelity prototypes — not making assumptions.

GenAI Architect

Your analysis focuses on business processes at operational and tactical levels: data flows, bottlenecks, integration points, data quality. Hands-on means mapping how things actually work — talking to stakeholders across the organization, not just reading documentation.

4. **Learning outcome:** Advising

You guide stakeholders toward informed decisions about GenAI solutions by adapting your communication to their perspective, drawing traceable conclusions from what you've learned, translating trade-offs into terms they can act on, and showing how each choice connects to the organizational context — including ethical implications when relevant. Your stakeholders may have different priorities, concerns, or levels of technical understanding, and may respond in unexpected ways.

Explanation: GenAI Engineer

You advise on technical choices: model selection, capability limits, quality trade-offs. Your challenge is translating technical reality into language stakeholders can act on.

Experience Designer

You bridge stakeholder goals and user needs. Your challenge is showing why what users need also serves the business — especially when they conflict.

GenAI Architect

You advise on where GenAI fits in the organization, with coherent recommendations that address structure, process, and information provision together. Your challenge is bridging jargon gaps — aligning team, stakeholders, and business value.

5. **Learning outcome:** Designing

Within the organizational context, you design how a GenAI solution will work by producing designs that align with what you've learned and guide implementation, preventing wasted effort on the wrong solution. You make deliberate choices, show the reasoning behind them — reasoning that can be defended when challenged — and refine your design as you learn.

Explanation: GenAI Engineer

Your designs are technical: architecture diagrams, C4 models, sequence diagrams, API specifications. Each choice is explained — another engineer should be able to implement from your design without needing you.

Experience Designer

Your designs are interactions: user flows, wireframes, mockups, visual aesthetics. Your designs must be complete enough that those who build can implement without guessing — including edge cases, not just the happy path.

GenAI Architect

Your designs are system-level: process structures, data flows, integration architectures, application interfaces. You take into account security and privacy requirements. Your designs serve as input for the team — clear enough that they can build within the architecture without needing you to explain.

6. **Learning outcome:** Realising

Using tools and resources appropriate to the context, you build your final GenAI solution and refine it as you learn — including what didn't work, what you struggled with, and how you moved forward — through a visible building process, testing that the solution achieves what was intended, and making choices that can be defended when challenged and that allow the solution to be maintained.

Explanation: GenAI Engineer

You write working code, deploy to infrastructure, and connect resources. Your git history tells the story. When you use AI coding tools, be transparent — the code is yours to defend.

Experience Designer

You build working prototypes that users can actually interact with. Your iterations tell the story: sketches to wireframes to mockups, evolved through user testing. When you use AI design tools, be transparent — the design is yours to defend.

GenAI Architect

You build working proofs of your architecture: integration prototypes, workflow automations, code that demonstrates systems connect. You validate that your proof of concept works and help stakeholders accept the changes. When you use AI coding tools, be transparent — what you produce is yours to defend.

7. **Learning outcome:** Managing & Controlling

For as long as you are responsible, you take ownership of the GenAI solutions you've built, keeping them aligned with what was intended by tracking performance, identifying problems early, making improvements when needed, and acting on opportunities before being asked — and keep refining as you learn.

Explanation: GenAI Engineer

You monitor what you built: tests passing, services running, model performance. You also watch the landscape for new models and techniques, acting on opportunities before being asked.

Experience Designer

You monitor whether the experience still works for users — through ongoing testing and feedback. You watch the landscape for new interaction patterns and shifting user expectations, proposing improvements before stakeholders ask.

GenAI Architect

You own the architecture: monitoring whether the system design still serves the organization, using data to track how processes perform. You're the linking pin — communicating changes between stakeholders and team before misalignment becomes a problem.

32.3. Expected entry profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

32.4. Outflow profile

	Analysis	Advise	Design	Realise	Manage & control
User interaction	2	2	2	2	2
Organisational processes	2	2	2	2	2
Infrastructure					
Software	2	2	2	2	2
Hardware-interfacing					

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

32.5. Examination and final grading

32.5.1. Assessment: Holistic and Longitudinal

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The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available, including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

32.7. Learning activities

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33. INFORMATION ABOUT [MA-CSE] CYBER SECURITY ATTACK AND DEFEND (30EC)

Hacking & Defending Against Cyber Security Threats

33.1. Content

IT Security deals with protecting computer environments from security threats such as cyber criminals, hackers, malware, but also fraud. Multiple technical security aspects play an important role in this protection, like network segmentation, encryption, secure programming. Also non-technical actions, controls and considerations such as risk analysis, user awareness, security organisation, governance/compliance, privacy, security by design and business continuity management have to be taken into account.

To determine if an IT-environment is secure enough, security tests can be performed. The environment is checked as if a real hacker would try to break in. This is called "Ethical Hacking". An ethical hacker needs to have a lot of detailed knowledge of technologies that are common in ICT. These will include at least network technology, web technology and

system technology. Just having the knowledge is not enough; the large diversity of systems makes it necessary for the student to be skilful enough to also apply this knowledge in practical situations.

Hacking is illegal, and therefore punishable by law. You are only allowed to apply hacking techniques to a client's environment if this client has given you formal permission to do so. Only in that case we're talking about "Ethical" Hacking, meaning ethically responsible hacking. To behave ethically, it is important to have some knowledge about law, legislation and responsible disclosure.

All above-mentioned technologies and knowledge areas will be part of this semester's project and learning activities.

33.2. Learning outcomes

1. Learning outcome: Offensive Security

You analyse the security of diverse IT environments, such as a business environment, a consumer product, or a technology, by ethical hacking according to a methodical approach. You also analyse security threats and resulting business risks according to a common risk analysis method and advise a client on security improvements of an IT environment on a physical, technical and organisational level.

Explanation: * Analyse the security: A pen-test or penetration test is a vulnerability assessment or security analysis. In a security analysis all layers of an IT-environment can be analysed: the physical environment, the network layer, the system layers, the application software layer, procedural/human layer.

* Diverse IT environments: A business environment can be analysed by means of a pen-test or red teaming. Security of a consumer product or a technology can be referred to as vulnerability analysis, where having a formal client could be replaced by following a responsible disclosure model to be able to account for a responsible and ethical approach. Technology can be IT, IoT or OT (industrial).

* Ethical hacking according to a methodical approach: Each student must show for at least 2 different types of security analysis: pentesting and vulnerability, where one of the two has an actual client to align with and deliver to. The process part of security testing is part of the learning goals: determining the type and scope a pen-test, confidentiality contract, pen-test plan, pen-test report, and the presentation and delivery of findings and an advice to the client.

* Security threats come from threat actors such as cyber criminals, script kiddies, frauds and can be caused by user errors or calamities

Business risks indicate the impact of the threat to the business continuity of a customer in the case the threat really happens

* Common risk analysis method has to be at least a qualitative method of risk analysis that groups the risk in categories low - medium – high

* Advice on security improvements to mitigate the threats and risks must be based on standards, best-practices, and security principles. Security controls based on the security incident cycle security controls can be preventive, detective, repressive or corrective. Security controls based on layers in an IT environment, can be physical, technical (on network, system, application and software level) or organisational (procedural, human, legal).

2. Learning outcome: Defensive Security

You realise a secure IT environment, considering functional requirements as well as the following non-functional requirements: security, monitoring, ethics, compliance, and usability. You also realise procedural response for security incidents and analyse these in an efficient and methodical way.

Explanation: * Secure IT environment: IT components, interacting and communicating over a network, with connection to the Internet

* Functional & non-functional requirements: Determine (analyse) the need with respect to security, monitoring, ethics, compliance, usability

* Realise: Translate the requirements into a secure solution design and implement and build this design

* Procedural response: Incident response is a structured methodology for handling security incidents, breaches, and cyber threats. A well-defined incident response plan allows you to effectively identify, minimize the damage, and reduce the cost of a cyber attack, while finding and fixing the cause to prevent future attacks.

* Security incidents: A computer security incident is a violation or imminent threat of violation of computer security policies, acceptable use policies, or standard security practices. Examples: DDOS attack, Ransomware attack, Phishing, Data breach, Malware infection, etc.

* Analyse in an efficient and methodical way: It should address the response phases as defined by a standardised framework like NIST

3. Learning outcome: Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research*, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

Explanation: * Research: You collect requirements and approach a challenge from multiple perspectives.

* Advice: You show the skills to inform and advise stakeholders within your project about your projects' deliverables.

* Methodologies: Use frameworks and methodologies from the Cyber Security domain: OWASP top 10, OSINT, the Cyber Kill Chain, CVE and CVSS, the NIST framework and, next to that, the DOT Framework for researching.

4. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

Explanation: * Goals and Actions: You use this reflection and feedback to look for opportunities that further your development and act on those opportunities.

* Adjusting: You investigate and try out strategies to adapt to changes and setbacks. You align your professional ambitions with personal values. You adopt a pro-active, open-minded approach to learning, you embrace feedback, and you are committed to self-improvement.

33.3. Expected entry profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

33.4. Outflow profile

Domain description HBO-i 2023

	Manage & control	Realise	Design	Advise	Analysis
User interaction					
Organisational processes	1	1	1		
Infrastructure	2	2	2	2	2
Software		2			
Hardware-interfacing					

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
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The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available, including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

33.7. Learning activities

The following learning activities, and corresponding learning tasks, help the student in showing the learning outcomes:

- The **group project** will function as the overall authentic task for the semester. In the project, students will be looking for vulnerabilities in ICT environments (offensive security) and try to detect them and protect the environment against attacks (defensive security).
- For **individual learning**, the student has to acquire and train basic knowledge and skills with regard to the needed techniques and relevant knowledge topics for executing the project successfully. This set of techniques is called "Body of Knowledge" (BoK). Next to the BoK, there are personal learning activities considering research competences and internship preparation

33.8. Learning materials

The Canvas course contains the relevant study materials and references to, and ideas about, where to look for materials. For some subjects, you are also expected to look for relevant educational materials yourself.

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34. INFORMATION ABOUT [MA-FED] FRONT END DEVELOPMENT (30EC)

34.1. Content

Embark on a human-centered journey through modern web development with mentors who guide you as senior colleagues. Develop expertise in Node.js, Three.js, Progressive Web Applications and creating technology that serves people.

Let's face it – AI can churn out functioning code faster than you can say "npm install," but it still can't tell when a user is about to rage-quit your app because the UX is terrible. We need developers with both technical chops *and* human intuition.

Build secure applications with testing strategies that ensure quality experiences. Learn APIs, databases, and modern deployment practices – focused on solving human problems, not just technical puzzles.

All skills are taught through collaborative problem-solving rather than abstract exercises. Your mentors will challenge you with authentic scenarios while supporting your growth in debugging, project management, and technical communication.

This experience values technical rigor and personal growth equally. You'll become a full-stack professional who builds with cutting-edge tech while understanding the humans behind the screens.

Ready to become technically brilliant and emotionally intelligent?

34.2. Learning outcomes

1. Learning outcome: User Interaction

You design and implement accessible user interfaces based on user research, test them with representative users, and justify design decisions using established principles to create engaging user experiences in web applications.

Explanation: You apply user-centered design methods to research needs, create prototypes, and implement interfaces that follow accessibility guidelines. You conduct usability tests, document your iterations, and clearly explain your design rationale with reference to design principles and user feedback.

2. Learning outcome: Development

You develop user-friendly web applications using modern frameworks, implement appropriate testing and version control practices, and ensure secure deployment while writing maintainable code that follows industry standards.

Explanation: You write clean, structured code for frontend and backend components, use version control effectively, and implement testing strategies appropriate to the project. You configure secure deployment environments and address performance and security concerns, documenting your technical choices and their rationale.

3. Learning outcome: Iterative Process

You apply an iterative design and development approach with clearly defined goals for each cycle, systematically document your process, and demonstrate how each iteration improves upon previous findings.

Explanation: You work through multiple design and development cycles with specific objectives, collect and analyze feedback systematically, and make evidence-based improvements. You maintain comprehensive documentation showing your progression and decision-making through each iteration.

4. Learning outcome: Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research*, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

5. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

34.3. Expected entry profile

Students should have a solid foundation in web development fundamentals before enrolling in this topic. This means we expect you to have worked with HTML, CSS and some JS before. Not entirely sure? Don't be scared; we will have lots of experts available to help you through the struggles you face.

We encourage engineers to join the this more web focused topic to broaden their knowledge and learn more about user interaction. Students that have focused more on user interaction before this semester should be ready to be challenged and learn futureproof skills to boost their chance on an internship.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	1	1	1	1	1
Organisational processes					
Infrastructure					
Software	1	1		1	
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

34.4. Outflow profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	2	2	2	2	2

Organisational processes					
Infrastructure					
Software	2	2	1	2	1
Hardware-interfacing					

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

34.5. Examination and final grading

34.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

34.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
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- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations and are focused on continuous improvement. You will demonstrate the learning outcome at a more than satisfactory level if you continue your development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

34.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

34.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

34.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

AI is not a replacement or threat, but a collaborative actor in your learning. We use AI to strengthen creativity, deepen understanding, and expand educational possibilities. You are expected to learn to work with AI effectively and productively: a core professional competency for ICT professionals.

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34.7. Learning activities

You are expected to demonstrate the competencies in these learning outcomes multiple times using different forms of evidence. This can be accomplished through:

1. **Client Project:** Demonstrate your user interaction and web development skills through a real-world project with an external client, where you address actual user needs and business requirements.
2. **Showcase Portfolio:** Present a collection of your best work that demonstrates your growth and proficiency in both UI/UX design and web development. This portfolio serves as a base for securing an internship.
3. **Personal Project:** Apply user interaction principles and web development practices in a self-initiated project that reflects your personal interests and professional goals.

34.8. Learning materials

- Canvas
- Onsite Workshops

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35. INFORMATION ABOUT [MA-FSD] FULL STACK DEVELOPMENT (30EC)

Agile Development, Quality Assurance & Software Process Automation

35.1. Content

You will learn how to create a full-stack web application in which the front and back-end layers are separated. You will learn how to do this in a modern agile software method, applying new corresponding technologies, and discovering the advantages and disadvantages of the approach. For the front-end you will work with a dedicated frontend framework of

your choice and will pay explicit attention to achieve a good user experience of your product. For the back-end you will work with REST API's, application frameworks.

You will also learn how to automate your Agile software development and deployment process, and how Software quality plays a central role in it. You will discover how generative AI and static analysis tools can facilitate the process of creating good quality software. Last but not least, you will learn the value and role of IT systems for business.

35.2. Learning outcomes

1. **Learning outcome:** 1. Full-stack Application

Both individually and in teams, you develop full-stack web applications. You design user-friendly applications by following a user-centered design approach and best practices. You justify and validate relevant architectural choices for your applications.

Explanation: Full-stack web application: Developing separate front and backend.

Architectural choices: Choices about data persistency, communication protocols (request/response message exchanges, simultaneous two-way communication), and backend structure.

2. **Learning outcome:** 2. Agile Development

You analyze and design the business context of the group-project case and the stakeholders' needs. You choose and apply the most suitable agile software development methods depending on the project context.

Explanation: Agile software development methods: The underlying 12 agile principles and 4 values of agile methods.

Stakeholders' needs: Functional and non-functional

Business context: The software must support the processes of the business ensuring alignment with the customer's operational needs.

3. **Learning outcome:** 3. Software Development Process Automation

You analyze, design, and realize a continuous integration (CI) and continuous deployment (CD) development environment for a full-stack web application that facilitates the automation of the software development process based on best practices.

Explanation: Software Development Process: The process should address phases such as Analysis, Design, Realization, Testing, and Deployment. It incorporates tools for example for security, performance, and other relevant quality attributes.

Best Practices: Utilize industry-standard practices, possibly including use of generative AI during your development work.

4. **Learning outcome:** 4. Software Quality Assurance

You justify the usage of tools and techniques and apply them to visualize the software quality. You make use of different test techniques against a selected set of software quality criteria which reflect the project needs (advice).

Explanation: Software Quality: Quality is assured on different levels of the process and product and agreed upon with the customer of your project. It addresses analysis, design, realization and test phases of the software development lifecycle.

Software Quality Criteria: You address criteria like maintainability (e.g. static code analysis, code reviews, refactoring), security and performance.

5. **Learning outcome:** Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research*, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

Explanation: Involve stakeholders: You actively ask and apply feedback from project stakeholders and advise them on solutions.

Conduct applied research: You conduct and document research activities related to your projects to make and justify your decisions using research methods and strategies.

Intercultural, ethical, and sustainable: You analyze and manage the cultural, ethical, and sustainable aspects in your projects applying theoretical frameworks.

6. **Learning outcome:** Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development.

You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

Explanation: Aware of: You use tools to identify and understand your strengths and weaknesses.

Personal growth: You undertake activities to improve your weaknesses and seize opportunities using your strengths. You choose and prepare for your next semester in line with your growth ambitions.

Learning attitude: You are proactive in planning and executing your learning activities, in formulating feedback questions, requesting and processing feedback from stakeholders (incl. teachers, classmates).

35.3. Expected entry profile

Students need to have completed the Software Design & Engineering semester from the introduction phase or have knowledge covered in this semester. When entering from the semester Intelligent Technologies, you have a background that is sufficient to successfully follow this semester as long as you follow a small self-paced module on software analysis & advice.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes					
Infrastructure					
Software	1	1	1	1	1
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

35.4. Outflow profile

This semester prepares you for an internship in software and the advanced-level semester Complex Software Systems. Additionally, it provides you with a strong basis of software skills to be applied in any domain in ICT.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction			1	1	
Organisational processes	1		1	1	
Infrastructure	1		1	1	
Software	2	2	2	2	2
Hardware-interfacing					

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

35.5. Examination and final grading

35.5.1. Assessment: Holistic and Longitudinal

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35.7. Learning activities

The education is a mix of short presentations/workshops, project work, guest lectures from companies. Frequent interactions between students and coaches aim at supporting the learning process of students.

You will work on two projects. Both projects will be for developing a web application. The first project is your individual project which will be based on your own ideas and interests.

Next to it, you will work also on a group project which reflects a case presented by one of our partners in education (external company). In the group project you will demonstrate that you can function as a software engineer in a more complex context – multiple stakeholders, non-functional requirements like usability and security, a business context expecting from you professional behavior. The group and business context may introduce for example cultural, ethical, sustainability challenges. Good communication, coordination and management is needed for the success of the project.

35.8. Learning materials

You are free in choosing your learning materials and learning activities. We do provide starting (and for some new topics even extensive) learning materials which you can use if you like them. You will have also to search yourself for relevant educational materials on a number of topics. Information about this is provided in the Canvas courses.

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36. INFORMATION ABOUT [MA-GD] GAME DESIGN (30EC)

36.1. Content

Games often appeal to us because of one simple fact: they're fun! But why is that? In order to find that out, *Game Design* aims to transform you from avid gamers to daring and critical game creators.

To do this, we will give you the toolkit you'll need to identify the feelings you want to evoke in the player, the message you want to convey, and what kind of fun you want to provide them with. You'll use these insights to design your own games, and learn how the individual elements of a game come together to create something that's more than the sum of its parts.

You won't just be designing, of course: you'll also be developing your game. The goal is to turn your design into a playable demo using one of the industry-standard game engines like Unity, Godot or Unreal Engine. To do this, you'll receive technical support and workshops explaining core development and design practices, storytelling basics, sound design, and more. We will host various open playtesting events to validate your design.

You'll be creating worlds; players will traverse the great plains and climb the majestic mountains that you envisioned.

36.2. Learning outcomes

1. Learning outcome: Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research*, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

Explanation: Methodological approach: You identify relevant challenges of the client and use a variety of research strategies, methods, and activities (such as those in the DOT Framework) in a structured way to find justified answers to these challenges.

Project goals: You can formulate a design challenge that is appropriate to the assignment.

Involve stakeholders: You collect different perspectives such as target-user needs and stakeholder requirements and involve those parties in the process of your project.

Conduct applied research: You can logically chain research activities to create value towards the project goals.

Advice: You have the communication skills to inform and advice stakeholders within your project about your projects' insights.

Decisions: You can substantiate the choices in your project based on analyses and documented in designs.

Reports: You can document your applied research steps effectively in concise reports (such as devlogs).

Ethical, intercultural, and sustainable aspects: When applicable, you consider and act according to ethical standards, sustainable practices, and work towards responsible ICT solutions.

2. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

Explanation: Strengths and weaknesses: You reflect on your own learning ability and actions, using feedback that you asked and received.

Actions: You use this reflection and feedback to look for opportunities that further your development and act on those opportunities.

Personal growth: You investigate and try out strategies to adapt to changes and setbacks. You align your professional ambitions with personal values.

Learning attitude: You adopt a pro-active, open-minded approach to learning, you embrace feedback, and you are committed to self-improvement.

3. Learning outcome: Analysis

You are able to obtain desired results(1) by researching game design related subjects methodically(2) and thorough(3).

Explanation: 1. With desired results we mean that you can clearly define the relevant research question at hand to scope the problem at hand to a manageable scale. This includes being able to define a design challenge where you take the client, context and audience into account. 2. Methodically means you show that you use justified research methods such as the DOT research framework. 3. Thorough means that you use different methods of analysis and validate your insights.

4. Learning outcome: Advice

You are able to make recommendations(1) about important decisions in the game design process based on the conclusions and results from the analysis and clearly communicate(2) these recommendations to relevant stakeholders.

Explanation: 1. With recommendations we mean suggestions that you make that will help achieve desired outcomes. 2. With clearly communicate we mean that you communicate with the relevant stakeholders in an appropriate way. This includes written reporting and pitch presentations.

5. Learning outcome: Design

You show that you can design a game in an analytical(1) way using existing frameworks, theories, and models(2). You can validate(3) game systems for both functional and aesthetic requirements using industry standard methods and deliverables(4).

Explanation: 1. Analytical means that your work is based on prior analysis insights. You support your design choices with sources such as: Target User Research, Existing products analysis, available technology research, stakeholder analysis. 2. With using existing frameworks, theories, and models we mean frameworks such as: UML, ERD, Design Patterns, MDA framework and Self Determination Theory as well as concept development methods such as diverging, converging and brainstorming. 3. You validate your designed game systems in an early stage using efficient and quick prototyping methods such as paper prototyping, roleplaying and wizard of Oz and use the results to refine your design. 4. Common industry standard methods and deliverables are: Game Design document, Tech design document consisting of elements such as UML diagrams and software architecture diagram and prototypes such as paper prototypes, mock-ups, and user tests.

6. Learning outcome: Realization

You show that you can develop(1) and validate(2) game prototypes, based on a design(3) where you create, find or adapt(4) audio-visual and narrative assets(5) using innovative and relevant technology(6) in a complex(7) context.

Explanation: 1. With develop we mean that you are able to create technical prototypes based on prior work or from scratch. 2. You are able to validate these prototypes to verify the quality of the product using for example user tests and heuristics. 3. With based on designs we mean that the prototypes are based on prior designs. 4. With create, find or adapt we mean that you can create basic assets, but more realistically will find existing assets and alter them to suit your aesthetic and technical needs. The goal is that you seek out a balance in creating assets from scratch and finding existing work. 5. With audio-visual and narrative assets, we mean all 'art' assets for your games such as: 2d art, 3d art, synthesized and/or recorded sound effects, (dynamic) music, story, scripts (dialogue). 6. You use innovative and relevant technology that suit your needs such as VR headsets, Mobile devices, Game Controllers, Industry Standard Game Engines. 7. A complex context means that the scope and complexity of both the situation and the product are representative of a real-world industry setting.

7. Learning outcome: Manage and Control

You are able to work in a structured and methodical way both individually as well as in a group context(1) where you use relevant processes and tooling(2) to facilitate your iterative process(3).

Explanation: 1. With both individually as well as in a group context we mean that you show you can adjust the process accordingly. 2. With relevant processes and tooling we mean processes and tools that are industry standard in order to effectively communicate and cooperate such as Scrum, version control tools and build management tools. 3. With iterative process we mean that you show a process of cyclic repeating steps to improve quality.

36.3. Expected entry profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	1	1	1	1	1
Organisational processes					
Infrastructure					
Software			1	1	1
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

36.4. Outflow profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	2	2	2	2	2
Organisational processes					

Infrastructure				
Software		2	2	2
Hardware-interfacing				

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

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Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available, including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

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36.7. Learning activities

During the semester, lectures and practicals will be offered in the form of workshops. These workshops will deal with subjects related to technology and design and students will work on a special case or, preferably, their own concept. Students will report to a dedicated coach/teacher to present and discuss progress on a regular basis.

Teachers will provide students with feedback on their work. Students are expected to record and process this feedback. For the Industry Project Assignment, there will be a weekly tutor meeting to discuss the group progress. During this meeting, the professional development and group processes are also discussed. The student is expected to demonstrate Industry Project progress to the client at set intervals.

Blended education

The semester works with blended learning, whereby, in addition to the activities on location, online education can take place that matches content and didactics, or when there are planning and scheduling issues. The basic rule is to have optimal use of teaching time on location. The blended set-up for the semester is explained in the Canvas course. The decision to have parts of the education online is reserved for the degree programme.

36.8. Learning materials

Canvas course including all linked information.

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37. INFORMATION ABOUT [MA-II] INDUSTRY 4.0 & INTERNET OF THINGS (30EC)

Machine to Machine Interaction, Data Analytics & Machine Learning, Business Data Processing

37.1. Content

The domain of "Smart Industry" (also known as Industry 4.0) is booming. It is the further automation of the traditional manufacturing and process industry in which the most modern technologies, such as large-scale machine-to-machine communication, the Internet of Things (IoT), cloud computing, cognitive computing, data analysis and artificial intelligence are being integrated. These new technologies allow for further automation, better communication, increasing autonomy (self-monitoring) and production of smart machines that can analyze and diagnose without human intervention.

The fourth industrial revolution is about the "smart factory", with a modular structure in which cyber physical systems (CPS) monitor physical processes, make a virtual copy of the physical world and make decentralized decisions based on the collected and analysed data. Through the Internet of Things (IoT), "cyber physical systems" communicate and cooperate with each other and with people in synchronous timing, both internally and across organizational services offered and used by the participants in the "value chain".

The industry and business community currently have a very large and growing demand for professionals who not only understand the context, but also know what is needed in the 4th industrial revolution and can translate this into smart solutions in which these technologies are applied. However, the domain is so large that professionals within this domain must not only be able to operate as a generalist, but also must be able to specialize. This means that they not only have a general overview of the domain and its possibilities, but also have a focus and expertise in a specific sub-area. This duality of their role is necessary because the projects and systems are multidisciplinary and so these professionals need to thrive within multidisciplinary programs and projects, have strong research skills and are able to come up with smart solutions for problems which are not not completely known or clearly formulated.

37.2. Learning outcomes

1. **Learning outcome:** Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research*, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

2. **Learning outcome:** Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

3. **Learning outcome:** Analysis: You analyse problems, processes and/or challenges of an (existing) Smart Industry system or parts thereof. You advise stakeholders on improvements, possible solutions and extensions based on information from analysed data from said system.

Explanation: You perform an analysis on a problem, process and/or system in which you conduct a research within the domain and come up with solutions that are presented to the stakeholders in an advisory report. You use techniques, like internet of things, cloud, process design, process improvement, data analysis, transformation data into information, human-machine interaction, process control, machine-to-machine communication, edge computing. You consider quality aspects like usability, safety, security, robustness, testability, extendibility and maintainability aspects within your solution are met. You consider KPI's and added (business) value of your solutions to the stakeholders.

4. **Learning outcome:** Design: You design possible new solutions, improvements and extensions of existing solutions based on the given advice in the domain of Smart Industry.

Explanation: You design solutions according to the requirements of the stakeholders. These solutions consist of a mixture of software and/or embedded components, smart devices, IoT and M2M protocols, infrastructure, and cloud storage and computing. And the design solutions are in accordance to the given KPI's. They enable production, distribution, analysis and visualization of (real-time) data and processes, and as such serve for further optimization of the system and/or production. You design and/or enhance user system interaction in an iterative way that explores the technology potential and human factors. You show that quality aspects, like usability, safety, security, robustness, testability, extendibility and maintainability are met.

5. **Learning outcome:** Implementation: You implement possible new solutions, improvements and extensions of existing solutions based on the given design of a Smart Industry system.

Explanation: You implement parts of a Smart Industry system, such as process controllers, communication, IoT protocols, and/or you set up an infrastructure for data acquisition, storage and analysis, you perform data analysis, and provide data visualization and human interaction components proper for the target user group. You integrate the separate components in a working system and/or in an existing system. You drafting and executing a test plan for components, integration, system, usability testing and validate your results with the stakeholders.

37.3. Expected entry profile

The domain of Smart industry is very broad and, as suggested by the topics listed above, it is suitable and accessible for all students with any prior knowledge, with completed semester 2. The semester offers possibilities to extend and apply your previous knowledge, or to gain new knowledge and experience on new topics. This is achieved by the given freedom to select your own program within the semester.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	1			1	
Organisational processes	1	1			
Infrastructure	1			1	
Software	1		1	1	
Hardware-interfacing	1		1	1	

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

37.4. Outflow profile

The semester Industry 4.0 & IoT prepares for:

- a possible internship in the direction of Smart Industry
- advanced semester Industry 5.0 and Study program Smart Industry & IoT
- study programme Robotics

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	2			2	
Organisational processes	2	2			
Infrastructure	2			2	
Software	2		2	2	
Hardware-interfacing	2	2	2	2	2

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

37.5. Examination and final grading

37.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

37.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome at a satisfactory level if you continue your development in this way.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations and are focused on continuous improvement. You will demonstrate the learning outcome at a more than satisfactory level if you continue your development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

37.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

37.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

37.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

AI is not a replacement or threat, but a collaborative actor in your learning. We use AI to strengthen creativity, deepen understanding, and expand educational possibilities. You are expected to learn to work with AI effectively and productively: a core professional competency for ICT professionals.

The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

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37.7. Learning activities

Students participate in a mixed learning environment that is both project based and task-based. This means that an authentic question, challenge or problem from professional practice is the basis for all learning activities and learning tasks will be defined to facilitate learning in smaller steps. Coaching and guidance gradually change from teacher-driven to a more student-driven approach during the semester and will be offered whenever needed and relevant. Learning content is offered when required or requested in a form of instructions, workshops, or 1-to-1 feedback session. This way, differentiation in style, speed and level is possible. It is expected that students prepared themselves for a session with teachers or experts by studying the provided study material. During the session focus will be on clarifying the details of the theory and on applying the knowledge in a relevant context.

Blended education

The semester works with blended learning, whereby, in addition to the activities on location, online education can take place that matches content and didactics, or when there are planning and scheduling issues. The basic rule is to have optimal use of teaching time on location. The blended set-up for the semester is explained in the Canvas course. The decision to have parts of the education online is reserved for the degree programme.

37.8. Learning materials

A Canvas course including modules with (self) study materials, inspiration sessions and external sources is available.

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38. INFORMATION ABOUT [MA-ID] INTELLIGENT DEVICES (30EC)

Distributed Embedded Software, Object Oriented Design, Edge AI, Internet of Things

38.1. Content

The theme of this semester is “Intelligent Devices.”

In today's world, embedded systems are at the heart of our increasingly connected, data-driven society. This semester focuses on how these intelligent devices are shaping the future across various domains — from smart homes and cities to healthcare, mobility, and industry.

Key subtopics we will explore include:

- **Embedded Systems Programming**

Develop low-level programming skills that enable efficient use of hardware resources — a must for building high-performance, power-conscious embedded devices.

- **Distributed Systems and Connectivity**

Discover how modern embedded systems communicate within large-scale networks — including the Internet of Things — to form the digital backbone of our smart society.

- **Professional Software Design**

Learn to design robust, maintainable software using C++ and SOLID principles while following industry standards and best practices.

- **Real-Time Operating Systems (RTOS)**

Dive into the principles of RTOS, a cornerstone for applications requiring precise timing, responsiveness, and reliability, such as in edge computing and AI-enabled systems.

- **Future Technologies and Trends**

Explore the frontiers of embedded innovation, including **Edge AI**, which brings intelligent decision-making closer to the source of data; **embedded security**, essential for safe and resilient systems

This semester combines theory with hands-on experience to help you build the skills needed to design and implement intelligent embedded systems for the real world.

38.2. Learning outcomes

1. **Learning outcome:** Programming Techniques

You successfully implement maintainable, clear, and documented object-oriented software by applying relevant Software Development Life Cycle (SDLC, 1) phases, with a focus on Implementation and Testing. The process includes developing software solutions for both simple and complex programming tasks, utilising skills in build processes, mocking, and simulation, while adhering to industry standards.

2. **Learning outcome:** Software Design Techniques

You design robust, maintainable object-oriented systems by applying SOLID principles and following the phases of the Software Development Life Cycle (SDLC) using Agile methodologies. The focus is on Requirements Analysis, Design, Testing, Documentation, and Maintenance. The learner applies Object-Oriented Analysis (OOA) and Object-Oriented Design (OOD) using UML diagrams (class, state, sequence) relevant to embedded systems, effectively relating the complexity of these diagrams to the problem at hand.

3. **Learning outcome:** Embedded Systems

You design and implement low-level embedded systems, including configuring and controlling peripherals (e.g., timers, sensors, actuators) and managing interrupts through direct register manipulation. You deepen your knowledge in one of the advanced embedded topics like feedback control, edgeAI/tinyML, extended low-level programming or any other relevant embedded topic. You focus on the Design, Testing, and Implementation phases of the Software Development Life Cycle (SDLC).

4. **Learning outcome:** Communication

You design and implement scalable and extendable communication solutions for interconnecting components or subsystems in a distributed system. Using all phases of the Software Development Life Cycle (SDLC), a decentralized system is created, where event-based components have their own state behavior. The focus is on selecting the most suitable communication solution for the specific project context, ensuring the system's scalability and extendibility.

5. **Learning outcome:** Real-time Operating Systems

You apply basic real-time operating system (RTOS) concepts, including task/thread management, synchronization, and inter-task communication, to optimize resource use, task structuring, and prioritization. Proficiency in using RTOS is demonstrated for efficient resource allocation and effective code structuring, while following the Software Development Life Cycle (SDLC). Emphasis is placed on understanding the role of RTOS in optimizing system performance within real-time constraints.

6. **Learning outcome:** Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research*, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

7. **Learning outcome:** Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

38.3. Expected entry profile

To maximize success in this semester, prior knowledge from the following semesters is recommended:

- Intelligent Technologies
- Software Design & Engineering (with some affinity for and enthusiasm toward embedded world)

However, you are welcome to join from other semesters if you enjoy a challenge! Embedded systems can be a valuable addition to your knowledge in software, AI, infrastructure and others.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes					
Infrastructure			1	1	
Software			1	1	1
Hardware-interfacing	1	1	1	1	1

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

38.4. Outflow profile

After completing this semester, students are encouraged to further develop their expertise in:

- Industry 4.0 & the Internet of Things
- Industry 5.0
- High-Tech Embedded Software Engineering

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes					
Infrastructure			1	1	
Software	1	1	2	2	2
Hardware-interfacing	2	2	2	2	2

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

38.5. Examination and final grading

38.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

38.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

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38.5.4. Retakes and repair

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38.6. AI Usage Guidelines

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38.7. Learning activities

You will participate in workshops where you gain and apply new knowledge. This knowledge is then put into practice through hands-on challenges and projects, which will form part of your personal portfolio. While many inspiring example projects and challenges are provided, students are also encouraged to bring in their own ideas to demonstrate proficiency in the learning outcomes.

38.8. Learning materials

A Canvas course including workshops with links to online sources, hardware for inspiration projects and challenges.

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39. INFORMATION ABOUT [MA-ISP] INTERNSHIP (30EC)

39.1. Content

You will do an internship in the 5th semester of your studies. During that period, you demonstrate in (professional) practice that you have integrated the required professional and technical development and that you can also apply it independently. You also get the chance to get a taste of professional practice and thus get a better idea of what it has to offer.

Internships are about working and learning in professional practice. It is an assignment/work that:

- you perform at an institution or company;
- matches the choice of study and what you can do with it later;
- asks you to make the action very clear, so that it can be understood by others, why and for what purpose you did or did not do something;
- results in a portfolio with professional products.

Duration internship

The internship lasts a whole semester: from Fontys week 1 to Fontys week 18. The assessment takes place at the end of the semester (Fontys week 19 or 20). Holidays and public holidays that fall within the semester count as internship weeks. So you have no school holidays this semester. Days off that you accumulate (see internship agreement) can be taken in consultation with the company. Days off and/or company closures, which are mandatory within the company, are of course not counted as days off.

An internship must have a minimum of 90 days. You perform these 90 days between Monday Fontys week 1 and Friday Fontys week 18.

Sick leave

Let's hope not, but it is of course possible that you become ill during a (prolonged) period of your internship or miss days on your internship for some other reason. Because you basically continue to work on holidays, you have some leeway. Being ill or taking leave for another reason does not have to be a problem to achieve the minimum number of days.

What are internship days?

Internship days are all days on which you work on your internship and/or on your internship assignment. You can include visits that you make in the context of your internship or for your education, such as counseling moments at school or visiting a minor market.

Internship permission

In order to start arranging an internship, you must have internship permission from the examination board. Please note: before you can actually start the internship, the internship coordinator for your study route must approve the internship location you have found based on the project proposal that you have completed and handed in yourself.

Permission to go on an internship is given when you have completed one of the semesters of the second year of study. You will then first follow the other semester of the second year (which you have not yet completed) and do an internship immediately after that (regardless of whether you pass this other semester). Hence, you will always have one semester to find a suitable internship. It is recommended to start searching in time (i.e. as soon as you know you have permission). After your internship, you will have to retake any semester from the second year that you have not yet passed until it is completed.

39.2. Learning outcomes

1. Learning outcome: Professional duties

You carry out professional duties at a junior bachelor level, producing professional-grade work aligned with the IT subdomain (IT-area) you are working in.

Explanation: Professional duties on a junior bachelor level = All or a subset of the activities Analysis, Design, Realize, Advise, Manage&Control.

As a reference use:

- 1) the HBO-I framework on proficiency level 2,
- 2) the level as required in OE3 or OE4

Professional products = end products and intermediate products as a result of the professional duties

In line with the IT-area = You deliver professional products that are characteristic for the IT area of your project. As a reference use: 1) the HBO-I framework on proficiency level 2, 2) the professional products required in OE3 or OE4

2. Learning outcome: Situation-orientation

You apply your previously acquired knowledge and skills in an authentic context to deliver relevant results for the project and company.

Explanation: Apply = You work in a methodological and structured way, adapted to the processes and way of working of the company

Relevant = Your work is relevant for one or more persons

Authentic context = you are embedded in an IT environment and work on a given IT problem with multiple stakeholders

3. Learning outcome: Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research*, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

4. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

39.3. Expected entry profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

39.4. Outflow profile

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

39.5. Examination and final grading

39.5.1. Assessment: Holistic and Longitudinal

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39.7. Learning activities

Key moments of examination

Week 3-5	Projectplan/projectdocument. Formative assessment; the first assessor provides feedback during the first company visit in week 3-5.
Week 9/10	Additional feedback moment at school; formative assessment.
Week 15	
Week 17	Portfolio including proces description and achievements. Hand in at the beginning of week 17.
Week 18	Formative assessment during the second company visit in week 18. Final presentation and/or demonstration of the final product, combined with accountability about the execution and evaluation of the process. Feedback and advised grading from internship coach and company coach.
Week 19/20	Summative assessment bij first and second assessor. Criterion-based Interview on learning outcomes taking into account all aspects of your internship.

39.8. Learning materials

All available learning materials are available on Canvas. Here you will find, among other things, the research framework and support for creating your portfolio.

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40. INFORMATION ABOUT [MA-BDA] IT & PROCESS TRANSFORMATIONS (30EC)

Business Process Optimization, Dashboarding, Low Code Systems Development & Implementation

40.1. Content

The focus of this semester is on a medium to large sized organisation which is part of a supply chain. During this semester you show that you can **systematically analyse organisational processes** and **advise** how these **processes** can be **optimised**. In order to **analyse** these processes you will **create informal insight** through an **Exploratory Data Analysis** (EDA). Besides that, you will **create formal insight** in the **data** using **basic modelling**. Using these insights, you **realise** a suggested **IT system** (or part thereof) based on your **design**, **implement** this in the **organisation** using given techniques and **measure** and **monitor** the **usage** using given formats and/or **methods**.

Professional Task

The professional task, in which we work together with our Partners in Education (PiE's), offers you the opportunity to integrate and apply the above skills together with your fellow students in a group. It also offers you the opportunity to professionalise yourself with regard to several learning outcomes in the field of Personal Development (*professional skills*).

Personal Projects

Students carry out two personal projects on an individual basis: a Business personal project based on an authentic task to demonstrate their Business-related learning outcomes, and a Data personal project to similarly demonstrate their Data-related learning outcomes. Note that the professional skills also come into play here.

40.2. Learning outcomes

1. Learning outcome: Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research*, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

Explanation: Methodological approach:

Identify the problem, determine the aim of a solution and pick an appropriate approach that is sound and reproducible.

Formulate project goals:

Define clear and specific objectives that the project aims to achieve.

Stakeholders:

Collaborate effectively with the various groups of interested parties including the client, own team members and teachers.

Applied research:

Apply systematic and practical investigations aimed at addressing specific problems, challenges, or questions with the goal of applying the findings to solve real-world issues.

Advice:

Offer one or more well-founded recommendations or suggestions with the intention of guiding your stakeholder's actions or decisions to help him make informed choices and solve problems.

Decisions:

Reach conclusions or resolutions after considering various alternatives.

Reports:

Structured documents that present information, analysis, findings, or recommendations on a particular subject.

Ethical, intercultural, and sustainable aspects:

Know the ethical standards and involve social ethical themes in the judgement process. Recognise your own boundaries and those of others and act accordingly. Ensure that your solution retains its quality over time without adverse effects.

2. **Learning outcome:** Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

Explanation: Strengths and weaknesses:

Your internal characteristics that influence your abilities, behaviours, and overall performance. Recognizing and understanding these aspects and overcoming your weaknesses is crucial for your personal and professional development.

Actions:

Make a well-considered choice of study or personal improvements, enhance your own learning capacity, recognise a learning need in yourself and act accordingly through reflection, evaluation, actively demanding and giving feedback.

Personal growth:

Examine what type of professional you want to be in the long term, which field and type of positions you aspire to and how you can stand out from others in the branch.

Learning attitude:

Be aware, see opportunities and seize them, motivate yourself and others, be able to profile yourself, a team and others. Be aware of your development, show leadership and take responsibility.

3. **Learning outcome:** Business analysis; You show that you can systematically analyse organisational processes and advise how these processes can be optimized.

Explanation: The organisation is medium to large size and part of a supply chain and therefore processes are assessed on governance, risk and compliance.

Analysis consists of the correlation of the bottlenecks and cause-effect relationships of the organisation within the supply chain. In your analysis you emphasise the origin of bottlenecks in the supply chain using process diagrams. Based on this information you create a gap-fit analysis.

4. **Learning outcome:** Exploratory Data Analysis; You create informal insight through an Exploratory Data Analysis (EDA).

Explanation: Creating insight, comprises the definition of the right sources, collecting usable data (subsets) from those sources, joining, transforming and cleaning the data. In addition you evaluate and assess the quality of the data (completeness, consistency, conformity, accuracy, integrity and timeliness). Finally, you create visuals of the prepared data. The insight creation is in the visualisation of the data and the possible interpretation of it. Often, it is up to the business representative to come up with conclusions and actions based on these visualizations.

Exploratory Data Analysis (EDA) includes usage of SQL, python (pandas) and end user BI tools.

5. **Learning outcome:** Explanatory Data Analysis; You create formal insight in the data using basic statistical modelling.

Explanation: Explanatory Data Analysis continues where Exploratory Data Analysis stops: based on the data preparation and the insights gained from informal, visual analysis, formal models are applied to the data to get further insight.

Use of models in Explanatory Data Analysis in LE3 Business will be restricted to linear modelling.

6. **Learning outcome:** Business optimisation; You realise the suggested IT system (or part thereof) based on the design, implement this in the organisation and measure and monitor the usage.

Explanation: Realise the implementation and acceptance of procedures in correlation with new or adapted information provision and control.

Educate and train end users in the renewed processes and use of a new IT.

Build and validate the Proof of Concept.

Structure a standard application, for example, using PowerAutomate.

40.3. Expected entry profile

Highly recommended to do before this semester: "Business IT & Data Analysis Introduction". Knowledge of relational databases and data analytical tools (SQL, Excel, R/Python) is necessary for successful completion. A good grade record

is highly recommended if you want to do this semester without doing “Business IT & Data Analysis Introduction”.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	1	1			
Organisational processes	1	1	1	1	1
Infrastructure	1	1			1
Software			1	1	
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

40.4. Outflow profile

Possible semesters after this: “AI, Machine Learning & Data”, “Applied Generative AI” and “Industry 4.0 & Internet of Things” are popular options.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	1	1	1	1	
Organisational processes	2	2	2	2	2
Infrastructure					
Software	2	1	2	1	1
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

40.5. Examination and final grading

40.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

40.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome at a satisfactory level if you continue your development in this way.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations and are focused on continuous improvement. You will demonstrate the learning outcome at a more than satisfactory level if you continue your development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

40.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

40.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

40.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

AI is not a replacement or threat, but a collaborative actor in your learning. We use AI to strengthen creativity, deepen understanding, and expand educational possibilities. You are expected to learn to work with AI effectively and productively: a core professional competency for ICT professionals.

The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available, including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

40.7. Learning activities

Didactical Approach

Student-centred learning is the educational vision of Fontys. We provide you with opportunities to learn not only at school but also in the work field. As a student, you are the owner of your learning process and are expected to take up the responsibility of proving your skills ultimately to your teachers.

The learning activities as well as the order of carrying them out are already set out for you in advance. This provides structure and the assurance of knowing what to expect. You get frequent feedback on the work you show to your teachers. Once you have good theoretical knowledge, you will work on applying it to practice.

You are in charge of your learning process. Based on the information from the lessons and workshops, you look for knowledge and sharpen your skills (together with other students in your group) in order to be able to solve important real-world issues. To help you with this, the learning outcomes have already been set and these form the basic starting points of this semester.

Learning Activities

Workshops by teachers as well as partners in education, lectures, presentations, demonstrations, videos, assignments, challenges, quizzes, interviews, presentations to the customer, self-study and company visits are many different activities that we will carry out.

Blended learning:

Keeping in line with the 21st century trends, blended learning is used to offer a rich and dynamic learning experience. Some instances are

- online courses (Datacamp, Microsoft learning paths, etc.) as well as reference materials and in-house videos to supplement the lessons in the class
- online quizzes following in-person lessons
- flipped classroom blended learning where students first investigate the professional task problem individually through online resources, then work together in the classroom towards a solution under the guidance of the tutor, and then get to interview the client for alignment and feedback.
- occasional hybrid meetings, for example, while working on the project (Teams+face-to-face)

40.8. Learning materials

Books

Theme	ISBN	Title	Author(s)	Publisher	Price
IT	1615470522	Super Charge Power BI: Power BI Is Better When You Learn to Write DAX	Matt Allington	Holy Macro! Books; 1st edition (June 1, 2018)	€ 29
IT	-	Super Charge Power BI: Power BI Is Better When You Learn to Write DAX (Kindle Edition)	Matt Allington	Holy Macro! Books; 1st edition (June 1, 2018)	€ 14,99
IT	1484251962	Expert T-SQL Window Functions in SQL Server 2019: The Hidden Secret to Fast Analytic and Reporting Queries	Kathi Kellenberger, Clayton Groom, Ed Pollack	Apress, 2nd edition (October 22, 2019)	€ 37,44

Theme	Title	Url	Price
IT	Power BI Desktop	https://powerbi.microsoft.com/en-us/desktop/	Free
IT	PowerAutomate	https://www.linkedin.com/learning/search?entityType=LEARNING_PATH~COURSE&keywords=power%20automate	Free with LinkedIn Learning subscription from Student Square
Professional Development	Hogeschooltaal	https://www.hogeschooltaal.nl	± € 75
Business	LinkedIn Learning	https://www.linkedin.com/learning	Free

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41. INFORMATION ABOUT [MA-MC] MEDIA CREATION (30EC)

41.1. Content

In the Media Creation Main semester, students found their own simulated media agencies as a group. This way, students further develop their professional skills and knowledge in a real world environment. This mimics a real design studio, where teachers and students collaborate closely, akin to senior and junior professionals. Key aspects include:

1. Preparing for internships, aiming for proficiency in UX and media creation, suitable for small media agencies.
2. Gaining a comprehensive understanding of the study program's main semesters, internship opportunities, and professional possibilities.
3. Guidance from lecturers to prepare for successful internship completion and advice on achieving professional goals in ICT.

The media creation semester serves as a building block to fill a student's "toolbox" with creative and technical skills, tool knowledge, empathy, design principles and a deep understanding of the user's perspective. The semester is executed in conjunction with the Minor Virtual Production, with which it shares some activities and materials.

41.2. Learning outcomes

1. **Learning outcome:** Conceptualize, design, and develop professional media products
You create engaging concepts and translate them into professional validated media products by applying user-centered design principles, visual design techniques and by exploring emerging trends and developments in media, design and technologies.

Explanation: Engaging & validated media products = You find this out through testing, is the target audience attracted, triggered, satisfied or otherwise iConceptualize, design, and develop professional media products
You create engaging concepts and translate them into professional validated media products by applying user-centered design principles, visual design techniques and by exploring emerging trends and developments in media, design and technologies. ntrigued to use the concept or product of it.

User-centered design principles = Your digital products are designed and build in accordance with guidelines from the media-field, like the usability heuristics, persuasive design techniques, UI-patterns, using the professional toolset, etc...

Emerging technologies = You explore technologies that are relevant for the current age of ICT, like Artificial Intelligence, Mixed reality, internet of things, camera systems, video studio infrastructure etc...

2. **Learning outcome:** Transferable production
You create professional documentation and use version control for your products in a personal and team context. You communicate design decisions and recommendations to relevant stakeholders.

Explanation: Professional documentation = You create and validate logical structures as ways of documenting your

process for the stakeholders and users

Version control in personal and team context = Employing fitting version control systems like Git or others to manage and track different versions of your products, both in individual and group projects.

Design decisions and recommendations = Communicating with stakeholders using methods like storyboarding and brand guides to facilitate (re)production. Clearly expressing suggestions for usage and maintenance, improvements, new features, or problem solutions in the final concept and product.

3. **Learning outcome:** Creative iterations

You present the successive iterations of your creative process, and the connections between them, of your methodically substantiated, iterative design and development process.

Explanation: Successive iterations = Validated and explained improvements done on a previous version based on feedback, new insights or (user)-tests.

Creative process = Every step you take for creating products in your project guided by a design model like Double Diamond, Design thinking and/or the DOT-framework, CMD methods.

Connections = The logical relationships or linkages between successive iterations.

4. **Learning outcome:** Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research*, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

Explanation: Stakeholder = For example a company, an artist, a professional, ... from the industry with a media creation task that you work on with your team

Methodological approach = i.e. DOT Framework, Double Diamond, Agile / Scrum, ...

Ethical, intercultural and sustainable aspects = You keep in mind the numerous ways of impact of your creations (also for example the energy usage and societal impact of AI)

5. **Learning outcome:** Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

Explanation: Promote your personal growth = You produce your own personal portfolio to prove this and all the other learning outcomes. The form and technical system of your portfolio is your choice, but it should reflect your core values and authentic self. Examples: A thoughtfully designed magazine, a self-crafted website, a collection of showreels (with documentation), ...

41.3. Expected entry profile

Students who have finished their first year successfully in one of the introduction semesters or students coming from a different main semester.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	1	1	1	1	1
Organisational processes					
Infrastructure					
Software					
Hardware-interfacing					

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

41.4. Outflow profile

The semester media creation prepares the student for an internship and / or is an ideal fit for a following semester in Game Design, Front End Development, Mobile Apps Development and Applied Generative AI.

Domain description HBO-i 2023

	Manage & control	Realise Design	Advise Analysis	
User interaction	2	2	2	
Organisational processes	2			
Infrastructure				
Software	1	1		
Hardware-interfacing	1	1		

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

41.5. Examination and final grading

41.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

41.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

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41.5.3. Assessment Guidelines

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This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

41.5.4. Retakes and repair

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41.6. AI Usage Guidelines

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Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

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41.7. Learning activities

Project based work

The semester revolves around a company or stakeholder's project assignment (e.g., Partner in Education, relevant industry professional). Students collaborate in project groups (emulating the professional work of a media agency including their own workflows and branding) to work on the assignment. This educational block can span multiple projects. The projects must cover and facilitate the semester's learning outcomes.

Technical workshops

Technical workshops are optional educational blocks where teachers organize interactive teaching moments centered around a specific topic / category, such as media production or user experience design. These workshops help students develop all kinds of skills aligned with one or more learning outcomes and the project challenge. Besides a set of predetermined (and published on canvas) activities, teachers can organize their teaching materials and contents based on the student's needs (on demand) and receive sufficient time to do so. The workshops cater to specific skillsets that students can use to progress in projects and their personal learning journey, but they can also be adapted to emerging topics as needed.

Showcase Events

There will be three showcase events for the project groups ("Media agencies") throughout the semester. These events are arranged in week 7, 12, and 17. The first event is a poster presentation of the developed concept to address the project challenge. The second event is a demo event where work-in-progress solutions and minimal viable prototypes can be tested by stakeholders. The third event is a public conference-style event where the final solutions are presented to stakeholders, teachers, and interested third parties.

Feedback sessions - Official formative feedback / Portfolio review (at least 3x)

Students submit their portfolio in Canvas before the deadline set by the teacher team. Each portfolio review has its own 'hand-in assignment' with Rubrics attached to it. These rubrics contain the Developmental-scale grading per LO (Learning Outcome). A teacher duo invites the student for a review appointment: these are always on-site physical meetings. Teachers indicate grading per LO (based on scale) in Canvas. Teachers put feedback and feedforward per LO in the Rubric comment box per LO. Reviews take place approximately in week 8, 13 and 18. The on-site meeting with the student will be used to present a quick rundown of the student's products aligned with the LOs. Afterwards the teacher duo is available for questions and oral explanation of the written feedback. The assessors (teacher duo) will copy/paste the feedback from their own notes (in Excel/Notion) 1-on-1 in Canvas per LO. Teacher duos are encouraged to use part of their allocated time to prepare the feedback on site (not in the evenings or during weekends). The do's and don'ts document as well as the "bingo card" are encouraged to be used as support for teachers in this process. These resources are also shared with students on canvas to make the entire process transparent. Every teacher (assessor) duo should complement each other in skills/LOs. For the second review moment, the assigned teacher duo is rotated per student team to give the students a broader range of portfolio feedback.

Individual activities

This optional educational block is for individual student work. Some students may choose not to attend technical workshops and develop their learning outcomes independently. Before doing so, they must consult with a coach / teacher to ensure their execution aligns with this semester's goals. Individual activities must contribute to students' growth, learning outcomes, and project-based work.

41.8. Learning materials

Learning materials are published on canvas.

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42. INFORMATION ABOUT [MA-MAD] MOBILE APPS DEVELOPMENT (30EC)

42.1. Content

In the SM semester you will work on a wide variety of developing techniques to make (native) apps for mobile platforms. You will work on both the concept, design and the interaction design, as well as a native prototype demonstrator of your app. The idea is to apply the basic knowledge you acquired by independently creating a design based on your own idea, to turn it into a testable prototype with a (natively) programmed proof-of-concept with a minimal set of requirements.

The activities of app developers are often the following:

- Mapping the wishes of the client and/or user.
- Conducting consultations with, for example, functional designers, UX designers, UI designers and programmers.
- Design, develop and / or implement new apps and further develop existing apps (based on consultation with client and development team).
- Monitoring the relationship between the functional and technical side of the app development (in collaboration with the rest of the development team).
- Informing back-end developers and ensuring a good link between the front-end of the app and back-end development.
- Preparation and management of (process) documentation.

- Thinking about new possibilities and applications for existing apps and opportunities for new apps.

42.2. Learning outcomes

1. Learning outcome: Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research*, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

2. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

3. Learning outcome: Analysis: you show you can convert a problem or opportunity into multiple smart mobile concepts based on thorough analysis.

Explanation: You use common research methods (like the DOT-framework) to analyse a problem, trends in the market and relevant stakeholder interests. You process the feedback given in your work. Convert: Generate ideas using ideation methods and crystallise them into appealing and engaging mobile challenges. Mobile Concept: A crystallised idea based on a problem description with a distinguishable research component.

4. Learning outcome: Advice: Based on relevant analysis, you advice on suitable user interaction and technological design, improvement of designs, research and/or methodology.

Explanation: You consider the following aspects in your advice: iterative way of working, outcomes of user tests, create POC's. You process the feedback given in your work. Relevant analysis: outcomes of the analysis phase done by you, your team or other developers

5. Learning outcome: Design: You translate concepts into user friendly designs and validate these designs through user tested iterations.

Explanation: Translate: Using design theory and methodologies, you are capable of designing engaging solutions. Engaging: the outcome from research and user tests states that users find the app refreshing/appealing/intriguing and or interesting. User friendly: The designs meet usability and UCD quality standards. Designs: Products that describe the problem, target audience, interaction narrative, user interface and comply with software engineering standards. You process the feedback given in your work. You incorporate the output of previous design and analysis steps into every new design iteration in a meaningful way.

6. Learning outcome: Management & Control: You continuously apply techniques to manage a development process and to assure its quality in direct communication with all stakeholders.

Explanation: Stakeholders are all clients, (future) developers, all users. You process the feedback given in your work.

42.3. Expected entry profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	1	1	1	1	1
Organisational processes					
Infrastructure					
Software			1	1	1
Hardware-interfacing					

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

42.4. Outflow profile

Domain description HBO-i 2023

	Manage & control	Realise Design	Advise Analysis	
User interaction	2	2	2	2
Organisational processes				
Infrastructure				
Software	1	1	2	2
Hardware-interfacing				

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

42.5. Examination and final grading

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Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

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- *Orienting*: You have made a start and explored possibilities to demonstrate the learning outcome.
- *Beginning*: You have taken initial steps that contribute to demonstrating the learning outcome.
- *Proficient*: You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome at a satisfactory level if you continue your development in this way.

- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations and are focused on continuous improvement. You will demonstrate the learning outcome at a more than satisfactory level if you continue your development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

42.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

42.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

42.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

AI is not a replacement or threat, but a collaborative actor in your learning. We use AI to strengthen creativity, deepen understanding, and expand educational possibilities. You are expected to learn to work with AI effectively and productively: a core professional competency for ICT professionals.

The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available, including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

42.7. Learning activities

A new development, following demands and suggestions from the Mobile Development community, allows for either broadening and/or deepening of technical and design skills among students towards becoming a T-shaped professional. We therefore offer workshops and assignments based upon a students' choice. The platform for which we develop is chosen based upon a students' own preference and on the requirements of the prototype, however it is mandatory to show technical skills on both the Android and iOS platforms on at least a basic level. This level should be demonstrated through a (natively) programmed app running on a mobile device, accompanied by a design portfolio. The student should involve the teachers in the learning process via weekly talks and 3 formative portfolio assessments. The feedback and conclusions are noted and curated by the student and added to their portfolio to reflect on.

(dCentralize, Peercode etc), together with other students. Initially it is essential that you know how to come up with a strong concept within the framework/themes of the clients and that you convince this client of its feasibility. Of course the idea is to come up with an app as impressive as possible. First of all this means you make optimal use of everything a mobile device has to offer, such as GPS, Camera, web services, compass etc. Additionally, don't forget to think outside 'the App'. The concept may be grand in its setup, with an App as the proof of concept. The knowledge and skills acquired in the workshops should be clearly applied in the prof assignment. There is a research element taking place simultaneously with the prof assignment. Students will carry out research for the prof assignment in duos.

There will be a focus on either:

- Descriptive ethics and target group analysis, or,
- A proof of concept of an innovative feature that contributes to the group project.

Blended education

The semester works with blended learning, whereby, in addition to the activities on location, online education can take place that matches content and didactics, or when there are planning and scheduling issues. The basic rule is to have optimal use of teaching time on location. The blended set-up for the semester is explained in the Canvas course. The decision to have parts of the education online is reserved for the degree programme.

42.8. Learning materials

Theory and practice

Lectures and practicals will be offered in the form of workshops. These workshops will deal with subjects related to techniques and design and students will work on a special case or their own concept. The products the students hand will serve as proof in the portfolio.

Progress is shown and discussed on a weekly basis with the teachers, at the students' initiative and recorded in FeedPulse by the students. There will be a weekly tutor meeting to discuss the prof assignment and the duo research progress. During this meeting the professional development and group processes are also discussed.

With regard to the theoretical part of the course, the students will be able to attend (online) workshops in which they can work on their assignments together with the teacher or individually.

The result of these workshops will be handed in at the LMS Canvas the same day and students will be able to view their own progress. Furthermore the student needs a laptop, webcam, microphone and software. For the prof assignment the student is allowed to create/choose their own development and project management environment.

Individual working sessions

Students are expected to work on their mobile products all days of the work week, even when nothing is rostered or scheduled.

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43. INFORMATION ABOUT [MA-NCA] NETWORK & CLOUD AUTOMATION (30EC)

Deploying & Automating Intelligent IT Services, DevSecOps

43.1. Content

During this semester, you will explore network and cloud automation by designing and implementing automated IT infrastructure solutions. You will acquire your knowledge of orchestration, automation, security, monitoring, and supporting services to create efficient and scalable systems.

Throughout the semester, you will develop and manage automated solutions, ensuring they align with business requirements while maintaining security and reliability. You will focus on orchestration to automate system and network configurations, utilizing automation through scripting to streamline processes. You will enhance security through Security Orchestration Automation and Response (SOAR), ensuring proactive threat detection and response. Additionally, you will explore supporting services essential for the business side of infrastructure automation and implement monitoring strategies to maintain performance and system health.

43.2. Learning outcomes

1. **Learning outcome:** Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research*, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

2. **Learning outcome:** Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

3. **Learning outcome:** Enhanced Interoperability

After conducting methodical research, you are able to analyze the requirements, provide advice, create a design, and implement the rollout for a sustainable infrastructure for a large SME organization based on the applicable standards, taking into account the various types of data, integrations, protocols, communication, and applications.

4. **Learning outcome:** Enhanced Securing

After conducting methodical research, you are able to analyze the requirements, provide advice, create a design, and implement modern and traditional advanced security measures including measures to increase business continuity, across all infrastructure domains based on the analyzed requirements of a large SME.

5. **Learning outcome:** Enhanced Automation

After conducting methodical research, you are, based on the applicable standards, able to analyze the requirements of a large SME and orchestrate the deployment of an infrastructure, applications, and business processes using coding, tooling and version control.

6. **Learning outcome:** Manage & Control enhanced

After conducting methodical research, you are able to advise and design an architecture based on the analyzed requirements of a large SME that considers management aspects and -processes and complies with laws and regulations.

43.3. Expected entry profile

We recommend to finish Network & Cloud Basics semester which is a foundation in networking and cloud technologies, where you learn core concepts of designing, implementing, and managing IT solutions in a business context.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes					
Infrastructure	1	1	1	1	
Software		1	1	1	
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	1	1	1	1

43.4. Outflow profile

By the end of the semester, you will have a strong foundation in network and cloud automation, enabling you to design, implement, and manage modern IT infrastructures effectively.

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes	2	2			
Infrastructure	2	2	2	2	2
Software	2	1	1	1	
Hardware-interfacing					

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

43.5. Examination and final grading

43.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

43.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome at a satisfactory level if you continue your development in this way.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations and are focused on continuous improvement. You will demonstrate the learning outcome at a more than satisfactory level if you continue your development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

43.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

43.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

43.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

AI is not a replacement or threat, but a collaborative actor in your learning. We use AI to strengthen creativity, deepen understanding, and expand educational possibilities. You are expected to learn to work with AI effectively and productively: a core professional competency for ICT professionals.

The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

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43.7. Learning activities

Your learning will be supported by hands-on exercises, workshops, and real-world challenges, helping you build practical skills in infrastructure automation, security management, and system monitoring. Collaboration with peers and guidance from teachers and coaches will enhance both your technical and professional competencies.

43.8. Learning materials

A Canvas course featuring the necessary learning resources

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44. INFORMATION ABOUT [MA-OLD] OPEN LEARNING (30EC)

44.1. Content

ICT & Personalized Program is a route where you have a lot of freedom and flexibility to determine your personal focus and structure, within the boundaries of the HBO-I competence framework. In practice, this means that anything within these boundaries potentially can be used in ICT & Personalized Program. This gives great opportunities for starting multi-disciplinary projects. Projects should be IT related, based on relevant questions and developments from the professional field and where possible externally validated.

Lots of freedom, but remember:

“Freedom comes with responsibility”

This large degree of freedom does not mean that there are no obligations. A large amount of autonomy and entrepreneurial behaviour is expected.

44.2. Learning outcomes

1. Learning outcome: Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research*, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

2. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

44.3. Expected entry profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

44.4. Outflow profile

The student builds a competence profile based upon the basic architectural layers (level 1) from OE2, this is taken to the next step at level 2. In addition, a student can deepen or broaden and will touch other architectural layers. These will initially be linked at level 1. Generally speaking, the core architectural layer level grows with the phase of the students' study. Phase OE1,OE2 ~ level 1, phase OE3,4,5, ~ level 2 and phase OE6,7,8, ~ level 3.

A balanced competence profile does not have too many competencies (no focus) but also not too little (no widening). At least all 9 competences have to be covered on level 2

Within Open Learning, students define their own dynamic competence profile per semester (exit level OEx) under supervision. In doing so, they draw up their own criteria for challenges (projects) that contribute to growth within the HBO-i competence framework. A complete competence level is reached by successfully ending a phase. A learning outcome is a partial contribution to a competence step.

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

44.5. Examination and final grading

44.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your

teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

44.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
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- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations and are focused on continuous improvement. You will demonstrate the learning outcome at a more than satisfactory level if you continue your development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

44.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

44.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

44.6. AI Usage Guidelines

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44.7. Learning activities

The student will determine the general lines of the content in this programme him or herself. Basically, everything in IT should be possible. Because everyone has his own track, it is impossible to plan a curriculum that fits for everyone. Learning happens asynchronous, not everybody does the same things, with the same speed, ... This means all workshops, presentations, etc., that will provide content, will be planned and scheduled on-demand or when the coaches notice it is necessary.

To facilitate this the student will create his/her own personal profile within the HBO-i Competence Framework, this framework gives a systematic overview of the whole IT landscape including their professional skills. This profile will be created shortly after the start of the semester. Before enrollment, the student will be invited for an intake interview. This interview is about motivation and a possible context to use within open learning. In an intake document, it is outlined which topics the student would like to engage. This usually is a very rough outline. For example, "Artificial intelligence", "Encryption" or "extreme interaction". The intake interview will be an evaluation to see whether you really understand what open learning is about, so you know what you're up to.

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For example, workshops of more general applicability are offered to all students, like Scrum-101, Research, Concept validation, while more specific subjects are offered to a smaller group of students.

44.8. Learning materials

Apart of the Canvas LMS environment, anything which would be applicable can and will be used.

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45. INFORMATION ABOUT [MA-OLC] OPEN LEARNING (30EC)

45.1. Content

ICT & Personalized Program is a route where you have a lot of freedom and flexibility to determine your personal focus and structure, within the boundaries of the HBO-I competence framework. In practice, this means that anything within these boundaries potentially can be used in ICT & Personalized Program. This gives great opportunities for starting multi-disciplinary projects. Projects should be IT related, based on relevant questions and developments from the professional field and where possible externally validated.

Lots of freedom, but remember:

"Freedom comes with responsibility"

This large degree of freedom does not mean that there are no obligations. A large amount of autonomy and entrepreneurial behaviour is expected.

45.2. Learning outcomes

1. Learning outcome: Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research*, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

2. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

45.3. Expected entry profile

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

45.4. Outflow profile

The student builds a competence profile based upon the basic architectural layers (level 1) from OE2, this is taken to the next step at level 2. In addition, a student can deepen or broaden and will touch other architectural layers. These will initially be linked at level 1. Generally speaking the core architectural layer level grows with the phase of the students' study. Phase OE1,OE2 ~ level 1, phase OE3,4,5, ~ level 2 and phase OE6,7,8, ~ level 3.

A balanced competence profile does not have too many competencies (no focus) but also not too little (no widening). At least all 9 activities (including professional development) have to be covered on level 2.

Within Open Learning, students define their own dynamic competence profile per semester (exit level OEx) under supervision. In doing so, they draw up their own criteria for challenges (projects) that contribute to growth within the HBO-i competence framework. A complete competence level is reached by successfully ending a phase. A learning outcome is a partial contribution to a competence step.

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

45.5. Examination and final grading

45.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

45.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

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45.5.3. Assessment Guidelines

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- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

45.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

45.6. AI Usage Guidelines

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45.7. Learning activities

The student will determine the general lines of the content in this programme him or herself. Basically everything in IT should be possible. Because everyone has his own track, it is impossible to plan a curriculum that fits for everyone. Learning happens asynchrone, not everybody does the same things, with the same speed, ... This means all workshops, presentations, etc., that will provide content, will be planned and scheduled on-demand or when the coaches notice it is necessary.

To facilitate this the student will create his/her own personal profile within the HBO-i Competence Framework, this framework gives a systematic overview of the whole IT landscape including their professional skills. This profile will be created shortly after the start of the semester. Before enrollment, the student will be invited for an intake interview. This interview is about motivation and a possible context to use within open learning. In an intake document, it is outlined which topics the student would like to engage. This usually is a very rough outline. For example, "Artificial intelligence", "Encryption" or "extreme interaction". The intake interview will be an evaluation to see whether you really understand what open learning is about, so you know what you're up to.

Because everyone has his own track, it is impossible to plan a curriculum that fits for everyone. Learning happens asynchronous, not everybody does the same things, with the same speed, ... This means all workshops, lessons, .. that will provide content, will be planned and scheduled on-demand or when the teachers notice it is necessary.

For example, workshops of more general applicability are offered to all students, like Scrum-101, Research, Concept validation, while more specific subjects are offered to a smaller group of students.

45.8. Learning materials

Apart of the Canvas LMS environment, anything which would be applicable can and will be used.

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46. INFORMATION ABOUT [O-MA-OLD] OPEN LEARNING (30EC)

46.1. Content

We have Open Education for 'Education for professionals' from the second year of your study. This means that you will define your own personal learning route. Do you want to become a real ICT & Business analyst? Or are you the next gen Cyber Security, Artificial Intelligence or Game Design specialist? Do you want to specialize in a particular technology? It's up to you!

You will work as much as possible in a context that is most supportive of your personal learning trajectory. This means it is rather impossible to determine a concrete challenge context in advance. This context is facilitated by relevant challenges which are provided by real case stakeholders, that are determined at the beginning of the process.

With this in mind, Education for professionals can be defined as follows:

Educating responsive professionals for all possible careers in IT. This is done by focusing on talent development and personalized competence profiles.

Students who have successfully concluded OE3 can start in OE4. Starting level would be the learning outcomes output of the OE3. The output level is based on the achieved growth on the set competence indicators of the student's personal competence profile. This should connect and align with the next OE a student is pursuing.

The output level is based on the achieved growth on the set competence indicators of the student's personal competence profile.

The Open Learning program will run as a part time course with at least two contact moments a week:

- Tuesday evening: 18u-21u: Fontys Location Eindhoven: TQ
- Thursday evening: 18u-19.30u: Online contact moment

The student will determine the general lines of the content in this program him or herself. Basically, everything in IT should be possible. Because everyone has his own track, it is impossible to plan a curriculum that fits for everyone. Learning happens asynchronously, not everybody does the same things, with the same speed, in the same manner, etc. This means all workshops, presentations, etc., that will provide content, will be planned and scheduled on-demand or when the coaches notice it is necessary.

To facilitate this, the student will create his/her own personal profile within the HBO-i Competence Framework, this framework gives a systematic overview of the whole IT landscape including their professional skills. This profile will be created shortly after the start of the semester.

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Blended Learning

At Education for Professionals, we apply the design principles of Blended Learning. This means that we learn with each other online and face-to-face. We have the following principles to concrete this:

- Face to face contact (18:00-21:00)

Every Tuesday evening from 18:00 to 21:00, we learn in groups or individually at FICT. The following activities could be scheduled:

1. Status updates: Give your group members and coaches insight in your learning process. What have you done last week? Show your plan for the coming weeks? Are there topics you would like to discuss? Etc.
2. Discuss about a topic:

You will zoom in into a topic which is introduced by a student (or coach).

1. Self-work/Collaboration: Students are given the time to work together on a challenge or a smaller task which is related to the discussion topic talk
2. Inspiration sessions / Workshops: Inspiration sessions and workshops could be scheduled.
 - Online contact (18:00-19:30)

Of course, students can do their own study activities online in all sorts of ways. For example, the student asks fellow students to discuss deliverables or issues with others. A meeting with the coach or expert could be scheduled. Workshops could be scheduled online as well.

46.2. Learning outcomes

1. Learning outcome: Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research*, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

Explanation: Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

2. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

Explanation: You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

3. Learning outcome: Analysis:

You can obtain desired results by analyzing the processes, products and information flows in a methodical and thorough way.

Explanation: You know what you are researching and why, taking into account the client, target audience & market. Using several justified research methods based on the DOT-framework
Applying multiple different methods of analysis or focusing on several different aspects. For example: client needs, audience needs, other related products, expert knowledge, literature etc.

4. Learning outcome: Advice:

You can give advice about the organization, processes and/or information important for decisions related to your challenge.

Explanation: Your advice is based on the conclusions and results from the analysis.
Your advice can be based on cost aspects and quality properties such as: availability, performance, security and scalability.

5. Learning outcome: Analysis

You can obtain desired results by analyzing the processes, products and information flows in a methodical and thorough way.

You know what you are researching and why, taking into account the client, target audience & market.
Using several justified research methods based on the DOT-framework
Applying multiple different methods of analysis or focusing on several different aspects. For example: client needs, audience needs, other related products, expert knowledge, literature etc.

Advice

You can give advice about the organization, processes and/or information important for decisions related to your challenge.

Your advice is based on the conclusions and results from the analysis.
Your advice can be based on cost aspects and quality properties such as: availability, performance, security and scalability.

Design:

You are able to design (parts of) systems, communicate and validate about it for both functional/technical and aesthetic requirements using industry standard methods, related to your challenge.

Explanation: With design we mean that you can use a structured and methodical approach in your design. You use a set of concept development methods such as diverging/converging, brainstorming and existing theories as well as structured technical design methods such as UML and RUP.

The design covers both the functional/technical and aesthetic design of your products. You communicate your design in a well presented and structured way to relevant stakeholders.

You validate your designed systems in an early stage using efficient and quick prototyping methods such as paper prototyping and use the results to refine your design.

6. Learning outcome: Realization:

You can implement and validate a product based on a design in a structured way using innovative and/or relevant technology adhering to quality criteria in the context of your challenge

Explanation: We expect that you validate your product and the quality of the product in a structured way using methods such as: User testing, expert reviews, code/peer reviews, (unit) tests.

Structured way means using industry standard development processes such as Scrum/Agile and Kanban.

With innovative and/or relevant technology we mean that you experiment and use upcoming and promising technology that suits your needs.

Your product adheres to previously defined quality criteria in accordance with stakeholders. Many non-functional requirements such as stability, performance, security etc. could be used to define these criteria.

7. Learning outcome: Manage and Control:

You can perform activities that focus on control, monitoring and optimization of the development, commissioning and use of ICT-systems. These activities are related to your challenge.

Explanation: We expect that you build your controlling, monitoring and optimisation of the development in a structured way.

We expect that you can communicate with stakeholders about your activities who are related to manage and control.

46.3. Expected entry profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

46.4. Outflow profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

46.5. Examination and final grading

46.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

46.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome at a satisfactory level if you continue your development in this way.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations and are focused on continuous improvement. You will demonstrate the learning outcome at a more than satisfactory level if you continue your development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

46.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

46.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

46.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

AI is not a replacement or threat, but a collaborative actor in your learning. We use AI to strengthen creativity, deepen understanding, and expand educational possibilities. You are expected to learn to work with AI effectively and productively: a core professional competency for ICT professionals.

The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available, including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

46.7. Learning activities

The teachers will provide you with formative feedback on deliverables and demonstrations. You are expected to record the feedback in FeedPulse and it is expected that this feedback will be processed in further deliverables, within reason. Your work will be assessed on all the competences, the learning outcomes (Analysis, Advice, Design, Realisation, Manage and Control, Future oriented organization, Investigative Problem Solving, Personal Leadership and Targeted Interaction). During the semester there are at least three sprint demo's where you can show your progress.

At these demo's the competences are evaluated based on state of the deliverables and your personal development at that point. This will lead to a grade which is a formative indicator of mastery of learning outcomes that moment and don't give any guarantee for further assessments. It is expected that you keep on progressing on the competences, not doing so might lead to a lower grade at the next formative assessment such as beginning or orienting.

46.8. Learning materials

All information about the education program Education for Professionals OE4 could be found in the CANVAS course. Select your own study materials you need during the semester. Everything you need is fine but keep up a good reference list!

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47. INFORMATION ABOUT [O-MA-OLE] OPEN LEARNING (30EC)

47.1. Content

We have Open Education for PRO from the second year of your study. This means that you will define your own personal learning route. Do you want to become a real ICT & Business analyst? Or are you the next gen Cyber Security, Artificial Intelligence or Game Design specialist? Do you want to specialize in a particular technology? It's up to you!

You will work as much as possible in a context that is most supportive of your personal learning trajectory. This means it is rather impossible to determine a concrete challenge context in advance. This context is facilitated by relevant challenges which are provided by real case stakeholders, that are determined at the beginning of the process.

With this in mind, Education for professionals can be defined as follows:

Educating responsive professionals for all possible careers in IT. This is done by focusing on talent development and personalized competence profiles.

Students who have successfully concluded OE4 can start in OE5. Starting level would be the learning outcomes output of the OE4. The output level is based on the achieved growth on the set competence indicators of the student's personal competence profile. This should connect and align with the next OE a student is pursuing.

The Open Learning program will run as a part time course with at least two contact moments a week:

- Tuesday evening: 18u-21u: Fontys Location Eindhoven: TQ
- Thursday evening: 18u-19.30u: Online contact moment

The student will determine the general lines of the content in this program him or herself. Basically, everything in IT should be possible. Because everyone has his own track, it is impossible to plan a curriculum that fits for everyone. Learning happens asynchronously, not everybody does the same things, with the same speed, in the same manner, etc. This means all workshops, presentations, etc., that will provide content, will be planned and scheduled on-demand or when the coaches notice it is necessary.

To facilitate this, the student will create his/her own personal profile within the HBO-i Competence Framework, this framework gives a systematic overview of the whole IT landscape including their professional skills. This profile will be created shortly after the start of the semester.

A complete explanation of the structure of the framework and how it functions can be found in the Canvas course.

Blended Learning

At Education for Professionals, we apply the design principles of Blended Learning. This means that we learn with each other online and face-to-face. We have the following principles to concrete this:

- Face to face contact (18:00-21:00)

Every Tuesday evening from 18:00 to 21:00, we learn in groups or individually at FHICT. The following activities could be scheduled:

1. Status updates: Give your group members and coaches insight in your learning process. What have you done last week? Show your plan for the coming weeks? Are there topics you would like to discuss? Etc.
2. Discuss about a topic:

You will zoom in into a topic which is introduced by a student (or coach).

1. Self-work/Collaboration: Students are given the time to work together on a challenge or a smaller task which is related to the discussion topic talk
2. Inspiration sessions / Workshops: Inspiration sessions and workshops could be scheduled.

- Online contact (18:00-19:30)

Of course, students can do their own study activities online in all sorts of ways. For example, the student asks fellow students to discuss deliverables or issues with others. A meeting with the coach or expert could be scheduled. Workshops could be scheduled online as well.

47.2. Learning outcomes

1. Learning outcome: Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research*, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

Explanation: Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

2. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

Explanation: You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

3. Learning outcome: Analysis:

You can obtain desired results by analyzing the processes, products and information flows in a methodical and thorough way.

Explanation: You know what you are researching and why, taking into account the client, target audience & market. Using several justified research methods based on the DOT-framework
Applying multiple different methods of analysis or focusing on several different aspects. For example: client needs, audience needs, other related products, expert knowledge, literature etc.

4. Learning outcome: Advice:

You can give advice about the organization, processes and/or information important for decisions related to your challenge.

Explanation: Your advice is based on the conclusions and results from the analysis.
Your advice can be based on cost aspects and quality properties such as: availability, performance, security and scalability.

5. Learning outcome: Design:

You are able to design (parts of) systems, communicate and validate about it for both functional/technical and aesthetic requirements using industry standard methods, related to your challenge.

Explanation: With design we mean that you can use a structured and methodical approach in your design. You use a set of concept development methods such as diverging/converging, brainstorming and existing theories as well as structured technical design methods such as UML and RUP.

The design covers both the functional/technical and aesthetic design of your products. You communicate your design in a well presented and structured way to relevant stakeholders.

You validate your designed systems in an early stage using efficient and quick prototyping methods such as paper prototyping and use the results to refine your design.

6. Learning outcome: Realization:

You can implement and validate a product based on a design in a structured way using innovative and/or relevant technology adhering to quality criteria in the context of your challenge.

Explanation: We expect that you validate your product and the quality of the product in a structured way using methods such as: User testing, expert reviews, code/peer reviews, (unit) tests.

Structured way means using industry standard development processes such as Scrum/Agile and Kanban.

With innovative and/or relevant technology we mean that you experiment and use upcoming and promising technology that suits your needs.

Your product adheres to previously defined quality criteria in accordance with stakeholders. Many non-functional requirements such as stability, performance, security etc. could be used to define these criteria.

7. Learning outcome: Manage and Control:

You can perform activities that focus on control, monitoring and optimization of the development, commissioning and use of ICT-systems. These activities are related to your challenge.

Explanation: We expect that you build your controlling, monitoring and optimisation of the development in a structured way.

We expect that you can communicate with stakeholders about your activities who are related to manage and control.

47.3. Expected entry profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

47.4. Outflow profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

47.5. Examination and final grading

47.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

47.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome at a satisfactory level if you continue your development in this way.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations and are focused on continuous improvement. You will demonstrate the learning outcome at a more than satisfactory level if you continue your development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

47.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

47.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

47.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

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The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

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Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available, including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

47.7. Learning activities

The teachers will provide you with formative feedback on deliverables and demonstrations. You are expected to record the feedback in FeedPulse and it is expected that this feedback will be processed in further deliverables, within reason. Your work will be assessed on all the competences, the learning outcomes (Analysis, Advice, Design, Realisation, Manage and Control, Future oriented organization, Investigative Problem Solving, Personal Leadership and Targeted Interaction). During the semester there are at least three sprint demo's where you can show your progress.

At these demo's the competences are evaluated based on state of the deliverables and your personal development at that point. This will lead to a grade which is a formative indicator of mastery of learning outcomes that moment and don't give any guarantee for further assessments. It is expected that you keep on progressing on the competences, not doing so might lead to a lower grade at the next formative assessment such as beginning or orienting.

47.8. Learning materials

All information about the education program Education for Professionals OE5 could be found in the CANVAS course. Select your own study materials you need during the semester. Everything you need is fine but keep up a good reference list!

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48. INFORMATION ABOUT [O-MA-OLC] OPEN LEARNING (30EC)

48.1. Content

We have Open Education for PRO from the second year of your study. This means that you will define your own personal learning route. Do you want to become a real ICT & Business analyst? Or are you the next gen Cyber Security, Artificial Intelligence or Game Design specialist? Do you want to specialize in a particular technology? It's up to you!

You will work as much as possible in a context that is most supportive of your personal learning trajectory. This means it is rather impossible to determine a concrete challenge context in advance. This context is facilitated by relevant challenges which are provided by real case stakeholders, that are determined at the beginning of the process.

With this in mind, Education for professionals can be defined as follows:

Educating responsive professionals for all possible careers in IT. This is done by focusing on talent development and personalized competence profiles.

Students who have successfully concluded OE2 can start in OE3. Starting level would be the learning outcomes output of the OE2. The output level is based on the achieved growth on the set competence indicators of the student's personal competence profile. This should connect and align with the next OE a student is pursuing.

The output level is based on the achieved growth on the set competence indicators of the student's personal competence profile.

The student will determine the general lines of the content in this program him or herself. Basically, everything in IT should be possible. Because everyone has his own track, it is impossible to plan a curriculum that fits for everyone. Learning happens asynchronously, not everybody does the same things, with the same speed, in the same manner, etc. This means all workshops, presentations, etc., that will provide content, will be planned and scheduled on-demand or when the coaches notice it is necessary.

To facilitate this, the student will create his/her own personal profile within the HBO-i Competence Framework, this framework gives a systematic overview of the whole IT landscape including their professional skills. This profile will be created shortly after the start of the semester.

A complete explanation of the structure of the framework and how it functions can be found in the Canvas course.

Blended Learning

At Education for Professionals, we apply the design principles of Blended Learning. This means that we learn with each other online and face-to-face. We have the following principles to concrete this:

- Face to face contact (18:00-21:00)

Every Tuesday evening from 18:00 to 21:00, we learn in groups or individually at FICT. The following activities could be scheduled:

- 1. Status updates: Give your group members and coaches insight in your learning process. What have you done last week? Show your plan for the coming weeks? Are there topics you would like to discuss? Etc.

2. Discuss about a topic:

You will zoom in into a topic which is introduced by a student (or coach).

1. Self-work/Collaboration: Students are given the time to work together on a challenge or a smaller task which is related to the discussion topic talk
 2. Inspiration sessions / Workshops: Inspiration sessions and workshops could be scheduled.
- Online contact (18:00-19:30)

Of course, students can do their own study activities online in all sorts of ways. For example, the student asks fellow students to discuss deliverables or issues with others. A meeting with the coach or expert could be scheduled. Workshops could be scheduled online as well.

48.2. Learning outcomes

1. Learning outcome: Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research*, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

Explanation: Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

2. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

Explanation: You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

3. Learning outcome: Analysis:

You can obtain desired results by analyzing the processes, products and information flows in a methodical and thorough way.

Explanation: You know what you are researching and why, taking into account the client, target audience & market. Using several justified research methods based on the DOT-framework
Applying multiple different methods of analysis or focusing on several different aspects. For example: client needs, audience needs, other related products, expert knowledge, literature etc.

4. Learning outcome: Advice:

You can give advice about the organization, processes and/or information important for decisions related to your challenge.

Explanation: Your advice is based on the conclusions and results from the analysis.
Your advice can be based on cost aspects and quality properties such as: availability, performance, security and scalability.

5. Learning outcome: Design:

You are able to design (parts of) systems, communicate and validate about it for both functional/technical and aesthetic requirements using industry standard methods, related to your challenge.

Explanation: With design we mean that you can use a structured and methodical approach in your design. You use a set of concept development methods such as diverging/converging, brainstorming and existing theories as well as structured technical design methods such as UML and RUP.
The design covers both the functional/technical and aesthetic design of your products. You communicate your design in a well presented and structured way to relevant stakeholders.
You validate your designed systems in an early stage using efficient and quick prototyping methods such as paper prototyping and use the results to refine your design.

6. Learning outcome: Realization:

You can implement and validate a product based on a design in a structured way using innovative and/or relevant technology adhering to quality criteria in the context of your challenge.

Explanation: We expect that you validate your product and the quality of the product in a structured way using methods such as: User testing, expert reviews, code/peer reviews, (unit) tests.

Structured way means using industry standard development processes such as Scrum/Agile and Kanban. With innovative and/or relevant technology we mean that you experiment and use upcoming and promising technology that suits your needs. Your product adheres to previously defined quality criteria in accordance with stakeholders. Many non-functional requirements such as stability, performance, security etc. could be used to define these criteria.

7. Learning outcome: Manage and Control:

You can perform activities that focus on control, monitoring and optimization of the development, commissioning and use of ICT-systems. These activities are related to your challenge.

Explanation: We expect that you build your controlling, monitoring and optimisation of the development in a structured way.

We expect that you can communicate with stakeholders about your activities who are related to manage and control.

48.3. Expected entry profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development			1	1

48.4. Outflow profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

48.5. Examination and final grading

48.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

48.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- *Undefined:* You have not yet undertaken any activities to demonstrate the learning outcome.

- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome at a satisfactory level if you continue your development in this way.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations and are focused on continuous improvement. You will demonstrate the learning outcome at a more than satisfactory level if you continue your development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

48.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

48.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

48.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

AI is not a replacement or threat, but a collaborative actor in your learning. We use AI to strengthen creativity, deepen understanding, and expand educational possibilities. You are expected to learn to work with AI effectively and productively: a core professional competency for ICT professionals.

The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available, including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

48.7. Learning activities

The teachers will provide you with formative feedback on deliverables and demonstrations. You show the work, alone or in a group context, but will be given feedback on an individual base. Your work will be assessed on all the competences (the learning outcomes (Analysis, Advice, Design, Realisation, Manage and Control, Future oriented organization, Investigative Problem Solving, Personal Leadership and Targeted Interaction). Besides the continuous feedback, there are three formal sprint demo's planned. For these demo's, you'll have to submit your portfolio to Canvas.

At these assessments the competences are evaluated based on state of the deliverables and your personal development at that point. This will lead to a grade which is a formative indicator of mastery of learning outcomes that moment and don't give any guarantee for further assessments. It is expected that you keep on progressing on the competences, not doing so might lead to a lower grade at the next formative assessment such as beginning or orienting.

The first and leading collection to show progress is your professional portfolio. The portfolio is built from the project deliverables you work on in each challenge.

The second type of deliverables is proof. All documents and artifacts that you create as a means to design, compare or research can serve as proof. It can contain for example a design rationale, sketches to explore designs, a stakeholder analysis matrix, code snippets (git-branches), a prototype etc.

Blended education

The semester works with blended learning, whereby, in addition to the activities on location, online education can take place that matches content and didactics, or when there are planning and scheduling issues. The basic rule is to have optimal use of teaching time on location. The blended set-up for the semester is explained in the Canvas course.

48.8. Learning materials

All information about the education program Education for Professionals OE3 could be found in the CANVAS course. Select your own study materials you need during the semester. Everything you need is fine but keep up a good reference list!

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49. INFORMATION ABOUT [AV-CSS] COMPLEX SOFTWARE SYSTEMS (30EC)

Development of Complex Software Environments, Non-Functional & Data Requirements

49.1. Content

A complex software system is distinguished by its elaborate architecture. This typically involves a multitude of interacting components, often dispersed across various platforms and utilizing diverse technologies. Such systems are designed to manage large datasets, intricate logical processes, and necessitate a high degree of scalability, dependability, and ease of maintenance.

Several factors contribute to this system complexity. Components within the system are often highly interdependent, meaning they rely on each other. This interconnectedness can make implementing changes risky and troubleshooting problems challenging. The scale of the system is also a key factor, often operating with a large number of users, transactions, or data points. Furthermore, the integration of diverse technologies and platforms creates potential compatibility issues. Finally, complex systems must adapt to evolving requirements, constantly changing to meet new business needs.

It is important to differentiate between a complex software system and a full-stack development. While a full-stack developer possesses expertise across a range of technologies (front-end, back-end, database), a complex software system developer is focussed on qualities of the system and puts more effort into the overall architecture and functionality of a software system. A full-stack developer might contribute to a complex system, but the complexity itself is a characteristic of the system's design and operation, not the specific skillset of the individual developer.

In this software systems the *quality of the system* takes the most attentions of the developers. Such software should be developed with security in mind and should be performant and maintainable. Many of the mentioned software qualities are leading to a design trade-off and require following of the industry best practices to resolve them.

In this semester we will learn how to design, implement and validate a high quality complex software systems according to the industry best practices. The measurable qualities of the system will be expressed in non-functional requirements as set by stakeholders.

In this semester, we will use a strategy of research based learning. Students will be challenged to solve a real world problem with real world stakeholders and have to perform research in order to tackle this problem. The complex context of the problem is essential to learn how to solve real world problems based on industry best practices.

49.2. Learning outcomes

1. Learning outcome: Software Engineering

You apply an engineering approach in all your activities to create software systems in progressive steps, while applying system thinking during informed decision making.

Explanation: Engineering approach

You demonstrate measurable feedback cycles during your work, to validate and guide your decisions and technical solutions.

You apply an engineering approach to aim at improving your software engineering skills.

You actively look for concepts, tools & technologies that potentially offer solutions for challenges in your software systems. You evaluate feasibility and decide upon integrations of these solutions by using small proof-of-concepts (experiments).

Progressive steps

You demonstrate progress in taking small steps towards your goal. This could be done e.g. by evidence-based experiments with concepts relevant to your project, including the use of GenAI-assisted prototyping and validation, and decide, based on outcomes, whether an experiment might add value to the project.

System thinking

You analyze interdependencies between components, processes, data and stakeholders, focusing on architectural connections such as data flows, communication patterns, system boundaries, and the impact of data complexity on design trade-offs.

Informed decision making

You motivate your design choices based on the business context, system landscape, and project requirements, ensuring your environment is fit-for-purpose.

2. Learning outcome: Software Quality

You demonstrate sustainable software quality in all your work, by systematically addressing both functional and non-functional requirements of a software system, and continuously validating system behavior against well-defined quality benchmarks.

Explanation: Sustainable software quality

You apply practices that support long-term sustainability of software, while adapting to the newest industry practices (eg. In fields of AI specific requirements, UX).

You aim at sharpening your software change skills

Continuous validation

You apply best practices to measure system qualities over time in a setup that is effective in reproducibility, diagnostics, and supports quality improvement cycles.

Well-defined quality

You use quality frameworks (e.g. ISO/IEC 25010 inspired) to identify and translate quality attributes such as scalability, maintainability, performance and security into software products of measurable quality.

3. Learning outcome: Software Maintenance

You apply, in various contexts, a maintainable software delivery process, which is compatible with the current stakeholder's IT organization, and continuously aim for improving the software development process itself by applying best practices.

Explanation: Maintainable

You design, use and improve an environment and processes that support continuous integration and an 'automate all' approach.

You deploy your software in a running environment and continuously monitor the desired behavior of the system and its integrations which are based on project specifications.

Your environment supports team collaboration.

You apply a process which manages system changes.

Development process improvement

You apply best practices to improve the software development process within a team

You verify the non-functional requirements of your infrastructure

You advise on potential improvements of stakeholder infrastructure to match project needs

You demonstrate the feasibility to deploy your system at stakeholder organization

You integrate practices (e.g., DORA metrics) that improve efficiency and reliability of software delivery.

4. **Learning outcome:** Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide advice to your stakeholders even in complex and uncertain contexts. You substantiate and validate future-oriented choices by use of ethical, intercultural, and sustainable arguments.

Explanation: Take responsibility

You identify, coordinate and optimise subtasks within your projects.

You are transparent about the impact of the solutions you create.

You ensure your work is effectively transferred to the stakeholders.

Applied research:

You formulate appropriate subjects to investigate within the context and can prioritise them.

You justify your choices of research patterns, strategies and methods used.

You demonstrate the reliability, validity and relevance of your research.

You reflect critically on the followed research process.

Complex and uncertain contexts:

You learn the domain of the project, speak the language of the domain to communicate with stakeholders.

You differentiate between essential and accidental complexity and take actions to understand the essential complexity in the context.

You stay updated on technology trends by contextualising it and share your knowledge in a meaningful way

Law, ethical, intercultural and sustainable:

You recognise and respect ethical boundaries — your own and those of others — and act with integrity and accountability.

You apply ethical standards and norms in the field of expertise, e.g. the GDPR or the AI Act.

You independently perform an impact and stakeholder analysis and provide your client with sound ethical and sustainable advice.

You collaborate in diverse and intercultural teams. You effectively manage cultural differences in your interactions with stakeholders.

5. **Learning outcome:** Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show that you are capable of carrying out these actions and achieving your goals, adjusting them as necessary.

Explanation: Goals and actions

You proactively set goals to match your growth aspirations.

You take actions that contribute to achieving your goals.

You facilitate learning within the others (eg. stakeholders, fellow developers).

You actively search for feedback, reflect and act upon it to improve your skills.

You validate whether you are getting closer to these goals and adjust, if necessary.

Long-term development

You are aware of developments in the field of software engineering and the multiple career paths.

You determine what kind of ICT professional you want to be and adjust your personal development plans, if necessary.

Based on this, you choose and prepare for your future.

49.3. Expected entry profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction			1	1	
Organisational processes	1		1	1	
Infrastructure	1		1	1	
Software	2	2	2	2	2
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

49.4. Outflow profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction			2	1	
Organisational processes	1		1	2	
Infrastructure	1	1	2	2	1
Software	3	3	3	3	3
Hardware-interfacing					

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	3	3	3	3

49.5. Examination and final grading

49.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

49.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- *Undefined*: You have not yet undertaken any activities to demonstrate the learning outcome.
- *Orienting*: You have made a start and explored possibilities to demonstrate the learning outcome.
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49.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

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49.7. Learning activities

Advanced Software is a research-based semester. By doing research, you will learn methods and technology that are common in a daily practice of the software industry. You can make your own choices based on your personal preferences and context. These choices can be related to technology selection, preferred project methodology and planning. There is a flexible learning path in which focus on earning outcomes is determined and you can choose from various learning and research activities.

You will spend your time on your Group Project (including knowledge exchange among students) and a bit of your time on your experiments.

Typically, the group project is part of a long-lasting research project initiated by a Partner in Innovation or professorship (dutch: Lectoraat), named hereafter: *client*. A client representative will discuss the assignment with the students. The assignment will be aligned with the semester learning outcomes and will trigger you into doing research, learning, and applying your knowledge in the context of an authentic project. Examples of projects that students have worked on in the past can be found here. The authenticity of this project mainly is enabled by the professional environment the client provides. Moreover, the problem that the client ask you to solve is authentic: the client doesn't have an answer to this problem themselves and very interested in further development of the project results that you deliver. By providing a professional environment, you will further develop your professional skills and your technical skills in the context of a large software project. It may also be possible to create a multidisciplinary environment by joining projects with students from different profiles.

The role of the experimentation is a playground for the technical skills required in the group projects – you learn technology and practice the skills needed for the group project. There will also be room for some experimentation of technologies that fit your personal interests (*personal flavors*).

The semester will be divided in sprints of 3 weeks. You learn by acquiring knowledge and skills through (group) investigations, workshops and educational materials, exchange knowledge with other students, and by applying new knowledge and skills in your projects. In addition to a collaborative learning, you can choose topics for further study and define your own learning path.

During this semester, you will be part of a group of 5 or 6 students. You will learn by means of an applied research in a group project. In addition, lectures and hands-on practices will be provided by Partners in Education/Innovation and you can attend lectures and workshops on graduation preparation, societal impact, and the research framework.

You will be coached and judged by teachers. One of them is your tutor for the group project and your semester coach. The other will guide you with topics related to software development. With each teacher you will have two dayparts scheduled per week.

First Week

The semester starts with a kick-off explaining the semester structure & processes. This first day you will be assigned to the case for your group project. During the first week you will be introduced to your teachers & contactperson of the group project. This week you will be orienting yourself on the domain of the group project & the challenge presented. You will

start by generating ideas and concretize these ideas in prototype products, that envision your potential future solution. Your results will be presented at the end of the week, and are the starting point for further exploration with your customer to define functional and non-functional criteria in the first sprint, starting in week two.

49.8. Learning materials

During the semester you will have access to the learning materials provided through the learning management system Canvas and the internet.

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50. INFORMATION ABOUT [AV-CSP] CYBER SECURITY PROFESSIONAL (30EC)

Security Innovation & Red, Blue or Yellow Specialisation

50.1. Content

The **Cyber Security Professional semester** prepares students to take on a specialised role in cybersecurity by working on real-world security challenges. Students choose a focus area—**Red Teaming (ethical hacking)**, **Blue Teaming (defensive security)**, or **Yellow Teaming (governance, risk & compliance)**—and build a portfolio that reflects their expertise in that role. For example, a Red Teamer may conduct a penetration test and report the results to a simulated client, while a Blue Teamer may design and implement a SIEM use case and monitoring strategy. Yellow Teamers might perform a risk assessment or develop an incident response plan based on relevant frameworks (e.g., ISO 27001 or NIST).

Throughout the semester, students are expected to define and carry out their own research or project work using structured methodologies—such as threat modeling, MITRE ATT&CK analysis, or security baselining—and provide well-substantiated advice to internal or external stakeholders. They work in small teams or individually, often in open-ended or changing scenarios, where they must analyse and respond to both technical and organisational security needs.

In addition to technical development, students plan and reflect on their personal growth as ICT professionals. This includes setting learning goals, keeping track of their progress, and adjusting their actions based on feedback and outcomes. They are expected to not only deliver technically sound solutions, but also consider broader ethical, sustainable, and intercultural impacts of their work, ensuring their solutions are relevant in a real-world professional context.

50.2. Learning outcomes

- 1. Learning outcome:** Professional standard
You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide advice to your stakeholders even in complex and uncertain contexts. You substantiate and validate future-oriented choices by use of ethical, intercultural, and sustainable arguments.
- 2. Learning outcome:** Personal leadership
You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show that you are capable of carrying out these actions and achieving your goals, adjusting them as necessary.
- 3. Learning outcome:** The Security Specialist
The student demonstrates expertise and competence in a self-chosen professional security role. This is evidenced through a portfolio containing professional products and outcomes of learning activities, accompanied by explanations of key aspects, as well as personal evaluation and reflection.

Explanation: Expertise and competence refer to the student's ability to perform, explain, and evaluate typical tasks and deliverables associated with the chosen role. This includes:

- * Applying technical skills using relevant technologies and tools,
- * Selecting and following appropriate standards and procedures,
- * Producing professional reports,
- * Collaborating within and outside the organization,
- * Addressing ethical considerations,
- * Sharing knowledge and advising peers.

The chosen professional role must reflect an existing cybersecurity profession with defined responsibilities and

tasks. Pre-defined options include: red teamer, blue teamer, security developer, and security officer or consultant.

4. Learning outcome: The Researcher and Developer

The student demonstrates the ability to research and develop (R&D) innovative solutions within an open and complex context. This is shown through a portfolio containing professional products and results of learning activities, including explanations of key aspects, as well as personal evaluation and reflection.

Explanation: Research involves:

- * Defining a structured research approach with clear research questions, using a combination and triangulation of diverse research strategies, and applying a methodical process to answer those questions;
- * Justifying choices and considering alternatives during research or project execution;
- * Planning and carrying out the transfer of R&D outcomes to clients, stakeholders, target groups, the broader community, and society.

Development includes:

- * Applying technical development skills;
- * Learning and implementing new technologies or technological concepts;
- * Demonstrating project management skills in your role as a developer and innovator, in collaboration with peers, stakeholders, and experts.

Innovative solutions are those that clearly contribute in a new way to the needs or challenges of stakeholders or target groups. This includes addressing technical, functional, organizational, societal, sustainable, and ethical considerations.

An open and complex setting refers to the student's ability to handle situations that begin with an undefined scope or uncertain outcomes, and to contribute to the development of a structured yet flexible process leading to a clearly defined, innovative solution.

50.3. Expected entry profile

The standard preparation semester is main Cyber Security Essentials.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes					
Infrastructure	2	2	2	2	2
Software					
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

50.4. Outflow profile

The outflow is level 3 competences in analysis, design, realization, advice, and manage and control. The semester prepares you for a graduation internship.

Domain description HBO-i 2023

						Manage & control
						Realise
						Design
						Advise
						Analysis
User interaction						
Organisational processes	2	2	2	2		
Infrastructure	3	3	3	3	3	
Software						
Hardware-interfacing						

Professional Development BSc 2020

						Investigative problem solving
						Targeted interaction
						Personal leadership
						Future oriented organisation
Professional development	3	3	3	3		

50.5. Examination and final grading

50.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

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50.7. Learning activities

The global learning activities are:

- a security specialisation, in which a student learns and show to be able to perform the typical tasks and responsibilities of a self-chosen security professional role.
- a group project, in which the student learns and shows research and development skills, from innovation and working with stakeholders to technical research and development.
- a personal project, in which the student has maximum freedom to choose personal learning goals.

50.8. Learning materials

The Canvas course contains the relevant study materials and references to and ideas about where to look for materials. The students are expected to also look for relevant educational materials on a number of topics themselves.

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51. INFORMATION ABOUT [AV-EAI] ENHANCED AI TECHNIQUES (30EC)

Artificial & Convolutional Neural Networks, Deep Learning, Reinforcement Learning & NLP

51.1. Content

This document describes the semester Enhanced AI Techniques (which was until summer 2025 termed semester "AI-Advanced"). Because the field of AI is constantly evolving, so does this semester. But the emphasis is always on **deep learning** algorithms and techniques. From students who choose this semester, a profound basic knowledge of AI techniques and principles is therefore expected at the start.

51.2. Learning outcomes

1. **Learning outcome:** (1) Data preparation; You are able to prepare and store a given dataset in such a way that it can be used in your data analysis and/or modelling.

Explanation: Preparing a dataset consists of determining the right sources, collecting the useful data (subsets) from those sources, merging them together and cleaning the data according to theories of data quality, in such a way that the process of cleaning and preparing those data is repeatable, transparent to others, and the results are suitable for data analysis and/or modelling. Storing a dataset includes investigating cloud solutions and arguing whether they should be used taking stakeholders' interests (including availability and costs) into account.

2. **Learning outcome:** (2) Data analysis & model engineering; You are able to reliably apply machine (including deep) learning algorithms and other data analysis techniques to a prepared dataset.

Explanation: Reliably refers to the process being reproducible, results being explainable to humans and based on decision making that is considered fair, whilst eliminating bias. Deep learning algorithms include Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Reinforcement Learning (RL) and Natural Language Processing (NLP). Applying consists of the following iterative steps: 1) exploratory data analysis; 2) feature engineering including normalization 3) select the most appropriate data analysis techniques; 4) model tuning with respect to recall, precision and accuracy using grid search and cross-validation.

3. **Learning outcome:** Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide advice to your stakeholders even in complex and uncertain contexts. You substantiate and validate future-oriented choices by use of ethical, intercultural, and sustainable arguments.

Explanation: Appropriate communication means reporting and/or presenting the approach, process or outcomes of a data analysis in a methodologically sound way. You can reflect on the effect of your communication and based on that reflection you define steppingstones ahead on the task, on the role and on the projected results. You effectively manage cultural differences in your interactions with all involved. Interaction about data analysis means that based on your message, your position and the different types of collaboration partners you address, you choose the right channel and format to communicate results, including appropriate data visualizations (such as data stories, infographics, or (a set of) static or interactive plot(s)).

Assess the context means the organizational and societal environment in which the data analysis project takes place. You show that you can identify the hallmarks and roles of the environment of the assignment and have a keen eye for a future-oriented, sustainable embedding of your work in an organization and society. Multiple perspectives include social and ethical considerations, law compliance, organizational data maturity, alignment with sustainable development goals, recognizing own boundaries and those of others and acting accordingly. Reflecting on ethics and governance of AI-based automated decision-making will be an important and integral part of your learning process. A viable business case adds value for the stakeholder (instead of simply using some algorithm). For machine learning, this means that the data analysis should have a predictive character (not hindsight but foresight). Sustainable manner means you take long-term effects (people, planet, profit, purpose) of your solution into account.

Identifying the problem means you can formulate research questions in order to determine the aim of your solution in an unstructured and multidisciplinary context. Effective approach means you use a systematic approach (such as the IBM Data Science Methodology) that comprises 4 phases/components: 1) project proposal 2) data preparation 3) data analysis/modelling and evaluation 4) results and deployment. Coming up with evidence-based solutions means you compose and pursue applied research methods (such as the DOT framework) and approaches based on reliable and verifiable sources. You are able to methodically and creatively find answers to applied research questions, considering alternatives and critically analysing your own and others' line of reasoning.

4. **Learning outcome:** Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show that you are capable of carrying out these actions and achieving your goals, adjusting them as necessary.

Explanation: Entrepreneurial mindset includes being aware, seeing opportunities and seizing them, motivating oneself and others, being able to profile oneself, a team and others. Learning capacity means guiding your own development and study progress, showing leadership and taking responsibility, enhancing ones' own learning capacity, demanding and giving active feedback, all with respect to the learning outcomes. Professional ambitions

means you are examining what type of professional you want to be in the long term, which field and type of position you aspire to and how you can stand out from others in field of AI/data.

5. **Learning outcome:** (5) Graduation preparation; You create chances to acquire and define an internship assignment based on a match between your ambitions, the school's requirements and the field of expertise related to your profile or specialization.

51.3. Expected entry profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction			1	1	
Organisational processes				1	
Infrastructure				1	
Software	2	2	2	2	2
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

51.4. Outflow profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction			2	2	
Organisational processes				2	
Infrastructure				2	
Software	3	3	3	3	3
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	3	3	3	3

51.5. Examination and final grading

51.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

51.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

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- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
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This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

51.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

51.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

51.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

AI is not a replacement or threat, but a collaborative actor in your learning. We use AI to strengthen creativity, deepen understanding, and expand educational possibilities. You are expected to learn to work with AI effectively and productively: a core professional competency for ICT professionals.

The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available, including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

51.7. Learning activities

This (S6/7) semester "Enhanced AI Techniques" focusses strongly on more advanced machine learning (ML) algorithms and techniques than were provided in the (S2) semester "AI Basics" and the second year (S3/S4) semesters "Applied Generative AI" and "AI, Machine Learning and Data".

The focus in this semester lies on such deep learning techniques as:

- Artificial Neural Networks (ANN)
- Convolutional Neural Networks (CNN)
- Reinforcement Learning (RL)
- Natural Language Processing (NLP)

Within this course, the terms 'machine learning' and 'deep learning' are understood as "a set of modern computer algorithms with which a structured data analysis can be conducted". Deep learning refers to neural network models in particular. Just enough mathematics for understanding these algorithms will be provided where needed. We consider mathematics as a useful means, which can provide a deeper insight in the underlying principles of an algorithm, rather than a goal on itself.

Students are expected to clean data in such a way that the process of cleaning is structured, explainable and repeatable. In order to explore and discover what data quality means, students are encouraged to go beyond the standard data visualisation techniques taught in the earlier (above mentioned) AI semesters. The focus of this Enhanced AI Techniques semester is on unstructured data in the format of images, video, speech, and or written text.

Moreover, the 'data practitioner' in this semester should be able to reflect on their personal professional development, including but not limited to: applying correct research methods & skills, being able to collaborate well with other people in the field of artificial intelligence / applied data science, communicate effectively verbally and in writing about research ideas and findings, and skills that challenge the student in these fields as to ensure an environment in which the student can learn how to reflect on them, and how to improve in them.

Students are strongly encouraged to use the data science methodology, as created by IBM Analytics and to use the steps and deliverables as described in CRISP-DM.

The semester is roughly divided in 3 parts that are intertwined through the course of the semester:

First, the **Core program** of this semester is composed of a recap of the basic machine learning algorithms from the second year of the study, workshops in the four above mentioned fields of deep learning (ANN, CNN, RL, and NLP), as well as workshops in the fields of:

- Data Visualisation
- Ethics & Law

Second, around this Core programme is a (group)**Project** in which students approach a relevant real world case that requires the application of the theories and skills acquired in the before mentioned workshops. This Project forms the heart of the semester and should provide the student with abundant opportunities to work on their learning outcomes.

Third, starting about halfway the semester, is a smaller **Data Challenge**, in which students work either alone or in duo's on an AI topic of their own choice. Especially in case the (group)project doesn't challenge the student enough to investigate certain artificial intelligence topics (e.g., like Natural Language Processing), this Data Challenge provides ample opportunity to show the learning outcomes at the appropriate level. The choice for the challenge topic could be one of the fields of the Core programme that they intend to study more thorough, or a completely different field that they intend to explore. The course provides some starting points with references to study materials in other interesting AI topics and the goal is to expand that 'library' as we go.

With regard to support from the tutors:

For the group project, each student group will get a **technical tutor** and a **process tutor**, which they will meet by default

every week. The latter also functions as the **semester coach**, and helps the student in planning and managing the semester demands. For the Data Challenge, each student will have **2 challenge tutors**, guiding their work. Before starting the Data Challenge, the student formulates a plan and discussed this plan with their tutors. As mentioned above, in case the student had not reached sufficient proficiency in one or more of the semester learning outcomes, the Data Challenge should include the possibility for the student to become proficient in them before the end of the semester.

The didactic form used in the Enhanced AI Techniques semester is **Research-Based Learning** (see Canvas). The basic rule is to have optimal use of teaching time **on location**. Therefore, by default, all the workshops will be given physically in building TQ and **not online**.

51.8. Learning materials

A Canvas course including modules with slides, assignments, and references to external sources and free online study materials (for example, MOOC) is available.

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52. INFORMATION ABOUT [AV-GP] GAME PRODUCTION (30EC)

52.1. Content

Game Production builds upon your *Game Development* and *Game Design* experience, elevating it to a more professional level. You learn to work together in a larger studio context and work with external visual design interns. You'll prepare for a Dragons' Den event, pitching your game concept to industry experts. This demands a more professional, structured approach and project scoping to produce a *vertical slice*—a polished yet limited segment of your game—in 10 weeks. Alongside a compelling pitch, a playable demo, and some market research, you might just persuade the Dragons to support or pivot the continuation of your game's development to an alpha release at the end of the semester.

The semester also offers an opportunity for an in-depth exploration of a chosen topic, allowing you to specialize and contemplate your focus for the rest of your studies at Fontys ICT.

Ambitious students also get the chance to start a real indie game studio, assisted by a network of game industry professionals (restrictions apply).

52.2. Learning outcomes

1. Learning outcome: Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide advice to your stakeholders even in complex and uncertain contexts. You substantiate and validate future-oriented choices by use of ethical, intercultural, and sustainable arguments.

Explanation: Responsibility: You manage planning and monitoring of your project effectively, adapting to the stakeholders' way of working and context. You take a critical look at your project from different perspectives such as target-user needs and stakeholder requirements, and find appropriate solutions.

Relevant methodologies: You identify relevant challenges of the client and use a variety of research strategies, methods, and activities (such as those in the DOT Framework) in a structured way to find justified answers to these challenges.

Advice: You describe and document your work effectively, so that your work is reproducible and transparent, and your outcomes can be used by others. You have the communication skills to inform and advice stakeholders within your project.

Future-oriented choices: When applicable, you consider and act according to ethical standards, sustainable practices, and work towards responsible ICT solutions. Developed solutions are validated in appropriate settings with the stakeholders.

2. Learning outcome: Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show that you are capable of carrying out these actions and achieving your goals, adjusting them as necessary.

Explanation: Independently formulate goals and actions: You reflect on your own learning ability and actions, using feedback that you asked and received. You use this reflection and feedback to look for opportunities that further your development.

Leadership: You can take the lead in planning and setting goals for your own project and development. You take responsibility for your role in a group project.

ICT professional: You know which role you envision in game development or the wider IT-landscape and what role you play in a team. You can justify why this role is a good fit for you as a professional.

Achieving your goals: You can prepare for your graduation internship or choice of minor that suits your personal development goals. This means exploring possible positions, communicating with the correct parties and analyzing the scope and fit of the assignment.

3. Learning outcome: Analysis

You are able to obtain desired results(1) by researching game related subjects (design and technical) methodically(2) and thorough(3) using the trends and developments(4) in the industry.

Explanation: 1. With desired results we mean that you can clearly define the relevant research question at hand to scope the problem at hand to a manageable scale. This includes being able to define a design challenge where you take the client, context and audience into account. 2. Methodically means you show that you use justified research methods such as the DOT research framework. 3. Thorough means that you triangulate using different methods of analysis and validate your insights. 4. Trends and developments: You are up to date with new trends and developments within the game industry such as: software algorithms, engines, game genres, new hardware, business models. You know when and how to use these in your products. Contribution to competence growth: UI.AN.3, SE.AN.2

4. Learning outcome: Advice

You are able to make recommendations(1) about the game design process and important decisions therein based on the conclusions and results from the analysis and clearly communicate(2) these recommendations to relevant stakeholders.

Explanation: 1. With recommendations we mean suggestions that you make that will help achieve desired outcomes and long-term goals. 2. With clearly communicate we mean that you communicate with the relevant stakeholders in an appropriate way. This includes written reporting and pitch presentations. Contribution to competence growth: UI.AD.3, SE-AD.2

5. Learning outcome: Design

You show that you can design a complex game in an analytical(1) way using existing frameworks, theories, and models(2). You can validate(3) game systems for both functional and aesthetic requirements using industry standard methods and deliverables(4).

Explanation: 1. Analytical means that your work is based on prior analysis insights. You support your design choices with sources such as: Target User Research, Existing products analysis, available technology research, stakeholder analysis. 2. With using existing frameworks, theories, and models we mean frameworks such as: UML, ERD, Design Patterns, MDA framework and Self Determination Theory as well as concept development methods such as diverging, converging and brainstorming. 3. You validate your designed game systems in an early stage using efficient and quick prototyping methods such as paper prototyping, roleplaying and wizard of Oz and use the results to refine your design. 4. Common industry standard methods and deliverables are: Game Design document, Tech design document consisting of elements such as UML diagrams and software architecture diagram and prototypes such as paper prototypes, mock-ups, and user tests. Contribution to competence growth: UI.DE.3, SE.DE.2

6. Learning outcome: Realisation

You show that you can develop(1) and validate(2) game prototypes, based on a design(3) where you create, find or adapt(4) audio-visual and narrative assets(5) using innovative and relevant technology(6) in a complex context.

Explanation: 1. With develop we mean that you are able to create technical prototypes based on prior work or from scratch. 2. You are able to validate these prototypes to verify the quality of the product using for example user tests and heuristics. 3. With based on designs we mean that the prototypes are based on prior designs. 4. With create, find or adapt we mean that you can create basic assets, but more realistically will find existing assets and alter them to suit your aesthetic and technical needs. The goal is that you seek out a balance in creating assets from scratch and finding existing work. 5. With audio-visual and narrative assets, we mean all 'art' assets for your games such as: 2d art, 3d art, synthesized and/or recorded sound effects, (dynamic) music, story, scripts (dialogue). 6. You use innovative and relevant technology that suit your needs such as VR headsets, Mobile devices, Game Controllers, Industry Standard Game Engines. Contribution to competence growth: UI.RE.3, SE.RE.3

7. Learning outcome: Manage and Control

You are able to work in a structured and methodical way both individually as well as in a group context(1) where you use relevant processes(2) and tooling to facilitate your iterative process in order to achieve a desired level of quality(3).

Explanation: 1. With both individually as well as in a group context we mean that you show you can adjust the process accordingly. 2. With relevant processes and tooling we mean processes and tools that are industry standard in order to effectively communicate and cooperate such as Scrum, version control tools and build management tools. 3. With quality we mean that you are able to define a level of quality to adhere to in accordance with stakeholders. Non-functional requirements such as stability, performance, security could be used to define these criteria. This includes elements such as automated tests, code reviews and continuous integration and building setups. Contribution to competence growth: UI.MC.3, SE.MC.3

52.3. Expected entry profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	2	2	2	2	2
Organisational processes					
Infrastructure					
Software			2	2	2
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

52.4. Outflow profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	3	3	3	3	3
Organisational processes					
Infrastructure					
Software	2	2	2	3	3
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	3	3	3	3

52.5. Examination and final grading

52.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

52.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- *Undefined*: You have not yet undertaken any activities to demonstrate the learning outcome.
- *Orienting*: You have made a start and explored possibilities to demonstrate the learning outcome.
- *Beginning*: You have taken initial steps that contribute to demonstrating the learning outcome.
- *Proficient*: You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome at a satisfactory level if you continue your development in this way.
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This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

52.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

52.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

52.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

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52.7. Learning activities

Game Studio Project (with industry partner)

Students form their own group with the goal to start a multidisciplinary game design team, in essence their own game company (including company name, logo, website). The game studio offers their services to an industry partner. The group will define an overall design challenge in accordance with the client and design, build, and validate a transferable prototype.

In addition, the group can define, design, build, and publish an entertainment game based on their own ideas. This structure allows students to prepare for possible incubator and graduation projects that can continue after the semester. The group is expected to find an external stakeholder/critical friend who can critically review the quality of the project.

Deep Dive (individual)

In order to allow students to dive into a topic that falls outside their role or scope of the Game Studio project, they work on a two-week project called the Deep Dive. Students can schedule their own time on the project. They present their insights of the Deep Dive in an in-depth transferrable product.

Workshops

During the semester a range of workshops on both game design and game technology as well as professional development skills are offered in the form of workshops. Students are expected to incorporate elements of these workshops in the deliverables for both group projects and show their individual contribution on their portfolio.

Way of working

The entire semester is organized as a project using scrum. In daily standups each project is discussed on process level. In bi-weekly sprint demo's the projects are evaluated.

Every scheduled day there will be experts and tutors available. There will be regular community meetings to learn from others and exchange solutions to problems. Also there will be events, like conferences and meetups where students are invited to join.

Blended education

The semester works with blended learning, whereby, in addition to the activities on location, online education can take place that matches content and didactics, or when there are planning and scheduling issues. The basic rule is to have optimal use of teaching time on location. The blended set-up for the semester is explained in the Canvas course. The decision to have parts of the education online is reserved for the degree programme.

52.8. Learning materials

Canvas course including all linked information.

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53. INFORMATION ABOUT [GRAA] GRADUATION FALL (30EC)

53.1. Content

Your last semester of your bachelor study is the graduation internship. During that period, you demonstrate in (professional) practice that you have integrated the required professional and technical IT development on bachelor level and that you can apply it independently. By getting to know the company you are graduating at, it is also the ideal opportunity to know at which company to quickly start your professional career.

Internships are about working and learning in a professional and authentic practice. It is an assignment/work that:

- you perform at an institution or company;
- is in an IT field that you already have knowledge about.

- is acquired by you and fulfilled by you in an independent manner, you are the projectleader.
- results in a final portfolio including a presentation and an IT product.

Duration internship

The internship lasts a whole semester: regularly from Fontys week 1 to Fontys week 18. The assessment takes place at the end of the semester (Fontys week 19 or 20). Holidays and public holidays that fall within the semester count as internship weeks. So you have no school holidays this semester. Days off that you accumulate according to your internship agreement can be taken in consultation with the company, including company closures and mandatory holidays.

Sick leave

Let's hope not, but it is of course possible that you become ill during your internship or miss days on your internship for some other reason. Because you basically continue to work on holidays, you have some leeway. Being ill for a few days or taking leave a few days for another reason does not have to be a problem to achieve the goals of your graduation internship.

Internship permission

In order to start the graduation internship, you must meet the admission criteria. This means you must have obtained all study points preceding the graduation semester. Also you must have a graduation project proposal that is reviewed by two reviewers and that you have aligned with the company/institution of your choice.

It is recommended to start searching for a graduation project in time. This would normally be at the beginning of the semester prior to your graduation semester.

53.2. Learning outcomes

1. Learning outcome: Professional duties

You carry out professional duties at a bachelor level, producing professional-grade work aligned with the IT subdomain (IT-area) you are specialising in.

2. Learning outcome: Situation-orientation

You apply your previously acquired knowledge and skills in a new, authentic context to deliver valuable and relevant results for both the project and the company.

3. Learning outcome: Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide advice to your stakeholders even in complex and uncertain contexts. You substantiate and validate future-oriented choices by use of ethical, intercultural, and sustainable arguments.

4. Learning outcome: Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show that you are capable of carrying out these actions and achieving your goals, adjusting them as necessary.

53.3. Expected entry profile

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	3	3	3	3

53.4. Outflow profile

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	3	3	3	3

53.5. Examination and final grading

53.5.1. Assessment: Holistic and Longitudinal

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53.5.4. Retakes and repair

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AI as a Collaborative Partner

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The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

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53.7. Learning activities

Key moments of examination

Week 3-5	Projectplan/projectdocument and company visit by first assessor (onsite or online). The first assessor provides feedback during the first company visit in week 3-5, and the assessors give feedback on the projectplan/project document. (Formative assessment)
Week 9/10	Intervision moment at school, sharing results, status and plan so far and receiving and reflecting on feedback. Company supervisor and assessors give feedback and feedforward, also based upon judgment from the student on status and progress. (Formative assessment)
Week 10-17	Hand in portfolio products with a request for feedback; formative assessment.
Week 17	Hand in portfolio. (Deadline Tuesday in week 17).
Week 19-20	Final presentation and/or demonstration of the final product, combined with accountability about the execution and evaluation of the process. Summative assessment by first and second assessor after advice from the other jury members (external expert and company supervisor). The session is public. Interested public do not have to register in advance. In case the student prefers a private session due to personal circumstances or company confidentiality reasons, the student must contact the first assessor no later than week 10 of the school's schedule to register the request in Canvas.

The internship is largely independently acquired by you as a student and carried out at an external client. A Fontys lectorate, staff-department or other Fontys organization is viewed as external as well. Also, a graduation internship at your own company or startup is allowed if it involves external stakeholders and a qualified external company supervisor. The full details on the criteria for the assignment, the company and company supervisor can be found on the FHICT policy wiki. Acquiring an interesting and relevant internship assignment requires entrepreneurship on your part; we expect your own initiative when formulating an assignment and executing it.

During your internship you will be supervised by someone from the company; your company supervisor. In addition, you will receive guidance from school by an internship teacher (the 1st assessor). He mainly guides you process-wise and is co-evaluator at the end. You will also be assigned a 2nd assessor. This is the chairperson during the criteria based interview and also co-assessor. During your internship, your internship teacher is your first point of contact from school and also acts as a semester coach.

53.8. Learning materials

There is no standard teaching material available for the graduation internship. You can, however, use the documents that you can find on Canvas. Here you will find, among other things, the research framework and support for writing your portfolio.

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54. INFORMATION ABOUT [GRAB] GRADUATION SPRING (30EC)

54.1. Content

The graduation internship marks the final phase of your professional education as Bachelor. This is your chance to step into the role of project leader for a graduation project of your choice. During this phase, you will showcase your ability to work as a junior professional in your chosen IT field, using the experience and knowledge from previous semesters.

Key objectives:

- **Independent project execution:** Apply your skills to independently manage and complete a significant project within a professional IT environment.
- **Learning and growth:** Discover and implement new methods and technologies relevant to your IT specialization and the specific domain you are working in. Use the interactive community of the graduation table and formative assessments to improve your work.
- **Guidance and support:** Receive assistance from your company supervisor (focusing on domain and content) and your university mentor (focusing on process and practice-based research). As member of one of the graduation tables you participate in plenary and bi-weekly meetings to gather feedback from assessors and peers.

Working with graduation tables

During this graduation period, we use the concept of graduation tables: a group of people who have a common focus. This group of people consists of students graduating around the same theme and teachers who are involved in coaching the students. The 'table' meets several times during the semester to share progress using various working methods. In this way, (peer) supervision takes place and calibration among coaches is also encouraged. While working on your individual project, use the table meetings to learn from each other and help each other, because together you get further than alone.

54.2. Learning outcomes

1. Learning outcome: Professional duties

You create professional products at bachelor's level within your IT subdomain, in line with the standards and working methods customary in the sector, and you will act accordingly in relevant complex professional situations.

Explanation:

Professional products: tangible results of professional IT work, such as software, analyses, designs, prototypes, or technical documentation.

Bachelor's level: Acting in unfamiliar, changing situations, with substantiated choices, integration of knowledge and solving more complex issues.

IT subdomain: the chosen area of specialisation within ICT, such as software engineering, AI, security, data, UX, etc.

Standards and working methods: common standards and methodologies in the work field of the relevant subdomain (e.g. coding standards, architectural principles, security guidelines). Complex professional situations: situations involving multiple stakeholders, uncertainties or dependencies in which considerations and professional judgement are required.

2. Learning outcome: Situation-orientation

You apply your knowledge and skills in an authentic professional context, where your activities and choices are aligned with the needs, processes and standards of the organisation and lead to demonstrable value for stakeholders.

Explanation: Explanation:

Authentic professional context: a real working environment (e.g. company, organisation, research group).

Needs, processes and standards of the organisation: what the organisation needs, how it works and what internal quality standards or procedures apply.

Demonstrable value: an effect that is visible and recognised by stakeholders (e.g. a validated improvement, prototype or recommendation).

Stakeholders: individuals or groups who have an interest in the outcomes of your project, such as customers, users, colleagues or clients.

3. Learning outcome: Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide advice to your stakeholders even in complex and uncertain contexts. You substantiate and validate future-oriented choices by use of ethical, intercultural, and sustainable arguments.

Explanation: Explanation

ICT issue: A problem with multiple causes, uncertainties, dependencies or possible solutions.

Relevant method: applied research to investigate an issue in a practical and structured manner, appropriate to the context.

Advice: a detailed recommendation for follow-up steps, solution or decision-making.

Choices: professional considerations based on evidence and arguments.

Ethical, intercultural and sustainable arguments: reflections on the impact on people, society, safety, the environment, privacy and long-term consequences.

4. Learning outcome: Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show that you are capable of carrying out these actions and achieving your goals, adjusting them as necessary.

Explanation: Explanation:

Goals: goals focused on your growth as an IT professional, such as skills, knowledge or professional behaviour.

Demonstrating leadership in development: independently planning, monitoring and implementing activities that contribute to your development and project results.

Making adjustments: reviewing or adjusting your plans based on reflection, feedback or changed circumstances (e.g. changing scope, new insights, unexpected obstacles).

54.3. Expected entry profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	3	3	3	3

54.4. Outflow profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	3	3	3	3

54.5. Examination and final grading

54.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

54.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome at a satisfactory level if you continue your development in this way.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations and are focused on continuous improvement. You will demonstrate the learning outcome at a more than satisfactory level if you continue your development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

54.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

54.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

54.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

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54.7. Learning activities

Part	Week	Student Activities	Teacher Activities	Key Milestones	Chairperson Activities
PART 1	0	Participate in MS Teams Kick-off. Introduction to graduation process	Familiarise with Canvas course and semester outline.	Kick-off meeting	Organise and host MS Teams Kick-off
	1	Start graduation project	Supervision starts. Assessors meet with each student.	Orientation	Organise dates for Preparation students QA and share details in prep-course (via TO).
	2	Project work	Supervision	—	-
	3	Submit Snapshot 1 and Prepare for Table Meeting 1 + Plan alignment meeting with company mentor & assessor (in wk 3-5)	Review and duo-calibrate portfolios, write feedback. Communicate preparation guide for Plenary Meeting.	Snapshot 1	-
	4	Participate in Table Meeting 1	Conduct Table Meeting 1	Table meeting 1	Open and close speech in meeting
PART 2	5	Project development	Supervision	—	Review incoming messages from prep students and if suitable for table, provide quickscan on their proposal
	6	Project development	Supervision	—	Host 1st MS Teams meeting for prep students. Format is 60min QA session.
	7	Project development	Supervision	—	-
	8	Submit Snapshot 2 (incl. midterm review) and Prepare for Table Meeting 2	Review and duo-calibrate portfolios, write feedback. Communicate preparation guide for Plenary Meeting.	Snapshot 2	-
	9	Participate in Table Meeting 2	Conduct Table Meeting 2	Table meeting 2	Open and close speech in meeting
	10	Project development	Supervision	—	Organise and host 2nd MS Teams meeting for prep students. Format is 60min QA session and inform SD which students (with 1st and 2nd assessor) are starting graduation next semester (of week 11)
	11	Project development	Supervision	—	-
	12	Submit information for conference. Project development	Supervision	—	Review incoming messages from prep students and if suitable for table, provide quickscan on their proposal
	13	Submit Snapshot 3 and Prepare for Table Meeting 3	Review and duo-calibrate portfolios, write feedback. Communicate preparation guide for Table Meeting.	Snapshot 3	-
	14	Participate in Table Meeting 3	Conduct Table Meeting 3	Table Meeting 3	Open and close speech in meeting
PART 3	15	Submit final portfolio version + concept poster. Plan alignment	Review portfolios and posters	Final submission (cut-off)	Review incoming messages from prep students and if suitable for table, provide quickscan on their proposal

Part	Week	Student Activities	Teacher Activities	Key Milestones	Chairperson Activities
		meeting 2 with company mentor & assessor (before portfolio conversation)			
	16	Participate in portfolio conversation.	Conduct portfolio conv. with student and write feedback (2 assessors).	Portfolio assessment	-
	17	Participate in portfolio conversation.	Conduct portfolio conv. with student and write feedback (2 assessors).	Portfolio assessment	Connect prep students with assessors and hold a 30min calibration meeting with assessors online.
	18	Prepare for graduation conference	Conference preparation	—	-Check if all prep assignments have been approved by assessors in Onstage. -Create Excel sheet with overview of student grades.
	19	Present poster at Graduation Conference. Submit archived Snapshot Portfolio (Week 15 version). Submit final poster in GradeWork	Visit conference. Support students. Confirm final grades in Table Assessors Meeting	Conference & final grading	-Provide short 10min opening speech during Graduation Conference for own table group. - Hold 30min talk with present external experts about the quality of assessed work globally. - Hold a short online calibration meeting with all assessors to confirm summative grading and fill the grades in Gradework. Inform SD which students (with 1st and 2nd assessor) are starting graduation next semester (of week 11)

54.8. Learning materials

All teaching material can be found in the graduation Canvas course.

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55. INFORMATION ABOUT [AV-HIT] HI-TECH EMBEDDED SOFTWARE ENGINEERING (30EC)

Hi-Tech Embedded Software Design & Solutions

55.1. Content

The unit of learning outcomes Hi-Tech Embedded Software Engineering, is the final preparation for graduation.

The semester focuses on performing authentic, methodically structured, professional tasks, which contribute to new insights for the professional field. The focus is on learning, by carrying out research and development activities, within practice-oriented research. Students are (mainly) responsible for setting up and monitoring their own learning path.

The research projects of this semester are based on practical problems from our Partners in Innovation and from the lectorates and fit the theme of this semester: "High Tech Embedded Software Engineering".

During the semester you will work on a research project from a Partner or Lectorate, and a personal project.

The semester is concluded by the Innovations Insight Fair in week 18, at which the results of all research projects are presented.

55.2. Learning outcomes

1. Learning outcome: Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide advice to your stakeholders even in complex and uncertain contexts. You substantiate and validate future-oriented choices by use of ethical, intercultural, and sustainable arguments.

2. **Learning outcome:** Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show that you are capable of carrying out these actions and achieving your goals, adjusting them as necessary.

3. **Learning outcome:** Analysis

You specify a distributed computer system including timing, resource use and performance, taking into account safety aspects. And you compile an acceptance test plan and an integration test plan.

Explanation: Based on a practical problem, you identify the direction of the solution and an appropriate approach.

A distributed computer system means: a system that consists of multiple hardware and software components, including networks, that work together in a coordinated way for a common goal.

Acceptance test plan means that you define a plan and methods in advance for validating your solution strategy and implementation (for sub-problems).

Integration test plan means that you define in advance a plan and methods that ensure that subsystems connect seamlessly during design and realization.

4. **Learning outcome:** Advise

Based on your analysis, you provide technical advisory on the (distributed) computer system that is to be realised, including hardware and software components and links.

Explanation: You can justify your advice for the direction of the solution towards your stakeholders. Together with a plan of action to come to solutions.

Technical advice means: a substantiated advice for solving the problems identified during the analysis and design.

5. **Learning outcome:** Design

You design a distributed computer system including determining actuators, sensors, timing, resource usage and performance.

Explanation: You design a system consisting of hardware and software components based on your advice, and substantiate this design. You use design standards, methods, techniques and quality standards that are relevant in the work field. When designing, you take into account correct decomposition and abstraction (for example, design by contract) with the aim of making reusable components, and a correct interaction between components. The software design includes both structural (static) and behavioural (dynamic) aspects. A distributed computer system means: a system that consists of multiple hardware and software components, including networks, that work together in a coordinated way for a common goal.

6. **Learning outcome:** Realisation

You realise a complete computer system (or parts thereof) including network, hardware and system software based on your own design. And you compile and carry out an acceptance procedure to validate the implementation.

Explanation: You translate your advice and your design into an implementation. You take care of the quality of the implementation. You create code that is reliable, reusable and transferable. You also take into account aspects such as timing, resource use and performance. When implementing, you make use of standards, methods and techniques relevant to the work field. And you ensure proper integration with existing code.

7. **Learning outcome:** Manage & Control

You set up and make use of a system for version management, release management, teamwork support and automated testing for hard- and software systems.

Explanation: Version control means: managing various versions of system components and changing components in a controlled manner using tooling (e.g. GitLab), with the aim of ensuring that changes made to components by developers do not interfere.

Release management means: the controlled and planned preparation of software for a release. And tracking component and system versions released for customer use.

Automated testing means: the application of tools in a software development process for continuous and automated software testing and reporting of test results.

55.3. Expected entry profile

To prepare you for this semester, you need to have finished a main semester and an internship, that both cover a good balance of hardware interfacing and software competences.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes					
Infrastructure			1	1	
Software	1	1	2	2	2
Hardware-interfacing	2	2	2	2	2

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

55.4. Outflow profile

This semester prepares you for the graduation semester.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes					
Infrastructure			1	1	
Software	2	2	3	3	3
Hardware-interfacing	3	3	3	3	3

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	3	3	3	3

55.5. Examination and final grading

55.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

55.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

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55.7. Learning activities

With this semester, students choose a learning environment with a flexible learning pathway, and a high degree of self-management and learner agency. Students make choices for their own learning pathway and choose research activities, development activities and study materials which fit the chosen research project. The learning content is strongly dependent on the selected projects as well. Because everyone has his own track, it is impossible to plan a curriculum that fits for everyone.

Learning happens asynchronously for these reasons: not every student does the same things, and the pace of students also differs. The result of this is that (almost) all workshops, lectures, etc., that offer content are scheduled on-demand or when coaches detect it is necessary.

Workshops of more general applicability are offered to all students, such as workshops about the student journey, the scrum process, making substantiated choices and validation of learning and delivered work (research skills), design, code quality and transferability.

Blended education

The semester works with blended learning, whereby, in addition to the activities on location, online education can take place that matches content and didactics, or when there are planning and scheduling issues. The basic rule is to have optimal use of teaching time on location. The blended set-up for the semester is explained in the Canvas course. The decision to have parts of the education online is reserved for the degree programme.

55.8. Learning materials

Apart of the Canvas LMS environment, the InnovationLab's GitLab and MS Teams anything which would be applicable can and will be used.

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56. INFORMATION ABOUT [AV-HC] HUMAN CENTERED DESIGN (30EC)

56.1. Content

In the Human Centered Design semester, you learn to solve complex issues in an investigative and applied way where, searching for the best solution, you focus on improving user experience and user interaction. You do this by developing your technical skills to create digital interactive products with cutting edge technologies, assisted by artificial intelligence. This includes front end programming languages and frameworks to create prototypes, MVP's and your showcase portfolio.

The emphasis is on performing tasks independently (even when working together in a group), showing leadership and taking responsibility. In all these projects, the guiding principles are "user interaction", "iterative working" and "applied practical research".

You choose your own projects and bear responsibility for its success!

56.2. Learning outcomes

1. Learning outcome: User interaction (analyse & advise):

You analyse the user, the user interaction and the user experience, also taking state of the art interactive technologies into account. You select a suitable design process to be able to advise on UX interventions based on a validated UX design.

Explanation: By analyse, we mean that you carry out research and record the conclusions. By state of the art

interactive technologies, we mean that you investigate the latest technologies that may be relevant to the project. By a suitable design process, we mean that you make a justified choice for your design process and describe the different phases of this process for your project. By advise we mean that you communicate and document your decisions and choices, as well as how you substantiated them.

2. Learning outcome: User interaction (execution & validation):

You execute and evaluate the user experience of an interactive product. You document the development process for the stakeholders.

Explanation: By execute, we mean that you produce a range of prototype(s) and/or MVP(s) you can present to stakeholders and users. By evaluate user experience, we mean that you not only put forward the user -friendliness of a product, but map the total user experience. By document, we mean that you can substantiate and validate the choices you make in relation to the design process and the user experience (including its interactions). By stakeholders, we mean anyone involved in the project with an interest in staying informed.

3. Learning outcome: Realisation of technical products:

You create & design technical products with existing components or libraries using predetermined quality criteria and version control.

Explanation: By you create, we mean that during the semester you will independently build your (showcase) portfolio and realise custom-made technical products. By you design, we mean that you provide a schematic representation of every (software) product/system you create during the semester. By existing components or libraries, we mean that you have investigated which program language, framework, or library is suitable for the projects you are working on. By quality criteria, we mean that you have predetermined the criteria the system must meet in consultation with stakeholders. For example, scalability, user-friendliness, security aspects and different types of devices. By version control we mean that you use a system to store and back up your projects, track changes, control collaboration, recover previous versions, and streamline the development processes.

4. Learning outcome: Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide advice to your stakeholders even in complex and uncertain contexts. You substantiate and validate future-oriented choices by use of ethical, intercultural, and sustainable arguments.

5. Learning outcome: Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show that you are capable of carrying out these actions and achieving your goals, adjusting them as necessary.

56.3. Expected entry profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	2	2	2	2	2
Organisational processes					
Infrastructure					
Software	2	2	1	2	1
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

56.4. Outflow profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	3	3	3	3	3
Organisational processes					
Infrastructure					
Software	2	2	2	2	1
Hardware-interfacing					

Professional Development BSc 2020

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56.5. Examination and final grading

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- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome at a satisfactory level if you continue your development in this way.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations and are focused on continuous improvement. You will demonstrate the learning outcome at a more than satisfactory level if you continue your development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

56.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

56.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

56.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

AI is not a replacement or threat, but a collaborative actor in your learning. We use AI to strengthen creativity, deepen understanding, and expand educational possibilities. You are expected to learn to work with AI effectively and productively: a core professional competency for ICT professionals.

The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available, including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

56.7. Learning activities

On-site teaching: The semester is organised so that students can work at least 2.5 half-day sessions on location. On the other days, all scheduled activities take place online or students are allowed to decide where and how they work.

Blended education: Blended learning can be used during the semester where, in addition to on-site teaching activities, online teaching can take place in line with content and didactics or with planning and scheduling issues. The starting point is the optimal use of teaching time on location. The decision to offer parts of the education online is reserved for the course.

Individual contribution: Students work with fellow students, stakeholders, and lecturers throughout the semester, but are ultimately assessed individually. It is therefore important that a student can demonstrate at all times, within a collaboration with others, what constitutes individual work. This applies to the chosen study cases (which may be worked on in pairs) but also to the group project.

Group project(s): Projects for a group of students with a client/ stakeholder. The project runs for an average of 6-10 weeks (this may vary). Students prepare a project plan and decide strategies and methods to manage the project. For these projects, a Partner in Innovation & or a research group of the lectorate may be the client or fulfil the role of expert and/or consultant. In these projects students usually work with by 4 to 6 students but also this may vary.

International collaboration: During the semester, there are several opportunities for international and intercultural collaboration. This contributes to the professionalisation and personal development of the student. Themes, topics, and locations of these projects are communicated and developed in advance, but also during the semester.

Workshops and inspiration sessions: A series of optional, interactive sessions on topics from the different "focus areas" and professional skills in which you can practice various relevant techniques. You will work individually or in smaller groups. These workshops can either be prepared and planned by teachers or initiated by you

56.8. Learning materials

CANVAS contains all the teaching materials and is also where you can submit your assignments. In addition to verbal feedback, you will also receive written feedback on the assignments you have submitted and your learning report (portflow) via CANVAS. You will always have access to the assignments you have submitted, your semester report and the feedback you have received.

CANVAS also contains exploratory resources. These resources may include information sites, online courses, videos, articles, books, etc. These resources help students orient themselves to underlying theory, common methods and/or concepts.

In addition to CANVAS, we also use MS Teams as an online work environment for communication and announcements. If necessary, communication can also take place via email. This will always be done using your Fontys email account.

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57. INFORMATION ABOUT [AV-IE] IMMERSIVE EXPERIENCES (30EC)

57.1. Content

During this semester, you'll explore how to create powerful experiences through a blend of storytelling, (digital) craftsmanship and emotional engagement. Based on your own interest, you'll work with a toolkit that spans both traditional and digital methods. Think of spatial computing, AI, interactive lighting, XR-technologies, photography, lasercutting, 3D-printing, etc. You'll learn how to transform ideas into meaningful experiences, whether through a gesture-controlled installation, a mixed reality experience or a multisensory environment.

The program emphasizes both the visible and hidden aspects of immersive experience design. You'll learn how to:

- Create emotional resonance
- Develop concepts and narratives
- Design and/or develop meaningful interactions
- Integrate physical and digital elements
- Work with immersive technology
- Use data to enhance and personalize experiences

This unique combination allows you to create experiences that deeply connect with people, whether through high-tech immersive environments, tactile interactive installations or hybrid experiences that bridge multiple realities.

57.2. Learning outcomes

1. Learning outcome: LO1: Analysing the Experienter –User

You analyze and apply insights into how people perceive, feel, and remember experiences in order to design emotionally engaging and meaningful immersive interactions that are grounded in user psychology, context, and societal relevance.

Explanation: • User-Centered Analysis — You analyze user behavior by considering their physical, psychological, emotional, and perceptual perspectives

- Contextual Awareness — You Integrate user insights into broader social and ethical contexts ensuring the experience is relevant, inclusive, and socially meaningful
- Design with behavioral and sensory insight — You understand and apply knowledge of human behavior, sensory perception, memory, and emotional impact to shape the quality and resonance of immersive experiences
- Iteratively evaluate and adapt — You continuously reflect and adapt based on user feedback and emotional response and translating these experience-related insights into actionable recommendations for concept development, interaction design, and design process planning.

2. Learning outcome: LO2: Creating the Interaction – Design

You create value through innovative concept development and interaction design, by translating user insights and contextual understanding into immersive, meaningful, and testable experiences.

Explanation: • Conceptual Innovation. You develop original, relevant, and experience-driven concepts that respond to user needs and project context — pushing the boundaries of traditional interaction by integrating physical, digital, and sensory dimensions

- Interaction Design. You design immersive interactions that are technically sound, emotionally resonant, and

appropriate for the project's goals — whether through spatial interaction, gesture-based control, or multi-sensory input

- **Strategic Design Process.** You apply a structured and reasoned design process, supported by user research and context analysis, to translate abstract ideas into detailed design solutions that serve both user and stakeholder goals
- **Iterative Development and Value Creation.** You prototype, test, and refine your design to deliver measurable value and impact.

3. **Learning outcome:** LO3: Realising the Experience –Technology

You realize and evaluate immersive experiences by applying appropriate tools and technologies, ensuring the solution aligns with user needs, project objectives, and professional standards.

Explanation: • **Technical Proficiency.** You apply immersive technologies creatively (e.g. XR, AI, spatial computing) to build impactful prototypes or experiences that are appropriate to the project phase and user goals

- **Tool-Based Creativity.** You integrate both digital and physical tools in creative and meaningful ways to develop hybrid and multisensory experiences that go beyond conventional interfaces
- **Phased Realization.** You realizes experiences using phase-appropriate methods —from early mockups to functional prototypes — ensuring alignment with the iterative nature of design processes
- **Professional Delivery.** You deliver immersive products with appropriate documentation, usability, and transferability —reflecting professional standards and stakeholder readiness.

4. **Learning outcome:** Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show that you are capable of carrying out these actions and achieving your goals, adjusting them as necessary.

Explanation: You demonstrate personal leadership by

- **Taking initiative.** You'll identify opportunities, set realistic goals, and pursue them, demonstrating enterprise and a forward-thinking mindset
- **Reflection and growth.** You'll reflect on your own progress, gather feedback, and make well-informed decisions about your future direction and professional role in the field of immersive experience
- **Profiling yourself.** You choose at least one field of expertise within the domain of Immersive Experience and clarify your ambitions, your unique skills within this field, then align your actions toward these goals in your field of expertise.

5. **Learning outcome:** Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide advice to your stakeholders even in complex and uncertain contexts. You substantiate and validate future-oriented choices by use of ethical, intercultural, and sustainable arguments.

Explanation: You demonstrate professional standard by:

- **Taking responsibility and solve problems.** You'll identify, analyze, and solve complex problems within your project's context
- **Considering ethics and context.** You'll account for social, ethical, and sustainable aspects, reflecting this in your decisions and designs
- **Collaboration and expertise sharing.** You collaborate with stakeholders, validate your choices, and proactively share your expertise to foster strong team solutions
- **Organising and execution.** You'll manage your project's phases (timing, budget, delivery) while honoring ethical standards and ensuring sustainable implementation
- **Applying methodological research approach.** You use structured design processes grounded in methodical research.

57.3. Expected entry profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	2	2	2	2	2
Organisational processes	1				

Infrastructure				
Software	1		1	
Hardware-interfacing	1			

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

57.4. Outflow profile

Domain description HBO-i 2023

	Manage & control	Realise	Design	Advise	Analysis
User interaction	3	3	3	3	3
Organisational processes	2				
Infrastructure					
Software	2	2			
Hardware-interfacing	2				

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	3	3	3	3

57.5. Examination and final grading

57.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

57.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
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- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome at a satisfactory level if you continue your development in this way.
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57.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

57.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

57.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

AI is not a replacement or threat, but a collaborative actor in your learning. We use AI to strengthen creativity, deepen understanding, and expand educational possibilities. You are expected to learn to work with AI effectively and productively: a core professional competency for ICT professionals.

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Ethical Boundaries

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57.7. Learning activities

On-site teaching: The semester is organised so that students can work at least 2.5 half-day sessions on location. On the other days, all scheduled activities take place online or students are allowed to decide where and how they work.

Blended education: Blended learning can be used during the semester where, in addition to on-site teaching activities, online teaching can take place in line with content and didactics or with planning and scheduling issues. The starting point is the optimal use of teaching time on location. The decision to offer parts of the education online is reserved for the course.

Individual contribution: Students work with fellow students, stakeholders, and lecturers throughout the semester, but are ultimately assessed individually. It is therefore important that a student can demonstrate at all times, within a collaboration with others, what constitutes individual work. This applies to the chosen study cases (which may be worked on in pairs) but also to the group project, which is usually worked on by 4 to 6 students.

Group project(s): Projects for a group of students with a client/ stakeholder. The project runs for an average of 8-12 weeks (this may vary). Students prepare a project plan and decide strategies and methods to manage the project. For these projects, a Partner in Innovation & or a research group of the lectorate may be the client or fulfil the role of expert and/or consultant.

International collaboration: During the semester, there are several opportunities for international and intercultural collaboration. This contributes to the professionalisation and personal development of the student. Themes, topics, and locations of these projects are communicated and developed in advance, but also during the semester.

Workshops and inspiration sessions: A series of optional, interactive sessions on topics from the different "focus areas" and professional skills in which you can practice various relevant techniques. You will work individually or in smaller groups. These workshops can either be prepared and planned by teachers or initiated by you.

57.8. Learning materials

CANVAS contains all the teaching material and this is also the hand-in place for the elaboration of problems. In addition to oral feedback, written feedback on the submitted assignments and the learning report will be given and registered via CANVAS. The student always has access to the submitted assignments, the semester report and the feedback received.

There are also orientating sources in CANVAS. The exploratory sources can be information sites, online courses, videos, articles, books, etc. These sources help the student to orient himself on underlying theory, common methods and / or concepts.

For communication and announcements we will also use MS Teams as an online working environment in addition to CANVAS. Should it be necessary, communication can also take place via email. This is always done with and to your Fontys mail account.

Portflow is an extension of Canvas and will be used to assess your portfolio. In Portflow, students can visualise and self-account for their development progress by collecting evidence, requesting feedback from teachers, fellow students and (external) experts and creating an assessment file. See the full explanation on how to use Portflow: <https://beleidswiki.fhict.nl/doku.php?id=en:beleid:portflow>

Feedpulse is an extension of Canvas and is used within this semester to support the feedback process (capturing development-oriented feedback on several occasions during the semester). Unambiguous use of FeedPulse within the institute is of great importance to the student. See the full explanation on how to use Feedpulse: <https://beleidswiki.fhict.nl/doku.php?id=en:beleid:feedpulse>

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58. INFORMATION ABOUT [AV-IV] INDUSTRY 5.0 (30EC)

AI & Robotics, Digital Twins, Swarm Computing, Smart Cities, Supply Chain Ecosystems

58.1. Content

In this semester students will learn by setting out relevant challenges and address these in a structured and methodical manner. Challenges should always be defined SMART and create a set of clear objectives and attributes that creates the student's need to deepen his/her skills in enabling technologies and understanding of processes while setting these skills in the perspective of what they mean in terms of solving the challenge (creating or adding value). Students should also become aware that challenges might change during projects and should be able to handle these changes.

For the 'safe' Single and Dual projects the challenges are set by the combination of context and the Body of Knowledge for Industry 5.0 (Advanced Smart Industry). For the individual project the challenge is defined based on the student's self

diagnose and orienting profile while the industry project's challenge is based on the diagnose of the cases of external stakeholders.

Portfolio

As a result of his/her methodical working and attitude, during the semester the student will create a set of professional (half)products that show the

- () rigor of the processes,
- () the compliance to professionalism and professional standards, () the relevance to the context,
- () the quality of outcomes based on SMART challenge attributes and () student's created or added value.

The set of products with a reading guide is the portfolio. The portfolio is what the student uses to show how he/she has reached the learning outcomes for the semester. These learning outcomes are connected to the levels 2 and 3 of the BA in ICT domain description (HBO-i).

58.2. Learning outcomes

1. Learning outcome: Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide advice to your stakeholders even in complex and uncertain contexts. You substantiate and validate future-oriented choices by use of ethical, intercultural, and sustainable arguments.

2. Learning outcome: Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show that you are capable of carrying out these actions and achieving your goals, adjusting them as necessary.

3. Learning outcome: Analysis and Advise - You are able to find, interpret and utilize information from the (smart industry) domain and the context of your challenge(s) and/or project(s) in a methodical manner (research cycle).

Explanation: - This means that, with the collected information, you are able to identify what is relevant and useful for your challenge(s) and project(s) (context mapping). - You are able to summarize and present your findings, processes/methods and conclusions in such a manner that your stakeholders understand your added value. NOTE – summarize and present (advise) can be applied to other states (design, realisation and management as well).

4. Learning outcome: Design - Based on the results of rigor analysis you are able to create, present and defend a plan that holds a promise to solve the challenge(s) within your project and gives insight in the professional product(s) (portfolio) need for the project(s).

5. Learning outcome: Realization - You are able to bring your plan to execution while proactively monitoring and preserving the compliance ('to what extent is my work an example of professionalism?') and added value ('to what extent does my work solve the project's challenge(s)?') of portfolio products through validation and testing.

6. Learning outcome: Manage & Control: You set up and make use of a system for version management, release management, teamwork support and automated testing for hard- and software systems. Your are able to hand-over your project's outcomes (portfolio) so others can co-operate and/or work further on the project.

Explanation: Version control means: managing various versions of system components and changing components in a controlled manner using tooling (e.g. GitLab), with the aim of ensuring that changes made to components by developers do not interfere. Release management means: the controlled and planned preparation of software for a release. And tracking component and system versions released for customer use. Automated testing means: the application of tools in a software development process for continuous and automated software testing and reporting of test results.

58.3. Expected entry profile

The standard preparation semester for this advanced semester is Industry 4.0 & IoT. If you have not previously completed this semester but are interested in the topics covered, you can take the Minor Smart Industry Technologies (see the Free Study Area under Minors).

Upon completion of Industry 5.0 semester, you will achieve Level 3 competencies in analysis, design, realization, advise and management & control.

Domain description HBO-i 2023

	Manage & control								
	Realise								
	Design								
	Advise								
	Analysis								
User interaction	2								
Organisational processes	2	2							
Infrastructure	2				2				
Software	2		2	2	2				
Hardware-interfacing	2	2	2	2	2			2	

Professional Development BSc 2020

Investigative problem solving	2
Targeted interaction	2
Personal leadership	2
Future oriented organisation	2
Professional development	

58.4. Outflow profile

This semester prepares you for:

- A graduation in the area of Smart Industry & IoT.
- Further specialisation in the area of Robotics.
- Other ICT advanced programs where the gained knowledge and skills can be applied, such as, embedded software engineering, AI, software engineering, infrastructure, ERP and business processes.

Domain description HBO-i 2023

						Manage & control
						Realise
						Design
						Advise
						Analysis
	User interaction	2				
	Organisational processes	2	3			
	Infrastructure	2			2	
	Software	2		2	2	
	Hardware-interfacing	3	3	3	3	3

Professional Development BSc 2020

Investigative problem solving	3
Targeted interaction	3
Personal leadership	3
Future oriented organisation	3
Professional development	

58.5. Examination and final grading

58.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

58.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

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Developmental Feedback Scale

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58.5.3. Assessment Guidelines

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- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

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58.6. AI Usage Guidelines

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Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

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Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

58.7. Learning activities

Students build their own individual portfolio. Within this semester, students choose a learning environment with a flexible learning path, and act on a high degree of self-management and learner agency. Students make choices for their own learning path strengthening their individual body of knowledge in the Triple Iteration Project (TriP) and the individual project. They are challenged to explore new knowledge in a methodical and structured manner based on a design challenge (research question) set out at the beginning of each project. Expert talks, open EDU resources, expert consults and coaching (process oriented and professional development oriented) support all of the projects.

During the Industry project students apply and extend their gained knowledge and insights in the more complex context of a project with an external product owner.

Next to the project's focus on quality and applied knowledge the transferability is an explicit aspect of each project.

Blended education

The semester works with blended learning, whereby, in addition to the activities on location, online education can take place that matches content and didactics, or when there are planning and scheduling issues. The basic rule is to have optimal use of teaching time on location. The blended set-up for the semester is explained in the Canvas course. The decision to have parts of the education online is reserved for the degree programme.

58.8. Learning materials

A Canvas course including modules with (self) study materials, inspiration sessions and external sources is available.

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59. INFORMATION ABOUT [AV-NCI] NETWORK & CLOUD ARCHITECTURE INNOVATION (30EC)

Interoperability, Agility & Innovation of Enterprise Architecture, DevSecOps

59.1. Content

The Network and Cloud Architecture Innovation semester helps you deepen your knowledge of designing, managing, and improving IT infrastructures. You'll work on real-world projects, using research-based methods to develop network and cloud solutions. The focus is on architecture, analytics, implementation, and management, preparing you for complex IT environments.

You'll apply structured methodologies to create secure and efficient infrastructures while aligning with industry best practices. Frameworks like TOGAF and ArchiMate will help you structure and visualize your designs. You'll also conduct applied research to ensure your solutions are relevant to industry needs.

Throughout the semester, you'll receive continuous feedback from coaches and peers. Two coaches will regularly check in to support your progress and help you maximize your learning experience.

59.2. Learning outcomes

1. Learning outcome: Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide advice to your stakeholders even in complex and uncertain contexts. You substantiate

and validate future-oriented choices by use of ethical, intercultural, and sustainable arguments.

2. Learning outcome: Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show that you are capable of carrying out these actions and achieving your goals, adjusting them as necessary.

3. Learning outcome: Architecture of an enterprise infrastructure

You can methodically translate the requirements for an enterprise infrastructure into different alternatives for an infrastructure design and justify these alternatives with clear advantages and disadvantages.

Explanation: When drawing up the various alternatives for the infrastructure architecture, the applications, data storage and performance such as speed, capacity, scalability, security, flexibility, and manageability of the various alternatives are accounted for.

4. Learning outcome: Realization of enterprise infrastructure solutions

You can realize an enterprise infrastructure in a methodical way and demonstrate with an extensive test report and demos that the infrastructure meets the specifications drawn up.

Explanation: When realizing the infrastructure, the student justifies the choices made and demonstrates how the infrastructure works with tests (demos) and reports. (Reliability, secure, tested, scalable)

5. Learning outcome: Analytics for enterprise infrastructure

You can collect data from various stakeholders / interest groups in a methodical way to arrive at requirements for the design of an enterprise infrastructure.

Explanation: When preparing requirements for the infrastructure, the application landscape, data storage, accessibility and performance such as speed, capacity, scalability, security (BIV) and management of the infrastructure are considered.

6. Learning outcome: Setting up management for an enterprise Infrastructure

You can in a methodical way the proactive management of an enterprise designing infrastructure, designing, testing and show that with monitoring incidents and events adequate it can handle.

Explanation: When realizing proactive management, you continuously have the status of the infrastructure in view (monitoring & analyses) and you can act quickly and effectively on incidents and events.

7. Learning outcome: Innovations and Trends

You have insight into innovations/trends for the infrastructure and/or the management of the infrastructure and you can research these innovations in a methodical way, report on them and indicate what the innovation/trend will mean for existing enterprise infrastructures.

Explanation: There are constant new developments within the infrastructure. By extensively researching an innovation and/or trends, you show that you master the research techniques and can place the innovations and/or trends in the future.

59.3. Expected entry profile

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction					
Organisational processes	2	1	1	1	2
Infrastructure	2	2	2	2	2
Software	1	1		1	1
Hardware-interfacing					

	Investigative problem solving	2	2	2	2
	Targeted interaction	2	2	2	2
	Personal leadership	2	2	2	2
	Future oriented organisation	2	2	2	2
Professional development		2	2	2	2

59.4. Outflow profile

Domain description HBO-i 2023

	Manage & control				
	Realise				
	Design				
	Advise				
	Analysis				
User interaction					
Organisational processes	2	2	1	2	2
Infrastructure	3	3	3	3	3
Software	1	1		1	1
Hardware-interfacing					

Professional Development BSc 2020

	Investigative problem solving	3	3	3	3
	Targeted interaction	3	3	3	3
	Personal leadership	3	3	3	3
	Future oriented organisation	3	3	3	3
Professional development		3	3	3	3

59.5. Examination and final grading

59.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

59.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- *Undefined*: You have not yet undertaken any activities to demonstrate the learning outcome.
- *Orienting*: You have made a start and explored possibilities to demonstrate the learning outcome.
- *Beginning*: You have taken initial steps that contribute to demonstrating the learning outcome.
- *Proficient*: You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome at a satisfactory level if you continue your development in this way.

- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations and are focused on continuous improvement. You will demonstrate the learning outcome at a more than satisfactory level if you continue your development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

59.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

59.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

59.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

AI is not a replacement or threat, but a collaborative actor in your learning. We use AI to strengthen creativity, deepen understanding, and expand educational possibilities. You are expected to learn to work with AI effectively and productively: a core professional competency for ICT professionals.

The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available, including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

59.7. Learning activities

This semester is structured around agile, self-directed learning, with multiple iterations. You are responsible for planning your projects while receiving structured support through:

- Group Project – Work with industry partners through Partners in Innovation on real network and cloud architecture challenges.
- Individual Project – Choose your own research topic and develop a project that fits your professional interests and learning goals.

You'll conduct research, ensuring your findings are valuable for both stakeholders and industry. You'll also collaborate with classmates, industry professionals, and stakeholders to refine solutions and make an impact. Your progress will be documented in a structured portfolio, showcasing your skills and achievements.

59.8. Learning materials

The Canvas course contains the relevant study materials and references to and ideas about where to look for materials. The students are expected to also look for relevant educational materials on a number of topics themselves.

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60. INFORMATION ABOUT [AV-OLF] OPEN LEARNING (30EC)

60.1. Content

ICT & Personalized Program is a route where you have a lot of freedom and flexibility to determine your personal focus and structure, within the boundaries of the HBO-I competence framework. In practice, this means that anything within these boundaries potentially can be used in ICT & Personalized Program. This gives great opportunities for starting multi-disciplinary projects. Projects should be IT related, based on relevant questions and developments from the professional field and where possible externally validated.

Lots of freedom, but remember:

“Freedom comes with responsibility”

This large degree of freedom does not mean that there are no obligations. A large amount of autonomy and entrepreneurial behaviour is expected.

60.2. Learning outcomes

1. Learning outcome: Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide advice to your stakeholders even in complex and uncertain contexts. You substantiate and validate future-oriented choices by use of ethical, intercultural, and sustainable arguments.

2. Learning outcome: Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show that you are capable of carrying out these actions and achieving your goals, adjusting them as necessary.

60.3. Expected entry profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

60.4. Outflow profile

The output level is based on the achieved growth on the set competence indicators of the students personal competence profile. This should connect and align with the next OE a student is pursuing.

A balanced competence profile does not have too many competencies (no focus) but also not too little (no widening). At least all 9 competences (including professional development) have to be covered on level 3. When this is your first advanced semester you need an addition 4 other competences on at least level 2.

Within Open Learning, students define their own dynamic competence profile per semester (exit level OEx) under supervision. In doing so, they draw up their own criteria for challenges (projects) that contribute to growth within the HBO-i competence framework. A complete competence level is reached by successfully ending a phase. A learning outcome is a partial contribution to a competence step.

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	3	3	3	3

60.5. Examination and final grading

60.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

60.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- *Undefined*: You have not yet undertaken any activities to demonstrate the learning outcome.
- *Orienting*: You have made a start and explored possibilities to demonstrate the learning outcome.
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This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

60.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

60.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

60.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

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Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

60.7. Learning activities

The student will determine the general lines of the content in this programme him or herself. Basically everything in IT should be possible. Because everyone has his own track, it is impossible to plan a curriculum that fits for everyone. Learning happens asynchrone, not everybody does the same things, with the same speed, ... This means all workshops, presentations, etc., that will provide content, will be planned and scheduled on-demand or when the coaches notice it is necessary.

To facilitate this the student will create his/her own personal profile within the HBO-i Competence Framework, this framework gives a systematic overview of the whole IT landscape including their professional skills. This profile will be created shortly after the start of the semester. Before enrollment, the student will be invited for an intake interview. This interview is about motivation and a possible context to use within open learning. In an intake document, it is outlined which topics the student would like to engage. This usually is a very rough outline. For example, "Artificial intelligence", "Encryption" or "extreme interaction". The intake interview will be an evaluation to see whether you really understand what open learning is about, so you know what you're up to.

Because everyone has his own track, it is impossible to plan a curriculum that fits for everyone. Learning happens asynchronous, not everybody does the same things, with the same speed, ... This means all workshops, lessons, .. that will provide content, will be planned and scheduled on-demand or when the teachers notice it is necessary.

For example, workshops of more general applicability are offered to all students, like Scrum-101, Research, Concept validation, while more specific subjects are offered to a smaller group of students.

60.8. Learning materials

Apart of the Canvas LMS environment, anything which would be applicable can and will be used.

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61. INFORMATION ABOUT [AV-OLG] OPEN LEARNING (30EC)

61.1. Content

ICT & Personalized Program or Open Learning is a route where you have a lot of freedom and flexibility to determine your personal focus and structure, within the boundaries of the HBO-I competence framework. In practice, this means that anything within these boundaries potentially can be used in ICT & Personalized Program. This gives great opportunities for starting multi-disciplinary projects. Projects should be IT related, based on relevant questions and developments from the professional field and where possible externally validated.

Lots of freedom, but remember:

“Freedom comes with responsibility”

This large degree of freedom does not mean that there are no obligations. A large amount of autonomy and entrepreneurial behaviour is expected.

61.2. Learning outcomes

1. Learning outcome: Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide advice to your stakeholders even in complex and uncertain contexts. You substantiate and validate future-oriented choices by use of ethical, intercultural, and sustainable arguments.

2. Learning outcome: Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show that you are capable of carrying out these actions and achieving your goals, adjusting them as necessary.

61.3. Expected entry profile

Any finished combination in the Main phase will suffice.

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

61.4. Outflow profile

The output level is based on the achieved growth on the set competence indicators of the students personal competence profile. This should connect and align with the next OE a student is pursuing.

A balanced competence profile does not have too many competencies (no focus) but also not too little (no widening). At least all 9 competences (including professional development) have to be covered on level 3. When this is your first advanced semester you need an addition 4 other competences on at least level 2.

Within Open Learning, students define their own dynamic competence profile per semester (exit level OEx) under supervision. In doing so, they draw up their own criteria for challenges (projects) that contribute to growth within the HBO-i competence framework. A complete competence level is reached by successfully ending a phase. A learning outcome is a partial contribution to a competence step.

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	3	3	3	3

61.5. Examination and final grading

61.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

61.5.2. Feedback: Confirmation and Guidance

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61.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

61.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

61.6. AI Usage Guidelines

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61.7. Learning activities

The student will determine the general lines of the content in this programme him or herself. Basically everything in IT should be possible. Because everyone has his own track, it is impossible to plan a curriculum that fits for everyone. Learning happens asynchrone, not everybody does the same things, with the same speed, ... This means all workshops, presentations, etc., that will provide content, will be planned and scheduled on-demand or when the coaches notice it is necessary.

To facilitate this the student will create his/her own personal profile within the HBO-i Competence Framework, this framework gives a systematic overview of the whole IT landscape including their professional skills. This profile will be created shortly after the start of the semester. Before enrollment, the student will be invited for an intake interview. This interview is about motivation and a possible context to use within open learning. In an intake document, it is outlined which topics the student would like to engage. This usually is a very rough outline. For example, "Artificial intelligence", "Encryption" or "extreme interaction". The intake interview will be an evaluation to see whether you really understand what open learning is about, so you know what you're up to.

Because everyone has his own track, it is impossible to plan a curriculum that fits for everyone. Learning happens asynchronous, not everybody does the same things, with the same speed, ... This means all workshops, lessons, .. that will provide content, will be planned and scheduled on-demand or when the teachers notice it is necessary.

For example, workshops of more general applicability are offered to all students, like Scrum-101, Research, Concept validation, while more specific subjects are offered to a smaller group of students.

61.8. Learning materials

Apart of the Canvas LMS environment, anything which would be applicable can and will be used.

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62. INFORMATION ABOUT [O-AV-OLF] OPEN LEARNING (30EC)

62.1. Content

We have Open Education for PRO from the second year of your study. This means that you will define your own personal learning route. Do you want to become a real ICT & Business analyst? Or are you the next gen Cyber Security, Artificial Intelligence or Game Design specialist? Do you want to specialize in a particular technology? It's up to you!

You will work as much as possible in a context that is most supportive of your personal learning trajectory. This means it is rather impossible to determine a concrete challenge context in advance. This context is facilitated by relevant challenges which are provided by real case stakeholders, that are determined at the beginning of the process.

With this in mind, Education for professionals can be defined as follows:

Educating responsive professionals for all possible careers in IT. This is done by focusing on talent development and personalized competence profiles.

Students who have successfully concluded OE5 can start in OE6. Starting level would be the learning outcomes output of the OE5. The output level is based on the achieved growth on the set competence indicators of the student's personal competence profile. This should connect and align with the next OE a student is pursuing.

The student will determine the general lines of the content in this program him or herself. Basically, everything in IT should be possible. Because everyone has his own track, it is impossible to plan a curriculum that fits for everyone. Learning happens asynchronously, not everybody does the same things, with the same speed, in the same manner, etc. This means all workshops, presentations, etc., that will provide content, will be planned and scheduled on-demand or when the coaches notice it is necessary.

To facilitate this, the student will create his/her own personal profile within the HBO-i Competence Framework, this framework gives a systematic overview of the whole IT landscape including their professional skills. This profile will be created shortly after the start of the semester.

A complete explanation of the structure of the framework and how it functions can be found in the Canvas course.

Blended Learning

At Education for Professionals, we apply the design principles of Blended Learning. This means that we learn with each other online and face-to-face. We have the following principles to concrete this:

- Face to face contact (18:00-21:00)

Every Tuesday evening from 18:00 to 21:00, we learn in groups or individually at FHICT. The following activities could be scheduled:

1. Status updates: Give your group members and coaches insight in your learning process. What have you done last week? Show your plan for the coming weeks? Are there topics you would like to discuss? Etc.
2. Discuss about a topic:

You will zoom in into a topic which is introduced by a student (or coach).

1. Self-work/Collaboration: Students are given the time to work together on a challenge or a smaller task which is related to the discussion topic talk
2. Inspiration sessions / Workshops: Inspiration sessions and workshops could be scheduled.

- Online contact (18:00-19:30)

Of course, students can do their own study activities online in all sorts of ways. For example, the student asks fellow students to discuss deliverables or issues with others. A meeting with the coach or expert could be scheduled. Workshops could be scheduled online as well.

62.2. Learning outcomes

1. Learning outcome: Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide advice to your stakeholders even in complex and uncertain contexts. You substantiate and validate future-oriented choices by use of ethical, intercultural, and sustainable arguments.

Explanation: You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide advice to your stakeholders even in complex and uncertain contexts. You substantiate and validate future-oriented choices by use of ethical, intercultural, and sustainable arguments.

2. Learning outcome: Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show that you are capable of carrying out these actions and achieving your goals, adjusting them as necessary.

Explanation: You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show that you are capable of carrying out these actions and achieving your goals, adjusting them as necessary.

3. Learning outcome: Analysis:

You can obtain desired results by analyzing the processes, products and information flows in a methodical and thorough way.

Explanation: You know what you are researching and why, taking into account the client, target audience & market. Using several justified research methods based on the DOT-framework
Applying multiple different methods of analysis or focusing on several different aspects. For example: client needs, audience needs, other related products, expert knowledge, literature etc.

4. Learning outcome: Advice:

You can give advice about the organization, processes and/or information important for decisions related to your challenge.

Explanation: Your advice is based on the conclusions and results from the analysis.
Your advice can be based on cost aspects and quality properties such as: availability, performance, security and scalability.

5. Learning outcome: Design:

You are able to design (parts of) systems, communicate and validate about it for both functional/technical and

aesthetic requirements using industry standard methods, related to your challenge.

Explanation: With design we mean that you can use a structured and methodical approach in your design. You use a set of concept development methods such as diverging/converging, brainstorming and existing theories as well as structured technical design methods such as UML and RUP.

The design covers both the functional/technical and aesthetic design of your products. You communicate your design in a well presented and structured way to relevant stakeholders.

You validate your designed systems in an early stage using efficient and quick prototyping methods such as paper prototyping and use the results to refine your design.

6. Learning outcome: Realization:

You can implement and validate a product based on a design in a structured way using innovative and/or relevant technology adhering to quality criteria in the context of your challenge

Explanation: We expect that you validate your product and the quality of the product in a structured way using methods such as: User testing, expert reviews, code/peer reviews, (unit) tests.

Structured way means using industry standard development processes such as Scrum/Agile and Kanban.

With innovative and/or relevant technology we mean that you experiment and use upcoming and promising technology that suits your needs.

Your product adheres to previously defined quality criteria in accordance with stakeholders. Many non-functional requirements such as stability, performance, security etc. could be used to define these criteria.

7. Learning outcome: Manage and Control:

You can perform activities that focus on control, monitoring and optimization of the development, commissioning and use of ICT-systems. These activities are related to your challenge.

Explanation: We expect that you build your controlling, monitoring and optimisation of the development in a structured way.

We expect that you can communicate with stakeholders about your activities who are related to manage and control.

62.3. Expected entry profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

62.4. Outflow profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	3	3	3	3

62.5. Examination and final grading

62.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

62.5.2. Feedback: Confirmation and Guidance

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62.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

62.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

AI is not a replacement or threat, but a collaborative actor in your learning. We use AI to strengthen creativity, deepen understanding, and expand educational possibilities. You are expected to learn to work with AI effectively and productively: a core professional competency for ICT professionals.

The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available, including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

62.7. Learning activities

The teachers will provide you with formative feedback on deliverables and demonstrations. You are expected to record the feedback in FeedPulse and it is expected that this feedback will be processed in further deliverables, within reason. Your work will be assessed on all the competences, the learning outcomes (Analysis, Advice, Design, Realisation, Manage and Control, Future oriented organization, Investigative Problem Solving, Personal Leadership and Targeted Interaction). During the semester there are at least three sprint demo's where you can show your progress.

At these demo's the competences are evaluated based on state of the deliverables and your personal development at that point. This will lead to a grade which is a formative indicator of mastery of learning outcomes that moment and don't give any guarantee for further assessments. It is expected that you keep on progressing on the competences, not doing so might lead to a lower grade at the next formative assessment such as beginning or orienting.

62.8. Learning materials

All information about the education program Education for Professionals OE6 could be found in the CANVAS course. Select your own study materials you need during the semester. Everything you need is fine but keep up a good reference list!

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63. INFORMATION ABOUT [O-AV-OLG] OPEN LEARNING (30EC)

63.1. Content

We have Open Education for PRO from the second year of your study. This means that you will define your own personal learning route. Do you want to become a real ICT & Business analyst? Or are you the next gen Cyber Security, Artificial Intelligence or Game Design specialist? Do you want to specialize in a particular technology? It's up to you!

You will work as much as possible in a context that is most supportive of your personal learning trajectory. This means it is rather impossible to determine a concrete challenge context in advance. This context is facilitated by relevant challenges which are provided by real case stakeholders, that are determined at the beginning of the process.

With this in mind, Education for professionals can be defined as follows:

Educating responsive professionals for all possible careers in IT. This is done by focusing on talent development and personalized competence profiles.

Students who have successfully concluded OE6 can start in OE7. Starting level would be the learning outcomes output of the OE6. The output level is based on the achieved growth on the set competence indicators of the student's personal competence profile. This should connect and align with the next OE a student is pursuing.

The output level is based on the achieved growth on the set competence indicators of the student's personal competence profile. At the end of the semester, the student has shown the following rule:

- For all competences (Advising - Analysing - Designing - Realising and Manage & Control) and the PO-skills proficiency level 3 has been demonstrated and in 4 other cells proficiency level 2 has been demonstrated.

The student will determine the general lines of the content in this program him or herself. Basically, everything in IT should be possible. Because everyone has his own track, it is impossible to plan a curriculum that fits for everyone. Learning happens asynchronously, not everybody does the same things, with the same speed, in the same manner, etc. This means all workshops, presentations, etc., that will provide content, will be planned and scheduled on-demand or when the coaches notice it is necessary.

To facilitate this, the student will create his/her own personal profile within the HBO-i Competence Framework, this framework gives a systematic overview of the whole IT landscape including their professional skills. This profile will be created shortly after the start of the semester.

A complete explanation of the structure of the framework and how it functions can be found in the Canvas course

Blended Learning

At Education for Professionals, we apply the design principles of Blended Learning. This means that we learn with each other online and face-to-face. We have the following principles to concrete this:

- Face to face contact (18:00-21:00)

Every Tuesday evening from 18:00 to 21:00, we learn in groups or individually at FHICT. The following activities could be scheduled:

1. Status updates: Give your group members and coaches insight in your learning process. What have you done last week? Show your plan for the coming weeks? Are there topics you would like to discuss? Etc.
2. Discuss about a topic:

You will zoom in into a topic which is introduced by a student (or coach).

1. Self-work/Collaboration: Students are given the time to work together on a challenge or a smaller task which is related to the discussion topic talk
2. Inspiration sessions / Workshops: Inspiration sessions and workshops could be scheduled.

- Online contact (18:00-19:30)

Of course, students can do their own study activities online in all sorts of ways. For example, the student asks fellow students to discuss deliverables or issues with others. A meeting with the coach or expert could be scheduled. Workshops could be scheduled online as well.

63.2. Learning outcomes

1. Learning outcome: Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide advice to your stakeholders even in complex and uncertain contexts. You substantiate and validate future-oriented choices by use of ethical, intercultural, and sustainable arguments.

Explanation: You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide advice to your stakeholders even in complex and uncertain contexts. You substantiate and validate future-oriented choices by use of ethical, intercultural, and sustainable arguments..

2. Learning outcome: Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show that you are capable of carrying out these actions and achieving your goals, adjusting them as necessary.

Explanation: You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show that you are capable of carrying out these actions and achieving your goals, adjusting them as necessary..

3. Learning outcome: Analysis:

You can obtain desired results by analyzing the processes, products and information flows in a methodical and thorough way.

Explanation: You know what you are researching and why, taking into account the client, target audience & market. Using several justified research methods based on the DOT-framework
Applying multiple different methods of analysis or focusing on several different aspects. For example: client needs, audience needs, other related products, expert knowledge, literature etc.

4. Learning outcome: Advice:

You can give advice about the organization, processes and/or information important for decisions related to your challenge.

Explanation: Your advice is based on the conclusions and results from the analysis.
Your advice can be based on cost aspects and quality properties such as: availability, performance, security and scalability.

5. Learning outcome: Design:

You are able to design (parts of) systems, communicate and validate about it for both functional/technical and aesthetic requirements using industry standard methods, related to your challenge.

Explanation: With design we mean that you can use a structured and methodical approach in your design. You use a set of concept development methods such as diverging/converging, brainstorming and existing theories as well as structured technical design methods such as UML and RUP.

The design covers both the functional/technical and aesthetic design of your products. You communicate your design in a well presented and structured way to relevant stakeholders.

You validate your designed systems in an early stage using efficient and quick prototyping methods such as paper prototyping and use the results to refine your design.

6. **Learning outcome:** Realization:

You can implement and validate a product based on a design in a structured way using innovative and/or relevant technology adhering to quality criteria in the context of your challenge.

Explanation: We expect that you validate your product and the quality of the product in a structured way using methods such as: User testing, expert reviews, code/peer reviews, (unit) tests.

Structured way means using industry standard development processes such as Scrum/Agile and Kanban.

With innovative and/or relevant technology we mean that you experiment and use upcoming and promising technology that suits your needs.

Your product adheres to previously defined quality criteria in accordance with stakeholders. Many non-functional requirements such as stability, performance, security etc. could be used to define these criteria.

7. **Learning outcome:** Manage and Control:

You can perform activities that focus on control, monitoring and optimization of the development, commissioning and use of ICT-systems. These activities are related to your challenge.

Explanation: We expect that you build your controlling, monitoring and optimisation of the development in a structured way.

We expect that you can communicate with stakeholders about your activities who are related to manage and control.

63.3. Expected entry profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

63.4. Outflow profile

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	3	3	3	3

63.5. Examination and final grading

63.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your

teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

63.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome at a satisfactory level if you continue your development in this way.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations and are focused on continuous improvement. You will demonstrate the learning outcome at a more than satisfactory level if you continue your development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

63.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

63.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

63.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

AI is not a replacement or threat, but a collaborative actor in your learning. We use AI to strengthen creativity, deepen understanding, and expand educational possibilities. You are expected to learn to work with AI effectively and productively: a core professional competency for ICT professionals.

The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available,

including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

63.7. Learning activities

The teachers will provide you with formative feedback on deliverables and demonstrations. You are expected to record the feedback in FeedPulse and it is expected that this feedback will be processed in further deliverables, within reason. Your work will be assessed on all the competences, the learning outcomes (Analysis, Advice, Design, Realisation, Manage and Control, Future oriented organization, Investigative Problem Solving, Personal Leadership and Targeted Interaction). During the semester there are at least three sprint demo's where you can show your progress.

At these demo's the competences are evaluated based on state of the deliverables and your personal development at that point. This will lead to a grade which is a formative indicator of mastery of learning outcomes that moment and don't give any guarantee for further assessments. It is expected that you keep on progressing on the competences, not doing so might lead to a lower grade at the next formative assessment such as beginning or orienting.

63.8. Learning materials

All information about the education program Education for Professionals OE7 could be found in the CANVAS course. Select your own study materials you need during the semester. Everything you need is fine but keep up a good reference list!

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64. INFORMATION ABOUT [AV-BDA] STRATEGIC INFORMATION & DATA ARCHITECTURES (30EC)

IT Governance, IT Architecture, data and process modelling

64.1. Content

The Advanced Semester Strategic Information & Data Architectures is focused on developing professional skills together with relevant technical skills in Information, Data and Process Management. The student is expected to learn how to assess the needs of the business and to ensure that they are met by integrating appropriate technology components and solutions.

The work of the semester is designed for learning how to work effectively and reliably on real-world projects in collaborative groups. Each student will collaborate with a group of other students to complete project work that will allow them to achieve the learning outcomes of the curriculum, and to demonstrate that they have succeeded in gaining those learning outcomes.

At the start of the semester, projects will be presented by Fontys Partners in Innovation. There may also be projects from other Fontys partners and programmes. These projects are real and current, and the solutions and products arising from them will be used in the business environment of the Partner enterprise. The students will be assigned a selection of these projects to work on in their project groups.

The projects will require the students to first formulate research questions and then use modern research methodologies to develop answers to those questions, and then to show how to formulate plans to implement the solutions and ultimately demonstrate working solutions where possible.

After completion of this semester, students should be able to manage their own work together with a group, and to work together to turn problems into research questions, questions into plans, and plans into solutions.

The course will take place in Strijp TQ, with students being required to be present at specified times during the week. A substantial amount of time will also be spent on self-organised work in the project groups.

In addition to the core project work of the semester, participation in additional events and activities, such as Career Day, the AI & Data event and Innovation Insight, will be scheduled in the course calendar for the semester. There will also be a sequence of workshops and various explorations related to the research subjects and knowledge gaps encountered during the course.

64.2. Learning outcomes

1. Learning outcome: Professional standard

You take responsibility when solving an ICT issue. You define and carry out your research using relevant selected methodologies, and provide advice to your stakeholders even in complex and uncertain contexts. You substantiate and validate future-oriented choices by use of ethical, intercultural, and sustainable arguments.

Explanation: Responsibility: You have the obligation to fulfil tasks, duties, or expectations associated with your particular role or position. You take ownership of taking part in finding a solution for all customer projects that are assigned to your team. You cooperate with teammates, divide tasks and deliver the expected results.

Relevant selected methodologies: You select a methodology depending on the specific goals, context, and nature of the research or project. Select or combine methodologies based on the strengths and limitations of each approach.

Advice to your stakeholders: You consider the context, the nature of the relationship, the type of cooperative partner and the specific needs or challenges the stakeholder is facing while offering your advice.

Complex and uncertain contexts: Identify the hallmarks and roles of the environment of the assignment and know the business legitimization.

Future-oriented choices: You make decisions and take actions that are focused on long-term goals, outcomes, and sustainability while considering the potential impacts on the future rather than just addressing immediate concerns. You do this for the offered solution, but also for your own planning and task division.

2. Learning outcome: Personal leadership

You independently formulate goals and actions that demonstrate leadership in your own long term development as an ICT professional. You show that you are capable of carrying out these actions and achieving your goals, adjusting them as necessary.

Explanation: Formulate goals and actions: You define specific objectives and determine the steps or actions needed to achieve them. You take initiative to plan and log your progress based on the learning outcomes.

Leadership: Refers to the ability to take initiative and responsibility for your own actions.

Long term development as an ICT professional: You focus on continuous learning, skill enhancement, and adaptation to the evolving landscape of IT.

Carrying out these actions: Execute and implement specific tasks or activities to achieve a goal or desired (learning) outcome.

Achieving your goals: Successfully reach the desired (learning) outcomes or milestones that have been set intentionally.

3. Learning outcome: Analysing

You analyse the consequences of strategic changes for organisational processes and information provision, assess the current and future situation, and evaluate acceptance and resistance to new technologies and organisational processes.

Explanation: You show understanding of the, possibly unstructured, strategic and operational issues of IT governance in the enterprise. You analyse and assess the needs of the enterprise by taking those issues into account.

You analyse new or existing organisational processes and current and required data structures, while applying frameworks, rules and principles for good business and IT practices. You support your analysis with data insights.

4. Learning outcome: Advising

You advise on the alignment of business processes and ICT with organisational objectives and how this can be monitored with data and recommend a change management approach for introducing appropriate ICT solutions while considering risk mitigation, sustainability and ethical and legal aspects.

Explanation: You advise on how to develop the enterprise data and IT environment to meet the needs of the enterprise based on known best practices. You determine the aim of an evidence-based solution, and choose an appropriate approach. You advise a solution that is solid and adaptable in the future. You practice continuous awareness of the potential consequences for people, planet, profit and purpose.

5. Learning outcome: Designing

You design an enterprise architecture and an implementation strategy using relevant methods for strategic IT solutions, substantiating architectural choices based on business value, technology trends, and organisational constraints, and translate this into change processes supported by suitable interventions.

Explanation: You are able to apply IT within a given strategy, but also to develop a strategy for business development based on new technological possibilities.

You design new or existing organisational and data processes and structures in a sustainable manner based on frameworks, rules and principles for good business and IT practices.

You remain curious throughout the solution process and ask questions from various perspectives, you match these questions with a fitting approach that is pragmatic, critical and based on sources.

6. Learning outcome: Realising

You realise changes in organisational processes and information provision and actively create support among relevant stakeholders.

Explanation: You drive the implementation of the plans and apply a structured DataOps approach taking into account the current and future enterprise architecture and validate the plans with stakeholders.

For implementation, you apply known models for change management from the IT perspective, as well as from the organisational and human perspective.

7. Learning outcome: Managing & controlling

You formulate and update principles, business rules, and architecture models, monitor strategic goals using data, and proactively identify and initiate changes in organisational processes.

Explanation: You apply (data) governance frameworks with continuous compliance monitoring and policy enforcement mechanisms that enable proactive management driven by data insights which are based on a structured DataOps approach.

You are able to explain your approach methodically and creatively, you find alternatives and critically analyse your own line of reasoning alongside others.

64.3. Expected entry profile

You are able to analyse business processes, execute a gap analysis and gather and document (user)requirements and use cases.

You are able to extract data, do an EDA, clean and transform the data and create insightful visualisations.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	1	1		1	1
Organisational processes	2	2	2	2	2
Infrastructure	1	1			1
Software	1	2	2	1	1
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	2	2	2	2

64.4. Outflow profile

You are capable of working on multiple projects simultaneously. You can advice a company on strategic level. You can research to find the best solution and back this up by creating a prototype. You can support a company to work more data driven and create insights for the management to help them make data driven decisions.

Domain description HBO-i 2023

	Analysis	Advise	Design	Realise	Manage & control
User interaction	2	1	1	1	1
Organisational processes	3	3	3	3	3
Infrastructure	2	1	1		
Software	2	2	1	2	2
Hardware-interfacing					

Professional Development BSc 2020

	Future oriented organisation	Personal leadership	Targeted interaction	Investigative problem solving
Professional development	3	3	3	3

64.5. Examination and final grading

64.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

64.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- *Undefined:* You have not yet undertaken any activities to demonstrate the learning outcome.
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64.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

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64.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

64.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

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Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

64.7. Learning activities

Workshops, explorations, project work, self-study and research, presentations, interviews, presentations to the customer, support for professional task by lecturers / semester coaches, poster sessions, networking events.

There are certain regular events and activities that are part of the FICT semester calendar, such as the AI & Data Event and Innovation Insight, in which all students of the Strategic Information & Data Architectures semester are expected to participate as part of the work of the course.

The semester includes an International Week which is an opportunity for cultural interchange and exchange of ideas. During this week the students of the course either visit another partner institution, or students from that institution visit the course site in Eindhoven. In the latter case, there will normally be the opportunity for students of FICT to pay a return visit to the partner institutions involved.

64.8. Learning materials

Course materials and links will be made available through Canvas, and portflow will be the sole site for submission and assessment of work and assignments (etc).

There will also be a dedicated team in MS Teams for the course, which is to be used for the everyday processes of the course, including project team communication and announcements and advisories, among others.

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65. INFORMATION ABOUT [GA] GLOBAL ACTING IN IT (30EC)

65.1. Content

Globalization and digitalization kickstarted a revolution, changing everything from how we live to how we work. Students are not just witnessing this history; they have the power to (re)shape the future. In our program, students will visit five different universities and team up with peers from around the world to dive into a semester long project with real world impact. Guided by the UN Sustainable Development Goals, they will explore how future technology can shape a better tomorrow.

The three keystones of the program are:

- Intercultural awareness and practice
- International collaboration and experience
- Examining IT from a global perspective, with focus on having a positive impact on society, the environment and beyond

The objectives of the minor are:

- Become a responsive ICT professional: adaptive, resilient, innovative and responsible
- Global citizenship & intercultural awareness
- Communication and collaboration across borders
- You can study and work on real life projects in an international environment

The IT topics taught by the 5 partner institutes during this minor are:

- Data Science
- UX Design
- Internet of Things
- Artificial Intelligence
- Immersive Technology

65.2. Learning outcomes

1. Learning outcome: Professional duties in international context

You execute professional IT duties within an international, multicultural environment, delivering professional products that address global challenges. Working across five key IT domains (Immersive Technology, UX Design, IoT, AI, and Data Science), you analyze, design, realize, advise and manage solutions for an international project. Your professional products demonstrate technical competency while considering international perspectives, cross border implications and sustainable impact on society and environment.

2. Learning outcome: Situation orientation

You apply your previously acquired IT knowledge and skills in an authentic international context, working collaboratively with students from international universities. Through the project, you adapt your methodological approach to diverse cultural working styles and organizational processes across five universities. Your contributions remain relevant to international stakeholder(s) while navigating different educational systems, technological infrastructures and cultural communication patterns.

3. Learning outcome: Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to formulate project goals, involve stakeholders, conduct applied research*, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

4. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

Explanation: You actively develop intercultural competencies using theoretical frameworks and reflecting on your growth through methodological approaches. You continuously assess your learning ability across diverse international settings, seeking feedback from peers and clearly articulating your vision for becoming an internationally minded IT professional capable of working in global, cross cultural environments.

65.3. Expected entry profile

Students entering the Global Acting in IT minor should possess HBO-i Level 2 competencies within their chosen IT discipline (software development, cybersecurity, business, interaction design, etc.). This means that they already can work independently on defined IT problems, apply standard methodologies and tools within their field and collaborate in team settings. They should be comfortable with fundamental technical concepts, able to analyze requirements and design basic solutions.

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

65.4. Outflow profile

Through the international minor experience, students will advance to HBO-i Level 3 within their own IT discipline, where they can handle complex, multifaceted IT challenges that require integration across multiple domains (Immersive Technology, UX Design, IoT, AI, Data Science). At this level, students will demonstrate the ability to work autonomously in uncertain, cross cultural contexts, lead international project initiatives, conduct independent research using diverse methodologies and make strategic decisions that consider global implications.

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

65.5. Examination and final grading

65.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

The learning outcomes describe the desired level, but because these can be demonstrated independently of the learning path, there is room to shape part of your own learning process. That's why it's important to frequently engage your teachers to ensure you're doing the right things. This is the longitudinal aspect: throughout the entire semester, the assessors monitor your progress.

65.5.2. Feedback: Confirmation and Guidance

Teachers primarily guide and assess you through feedback. You will have weekly conversations with one or more teachers about your work. You record this feedback yourself in FeedPulse, after which teachers respond with information about your progress.

In addition, portfolio submissions take place at set moments during the semester. You receive feedback on these from your teachers based on the developmental feedback scale. This scale has five levels.

Developmental Feedback Scale

- **Undefined:** You have not yet undertaken any activities to demonstrate the learning outcome.
- **Orienting:** You have made a start and explored possibilities to demonstrate the learning outcome.
- **Beginning:** You have taken initial steps that contribute to demonstrating the learning outcome.
- **Proficient:** You have repeatedly shown that you've created a foundation to demonstrate the learning outcome. You will demonstrate the learning outcome at a satisfactory level if you continue your development in this way.
- **Advanced:** You have repeatedly shown that you've been working on this learning outcome with good results. You have performed above expectations and are focused on continuous improvement. You will demonstrate the learning outcome at a more than satisfactory level if you continue your development in this way.

This feedback is a snapshot of your current growth toward the learning outcomes. It's possible that you may go down in your development if you don't consistently demonstrate certain behaviors in practice.

Within the semester, there are at least three of these formative assessment moments where assessors provide feedback per learning outcome using the developmental feedback scale. The last of these three formative assessment moments is

no later than two weeks before the assessors' meeting. This gives you the opportunity to address and process the feedback received.

65.5.3. Assessment Guidelines

In all cases, the assessors decide on the assessment by mutual agreement. The only fixed rule for assessment is:

- If a student has achieved at least 'Proficient' for all learning outcomes, the assessment is a 'Satisfactory' or better. All other cases are at the discretion of the assessors' meeting.

This rule gives you the ability to focus on the most important areas for improvement. It should not be interpreted as the minimum requirement to achieve a Satisfactory assessment.

65.5.4. Retakes and repair

Retakes or remediation do not apply to this semester. You are expected to resubmit work for feedback an unlimited number of times.

65.6. AI Usage Guidelines

At Fontys ICT we have a vision on the use of Gen AI (AI), which can be found in the AI Manifesto. Herein we establish expectations for how you engage with AI throughout your learning journey.

AI as a Collaborative Partner

AI is not a replacement or threat, but a collaborative actor in your learning. We use AI to strengthen creativity, deepen understanding, and expand educational possibilities. You are expected to learn to work with AI effectively and productively: a core professional competency for ICT professionals.

The core principle: You are the architect, not the tool. AI accelerates your work, but you remain accountable for everything you deliver. When using AI, apply the transparency principle: be clear and honest about what AI contributed versus what represents your own analysis, judgment, or creation.

Validation and Critical Engagement

Working with AI means constantly evaluating outputs with a critical, curious mindset. AI can hallucinate facts, produce insecure code, or embed biases. Developing the ability to evaluate, question, and verify is essential. Your decisions matter so make sure you can explain them. For significant AI contributions, document your validation process

Ethical Boundaries

Responsible AI use requires respecting clear boundaries. Never input personal data, company secrets, or confidential information into public AI tools. Basic AVG/GDPR guidelines always apply. Use institutional AI tools when available, including the facilities Fontys ICT provides. Respect intellectual property: understand what licenses apply to AI-generated content and ensure compliance.

Your AI practices reflect not just on you, but on Fontys ICT and any professional partners you work with.

65.7. Learning activities

This minor consists of the following activities:

- A start-up week with introductions and start-up on:
 - Intercultural development & their professional development portfolio (PDP)
 - The overarching project. This project is related to the UN sustainable development goals. Students with work on this project throughout the semester in multicultural, multidisciplinary teams.
- Five 3-week modules organised by the 5 partner institutes. The students learn about the following IT topics, apply this to their project and keep track of their (intercultural) development in their PDP:
 - Data Science (Belgium Campus iTversity, South-Africa)
 - Ux Design (Universitat de Lleida, Spain)
 - Internet of Things (University of applied sciences Upper Austria)
 - Artificial Intelligence (PXL, Belgium)
 - Immersive Technology (FICT, Netherlands)
- Two weeks for a repair assignment if applicable.

65.8. Learning materials

Canvas is used as the central location for all learning materials. Each institute can use this to share any relevant learning materials with the students.

66. INFORMATION ABOUT [SIT] SMART INDUSTRY TECHNOLOGIES (30EC)

HBO-ICT Minor

66.1. Content

The overall goal of Smart Industry is to provide solutions so that the production, products and service systems become smart and interconnected, flexible and adjustable in real time to meet highly specific end-user demands. It is emerging by integrating new technical developments in IT, such as Internet of Things (IoT) with network technologies (5G), cloud/edge computing, data analytics and artificial intelligence (AI), smart supply chains, with manufacturing and product OT technologies. Enriched by these IT technologies the production, products and service systems become smart and interconnected, flexible and adjustable on the fly to meet highly specific end-user demands. It is not surprising that manufacturers and organisations are making radical changes into this direction. This on the other hand, requires for more professionals who can operate on the boundaries of engineering and ICT.

The semester focuses on performing authentic, methodically structured, professional tasks, which contribute to new insights for the professional field. The focus is on learning, by carrying out research and development activities, within practice-oriented research. Students are (mainly) responsible for setting up and monitoring their own learning path. The learning process will be supported by workshops, contacts with experts and coaching in various forms and whenever needed.

The research projects of this minor are based on real life problems from our company partners and from the lectorates. Students will work on one of these research projects, and on projects of their own. The variety of projects allows students to choose their development path—either as **generalists**, by broadening their knowledge and learning how new technologies are integrated into Smart Industry, or as **specialists**, by deepening their expertise and applying it to the Smart Industry domain.

The **Smart Industry Technologies Minor** aims at students from Fontys ICT and Fontys Engineering who are interested in enriching own knowledge, skills and experience in new technologies and integration of those. The most obvious attendants for this minor are students from any profile at ICT and students from Mechatronics at Engineering. But besides those, this minor explicitly invites students from other profiles at the Engineering department to attend this minor. As such, this minor is a joint effort of the ICT and Engineering departments at Fontys. This cross-disciplinary, ICT & Engineering learning environment makes this minor unique among other Smart Industry study programs in the Netherlands.

The minor emphasizes hands-on learning through real-world projects developed in collaboration with industry partners. These projects involve vertical system integration, covering a range of levels from field and control systems to production and operational processes. This approach provides students with a complete understanding of how technologies come together in real-world industrial environments.

To support these projects, students will have access to a range of resources, including specialized workshops, expert sessions, practical exercises, and coaching, as well as field lab equipment. These resources will be provided by both the ICT and Engineering departments, as well as by partner companies or other ICT/Engineering units, such as: Smart Industry and IoT Specialisation (ICT), Minor Adaptive Robotics (Engineering), and Technology Advanced semester (ICT), depending on the needs. The support will be tailored to meet the needs of each project, ensuring students receive the guidance necessary to apply theoretical knowledge to practical research challenges.

The support includes but is not limited to:

1. Dedicated workshops to cover the basics of Network, Security, Data Analytics, System Engineering, Industrial Automation.
2. Dedicated Advanced workshops on IoT, Infrastructure, Vision, AI, Digital Twin, Industrial Automation, Security.
3. Workshops/ expert sessions on demand from students in the range of the above topics, and broader on Robotics, Additive manufacturing, Embedded programming.

66.2. Learning outcomes

1. **Learning outcome:** Professional standard

Both individually and in teams, you apply a relevant methodological approach used in the professional field to

formulate project goals, involve stakeholders, conduct applied research*, provide advice, make decisions, and deliver reports. In doing so, you keep in view the relevant ethical, intercultural, and sustainable aspects.

2. Learning outcome: Personal leadership

You are aware of your own strengths and weaknesses, both in the field of ICT and in your personal development. You choose actions in line with your core values to promote your personal growth and develop your learning attitude.

3. Learning outcome: Smart Industry proficiency

The student is able to define and explain the core concepts, principles, and technologies underlying smart industry. The student is able to analyse the situation, can identify new technological opportunities, and develop adequate solutions that utilizes relevant tools commonly used in smart industry applications, and reflect on the impact of these solutions on the productivity and efficiency.

4. Learning outcome: Smart Industrial Integrated solutions

The student is able to design and implement solutions in which he/she integrates various technologies relevant in Smart Industry, such as:

- o Monitoring and controlling smart industrial processes and systems
- o Configuring and programming sensors, actuators, and controllers, as well as network and infrastructure, to enable communication and interoperability within smart systems
- o Collecting relevant data and utilizing them for decision making regarding improvements and optimizations
- o Addressing safety and security risks, and implementing best practices to safeguard data networks, and devices against threats and unauthorized access.

66.3. Expected entry profile

The admission and outflow profiles are somewhat loosened, due to the diverse admission group we are targeting. Students from the Institute of Engineering and ICT can directly participate in the minor under the condition that they have completed the 1st and 2nd years of own study program.

We expect students to have competences from the HBO-i domain description on level 1 (Analyse, Advice, Design, Realisation, Control & Manage) as well as the professional development competences (Future Oriented Organisation, Personal Leadership, Targeted Interaction and Investigative Problem Solving) on HBO-i level 1.

In the HBO-Engineering domain description there are equivalent competences: Analyse, Design, Realize, Control, and Manage, Advise, Research and Professionalize, divided in Technical competences and General Higher Professional Education competences. They map holistically one-to-one to the above given HBO-I competences and we expect students coming from the engineering domain to have these competences on level 1 of the HBO Engineering domain description.

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	1	1	1	1

66.4. Outflow profile

Domain description HBO-i 2023

	Manage & control	Realise	Design	Advise	Analysis
User interaction	1	1			
Organisational processes	1	1			
Infrastructure	2				2

Software	1		1	2	
Hardware-interfacing	2	2	2	2	2

Professional Development BSc 2020

	Investigative problem solving	Targeted interaction	Personal leadership	Future oriented organisation
Professional development	2	2	2	2

66.5. Examination and final grading

66.5.1. Assessment: Holistic and Longitudinal

Assessment in this semester is holistic and longitudinal. Holistic means that everything you do as a student counts toward the final grade: the amount of work, the quality of the work, your professionalism, and the demonstrated competencies.

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66.7. Learning activities

Authentic questions, challenges and problems from professional practice are the basis for all learning activities, and different smaller steps learning tasks will be defined to facilitate the learning process. Coaching and guidance gradually change from teacher-driven to a self-regulating student-driven approach during the semester. Coaching and guidance will be offered whenever needed and relevant. Learning content is offered when required or requested in a form of instructions, workshops, or feedback session. This way, differentiation in style, speed and level is possible. It is expected that students prepared themselves for instruction sessions with teachers or experts by studying the provided study material. During these session focus will be on clarifying the details of the theory and on discussing the details about applying the knowledge in a relevant practical context.

Learning happens asynchronously for these reasons: not every student does the same things, and the pace and entry level of students also differs. The result of this is that a significant number of workshops, lectures, and so, that offer in-depth content are scheduled on- demand or when coaches detect it is necessary. The support includes but is not limited to:

1. Dedicated workshops, mostly planned towards the beginning of the semester, that cover the basics of Network, Security, Data Analytics, System Engineering, Industrial Automation.
2. Dedicated Advanced workshops on IoT, Infrastructure, Vision, AI, Digital Twin, Industrial Automation, Security.
3. Workshops/ expert sessions on demand from students in the range of the above topics, and broader on Robotics, Additive manufacturing, Embedded programming.

The support will be tailored to meet the needs of each project, ensuring students receive the guidance necessary to apply theoretical knowledge to practical research challenges.

Blended education

The semester works with blended learning, whereby, in addition to the activities on location, online education can take place that matches content and didactics, or when there are planning and scheduling issues. The basic rule is to have optimal use of teaching time on location. The blended set-up for the semester is explained in the Canvas course. The decision to have parts of the education online is reserved for the degree programme.

66.8. Learning materials

A Canvas course including modules with (self) study materials, inspiration sessions and external sources is available.

